



UNIVERSIDADE FEDERAL DE SERGIPE  
CENTRO DE CIÊNCIAS EXATAS E TECNOLOGIA  
PROGRAMA DE PÓS-GRADUAÇÃO EM CIÊNCIA DA COMPUTAÇÃO

## **A Framework for Inverse Modeling Applied to Multi-objective Evolutionary Algorithms**

Dissertação de Mestrado

Artur Leandro da Costa Oliveira



São Cristóvão – Sergipe

2022

UNIVERSIDADE FEDERAL DE SERGIPE  
CENTRO DE CIÊNCIAS EXATAS E TECNOLOGIA  
PROGRAMA DE PÓS-GRADUAÇÃO EM CIÊNCIA DA COMPUTAÇÃO

Artur Leandro da Costa Oliveira

## **A Framework for Inverse Modeling Applied to Multi-objective Evolutionary Algorithms**

Dissertação de Mestrado apresentada ao Programa de Pós-Graduação em Ciência da Computação da Universidade Federal de Sergipe como requisito parcial para a obtenção do título de mestre em Ciência da Computação.

Orientador(a): André Britto de Carvalho  
Coorientador(a): Renê Pereira de Gusmão

São Cristóvão – Sergipe

2022

FICHA CATALOGRÉFICA ELABORADA PELA BIBLIOTECA CENTRAL  
UNIVERSIDADE FEDERAL DE SERGIPE

Oliveira, Artur Leandro da Costa  
O48f A framework for inverse modeling applied to multi-objective evolutionary algorithms / Artur Leandro da Costa Oliveira ; orientador André Brito de Carvalho. - São Cristóvão, 2022.  
142 f.; il.

Dissertação (mestrado em Ciência da Computação) - Universidade Federal de Sergipe, 2022.

1. Computação. 2. Aprendizado do computador. 3. UML (Computação). I. Carvalho, André Brito de orient. II. Título.

CDU 004



UNIVERSIDADE FEDERAL DE SERGIPE  
PRÓ-REITORIA DE PÓS-GRADUAÇÃO E PESQUISA  
COORDENAÇÃO DE PÓS-GRADUAÇÃO  
PROGRAMA DE PÓS-GRADUAÇÃO EM CIÊNCIA DA COMPUTAÇÃO

**Ata da Sessão Solene de Defesa da Dissertação do  
Curso de Mestrado em Ciência da Computação-UFS.  
Candidato: ARTUR LEANDRO DA COSTA OLIVEIRA**

Em 08 dias do mês de junho do ano de dois mil e vinte dois, com início às 14h00min, realizou-se na Sala de Reunião do PROCC da Universidade Federal de Sergipe, na Cidade Universitária Prof. José Aloísio de Campos, a Sessão Pública de Defesa de Dissertação de Mestrado do candidato **Artur Leandro da Costa Oliveira**, que desenvolveu o trabalho intitulado: **“A Framework for Inverse Modeling Applied to Multi-objective Evolutionary Algorithms”**, sob a orientação do Prof. Dr. **André Britto de Carvalho**. A Sessão foi presidida pelo Prof. Dr. **André Britto de Carvalho** (PROCC/UFS), que após a apresentação da dissertação passou a palavra aos outros membros da Banca Examinadora, Prof. Dr. **Renato Tinos** (USP) e, em seguida, o Prof. Dr. **Leonardo Nogueira Matos** (PROCC/UFS) e o Prof. Dr. **Renê Pereira de Gusmão** (PROCC/UFS). Após as discussões, a Banca Examinadora reuniu-se e considerou o mestrando (a) aprovado “(aprovado/reprovado)”. Atendidas as exigências da Instrução Normativa 05/2019/PROCC, do Regimento Interno do PROCC (Resolução 67/2014/CONEPE), e da Resolução nº 04/2021/CONEPE que regulamentam a Apresentação e Defesa de Dissertação, e nada mais havendo a tratar, a Banca Examinadora elaborou esta Ata que será assinada pelos seus membros e pelo mestrando.

Cidade Universitária “Prof. José Aloísio de Campos”, 08 de junho de 2022.

Documento assinado digitalmente  
 André Britto de Carvalho  
Data: 11/06/2022 18:22:32-0300  
Verifique em <https://verificador.iti.br>

**Prof. Dr. André Britto de Carvalho  
(PROCC/UFS)  
Presidente**

  
**Prof. Dr. Renato Tinos  
(USP)  
Examinador Externo**

Documento assinado digitalmente  
 Rene Pereira de Gusmao  
Data: 10/06/2022 20:13:56-0300  
Verifique em <https://verificador.iti.br>

**Prof. Dr. Renê Pereira de Gusmão  
(PROCC/UFS)  
Examinador Interno**

Documento assinado digitalmente  
 Leonardo Nogueira Matos  
Data: 09/06/2022 15:41:49-0300  
Verifique em <https://verificador.iti.br>

**Prof. Dr. Leonardo Nogueira Matos  
(PROCC/UFS)  
Examinador Interno**

Documento assinado digitalmente  
 ARTUR LEANDRO DA COSTA OLIVEIRA  
Data: 12/06/2022 09:04:02-0300  
Verifique em <https://verificador.iti.br>

**Artur Leandro da Costa Oliveira  
Candidato**

*This thesis is dedicated to my family and friends.  
For their endless love, support and encouragement.*

# Acknowledgements

I have taken efforts in this project. However, it would not have been possible without the kind support and help of many individuals. I would like to extend my sincere thanks to all of them.

First, I am indebted to my family, whose value to me only grows with age. Also, to my girlfriend Jéssica, that supports me with love and encouragement to the completion of this work.

To all my friends and the co-workers at MPSE for their kind cooperation which willingly helped me out with their abilities. A special thanks to João, Fernando, Italo, Denisson, and Marcos, who participated directly or indirectly in this project.

Finally, I would like to express my deep gratitude to Professor André Britto and Professor Renê Gusmão, for their guidance and constant supervision as well as for providing the necessary information and support in producing this master's thesis.

# Abstract

Many-Objective Optimization Problems (MaOPs) are a class of complex optimization problems defined by having more than three objective functions. Traditional Multi-Objective Evolutionary Algorithms (MOEAs) have shown poor scalability in solving this kind of problem. The use of machine learning techniques to enhance optimization algorithms applied to MaOPs has been drawing attention due to their capacity to add domain knowledge during the search process. One method of this kind is inverse modeling, which uses machine learning models to enhance MOEAs differently, mapping the objective function values to the decision variables. This method has shown a good performance in diverse optimization problems due to the ability to directly predict solutions closed to the Pareto-optimal front, among these methods, we can highlight the Decision Variable Learning (DVL). The strategies involving inverse models found, including the DVL, have some limitations such as the exploration of the performance of different machine learning models and the strategies in using the generated knowledge during the search. The main goal of this work is to create a framework that uses an inverse modeling approach coupled to any MOEA found in the literature. More precisely, three main steps were taken to achieve the goals. First, we perform a systematic review of the literature to identify the main uses of machine learning techniques enhancing optimization algorithms. Secondly, we analyze the performance of different machine learning methods in the DVL, seeking to understand the main characteristics of inverse modeling through the DVL algorithm. In the last step, we propose a framework that is an extension of the DVL algorithm, based on the knowledge obtained in the systematic review and our analysis of the DVL. This framework results in an algorithm for MaOPs recommended for situations that exist restrictions on the number of evaluations in the objective function.

**Keywords:** multi-objective optimization, machine learning, inverse models.

# List of Figures

|                                                                                                                                                                                                                                   |    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Figure 1 – Classification of works combining metaheuristics and machine learning. . .                                                                                                                                             | 26 |
| Figure 2 – Classification of enhancing metaheuristics with machine learning. . . . .                                                                                                                                              | 27 |
| Figure 3 – Evolutionary Computation taxonomy. . . . .                                                                                                                                                                             | 28 |
| Figure 4 – Comparison between Metamodeling methods in EAs and the Pareto Estimation (PE) method. . . . .                                                                                                                          | 30 |
| Figure 5 – Systematic review conduction. . . . .                                                                                                                                                                                  | 35 |
| Figure 6 – Countries that have papers published about the research field. . . . .                                                                                                                                                 | 37 |
| Figure 7 – Authors with at least three papers published in the research field. . . . .                                                                                                                                            | 38 |
| Figure 8 – Main venues where papers are published in the research field. . . . .                                                                                                                                                  | 39 |
| Figure 9 – Number of papers published per year. . . . .                                                                                                                                                                           | 40 |
| Figure 10 – Number of papers for each classification. (a) Time of hybridization (b) Aim of hybridization and (c) Involved Component. . . . .                                                                                      | 41 |
| Figure 11 – Main models and metaheuristics used in at least three papers. . . . .                                                                                                                                                 | 49 |
| Figure 12 – Metaheuristics used for comparison with at least three papers. . . . .                                                                                                                                                | 53 |
| Figure 13 – Metrics used for validation with at least three papers. . . . .                                                                                                                                                       | 54 |
| Figure 14 – Regression experiment workflow. . . . .                                                                                                                                                                               | 69 |
| Figure 15 – Visual comparisons for the DTLZ1 benchmark. (a) and (b) are the objectives from the predicted decision variables by the SVR model for 3 objectives and the RFR model for 10 objectives respectively. . . . .          | 71 |
| Figure 16 – Visual comparisons for the DTLZ2 benchmark. (a) and (b) are the objectives from the predicted decision variables by the SVR model for 3 objectives and the RFR model for 10 objectives respectively. . . . .          | 71 |
| Figure 17 – Visual comparisons for the DTLZ3 benchmark. (a) and (b) are the objectives from the predicted decision variables by the SVR model for 3 objectives and the RFR model for 10 objectives respectively. . . . .          | 72 |
| Figure 18 – Visual comparisons for the DTLZ4 benchmark. (a) and (b) are the objectives from the predicted decision variables by the RFR model for 3 and 10 objectives respectively. . . . .                                       | 72 |
| Figure 19 – Visual comparisons for the DTLZ5 benchmark. (a) and (b) are the objectives from the predicted decision variables by the SVR model for 3 objectives and the RFR model for 10 objectives respectively. . . . .          | 72 |
| Figure 20 – Visual comparisons for the DTLZ6 benchmark. (a) and (b) are the objectives from the predicted decision variables by the MO-MLP model for 3 objectives and the MO-Linear model for 10 objectives respectively. . . . . | 73 |



|                                                                                                                                                                                                   |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Figure 21 – Visual comparisons for the DTLZ7 benchmark. (a) and (b) are the objectives from the predicted decision variables by the MO-Linear model for 3 and 10 objectives respectively. . . . . | 73  |
| Figure 22 – DVL Framework. . . . .                                                                                                                                                                | 79  |
| Figure 23 – Vector of the phases of all intersections. . . . .                                                                                                                                    | 84  |
| Figure 24 – Impact of the DVL Framework over the MOEAs for 3 objectives. . . . .                                                                                                                  | 88  |
| Figure 25 – Impact of the DVL Framework over the MOEAs for 10 objectives. . . . .                                                                                                                 | 89  |
| Figure 26 – Traffic light synchronization scenario. . . . .                                                                                                                                       | 93  |
| Figure 27 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration. . . . .                                                            | 95  |
| Figure 28 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ1 with 3 objectives. . .                                 | 116 |
| Figure 29 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ2 with 3 objectives. . .                                 | 117 |
| Figure 30 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ3 with 3 objectives. . .                                 | 118 |
| Figure 31 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ4 with 3 objectives. . .                                 | 119 |
| Figure 32 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ5 with 3 objectives. . .                                 | 120 |
| Figure 33 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ6 with 3 objectives. . .                                 | 121 |
| Figure 34 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ7 with 3 objectives. . .                                 | 122 |
| Figure 35 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ1 with 10 objectives. . .                                | 123 |
| Figure 36 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ2 with 10 objectives. . .                                | 124 |
| Figure 37 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ3 with 10 objectives. . .                                | 125 |
| Figure 38 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ4 with 10 objectives. . .                                | 126 |
| Figure 39 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ5 with 10 objectives. . .                                | 127 |
| Figure 40 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ6 with 10 objectives. . .                                | 128 |
| Figure 41 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ7 with 10 objectives. . .                                | 129 |

|                                                                                                                                                                    |     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Figure 42 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ1 with 3 objectives. . .  | 131 |
| Figure 43 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ2 with 3 objectives. . .  | 132 |
| Figure 44 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ3 with 3 objectives. . .  | 133 |
| Figure 45 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ4 with 3 objectives. . .  | 134 |
| Figure 46 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ5 with 3 objectives. . .  | 135 |
| Figure 47 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ6 with 3 objectives. . .  | 136 |
| Figure 48 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ7 with 3 objectives. . .  | 137 |
| Figure 49 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ1 with 10 objectives. . . | 138 |
| Figure 50 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ2 with 10 objectives. . . | 139 |
| Figure 51 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ3 with 10 objectives. . . | 140 |
| Figure 52 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ4 with 10 objectives. . . | 141 |
| Figure 53 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ5 with 10 objectives. . . | 142 |
| Figure 54 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ6 with 10 objectives. . . | 143 |
| Figure 55 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ7 with 10 objectives. . . | 144 |

# List of Tables

|                                                                                                                                            |    |
|--------------------------------------------------------------------------------------------------------------------------------------------|----|
| Table 1 – The generic search string. . . . .                                                                                               | 33 |
| Table 2 – Inclusion and exclusion criteria. . . . .                                                                                        | 34 |
| Table 3 – Advanced search. . . . .                                                                                                         | 34 |
| Table 4 – Number of papers returned by each database. . . . .                                                                              | 35 |
| Table 5 – Number of papers based on the classification of Time of hybridization. . . . .                                                   | 42 |
| Table 6 – Number of papers based on the classification of Aim of hybridization. . . . .                                                    | 43 |
| Table 7 – Number of papers based on the classification of Involved Component. . . . .                                                      | 46 |
| Table 8 – Number of papers based on the classification of Machine Learning. . . . .                                                        | 48 |
| Table 9 – Papers classified by the use of benchmark functions. . . . .                                                                     | 52 |
| Table 10 – Machine learning models used in the experiment and each parameter configuration. . . . .                                        | 67 |
| Table 11 – Statistical results (mean and standard deviation) of the MSE values for 3 and 10 objectives. . . . .                            | 70 |
| Table 12 – Reference points used for each DTLZ problem. . . . .                                                                            | 74 |
| Table 13 – DVL parameter configuration for the optimization experiment. . . . .                                                            | 74 |
| Table 14 – Statistical results (mean and standard deviation) of the Hypervolume values for 3 objectives. . . . .                           | 75 |
| Table 15 – Statistical results (mean and standard deviation) of the Hypervolume values for 10 objectives. . . . .                          | 76 |
| Table 16 – Wins/Ties/Loses of the Statistical Comparison. . . . .                                                                          | 77 |
| Table 17 – Reference points used for each DTLZ problem. . . . .                                                                            | 84 |
| Table 18 – DVL parameter configuration for the optimization experiment. . . . .                                                            | 85 |
| Table 19 – Statistical results (mean and standard deviation) of the Hypervolume values for 3 objectives. . . . .                           | 90 |
| Table 20 – Statistical results (mean and standard deviation) of the Hypervolume values for 10 objectives. . . . .                          | 91 |
| Table 21 – Mean number of evaluations of each algorithm of the Traffic Lights Signaling Experiment. . . . .                                | 94 |
| Table 22 – Statistical results (mean and standard deviation) of the Hypervolume values of the Traffic Lights Signaling Experiment. . . . . | 95 |

# List of abbreviations and acronyms

|         |                                                                          |
|---------|--------------------------------------------------------------------------|
| MOP     | Multi-Objective Problem                                                  |
| MOEA    | Multi-Objective Evolutionary Algorithm                                   |
| GA      | Genetic Algorithm                                                        |
| PSO     | Particle Swarm Optimization                                              |
| EA      | Evolutionary algorithm                                                   |
| MaOP    | Many-Objective Optimization Problem                                      |
| EDA     | Estimation of Distribution Algorithm                                     |
| SAEA    | Surrogate-Assisted Evolutionary Algorithm                                |
| DM      | Decision-Maker                                                           |
| DVL     | Decision Variable Learning                                               |
| NSGA    | Non-dominated Sorting Genetic Algorithm                                  |
| PS      | Pareto-optimal set                                                       |
| PF      | Pareto Front                                                             |
| GRASP   | Greedy Randomized Adaptive Search Procedure                              |
| ML      | Machine Learning                                                         |
| SVM     | Support Vector Machine                                                   |
| PCA     | Principal Component Analysis                                             |
| MDP     | Markov Decision Process                                                  |
| ANN     | Artificial neural network                                                |
| EC      | Evolutionary Computation                                                 |
| LS      | Local Search                                                             |
| PE      | Pareto estimation                                                        |
| RBFNN   | Radial basis function neural network                                     |
| MAEA-gD | Many-objective evolutionary algorithm based on generalized decomposition |

|         |                                                                                                         |
|---------|---------------------------------------------------------------------------------------------------------|
| IM-MOEA | Inverse modeling-based Multi-Objective Evolutionary Algorithm                                           |
| LHS     | Latin Hypercubic Sampling                                                                               |
| RP      | Reference points                                                                                        |
| GECCO   | Genetic Evolutionary Computation Conference                                                             |
| CEC     | Congress on Evolutionary Computation                                                                    |
| ICML    | International Conference on Machine Learning                                                            |
| BRACIS  | Brazilian Conference on Intelligent Systems                                                             |
| MLP     | Multilayer perceptron                                                                                   |
| PBIL    | Population-based incremental learning                                                                   |
| GOMEA   | Gene-pool Optimal Mixing Evolutionary Algorithm                                                         |
| LT      | Linkage Tree                                                                                            |
| CMA-ES  | Covariance matrix adaptation evolution strategy                                                         |
| BCIs    | Brain Computer Interfaces                                                                               |
| MGPMDE  | Multi-fidelity Gaussian process and radial basis function-model assisted memetic differential evolution |
| MFO     | Multifidelity/Multilevel/Variable-fidelity optimization                                                 |
| OBL     | Opposition-based Learning                                                                               |
| SCA     | Sine cosine algorithm                                                                                   |
| DE      | Diferential Evolution                                                                                   |
| HV      | Hypervolume                                                                                             |
| IGD     | Inverted Generational Distance                                                                          |
| DT      | Decision Tree                                                                                           |
| CSSL    | Cotraining algorithm for semi-supervised learning                                                       |
| ZDT     | Zitzler, Deb and Thiele                                                                                 |
| DTLZ    | Deb, Thiele, Laumanns and Zitzler                                                                       |
| CF      | Constrained Functions                                                                                   |

|        |                                                              |
|--------|--------------------------------------------------------------|
| WFG    | Walking Fish Group                                           |
| FJSSP  | Flexible job-shop scheduling problem                         |
| PFSP   | Permutation flowshop problem                                 |
| CMT    | Christofides, Mingozzi, and Toth                             |
| CVRP   | Capacitated Vehicle Routing Problem                          |
| CFD    | Computational fluid dynamics                                 |
| EM     | Electromagnetic Model                                        |
| JSSP   | Job shop scheduling problem                                  |
| RFR    | Random forest regression                                     |
| MSE    | Mean-squared error                                           |
| ITS    | Intelligent Transportation Systems                           |
| SUMO   | Simulation of Urban Mobility                                 |
| RVEA   | Reference Vector Guided Evolutionary Algorithm               |
| MOEA/D | Multiobjective Evolutionary Algorithm Based on Decomposition |

# Contents

|           |                                                                                                                                                    |           |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Introduction</b>                                                                                                                                | <b>17</b> |
| 1.1       | Objectives of the research                                                                                                                         | 20        |
| 1.1.1     | Specific objectives                                                                                                                                | 20        |
| 1.1.2     | Methodology                                                                                                                                        | 20        |
| 1.2       | Document Structure                                                                                                                                 | 21        |
| <b>2</b>  | <b>Theoretical Background</b>                                                                                                                      | <b>23</b> |
| 2.1       | Overview                                                                                                                                           | 23        |
| 2.1.1     | Stochastic Optimization                                                                                                                            | 23        |
| 2.1.2     | Machine Learning                                                                                                                                   | 24        |
| 2.1.3     | Machine Learning Enhancing Metaheuristics                                                                                                          | 26        |
| 2.1.4     | Inverse Modeling                                                                                                                                   | 28        |
| 2.2       | Systematic Review                                                                                                                                  | 32        |
| 2.2.1     | Research Method                                                                                                                                    | 32        |
| 2.2.1.1   | Planning the review                                                                                                                                | 32        |
| 2.2.1.1.1 | Research Questions                                                                                                                                 | 33        |
| 2.2.1.1.2 | Search String                                                                                                                                      | 33        |
| 2.2.1.1.3 | Selection Criteria                                                                                                                                 | 33        |
| 2.2.1.2   | Conducting the Review                                                                                                                              | 34        |
| 2.2.2     | Results                                                                                                                                            | 36        |
| 2.2.2.1   | RQ1. How can the current research in Machine Learning applied in Metaheuristic be classified and what is the current state of this classification? | 38        |
| 2.2.2.1.1 | Time of hybridization                                                                                                                              | 39        |
| 2.2.2.1.2 | Aim of hybridization                                                                                                                               | 41        |
| 2.2.2.1.3 | Involved Component                                                                                                                                 | 44        |
| 2.2.2.2   | RQ2. What are the Machine Learning techniques used in Metaheuristic algorithms?                                                                    | 47        |
| 2.2.2.3   | RQ3. Which strategies were used to validate the algorithm?                                                                                         | 51        |
| 2.2.2.4   | RQ4. In which real-world optimization problems the algorithms were applied?                                                                        | 53        |
| 2.2.3     | Research agenda                                                                                                                                    | 55        |
| 2.2.3.1   | Summary                                                                                                                                            | 55        |
| 2.2.3.2   | Gaps                                                                                                                                               | 55        |
| 2.2.3.3   | Trends and research opportunities                                                                                                                  | 56        |

|           |                                                                 |           |
|-----------|-----------------------------------------------------------------|-----------|
| 2.2.3.3.1 | SAEAs                                                           | 56        |
| 2.2.3.3.2 | EDAs                                                            | 57        |
| 2.2.3.3.3 | Explore Machine learning techniques                             | 57        |
| 2.2.3.3.4 | Knowledge Discovery                                             | 58        |
| 2.2.4     | Threats to validity                                             | 59        |
| 2.2.4.1   | Construct validity                                              | 59        |
| 2.2.4.2   | Internal validity                                               | 59        |
| 2.2.4.3   | External validity                                               | 60        |
| 2.2.4.4   | Conclusion validity                                             | 60        |
| <b>3</b>  | <b>Validating DVL Performance</b>                               | <b>61</b> |
| 3.1       | Decision Variable Learning algorithm (DVL)                      | 62        |
| 3.1.1     | Dataset and environment variables preparation                   | 63        |
| 3.1.2     | Iterative process                                               | 64        |
| 3.1.2.1   | <i>FindClosestSolutions</i>                                     | 64        |
| 3.1.2.2   | <i>TrainModels</i>                                              | 65        |
| 3.2       | Experiments and Results                                         | 66        |
| 3.2.1     | Learning Models                                                 | 67        |
| 3.2.2     | Regression Experiment                                           | 68        |
| 3.2.3     | Optimization Experiment                                         | 71        |
| <b>4</b>  | <b>DVL Framework</b>                                            | <b>78</b> |
| 4.1       | Algorithms                                                      | 79        |
| 4.2       | Experiments and Results                                         | 82        |
| 4.2.1     | Optimization Problems                                           | 82        |
| 4.2.1.1   | Traffic Lights Signaling: A Many-Objective Optimization Problem | 82        |
| 4.2.2     | Algorithms, Parameters and quality metrics                      | 84        |
| 4.2.3     | Results and discussion                                          | 86        |
| 4.2.4     | Traffic Lights Signaling Experiment                             | 91        |
| <b>5</b>  | <b>Conclusion</b>                                               | <b>96</b> |
|           | <b>Bibliography</b>                                             | <b>98</b> |



|                                                                          |            |
|--------------------------------------------------------------------------|------------|
| <b>Appendix</b>                                                          | <b>114</b> |
| <b>APPENDIX A Heatmap plots of the optimization experiment. . . . .</b>  | <b>115</b> |
| <b>APPENDIX B Heatmap plots of the DVL Framework experiment. . . . .</b> | <b>130</b> |

# 1

## Introduction

Multi-Objective Problems (MOPs) are a class of optimization problems that involves more than one conflicting objective function to be optimized simultaneously ([COELLO et al., 2007](#)). Due to the conflict between the objective functions, there is not a single solution able to optimize all the objectives, but a resulting set is achieved with the best solutions. To obtain this solution set, we use the Pareto Optimality Theory ([COELLO et al., 2007](#)). Metaheuristics is the name given to the most general kind of optimization algorithms ([LUKE, 2013](#)) that are applied to a very wide range of problems. Usually, MOPs can be solved by Multi-Objective Evolutionary Algorithms (MOEAs), such as Genetic Algorithms (GAs), Particle Swarm Optimization (PSO), among others. MOEAs are based on Evolutionary algorithms (EAs), which is a class of population-based search metaheuristic.

MOEAs have been shown to perform well on a wide range of MOPs with two or three objective functions. However, with the growth of this area, several problems with a high degree of complexity have been found making most of MOEAs facing difficulties to solve them. Most of these complex problems are real-world problems and difficulties faced by the algorithms are often because some of these problems are discontinuous, non-differential, with many good locations, or because of the difficulty these problems present in evaluating a solution such as ambiguity, noise, and slowness to evaluate ([SHI; RASHEED, 2010](#)). Another well-known class of complex problems that MOEAs have experienced substantial difficulties to solve is the Many-Objective Optimization Problems (MaOPs), which have more than three objective functions ([SCHUTZE; LARA; COELLO, 2010](#)). The MaOPs have attracted increasing attention in the evolutionary optimization field ([LI et al., 2015](#)).

The poor scalability of traditional MOEAs in solving MaOPs has triggered a sizeable amount of research, aiming to provide alternative approaches that can properly handle this kind of problem ([MARTÍ et al., 2010](#)). A set of these approaches proposes to obtain knowledge of the domain during the search process or by external knowledge that could lead to enhance the search.

They are most motivated by the amount of available data created during the search, which leads to better handle the MaOPs. The objective of this approach is to extract information hidden in the data, which contains the behavior of the distribution of population individuals in the searching space, the landscape with a feasible area of convergence, and the relationship between the Pareto front and the Pareto optimally set (FAN et al., 2016). A way to extract this knowledge is by incorporating machine learning algorithms and techniques in the MOEAs.

The combination of optimization algorithms with machine learning techniques has been the subject of research in the last few years. The existing literature in this area can be divided into two main groups: the first one is concerned in improving the performance of machine learning techniques with metaheuristics, and the second group relies on the use of machine learning for enhancing metaheuristics (CALVET et al., 2017). Regarding the first group, different metaheuristic algorithms successfully improve the performance of machine learning techniques. This enhancement acts in optimizing machine learning models areas, such as parameters and architecture. The second group, which is the theme of this systematic review, can be viewed as knowledge extraction and integration into the optimization process. This knowledge can take different forms and result in different ways of integrating the knowledge into the metaheuristic (TALBI, 2016). Three different criteria are used to refine the classification of this kind of integration: Time of Hybridization, Aim of Hybridization, and Involved Component. Time of Hybridization relies on the time of extracting knowledge from the optimization process. The Aim of the Hybridization is about the result of the integration. The Involved Component classifies in which area of the algorithm the integration occurs.

Some known algorithms approaches that employs machine learning models to enhance metaheuristics are the Estimation of Distribution Algorithms (EDAs) (LARRAÑAGA; LOZANO, 2001) and Surrogate-Assisted Evolutionary Algorithms (SAEAs) (JIN, 2011). EDAs are variants of the evolutionary algorithms that at each iteration extract knowledge from the solutions found and build a probabilistic model, which replaces the common GAs operators of crossover and mutation. The motivation of this method is to understand the relationship between the variables and use it to explore the search domain. The SAEAs uses surrogates, also know as metamodels, to substitute the real objective function. Mainly motivated from reducing computational time in evolutionary optimization of expensive problems, such as aerodynamic design optimization or drug design, where complex simulations are involved. The idea behind SAEAs is that, if a surrogate model of the real expensive function can be created with relatively few samples, and also few evaluations, then this surrogate model can be used instead of the expensive objective function in the optimization process. Different strategies are driven from these ideas that lead to make SAEAs a good approach to solve real-world applications (JIN, 2011).

Motivated by the metamodel used in SAEAs algorithms, an inverse model of the problem is proposed in (GIAGKIOZIS; FLEMING, 2012; GIAGKIOZIS; FLEMING, 2014). Differently from the surrogate in the SAEAs, the inverse model maps the objective function values to the

decision variables, being capable of understanding the Pareto-optimal set and propose an optimal solution set. In their work, this inverse model is used to generate more non-dominated solutions for the decision-maker (DM), based on the approximated Pareto set obtained by MOEAs at the end of evolutionary optimization, increasing the density of the Pareto set found. Their method is further improved in many aspects, [Kudikala, Giagkiozis & Fleming \(2013b\)](#), [Kudikala, Giagkiozis & Fleming \(2013a\)](#) proposes the use of clustering methods to separate the search space for training the inverse model increasing the number of non-dominated solutions generated. The work in [\(YAN; GIAGKIOZIS; FLEMING, 2015\)](#) studies how the samples of the inverse model can improve the final solutions predictions and propose an evolutionary algorithm that provides better samples for training the inverse model. The inverse model can be also used as the probabilistic model in an EDA variant algorithm like the one proposed in [\(CHENG et al., 2015\)](#).

Recently, [Santos, Oliveira & Britto \(2019\)](#) proposed an algorithm denominated Decision Variable Learning (DVL), which uses the concept of the inverse model differently. Instead of using the model to generate more non-dominated solutions or to modeling the operators of an evolutionary algorithm as the EDAs, the inverse model in this work is used to directly predict the Pareto-optimal set. Their method runs standalone, without the presence of an optimization algorithm like the methods previously explained. In their experiments, the DVL surpassed the NSGA-III [\(DEB; JAIN, 2014\)](#), a classic algorithm from literature, in some benchmark problems, in situations where there is a restriction in the number of evaluations of the objective function and when there is a high number of objectives.

The strategies involving inverse models found have some limitations, one of them is the exploration of the performance of different machine learning models. In the works found, only a few different models were used, and no experimental comparison was made with well-known machine learning models in the literature to realize this kind of task. Another limitation is related to the strategy in using the generated knowledge during the search. In each work, the inverse modeling is applied in different ways, and despite the good results, there is no clarity on the best way to use the knowledge, as well as an understanding of the expected effects. For example, inverse modeling for generating more non-dominated solutions in [\(GIAGKIOZIS; FLEMING, 2014\)](#) has encountered a positive side effect. On many test instances, the estimated Pareto set turns out to be superior to the initial set. This assumption, coupled with the work developed in the DVL algorithm, shows that exists a research area to be developed on the estimation of the Pareto-optimal set. Further experimentation can be carried out to validate the hypothesis that it is possible to estimate the Pareto set with the decision variables. A well-defined set of benchmark test instances, and the use of different machine learning models and techniques could be used to validate the method. Also, the combination of the DVL with an MOEA could lead to mutual improvement between algorithms. While the DVL estimates the Pareto set with a low number of evaluations, the MOEA could lead the population predicted by the DVL to better convergence and divergence of the Pareto front. For this, some improvements in the design of the DVL algorithm have to be done to achieve this combination. Also, the specification of a

framework, for this reason, can abstract the method to be used with any MOEA benefiting from the use of different MOEAs for MOPs and MaOPs.

## 1.1 Objectives of the research

The main objective of this work is to create a framework that uses an inverse modeling approach coupled to any MOEA found in the literature. More precisely, three main steps were taken to achieve the goals. First, we perform a systematic review of the literature to identify the main uses of machine learning techniques enhancing optimization algorithms. Secondly, we analyze the performance of different machine learning methods in the DVL, seeking to understand the main characteristics of inverse modeling through the DVL algorithm. In the last step, we propose a framework that is an extension of the DVL algorithm, based on the knowledge obtained in the systematic review and our analysis of the DVL. This framework results in an algorithm for MaOPs recommended for situations that exist restrictions on the number of evaluations in the objective function. It uses a modified generic version of the DVL algorithm, which contains an inverse model of the problem inside to interact with the MOEA during the search. This DVL version needs to be adaptable in terms of machine learning models for a better understanding of different optimization problems, and the algorithm must have a computational complexity as close as possible to the MOEA coupled to it, to not slow the search process.

### 1.1.1 Specific objectives

The following specific objectives were defined to achieve the main objective:

1. Perform a systematic review of the literature of machine learning methods enhancing metaheuristics;
2. Validate the hypothesis that inverse models can be used to estimate the Pareto optimal set;
3. Propose a DVL variation that can be used in union with other MOEAs inside a framework;
4. Propose a framework that uses the DVL variation for generating new (and promising) solutions in multi-objective optimization.
5. Validate the framework in different scenarios of multi-objective optimization.

### 1.1.2 Methodology

This research work is characterized as applied research that contains exploratory and descriptive objectives, with a qualitative approach of bibliographic character and experimental, composed of five parts. The first part consists of a systematic review of the literature of machine learning methods enhancing metaheuristics. At this point, the general state of the art of the

problem will be analyzed and acquired the theoretical background necessary to develop this work. In the second part, we study the literature in depth to find possible approaches that use inverse models with metaheuristics. This part assists us in proposing this research and provides knowledge of the studied field used by some researchers.

The third part consists of deep experimentation over inverse models. Where diverse machine learning techniques are explored to validate their performance in estimating the Pareto optimal set. In the fourth step, we propose an algorithm that is a variation of the DVL, and a framework for the use of different strategies of the DVL inverse modeling in multi-objective optimization. This algorithm uses the inverse model to predict new solutions nearest to the Pareto optimal set and send these solutions to the MOEA to improve the search process. The framework correspond to the different ways of integration between the algorithms. The validation of our proposed framework is executed in the fifth step, where experiments with different MOEAs are made in several multi-objective optimization problems. The new proposed framework need to improve the performance of the MOEAs according to some metrics of the literature in optimization problems.

## 1.2 Document Structure

This document is organized into chapters and sections to facilitate understanding, provide better navigation and allow a better reading of the document. The main steps of this work were organized into three main chapters: chapter 2 - Theoretical Background with the Systematic Review, chapter 3 - DVL validation, and chapter 4 - DVL framework. Chapters 3 and 4 were organized separately, each one containing an explanation of the methods, the experiments, and the results obtained. Each chapter of the document is well described below:

1. Chapter 1 - Introduction: presents the preliminary definitions of the literature, problematic, arguments and hypothesis on the topic, in addition to the objectives, and the proposed methodology to achieve these objectives;
2. Chapter 2 - Theoretical Background: describes the theoretical context, with the revision of literature (Systematic Review) related to the proposed theme, as well as the results of this review and synthesis of the summary process of the studies, and finally a discussion about inverse models;
3. Chapter 3 - Validating DVL Performance: define the steps made to prove the hypotheses that inverse models can be used to estimate the Pareto optimal set;
4. Chapter 4 - DVL Framework: define the architecture of the framework and the reasons for such design, and validate the framework with different MOEAs in a set of multi-objective optimization problems.

5. Chapter 5 - Conclusion: resumes the hypothesis validations found and the results of the experiments. And also exposes the open research areas and future directions over the DVL.

# 2

## Theoretical Background

### 2.1 Overview

#### 2.1.1 Stochastic Optimization

The field of Stochastic Optimization represents optimization methodologies that employ some degree of randomness to find optimal solutions as possible to hard optimization problems. Usually, these kinds of problems cannot be solved optimally, or to any guaranteed bound, by any exact (deterministic) method within a “reasonable” time limit (BOUSSAÏD; LEPAGNOT; SIARRY, 2013). An important subset of optimization problems is the ones that present more than one objective function. They are normally called multi-objective optimization problems (MOPs). In general, MOPs can be defined as follows:

$$\begin{aligned} & \text{minimize} \quad F(\mathbf{X}) = (f_1(\mathbf{X}), \dots, f_m(\mathbf{X}))^T \\ & \text{subject to} \quad \mathbf{X} \in \Omega \end{aligned} \tag{2.1}$$

where  $\mathbf{X} = (x_1, x_2, \dots, x_n)^T \in \Omega$  constitutes the decision variable (vector),  $\Omega$  is the decision space,  $R^m$  is the objective space, and  $F : \Omega \rightarrow R^m$  is a mapping from the decision space to the objective space composing of  $m$  real-value evaluation functions (FAN et al., 2016). Generally, in the MOPs, no single point in the decision space can reach the optimum in each objective simultaneously. So the solutions set are found through the use of the Pareto Optimality Theory (COELLO et al., 2007). Where, considering a minimization problem, a decision vector  $\mathbf{X}_u \in \Omega$  is said to dominate other vector  $\mathbf{X}_v \in \Omega$  (denoted by  $\mathbf{X}_u < \mathbf{X}_v$ ) if and only if  $F(\mathbf{X}_u)$  is partially less than  $F(\mathbf{X}_v)$ , i.e.  $f_i(\mathbf{X}_u) \leq f_i(\mathbf{X}_v), \forall i \in \{1, \dots, m\}$  and  $F(\mathbf{X}_u) \neq F(\mathbf{X}_v)$ . The set of vectors that cannot be dominated by any other point from the decision space is defined as Pareto-optimal set (PS), the map of the PS to the objective space is defined as the Pareto Front (PF). Other well-known classes of complex problems are the Many-Objective Optimization



Problems (MaOPs), which have more than three objective functions. This kind of problem has attracted increased attention in evolutionary optimization by the difficulties faced in solving them (FLEMING; PURSHOUSE; LYGOE, 2005).

Metaheuristics are the most general kind of optimization algorithms (LUKE, 2013), they are applied to find satisfactory solutions for optimization problems. They can be classified as single-solution based metaheuristics, whereas are more exploitation-oriented, such as Simulated Annealing, Tabu Search, GRASP, Guided Local Search, Iterated Local Search, etc. And can be classified as Population-Based Metaheuristics that are more exploration-oriented: Evolutionary Algorithms (EAs) and Swarm Intelligence (BOUSSAÏD; LEPAGNOT; SIARRY, 2013). The use of Multi-Objective Evolutionary Algorithms (MOEAs) to solve optimization problems has been motivated mainly because of the population-based nature of EAs. Also, these algorithms use mechanisms based on Pareto optimality that avoids convergence to a single solution preserving the diversity of the PF. Some real-world optimization problems drive the use of non-traditional approaches, mainly due to large search spaces, uncertainty, noise, disjoint Pareto curves, etc (COELLO et al., 2007). It is known that MOEAs scale poorly, facing serious difficulties in solving MaOPs. Primary because of the exponential increase in the number of non-dominated solutions in each population, putting pressure towards the Pareto-optimal front. Consequently, the convergence property of MOEAs is deteriorated (SCHUTZE; LARA; COELLO, 2010).

### 2.1.2 Machine Learning

Machine Learning (ML) is a subfield of computer science and artificial intelligence that studies and design algorithms which learn from data, composed by observations, and make predictions based on these data (PROVOST; KOHAVI, 1998). ML simulates human learning by acquiring knowledge from the real world. Formally, (ANDERSON, 1990) defines ML as follows: “A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E”.

A large set of ML algorithms is proposed in the literature. They can be classified by the human supervision on training (DEY, 2016). There is four main types of this classification: Supervised Learning, Unsupervised Learning, Semisupervised Learning, and Reinforcement Learning. In Supervised Learning, the algorithm is provided with training data and correct answers. Its task is to learn with the training data and predict outputs, based on provided data, as close as possible to the expected answers. Typical tasks of Supervised Learning are classification and regression.

In Unsupervised Learning, the algorithm does not have correct answers for the training data. In this type of ML system, instead of predict expected values, the algorithm tries to capture some characteristics of the training data and the observed random variables. Typical tasks of the Unsupervised Learning are clustering, visualization and dimensionality reduction, and

association rule learning. Semisupervised Learning is when the algorithm is provided with a training set with missing information, some answers are labeled some do not have any information. Most Semisupervised learning algorithms are combinations of Unsupervised and Supervised algorithms.

Reinforcement Learning uses a learning system as an agent that interacts with the environment. Based on these interactions, the learning system can select and perform actions and get rewards or be punished for its choices. The goal in the Reinforcement Learning system is to train the agent to maximize the expected sum of future rewards, called the return. As an example, many robots use this kind of learning system to learn how to move.

Diverse algorithms and techniques detailed in the literature are capable of performing machine learning tasks (DEY, 2016). For Supervised Learning, a well-known algorithm is the Decision Tree, which uses trees as data structure, it starts with a root node and ends with a decision made by the leaves that generate the predicted result. The decision nodes group attributes by sorting them based on their values. Naïve Bayes, is a method that depends on the conditional probability to classify or cluster the dataset. Support Vector Machine (SVM), widely used in machine learning, works on the principle of margin calculation, it draws margins between the classes and uses these margins for minimizing the classification error. For Unsupervised Learning, the k-means algorithm is a widely used clustering method for this task that creates  $k$  distinct clusters around centroids calculated by the features. Principal Component Analysis (PCA) is another unsupervised method that has as principal function the reduction of the dimension of the data to make the computation faster and easier.

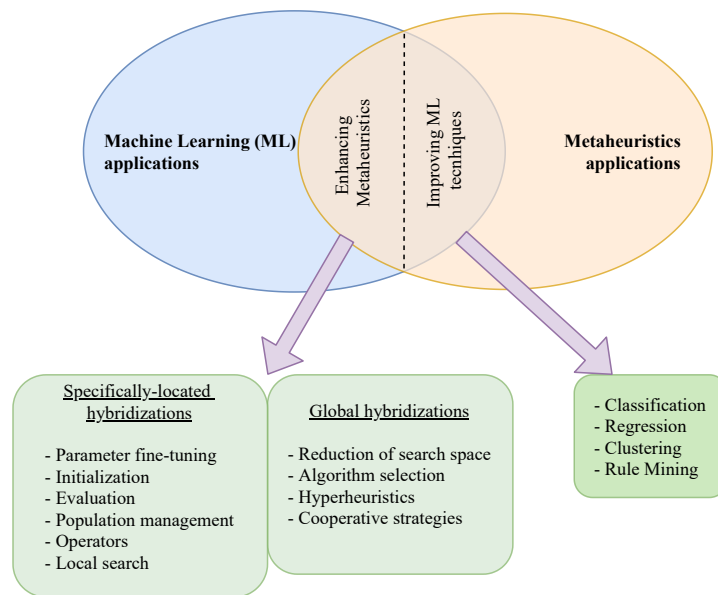
For Semisupervised Learning, the Generative models are one of the oldest semisupervised methods that use a mixed distribution e.g. Gaussian mixture models to identify the unlabeled data. Self-training is a technique that uses a classifier trained with a portion of labeled data, and afterward, this classifier is fed with unlabeled data to predict their labels. And the Transductive SVM, which is an extension of the SVM method to work in Semisupervised Learning (DEY, 2016). For Reinforcement learning, the Markov Decision Process (MDP) maps the process of the agent interacting with the environment. And Q-learning (WATKINS; DAYAN, 1992), a model-free algorithm that learns the value of an action in a particular state.

Another important method, widely used in the literature is the artificial neural network (ANN) (WANG, 2003). Inspired by the sophisticated functionality of human brains, it can be used in diverse machine learning tasks. ANN consists of an input layer, one or more hidden layers of neurons, and a final layer of output neurons. That construction can approximate any computable function to arbitrary precision. The numbers given to the input are independent variables, and the numbers to the output are dependent variables that map the function in question. The input and outputs of an ANN can be transformed to take different forms, such as number, binary, and label. Allowing the ability to be applied in a wide range of tasks.

### 2.1.3 Machine Learning Enhancing Metaheuristics

An overview of the interaction between Optimization and Machine learning can be found in (CALVET et al., 2017), which analyses and discusses the different areas where these two fields of computer science can combine. Figure 1 shows this interaction and also describes some relevant and representative topics. The intersection of the two areas can be divided into Enhancing Metaheuristics and Improve ML techniques. The first one is the theme of our systematic review, well detailed in section 2.2, while the second one references the improvement that metaheuristics can make to ML techniques.

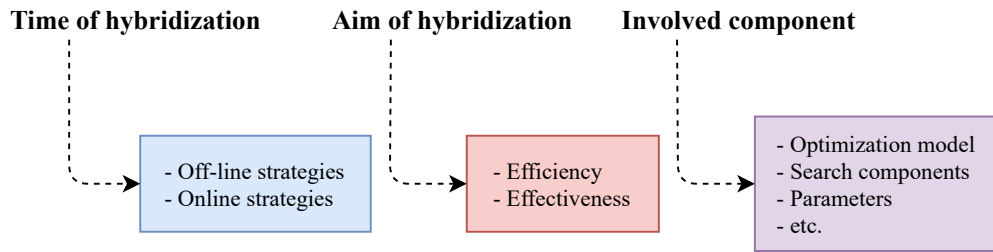
Figure 1 – Classification of works combining metaheuristics and machine learning.



Source: (CALVET et al., 2017)

The subarea of machine learning enhancing metaheuristics algorithms is well detailed in (JOURDAN; DHAENENS; TALBI, 2006; TALBI, 2016), where classifications of the area are well proposed by these papers. Besides the differences in terminologies of these works, the concepts remain the same. Figure 2 shows the three main types of this enhancement. Time of hybridization concerns to when the knowledge is extracted. Off-line strategies extract the knowledge before the search starts, while online strategies extract the knowledge dynamically during the search. The Aim of hybridization concerns to the impact of the result between the integration of the two areas. Integration based on Efficiency improves the search time by reducing the search space or by approximating the objective function. Effectiveness incorporates knowledge to the search, leading to a better quality of solutions. The last type is the Involved component, usually, metaheuristics can be divided into different search components. This classification corresponds to the place in the metaheuristic where the machine learning component was inserted.

Figure 2 – Classification of enhancing metaheuristics with machine learning.



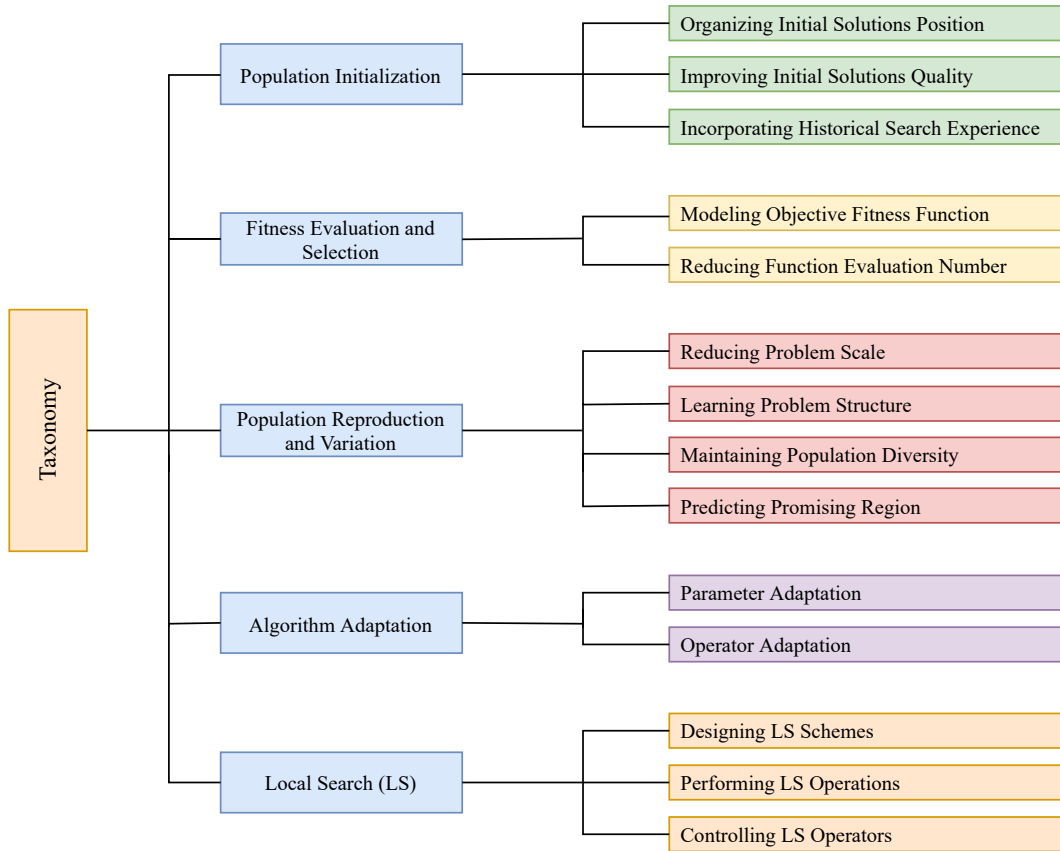
Source: (TALBI, 2016)

In our systematic review, the terms presented in Figure 2 will be used for the classification of the papers, except for the Involved component. About this criteria, we prefer to use the terms of the taxonomy proposed by (ZHANG et al., 2011), shown in Figure 3. Their taxonomy details all components involved in integration with machine learning on Evolutionary Computation (EC), which can be abstract to any kind of metaheuristic algorithm. The Evolutionary algorithm can be divided into 5 main areas: Population Initialization, Fitness Evaluation and Selection, Population Reproduction and Variation, Algorithm Adaptation, and Local Search (LS). Population Initialization is the first step on an Evolutionary algorithm, generally, metaheuristics produce the initial population randomly. The integration with ML can benefit this step by organizing the initial population and changing the decision vector to improve solutions quality. Other techniques use historical search experience to generate this initial set.

The Machine Learning method used in the Fitness Evaluation and Selection step aims to model the objective function or reduce the number of evaluations. Population Reproduction and Variation is a crucial step with much attention in the optimization literature. In this step, ML techniques can generate new solutions with embedded knowledge about the problem. This knowledge can reduce the problem scale, help in diversity and convergence of new solutions, and predict promising regions. Algorithm Adaptation refers to the algorithm as a whole, where ML is used to improve the optimization parameters and operators. The last taxonomy refers to the Local Search (LS), where the ML can be used to design, choose when to use, and control the different local search strategies.

In addition to the studies presented above, other papers found in the literature aggregated value to prepare our systematic review. The papers in (PELIKAN; GOLDBERG; LOBO, 2002; BENGIO; LODI; PROUVOST, 2020; FAN et al., 2016) contribute by identifying search terms for our research method and describe base algorithms that are used by other algorithms that enhance metaheuristics. Also, they show different application areas that this kind of algorithm can be used.

Figure 3 – Evolutionary Computation taxonomy.



Source: (ZHANG et al., 2011)

### 2.1.4 Inverse Modeling

Methods based on surrogates have gained prominence in recent years. In the context of MOPs, most of the related work focuses on constructing learning models to predict several objective functions, from the decision variable as an input. They provide approximate models to represent the original problem in a way that facilitates solving it by optimization algorithms, surrogates can also be called metamodels. There are several metamodeling methods, but they can be divided into two main categories (SHI; RASHEED, 2010): classic function transformation methods and surrogate modeling methods. The first category corresponds to the techniques that seek to represent the set of functions of the problem to a simpler one, such as transforming it into a linear function or using classical methods to approximate the problem functions by finding a weighted sum of simpler functions. The second category seeks a new expression for each objective function that is constructed based on the data obtained previously from the real objective function, called surrogates (SANTANA-QUINTERO; MONTANO; COELLO, 2010). There are three main ways to create surrogates, from instance-based learning models, machine learning, and statistical learning.

In (SHI; RASHEED, 2010), a study on methods of approaching function in optimization problems is made. In this work, a division is presented between the methods of approximation, which are the classical methods of function transformation and surrogate modeling methods. Also, the paper describes the types of surrogate-based methods that were separated into learning models based on instances, machine learning, and statistical learning. In (SANTANA-QUINTERO; MONTANO; COELLO, 2010), the authors seek to analyze several techniques to deal with functions with high computational costs. For this, they approach the concept of incorporation of information in optimization problems. According to the authors, there are three main ways to do this: problem approximation, which tries to replace the problem with one that is very close to it, but which is easier to solve; functional approximation (Metamodels); and evolutionary approximation, which is specific to evolutionary algorithms, this approach attempts to reduce function evaluations by estimating the adequacy of an individual from other similar individuals. This work also analyzes some real problems that were solved using some techniques studied in the paper.

However, despite the existence of related work on surrogates applied to multi- and many-objective optimization, they only explore learning from decision variables to model the objective functions. During our study of surrogates some questions arose:

1. Since in MaOPs we have objective vectors with higher dimensions, could it be possible to learn the behavior from objective functions to estimate the decision variables?
2. Since it is possible to learn the behavior from decision variables using surrogates, is it possible to learn inverse functions and from that functions, to predict sub-optimal solutions to the problem?

According to our research, the first attempt of constructing an inverse modeling method is made in Giagkiozis & Fleming (2012), Giagkiozis & Fleming (2014), called Pareto Estimation (PE). Where an inverse functional mapping from PF to PS was built based on the non-dominated solutions set obtained by the execution of an MOEA at the end of evolutionary optimization. This model is used to generate additional non-dominated solutions, thereby enhancing the density of the solutions set. Their motivation is that in many MOPs, principally in MaOPs, the decision-maker (DM) is not satisfied with the Pareto set (PS) obtained by the MAOE. This method alleviates these difficulties for continuous optimization problems by producing more and, usually, better distributed Pareto optimal solutions along the entire Pareto front (PF). In these works, it should be noted that the inverse model is not used during the optimization process.

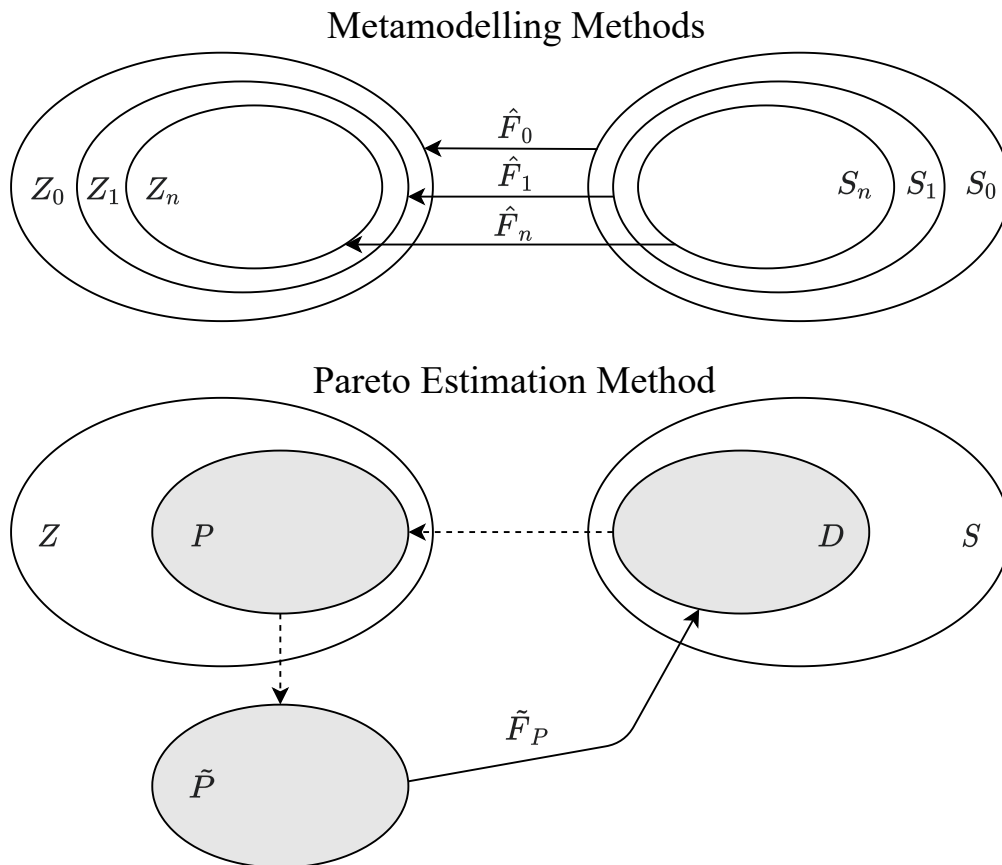
The general approach, when substituting the real objective function with a surrogate model for use in an EA, has the following steps:

1. Sample the real objective function.

2. Use the obtained samples to create a surrogate model.
3. Use the surrogate model in the optimization.
4. If the convergence criteria are met, stop; if not, go to Step 1.

An illustration of the metamodelling process in comparison with the PE method can be seen in Figure 4. Where  $S \in \mathbb{R}^n$  is the decision space,  $Z \in \mathbb{R}^m$  is the objective space, and  $F : S \rightarrow Z$  the objective function of the optimization problem. The function  $\hat{F}$  represents the surrogate or metamodel created a priori or in every iteration ( $\hat{F}_0, \hat{F}_1, \dots, \hat{F}_n$ ) of the optimization process. Different metamodel approaches are well detailed in (DEB et al., 2019), where a taxonomy of frameworks is proposed to help solve multi-objective optimization problems. They also present a comparative study of the results obtained by the frameworks.

Figure 4 – Comparison between Metamodeling methods in EAs and the Pareto Estimation (PE) method.



Source: Produced by authors.

Also, in Figure 4 the set  $D$  represents the non-dominated set and  $P$  the Pareto Front (PF) of the objectives from  $D$ . In the Pareto Estimation method is possible to verify that the inverse



model  $\tilde{F}_P : \tilde{P} \rightarrow D$  uses a set  $\tilde{P}$  as input for training. This set  $\tilde{P}$  is the result of the normalization of the set  $P$  scaled to the range  $[0, 1]$ . To perform the approximation of the inverse function the researchers use a Radial Basis Function Neural Network (RBFNN). In their experiments, the use of the PE method increases the number of non-dominated solutions along the surface, enhancing the ability of the DM to identify the best solutions. Also, a positive side effect is encountered in their experiments that were not in the intended purpose of the method, in some test instances, the generated solutions by the inverse model have superior values in comparison to the set provided by the MOEA.

Enhancements of the PE method can be obtained in the literature. In (KUDIKALA; GIAGKIOZIS; FLEMING, 2013b; KUDIKALA; GIAGKIOZIS; FLEMING, 2013a) the PE method is extended to improve the density of available non-dominated solutions in multi-objective multi-modal problems. This kind of problem has multiples decision vectors that map to identical objective vectors on the Pareto front. In this case, the authors propose to subdivide the objective space, by using a clustering algorithm, into different  $C_m$  clusterings. So an ANN is trained for each cluster of solutions to identify the map  $\tilde{F}_{\tilde{P}_{C_m}} : \tilde{P} \rightarrow C_m$ . In these works, it is noted that the quality of the resulting inverse model depends heavily on the training samples used. So in (YAN; GIAGKIOZIS; FLEMING, 2015) is proposed the MAEA-GD/RD an optimization algorithm based on the MAEA-gD with sample redistribution, which reallocates the samples in the current population in objective space for training the inverse model. Doing this, more samples are present in regions where higher quality topological information is more likely to be obtained and used to improve the identification of  $\tilde{F}_P$ .

Motivated by the PE method and the evolution of the EDAs, is proposed in (CHENG et al., 2015) the IM-MOEA algorithm, which provides a different way of using inverse models. Instead of a method used after the optimization process, the inverse models are inserted internally in the optimization algorithm and then used to create offspring solutions by sampling the objective space. They propose a decomposition of the inverse function, also reducing the number of inverse models, and a reproduction operator who benefits from the inverse model. The purpose of their method is to use the inverse models to increase the density of solutions in the preferred regions at a low computational cost. This approach has gained attention in the literature providing diverse algorithms and variants of the IM-MOEA. The work in (GHOLAMNEZHAD; BROUMANDNIA; SEYDI, 2020) proposes a variant where a random forest feature importance method is used to determine the best assignment of decision variables, improving the training performance of the inverse model. Also, the method proposed in (ZHANG et al., 2020) trains an ANN to perform the grouping strategy for training the inverse model. Different from IM-MOEA, the algorithm IM-MOEA/D proposed by (GEE; TAN; ALIPPI, 2016) is based on decomposition-based MOEA rather than domination-based MOEA. Their algorithm achieves good performance while resolving dynamic multi-objective optimization problems.



## 2.2 Systematic Review

To the best of our knowledge, systematic reviews or mappings that present an overview of the proposed research field of metaheuristics enhanced by machine learning were not found in the literature. Motivated by the lack of secondary studies in the area, a greater understanding in the study of inverse models, and the study of the interception between these areas. We contribute to the literature by providing, in this section, a systematic review of the research field. We followed a well-established protocol based on Kitchenham guidelines ([KITCHENHAM, 2004](#)) and analyses of 111 papers published in the last ten years. Based on the related works, we discuss some relevant aspects and the finds of different aspects not addressed, such as machine learning techniques used, validation strategies and kind of applications field. Also, we identify new research opportunities and trends.

### 2.2.1 Research Method

In this work, we use a systematic literature review, a methodology to review research results. Based on evidence, which is defined in this context as a synthesis of best quality scientific studies on a specific topic or research question. The systematic literature review assumes that the accumulation of results from scientific experiments is more reliable than the results from expert's opinion. This kind of research was initially developed in medicine and since other domains have adopted this approach, e.g. Criminology, Social policy, Economics, Nursing etc ([KITCHENHAM et al., 2009](#)).

We conduct this systematic literature review following the guidelines proposed by ([KITCHENHAM, 2004](#)) which adapts the methodology to the needs of software engineering researchers. This approach divides the review process into three main phases: Planning the Review, Conducting the Review, Reporting the Review. The first one, Planning the Review, is responsible for creating the protocol used in the research. The second phase, Conducting the Review, performs the planned tasks in the previous phase. The third one, named Reporting the Review, specifies the communication of the research. The last phase resulted in the elaboration of the present work. In this section, we describe the two initial phases: Planning the Review and Conducting the Review.

#### 2.2.1.1 Planning the review

This systematic review was motivated by the lack of secondary research in the area. In our preliminary studies, only a few secondary works were found, mainly surveys that outline the specific research area where Machine Learning enhances Metaheuristics algorithms. Instead, the number of primary research in this area has grown due to the motivation in solving expensive and hard optimization problems with multiple objectives. It makes a good opportunity to conduct a systematic review in the area.

### 2.2.1.1.1 Research Questions

Based on our preliminary studies and related works, and considering relevant aspects of the area, we formulated four different research questions (RQs) to be answered by the collected literature.

- RQ1. How can the current research in Machine Learning applied in Metaheuristics be classified, and what is the current state of this classification?
- RQ2. What are the Machine Learning techniques used in Metaheuristic algorithms?
- RQ3. Which strategies were used to validate the algorithm?
- RQ4. In which real-world optimization problems were the algorithms applied?

### 2.2.1.1.2 Search String

The selected key search terms in Table 1 were built through an improvement process. First, some initial terms were identified based on our preliminary studies and the reading of the surveys of the area. Then keywords were refined and extended during the search process. In this phase, we use Google Scholar and IEEE Explore to test and refine the search string. We identify that the studies in this area have several terms and keywords to describe their work. For example, the terms "hybridization", "adaptive learning", "incremental learning", "linkage learning", and some variants are present in the studies but do not necessarily identify our aim area. "Hybridization" can refer to the incorporation of any kind of method into the algorithm, not necessarily machine learning into optimization. This led to difficulty finding the key terms, an increase in the search string size, and an increase in the number of the result papers.

Table 1 – The generic search string.

---

"machine learning" AND ("optimization algorithm" OR metaheuristic  
OR meta-heuristic OR evolutionary OR genetic) AND (hybrid OR  
hybridization OR hybridized OR "probabilistic model" OR  
"adaptive learning" OR "adaptation learning" OR "incremental learning"  
OR "linkage learning" OR surrogate OR "response surfaces" OR  
metamodels OR "approximate models")

---

Source – Produced by authors.

### 2.2.1.1.3 Selection Criteria

Our goal was to include only relevant studies to strengthen the validity of the review. For this purpose, it is necessary to adopt inclusion and exclusion criteria for filtering the results. The papers were selected based on these filters presented in Table 2. Selection occurs firstly by reading the title and abstract screening. If there were doubts about a paper, it was reviewed in the next step, considering the full text.

Table 2 – Inclusion and exclusion criteria.

|                                                                                                       |
|-------------------------------------------------------------------------------------------------------|
| <b>Inclusion criteria</b>                                                                             |
| Publications that use machine learning to assist metaheuristic algorithms                             |
| <b>Exclusion criteria</b>                                                                             |
| Publications that have not been peer-reviewed and not published in a conference, journal, press, etc. |
| Duplicated papers                                                                                     |
| Secondary study papers                                                                                |
| Not available online                                                                                  |
| Publications written in languages other than English                                                  |
| Publications dated before 2011                                                                        |
| Out of scope                                                                                          |

Source – Produced by authors.

### 2.2.1.2 Conducting the Review

A list of sources was carefully chosen to execute the systematic review. These online repositories represent the major scientific publication sites where papers of different natures can be found. The resulting databases are IEEE Xplore, ACM Digital Library, Science Direct, Scopus, SpringerLink, Web of Science, and Wiley Online Library. Some databases have different limitations when executing the search string, for instance, in Web of Science, it is not possible to search for full text. Another example is the Science Direct, Scopus, and SpringerLink databases that have a limitation of search terms, so we split the search string used in these databases into different parts to perform the search. The searched content that was defined for each repository is shown in Table 3, where SS represents the search string presented in Table 1. Table 4 shows the number of papers returned by each source and the number of works considered for data extraction.

Table 3 – Advanced search.

| Database/Source      | Advanced search                                 |
|----------------------|-------------------------------------------------|
| IEEE Xplore          | String: SS<br>Content: Title and Abstract       |
| ACM Digital Library  | String: SS<br>Content: Title and Abstract       |
| Science Direct       | String: Split SS<br>Content: Any field          |
| Scopus               | String: Split SS<br>Content: Title and Abstract |
| SpringerLink         | String: Split SS<br>Content: Any field          |
| Web of Science       | String: SS<br>Content: Title and Abstract       |
| Wiley Online Library | String: SS<br>Content: Any field                |

Source – Produced by authors.

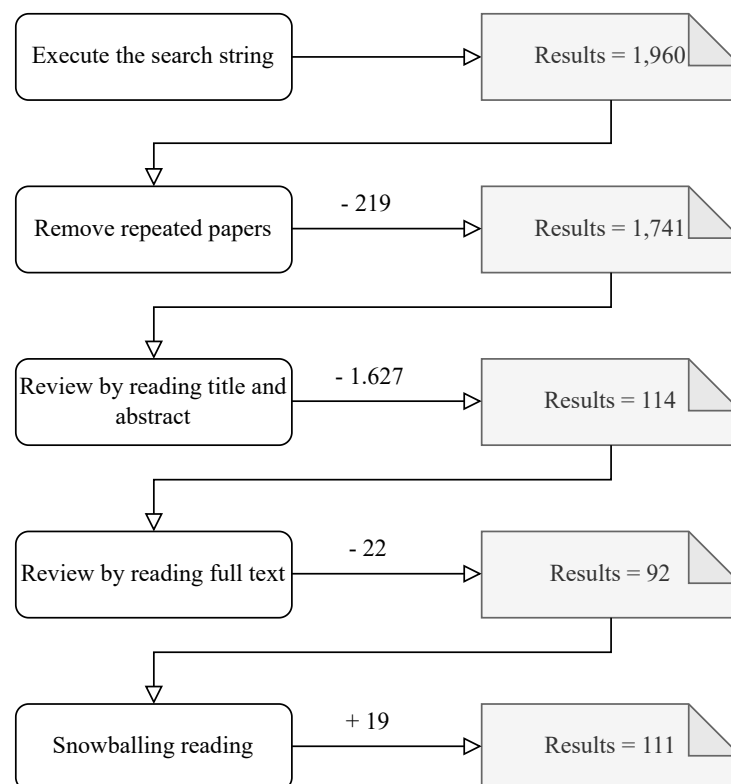
Table 4 – Number of papers returned by each database.

| Database             | URL                                                                             | Number of Papers | Selected Papers |
|----------------------|---------------------------------------------------------------------------------|------------------|-----------------|
| IEEE Xplore          | <a href="https://ieeexplore.ieee.org/">https://ieeexplore.ieee.org/</a>         | 184              | 21              |
| ACM Digital Library  | <a href="https://dl.acm.org/">https://dl.acm.org/</a>                           | 328              | 7               |
| Science Direct       | <a href="https://www.sciencedirect.com/">https://www.sciencedirect.com/</a>     | 368              | 20              |
| Scopus               | <a href="https://www.scopus.com/">https://www.scopus.com/</a>                   | 231              | 20              |
| SpringerLink         | <a href="https://link.springer.com/">https://link.springer.com/</a>             | 482              | 14              |
| Web of Science       | <a href="https://www.webofknowledge.com/">https://www.webofknowledge.com/</a>   | 367              | 10              |
| Wiley Online Library | <a href="https://onlinelibrary.wiley.com/">https://onlinelibrary.wiley.com/</a> | 0                | 0               |
| Total                |                                                                                 | 1.960            | 92              |

Source – Produced by authors.

The search was performed during January 2021, and a total of 1.960 papers returned as a result of the execution of the search string on the source list. The steps of the review are shown in Figure 5. Despite the high number of articles found, the result could be greater if we not adopted the exclusion criteria of publications older than ten years. This high number of papers found outlines the lack of existence of a term to represent the Machine Learning enhancing Metaheuristic algorithms. Different terms exist in the literature, and authors use different keywords for representing their studies in the area.

Figure 5 – Systematic review conduction.



Source: Produced by authors.

From the 1960 papers returned, 219 were removed by duplication, leaving 1.741 papers still to be analyzed. In the next phase, we proceeded with a title and abstract screening, achieving 114 papers to perform a full-text screening. Performing the selection criteria, 22 papers were removed, resulting in a total of 92. In these remaining papers, we performed a backward snowballing procedure (WOHLIN, 2014) by using the reference list of each selected paper to identify new studies to include that attend our selection criteria. The execution of the snowballing resulted in the addition of 19 papers to our resulting set. Finally, 111 papers were selected for the next phase, for data extraction.

Concerning the papers removed by the inclusion and exclusion criteria, we found papers that mention the terms of the search string but are out of the scope of this systematic review. For instance, studies that use optimization to support machine learning. We also found hybrid optimization algorithms that use two different kinds of metaheuristics or other techniques instead of machine learning. These kinds of papers and the secondary studies constitute a large number of papers in our search that were removed. Most of the removed primary studies, for example, concerns about the improvement of Artificial Neural Networks (ANN) by optimization algorithms. This specific topic could lead, in future work, to an execution of a systematic review to clarify and answer important questions in the field.

In the last phase, we extracted the data of the selected papers and stored it in a spreadsheet. We consider the title, publication year, venue, and some additional information, to answer our research questions. The results were synthesized, analyzed, and are discussed in the next section.

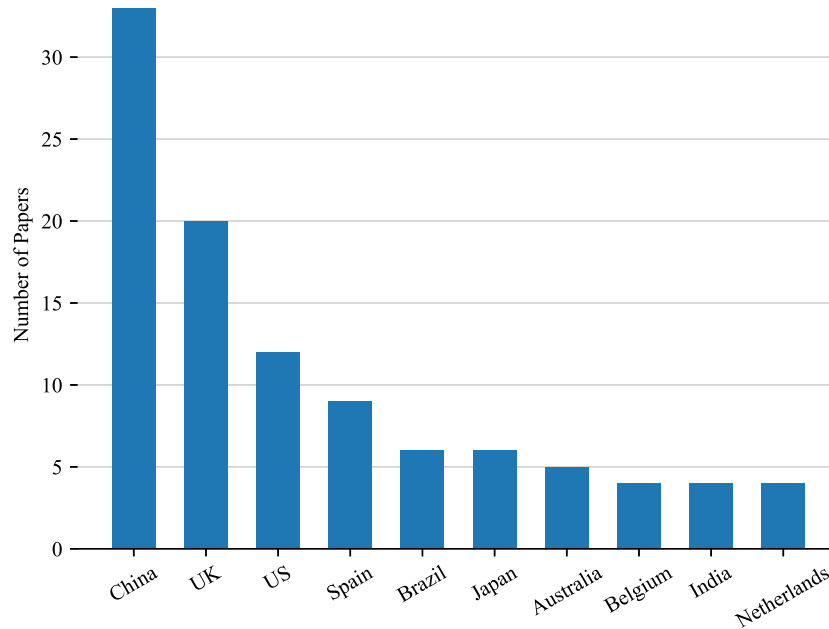
### 2.2.2 Results

This section aims to answers the research questions designed in this study. But firstly, we give a mapping of the research field by identifying and categorizing the selected studies accordingly with the extracted data. After this, each research question is analyzed and properly answered.

Mapping the main countries and the main authors of the area may contribute to identifying people and places of interest that have been working in the field. This allows to guide and unite researchers to find the best partnerships to work. Figure 6 shows the main countries that have the most number of papers published. China outperforms the other countries with a total of 33 published papers. The United Kingdom (UK) and the United States (US) appear right afterward with 20 and 12 publications each. We found a total of 35 different countries that published in the area. The countries that do not appear in Figure 6 have less than three publications. Some of the papers were published by authors from two or more countries, authors from China are the ones that have more collaboration with others countries.

Regarding the authors, we identified 364 that have been published about the research field. Figure 7 shows the authors that have at least three papers published. Bo Liu is the author

Figure 6 – Countries that have papers published about the research field.



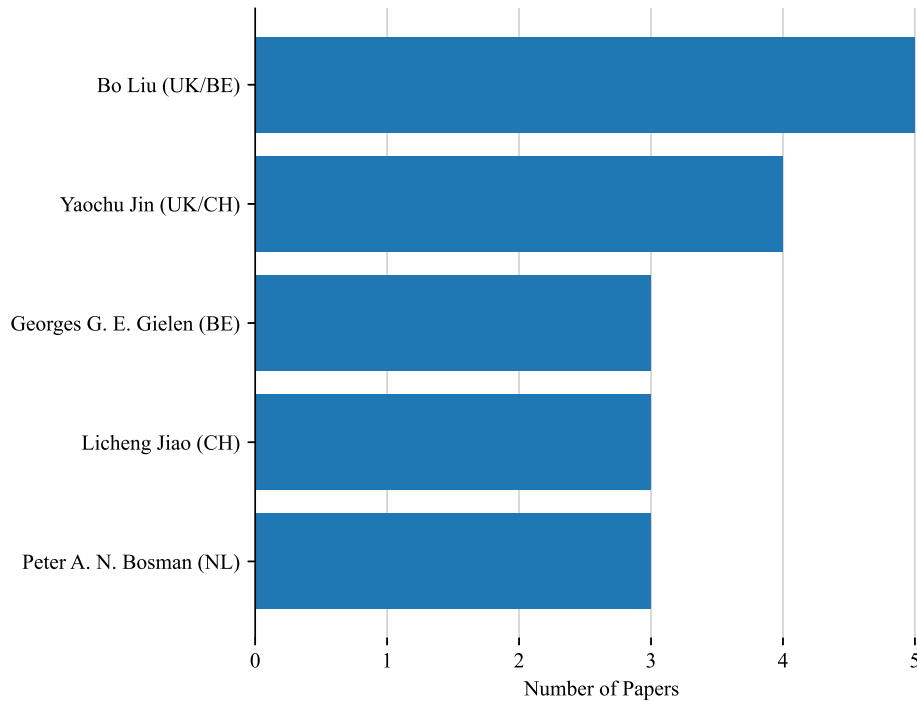
Source: Produced by authors.

with more publications, some from U.K. and others from Belgium. Three publications of Bo Liu are in common with George G. E. Gielen, this may lead to an existence of a place of interest in the area between U.K. and Belgium. Two authors from China appear on the plot representing the strength that China has in this research field. None US authors appear in this list, meaning a diversification of the contributions in the area.

We identify a total of 16 different conferences and 53 journals in the studies. In Figure 8 we only show the most relevant venues in relation to the number of papers found. The Genetic Evolutionary Computation Conference (GECCO) and the Congress on Evolutionary Computation (CEC) are the preferred conferences, each one with 9 papers found. The International Conference on Machine Learning (ICML) and the Brazilian Conference on Intelligent Systems (BRACIS) appears with 2 papers found in each. About the other conferences, only 1 paper was found in each. The papers are also published in journals of optimization and machine learning areas, where the preferred journal is Neurocomputing with 4 papers found. The remaining papers are well distributed between the journals. The number of different journals found was quite above the number of conferences. This difference shows the maturity of this research area since more detailed publications are made in journals.

Figure 9 presents the results obtained of papers published by year. We only consider studies published from 2011 onward. A significant increase could be seen until nowadays, mainly

Figure 7 – Authors with at least three papers published in the research field.



Source: Produced by authors.

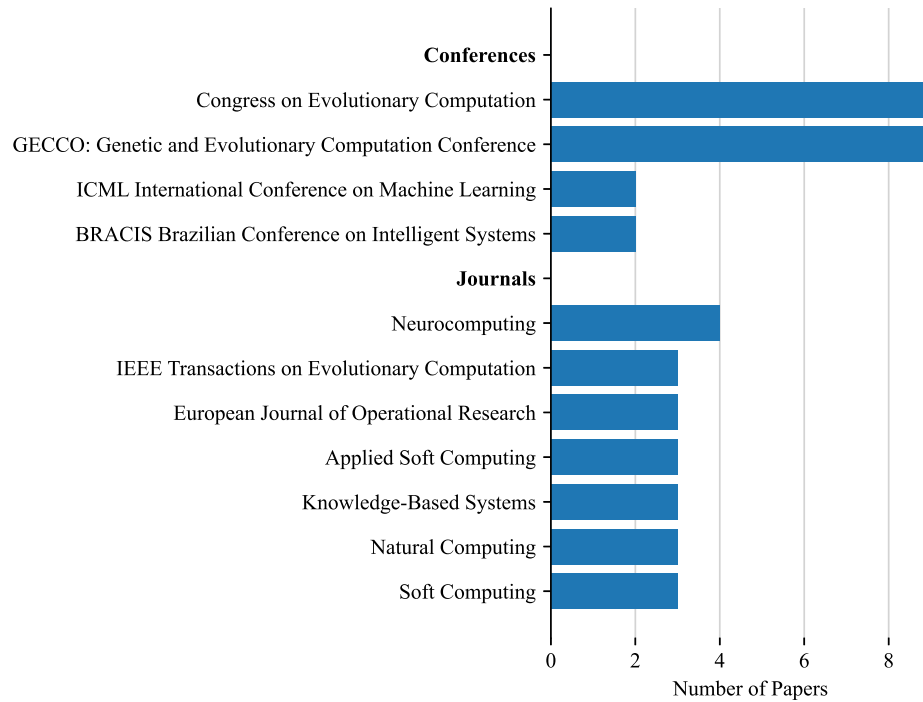
in 2019 and 2020 when the number of papers doubles in value. This means that a trend on machine learning supporting optimization has been developed nowadays, mainly due, as we will describe in this section, to the urge on solving expensive objective functions. The year 2021 presents few works because it is the year of the execution of this systematic review, representing only a short period of studies in this year.

The following subsections present our findings based on the research questions that were formulated to understand the approaches and elements of the field in literature. Each subsection represents a research question in order to better understand the data collected.

#### 2.2.2.1 RQ1. How can the current research in Machine Learning applied in Metaheuristic be classified and what is the current state of this classification?

The current classification for machine learning enhancing metaheuristic presented in [subsection 2.1.3](#), showed in Figure 2, and the taxonomy of Evolutionary Computation in Figure 3 are capable to answer the current research in the area. From the 111 resulting papers, no different strategies or enhancement ways that were not presented in the current classification were found. Figure 10 shows the overall number of papers found in each classification. The majority of papers found, with 86.5%, were classified as Online by the Time of hybridization, 63.1% were classified as Efficiency by the Aim of hybridization, and 63.2% were classified as Fitness

Figure 8 – Main venues where papers are published in the research field.



Source: Produced by authors.

### Evaluation and Selection by the Involved Component.

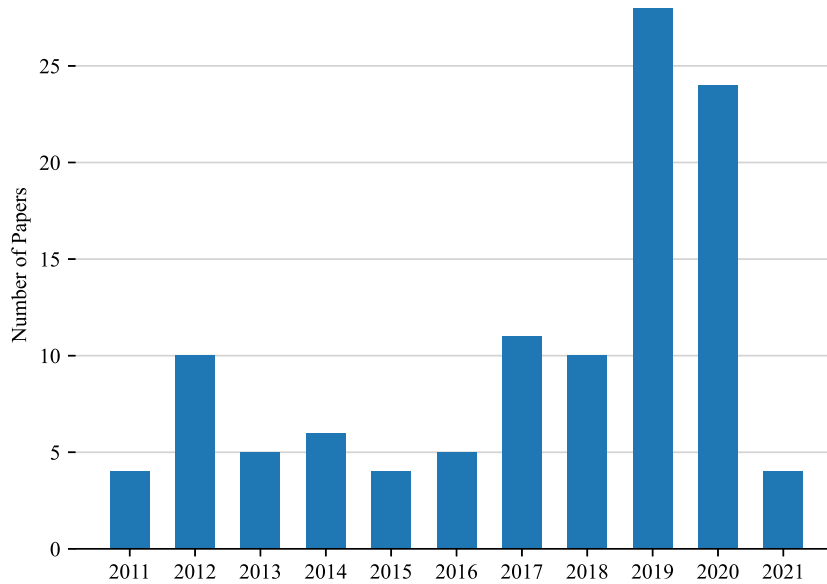
To answer the second part about the current state of this classification, we divide this subsection into three main parts. First, we analyze the papers by the classification of Time of hybridization. The second step is to analyze the classification of papers based on Aim of hybridization. At last, the taxonomy of Evolutionary Computation is used for the analysis of the resulting papers.

#### 2.2.2.1.1 Time of hybridization

From the 111 resulting papers, the majority, a total of 96 papers, are classified as online and only 15 as off-line, as is summarized in Table 5. The 96 papers that were classified as online dynamically discover the knowledge during the search. This strategy is used when the ML technique used does not violate any restrictions of the optimization problem on the search step, like restrictions on calling the objective function online. And also, when there is no need to incorporate experience knowledge from different executions of the algorithm. This online classification has diverse ML techniques used to learn from different phases of the algorithm that will be analyzed, detailed and discussed further in the following subsection. It is important to note that none of the resulting papers used both online and off-line strategies in combination.



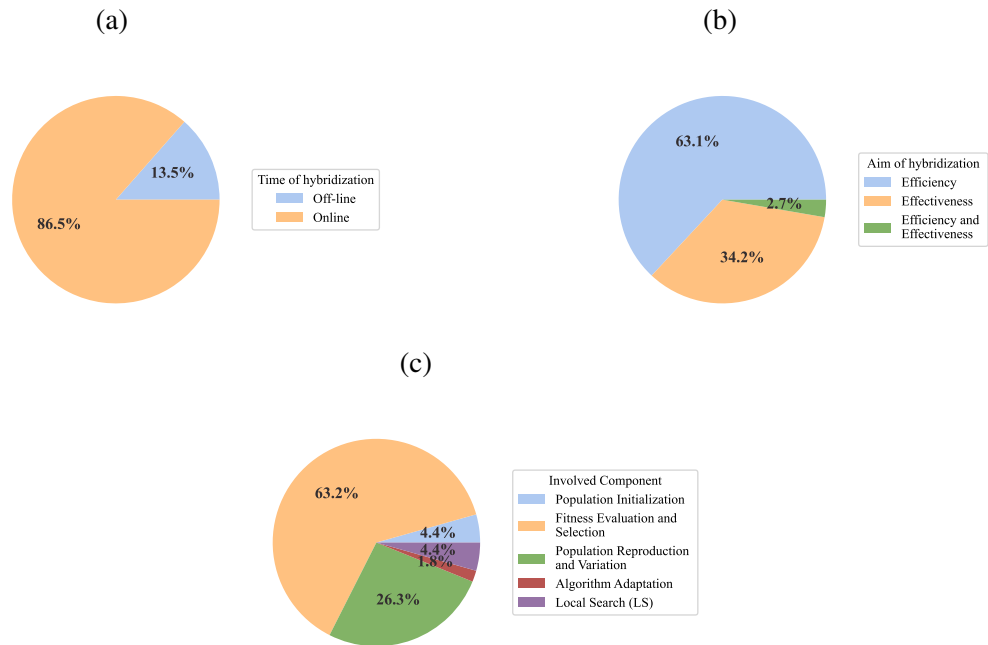
Figure 9 – Number of papers published per year.



Source: Produced by authors.

From the 15 off-line papers, 13 are implementations of Surrogate-Assisted Evolutionary Algorithms (SAEAs) (JIN, 2011). The SAEAs use computational models, also known as surrogates or metamodels, to approximate the objective function. Mainly motivated by reducing computational time while running the search on expensive problems. The papers in (YIN; TSAI, 2020; PROBST et al., 2019; TANG et al., 2020; PYLE; CHIMEH; RICHMOND, 2019; KACEM et al., 2020; WANG et al., 2018; PARQUE; MIYASHITA, 2017; SAIKIA; RAKSHIT et al., 2020; GHORBEL et al., 2019) have an expensive objective function calculated by a simulator. The result of diverse objective function evaluations is made off-line, before the search begins, by the simulator. In some cases, these simulations are executed separately in parallel or using a supercomputer which restricts function calls to be made online. The SAEA in (GARCÍA-ORTEGA; GARCÍA-SÁNCHEZ; MERELO-GUERVÓS, 2020), differently from the others, has an objective function that is not expressed mathematically or by simulators. The objective function of this problem is basically a machine-learning model, in particular, a Multilayer Perceptron (MLP) (GARDNER; DORLING, 1998), trained from data extracted and processed from film databases before the search starts.

Figure 10 – Number of papers for each classification. (a) Time of hybridization (b) Aim of hybridization and (c) Involved Component.



Source: Produced by authors.

### 2.2.2.1.2 Aim of hybridization

The Aim of hybridization classifies the purpose of the ML enhancement. The combination allows to improve the Efficiency of the algorithm by reducing the search time, or the combination is used to improve the Effectiveness of the algorithm leading to better quality of solutions. Table 6 demonstrates our results based on this classification. From the 70 papers classified as Efficiency, all of them are SAEAs implementations. Mainly due to the nature of this kind of optimization algorithm that is used to reduce the computational time by swapping the expensive objective function by a surrogate. Modeling the objective function can be seen as a supervised ML problem, where the solutions are used to train the model according to the real objective function values. Generally, the use of a metamodel to predict the objective function value is less expensive than using the real objective function, causing an increase in the Efficiency of the algorithm.

We have encountered different ways of enhancing Effectiveness in optimization problems. The studies in (LI et al., 2019b; ZANGARI et al., 2017a; MA et al., 2014; VIRGOLIN et al., 2018; HO et al., 2015; ZANGARI et al., 2017b; WANG et al., 2012; QI et al., 2012; CHENG et al., 2015; SHIM et al., 2013; VIRGOLIN et al., 2017; THIERENS; BOSMAN, 2011) are based on, or variations, of the Estimation of Distribution Algorithm (EDA) (LARRAÑAGA; LOZANO, 2001). Instead of using commonly mutation operators of evolutionary algorithms, it extracts statistical information from selected solutions and builds a posterior probability distribution model to predict promising solutions. Also, in these studies, we found algorithms based on

Table 5 – Number of papers based on the classification of Time of hybridization.

| Time of hybridization | Papers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Number of Papers |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Off-line              | (VIJAYACHANDRAN; DAVIDSON; WAAS, 2020; WENG et al., 2020; GARCÍA-ORTEGA; GARCÍA-SÁNCHEZ; MERELO-GUERVÓS, 2020; YIN; TSAI, 2020; PROBST et al., 2019; TANG et al., 2020; MICHALAK, 2018; PYLE; CHIMEH; RICHMOND, 2019; LI; XIANG; TAN, 2019; KACEM et al., 2020; DEY et al., 2019; WANG et al., 2018; PARQUE; MIYASHITA, 2017; SAIKIA; RAKSHIT et al., 2020; GHORBEL et al., 2019)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 15               |
| Online                | (LI et al., 2019b; ZANGARI et al., 2017a; CUI; TURAN; SAYER, 2012; MA et al., 2014; WANG; TANG, 2017; JØRGENSEN; GROVES; HAMMER, 2017; GUO; CHENG; WANG, 2017; ELAZIZ; OLIVA; XIONG, 2017; LI et al., 2019a; LI et al., 2020; SIMONS; SMITH; WHITE, 2014; FARFÁN; CEA, 2021; VOLZ, 2020; SONG et al., 2019; BLIEK; VERWER; WEERDT, 2020; STOEAN, 2020; HARADA; KAIDAN; THAWONMAS, 2020; PARK, 2019; SANTANA et al., 2012; MICHALAK, 2019; VIRGOLIN et al., 2018; HO et al., 2015; PANTULA; MITRA, 2019; DAI; WANG; YANG, 2020; PENNISI; RUSSO; PAPALARDO, 2018; AL-DUOLI et al., 2018; LE et al., 2020; PENG et al., 2019; WU; WANG; HONG, 2020; GAO et al., 2020; HOSSEINZADEH et al., 2020; CAO et al., 2019; STANDER; WOOLWAY; ZYL, 2020; GATOPOULOS et al., 2020; TOSCANO; DEB, 2016; ZHANG; SHAFI; ABBASS, 2016; DU; RADA, 2012; ABDESSALEM et al., 2018; LU; LI; JIN, 2013; JENNINGS et al., 2019; ZANGARI et al., 2017b; GARCÍA et al., 2019; HILDEBRANDT; BRANKE, 2015; PLASTINO et al., 2014; WANG; YIN, 2018; ZHOU; JIN; DING, 2020; NGUYEN; NGUYEN, 2021; SUN et al., 2018; JIANG et al., 2017; REMLI et al., 2017; TIAN et al., 2019; LINDMARK; SERVIN, 2018; BACIGALUPO et al., 2021; LIU; KOZIEL; ZHANG, 2016; XIE et al., 2021; MLAKAR et al., 2015; TIAN et al., 2020b; AL-JUBOORI; DATTA, 2019; GONZALEZ; JALALI; NIEUWENHUYSE, 2020; TIAN et al., 2020a; TENNE, 2012b; KOTINIS, 2014; NOURI; DRISS; GHÉDIRA, 2018; LIU et al., 2011; LIU et al., 2012; KUREK; BECKER; LUK, 2013; AZZOUZ; BECHIKH; SAID, 2014; RANA et al., 2017; OHTSUKA et al., 2018; MAHESWARANATHAN et al., 2019; MARTINEZ-CANTIN, 2018; GARRISON; PETTY, 2019; SHE et al., 2019; SANTOS; OLIVEIRA; BRITTO, 2019; SHAHANE et al., 2020; SONG et al., 2018; WANG et al., 2012; QI et al., 2012; WANG; WU; RAHNAMEYAN, 2011; CHIA et al., 2012; WANG; TANG, 2016; TENNE, 2012a; AKINSOLU et al., 2019; LE et al., 2021; LIU et al., 2014; ARIAS-MONTAÑO; COELLO; MEZURA-MONTES, 2012; ZHANG; SHAFI; ABBASS, 2014; CHENG et al., 2015; SHIM et al., 2013; ALI; SIARRY; PANT, 2012; ZULUAGA; KRAUSE; PÜSCHEL, 2016; VIRGOLIN et al., 2017; THIERENS; BOSMAN, 2011; SUN et al., 2013; TINÓS, 2020) | 96               |

Source – Produced by authors.

well-known EDAs variants such as Population-Based Incremental Learning (PBIL) (BALUJA; CARUANA, 1995) that achieve good results in solving different optimization problems. PBIL is one of the first EDAs found in the literature, that at each generation, updates a vector describing the univariate statistics of the best solutions. A comparison of different PBILs variants is demonstrated in (ZANGARI et al., 2017b), the researchers identify that the type of learning step

Table 6 – Number of papers based on the classification of Aim of hybridization.

| Aim of hybridization         | Papers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Number of Papers |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Efficiency                   | (VIJAYACHANDRAN; DAVIDSON; WAAS, 2020; WENG et al., 2020; GARCÍA-ORTEGA; GARCÍA-SÁNCHEZ; MERELO-GUERVÓS, 2020; YIN; TSAI, 2020; PROBST et al., 2019; GUO; CHENG; WANG, 2017; LI et al., 2019a; LI et al., 2020; SIMONS; SMITH; WHITE, 2014; FARFÁN; CEA, 2021; LI et al., 2019; TANG et al., 2020; VOLZ, 2020; SONG et al., 2019; BLIEK; VERWER; WEERDT, 2020; STOEAN, 2020; HARADA; KAIDAN; THAWONMAS, 2020; PARK, 2019; MICHALAK, 2019; DAI; WANG; YANG, 2020; PENNISI; RUSSO; PAPPALARDO, 2018; LE et al., 2020; PENG et al., 2019; PYLE; CHIMEH; RICHMOND, 2019; WU; WANG; HONG, 2020; GAO et al., 2020; LI; XIANG; TAN, 2019; CAO et al., 2019; STANDER; WOOLWAY; ZYL, 2020; GATOPOULOS et al., 2020; TOSCANO; DEB, 2016; DEY et al., 2019; LU; LI; JIN, 2013; JENNINGS et al., 2019; WANG et al., 2018; HILDEBRANDT; BRANKE, 2015; ZHOU; JIN; DING, 2020; NGUYEN; NGUYEN, 2021; TIAN et al., 2019; LINDMARK; SERVIN, 2018; BACIGALUPO et al., 2021; PARQUE; MIYASHITA, 2017; SAIKIA; RAKSHIT et al., 2020; XIE et al., 2021; MLAKAR et al., 2015; TIAN et al., 2020b; AL-JUBOORI; DATTA, 2019; GONZALEZ; JALALI; NIEUWENHUYSE, 2020; TIAN et al., 2020a; TENNE, 2012b; LIU et al., 2011; LIU et al., 2012; KUREK; BECKER; LUK, 2013; AZZOUZ; BECHIKH; SAID, 2014; RANA et al., 2017; OHTSUKA et al., 2018; MAHESWARANATHAN et al., 2019; MARTINEZ-CANTIN, 2018; GARRISON; PETTY, 2019; SHE et al., 2019; GHORBEL et al., 2019; SHAHANE et al., 2020; SONG et al., 2018; TENNE, 2012a; AKINSOLU et al., 2019; LE et al., 2021; LIU et al., 2014; ARIAS-MONTAÑO; COELLO; MEZURA-MONTES, 2012; ZULUAGA; KRAUSE; PÜSCHEL, 2016; SUN et al., 2013) | 70               |
| Effectiveness                | (LI et al., 2019b; ZANGARI et al., 2017a; MA et al., 2014; WANG; TANG, 2017; JØRGENSEN; GROVES; HAMMER, 2017; ELAZIZ; OLIVA; XIONG, 2017; SANTANA et al., 2012; VIRGOLIN et al., 2018; MICHALAK, 2018; HO et al., 2015; PANTULA; MITRA, 2019; AL-DUOLI et al., 2018; HOSSEINZADEH et al., 2020; ZHANG; SHAFI; ABBASS, 2016; DU; RADA, 2012; KACEM et al., 2020; ABDESSALEM et al., 2018; ZANGARI et al., 2017b; GARCÍA et al., 2019; PLASTINO et al., 2014; WANG; YIN, 2018; SUN et al., 2018; JIANG et al., 2017; REMLI et al., 2017; KOTINIS, 2014; NOURI; DRISS; GHÉDIRA, 2018; WANG et al., 2012; QI et al., 2012; WANG; WU; RAHNAMAYAN, 2011; CHIA et al., 2012; WANG; TANG, 2016; ZHANG; SHAFI; ABBASS, 2014; CHENG et al., 2015; SHIM et al., 2013; ALI; SIARRY; PANT, 2012; VIRGOLIN et al., 2017; THIERENS; BOSMAN, 2011; TINÓS, 2020)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 38               |
| Efficiency and Effectiveness | (CUI; TURAN; SAYER, 2012; LIU; KOZIEL; ZHANG, 2016; SANTOS; OLIVEIRA; BRITTO, 2019)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                |

Source – Produced by authors.

implementation plays a crucial role in the results of the algorithm in the search process. So the choice of the PBIL variant can lead to different results in different optimization problems. In (HO et al., 2015), a PBIL implementation for inverse Problems in continuous Space is proposed, where the default implementation is changed to adapt to the optimization problem. Another variant of EDA is proposed by (THIERENS; BOSMAN, 2011), called the Gene-pool Optimal Mixing

Evolutionary Algorithm (GOMEA). This variant mixes EDA with Genetic Algorithm (GA), using a linkage tree (LT) structure to identify the contributions of the variables. Also, another study found in (VIRGOLIN et al., 2017) improves the GOMEA with Genetic Programming (GP), implementing the algorithm GP-GOMEA that requires no parameter specification. In (VIRGOLIN et al., 2018) the GP-GOMEA is adapted to resolve real optimization problems. The GOMEA algorithm and its variants have proven to have good results in real optimization problems and benchmark problems.

Another algorithm found in our results is the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) (HANSEN; OSTERMEIER, 2001). Different from the EDA, it uses a covariance matrix distribution to sample new solutions in the population. In (SANTANA et al., 2012) the CMA-ES is adapted to resolve an optimization problem of classifying Brain Computer Interfaces (BCIs). In (JIANG et al., 2017), the CMA-ES is integrated with the Differential Evolution (DE), a well-known optimization algorithm. It achieves relevant results in benchmarking problems in comparison with other state-of-art algorithms.

Three different papers draw attention by reaching the Efficiency and Effectiveness mutually. In (CUI; TURAN; SAYER, 2012), a reinforcement learning approach is proposed, the method is used to reduce the computation time and also correct the search leading to better solutions. The algorithm MGPMDE proposed by (LIU; KOZIEL; ZHANG, 2016) is a Multi-fidelity/Multilevel/Variable-fidelity Optimization (MFO) method (EL-BELTAGY; KEANE, 1999). This technique uses two ML models, the first one is an approximation model that can be referenced as a surrogate model acting in the improvement of the Efficiency of the metaheuristic. The second one is a low-fidelity model that explores the space leading to an improvement in Effectiveness. The DVL algorithm proposed by (SANTOS; OLIVEIRA; BRITTO, 2019) uses the values found by the objective function evaluations for training a metamodel that predicts promising areas of the search space. The new solutions have better quality enhancing the Effectiveness, and also the algorithm reduces the number of evaluations improving the Efficiency.

### 2.2.2.1.3 Involved Component

The classification of the component involved, based on the taxonomy of EC showed in Figure 3, can be seen in Table 7. Only 5 papers were produced to improve the quality of the initial solutions. The main technique used to enhance this component is the opposition-based learning (OBL) (TIZHOOSH, 2005). Basically, it assumes that the initial solutions generated randomly can be far away or not far away from the optimal solutions. If an initial solution is close, it can converge fast to the optimal, but if it is far away, it can lead to a slow convergence of the Pareto Front. To reduce the performance impact due to the convergence of the distant solutions, the OBL proposes to calculate the opposite solutions from the solutions set. These new opposite solutions can lead the search in the opposite direction, improving it by realizing a fast convergence to the optimal set. In (ELAZIZ; OLIVA; XIONG, 2017; WANG; WU; RAHNAMAYAN, 2011;

[ALI; SIARRY; PANT, 2012](#)), the OBL shows to enhance different optimization algorithms such as the sine cosine algorithm (SCA) and the DE in many optimization problems. The other 2 papers improve the initial solutions in different ways. In ([KACEM et al., 2020](#)), an ANN is used to learning from the experience to improve initial solutions quality. And in ([LIU; KOZIEL; ZHANG, 2016](#)), a surrogate is used for validating the searching of initial solutions followed by a clustering method to choose the best group of solutions.

The most found component involved in the resulting papers is the Fitness Evaluation and Selection, mainly due to the high use of SAEAs algorithms. All papers, from a total of 71 are SAEAs implementations or variations that use in some way a metamodel to substitute the expensive objective function. Some of the papers found investigate the enhancement by selecting a portion of the population. In ([LIU et al., 2014](#)), the surrogate model is first trained with a population created by Latin Hypercubic Sampling (LHS) ([IMAN; DAVENPORT; ZEIGLER, 1980](#)), which generates samples on a multidimensional distribution. After this step, during the generations, the model is trained only with the best solution found. This strategy made it possible to achieve global optimization, support the search, and use a small number of objective function evaluations. Another strategy found for training the surrogate is proposed by ([SONG et al., 2018](#)), where it adaptively improves the model by using the Pareto optimal solutions in the local search. These solutions are found by some well-known metrics used in MOPs such as Hypervolume (HV) ([WHILE et al., 2006](#)) and Inverted Generational Distance (IGD) ([ISHIBUCHI et al., 2015](#)). The HV metric is also used by the SAEA in ([AZZOUZ; BECHIKH; SAID, 2014](#)). In the same way as the previous paper, it uses the metric to found the best solutions for train the model. The rest of the SAEAs found are original implementations that change the model used, the optimization problem, or the optimization algorithm. These changes will be detailed analyzed in the next subsections.

A total of 30 papers were found that acts on the component of Population Reproduction and Variation. A representative portion of these papers, 14 from 30, proposed by ([LI et al., 2019b](#); [ZANGARI et al., 2017a](#); [MA et al., 2014](#); [SANTANA et al., 2012](#); [VIRGOLIN et al., 2018](#); [HO et al., 2015](#); [ZANGARI et al., 2017b](#); [JIANG et al., 2017](#); [WANG et al., 2012](#); [QI et al., 2012](#); [CHENG et al., 2015](#); [SHIM et al., 2013](#); [VIRGOLIN et al., 2017](#); [THIERENS; BOSMAN, 2011](#)), are variations from the EDA and CMA-ES. Mainly because of the nature of the implementation of these algorithms that uses a probabilistic model and a covariance matrix to learn from the problem and assists the reproduction by predicting promise areas of the solutions that can guide the algorithm to better convergence. The 16 remaining papers apply different strategies to achieve good results, such as clustering, association rules, and supervised tasks to extract the knowledge and enhance the solutions. As an example, the work in ([TINÓS, 2020](#)) proposes a new method that uses a radial basis function network (RBFN) to generate crossover masks. The RBFN model is trained as a supervised task where incorporates the knowledge from successful recombination operations in the GA algorithm during the optimization process.



Table 7 – Number of papers based on the classification of Involved Component.

| Involved Component                    | Papers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Number of Papers |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Population Initialization             | (ELAZIZ; OLIVA; XIONG, 2017; KACEM et al., 2020; WANG; WU; RAHNAMAYAN, 2011; ALI; SIARRY; PANT, 2012; LIU; KOZIEL; ZHANG, 2016)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5                |
| Fitness Evaluation and Selection      | (VIJAYACHANDRAN; DAVIDSON; WAAS, 2020; WENG et al., 2020; GARCÍA-ORTEGA; GARCÍA-SÁNCHEZ; MERELO-GUERVÓS, 2020; YIN; TSAI, 2020; PROBST et al., 2019; CUI; TURAN; SAYER, 2012; GUO; CHENG; WANG, 2017; LI et al., 2019a; LI et al., 2020; SIMONS; SMITH; WHITE, 2014; FARFÁN; CEA, 2021; LI et al., 2019; TANG et al., 2020; VOLZ, 2020; SONG et al., 2019; BLIEK; VERWER; WEERDT, 2020; STOEAN, 2020; HARADA; KAIDAN; THAWONMAS, 2020; PARK, 2019; MICHALAK, 2019; DAI; WANG; YANG, 2020; PENNISI; RUSSO; PAPPALARDO, 2018; LE et al., 2020; PENG et al., 2019; PYLE; CHIMEH; RICHMOND, 2019; WU; WANG; HONG, 2020; GAO et al., 2020; LI; XIANG; TAN, 2019; CAO et al., 2019; STANDER; WOOLWAY; ZYL, 2020; GATOPOULOS et al., 2020; TOSCANO; DEB, 2016; DEY et al., 2019; LU; LI; JIN, 2013; JENNINGS et al., 2019; WANG et al., 2018; HILDEBRANDT; BRANKE, 2015; ZHOU; JIN; DING, 2020; NGUYEN; NGUYEN, 2021; TIAN et al., 2019; LINDMARK; SERVIN, 2018; BACIGALUPO et al., 2021; PARQUE; MIYASHITA, 2017; SAIKIA; RAKSHIT et al., 2020; LIU; KOZIEL; ZHANG, 2016; XIE et al., 2021; MLAKAR et al., 2015; TIAN et al., 2020b; AL-JUBOORI; DATTA, 2019; GONZALEZ; JALALI; NIEUWENHUYSE, 2020; TIAN et al., 2020a; TENNE, 2012b; LIU et al., 2011; LIU et al., 2012; KUREK; BECKER; LUK, 2013; AZZOUZ; BECHIKH; SAID, 2014; RANA et al., 2017; OHTSUKA et al., 2018; MAHESWARANATHAN et al., 2019; MARTINEZ-CANTIN, 2018; GARRISON; PETTY, 2019; SHE et al., 2019; GHORBEL et al., 2019; SHAHANE et al., 2020; SONG et al., 2018; TENNE, 2012a; AKINSOLU et al., 2019; LE et al., 2021; LIU et al., 2014; ARIAS-MONTAÑO; COELLO; MEZURA-MONTES, 2012; ZULUAGA; KRAUSE; PÜSCHEL, 2016; SUN et al., 2013) | 72               |
| Population Reproduction and Variation | (LI et al., 2019b; ZANGARI et al., 2017a; MA et al., 2014; WANG; TANG, 2017; JØRGENSEN; GROVES; HAMMER, 2017; SANTANA et al., 2012; VIRGOLIN et al., 2018; MICHALAK, 2018; HO et al., 2015; PANTULA; MITRA, 2019; AL-DUOLI et al., 2018; HOSSEINZADEH et al., 2020; ZHANG; SHAFI; ABBASS, 2016; DU; RADA, 2012; ZANGARI et al., 2017b; WANG; YIN, 2018; SUN et al., 2018; JIANG et al., 2017; REMLI et al., 2017; SANTOS; OLIVEIRA; BRITTO, 2019; WANG et al., 2012; QI et al., 2012; WANG; WU; RAHNAMAYAN, 2011; CHIA et al., 2012; ZHANG; SHAFI; ABBASS, 2014; CHENG et al., 2015; SHIM et al., 2013; VIRGOLIN et al., 2017; THIERENS; BOSMAN, 2011; TINÓS, 2020)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 30               |
| Algorithm Adaptation                  | (AL-DUOLI et al., 2018; KOTINIS, 2014)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2                |
| Local Search (LS)                     | (ABDESSALEM et al., 2018; GARCÍA et al., 2019; PLASTINO et al., 2014; NOURI; DRISS; GHÉDIRA, 2018; WANG; TANG, 2016)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5                |

Source – Produced by authors.

Only 2 papers are found that enhance the Parameter Adaptation of metaheuristics. Both improve the parameters dynamically during the search. The algorithm proposed by (KOTINIS, 2014), uses clustering with fuzzy controllers to adapt the DE parameters. While in (AL-DUOLI

et al., 2018), a decision tree (DT) obtains knowledge based on how the value of the parameter provides good solutions. The last taxonomy to analyze is the Local Search where 5 papers were found that enhances this component. The papers in (ABDESSALEM et al., 2018; GARCÍA et al., 2019; WANG; TANG, 2016; NOURI; DRISS; GHÉDIRA, 2018) use ML to determine where and when to perform a Local Search. This knowledge is extracted from the problem, and different local search operations are proposed to better the divergence and convergence of the algorithm. In (PLASTINO et al., 2014) the Local Search component is enhanced by refining solutions within the solution's neighborhood from patterns extracted of elite solutions.

### 2.2.2.2 RQ2. What are the Machine Learning techniques used in Metaheuristic algorithms?

For a better understand and organization, we separate this subsection according to the ML classification by the supervision on training, defined as Supervised Learning, Unsupervised Learning, Semi-supervised Learning, and Reinforcement Learning. The results found are shown in Table 8.

The majority of the papers, a total of 87 uses Supervised Learning to enhance the metaheuristic. The high presence of papers based on the SAEAs algorithm plays an important role in this statistic. In many cases, the metamodel used on an SAEA is modeled as regression, generally because the objective problem is a continuous value. Moreover, the vector of decision variables (or solution) is used as a feature vector for the model. A variety of well-known computational models in the literature are used in this task, including polynomials (also known as response surface methodologies), Gaussian processes (also known as Kriging), neural networks (JIN, 2011). The models and metaheuristics found that were used in more than three papers are shown in Figure 11. Kriging and ANNs were the most used models, and GA, PSO, and DE the metaheuristics for mono-objective and NSGA-II and MOEA/D for multi-objective.

Different strategies and techniques found were used to enhance the prediction of the models. The papers in (ARIAS-MONTAÑO; COELLO; MEZURA-MONTES, 2012; AKINSOLU et al., 2019; PENNISI; RUSSO; PAPPALARDO, 2018; HARADA; KAIDAN; THAWONMAS, 2020) differ from other implementations by using surrogates in parallelism. Their results showed that this strategy can benefit some optimization problems. In (YIN; TSAI, 2020; LU; LI; JIN, 2013; WANG et al., 2018; AL-JUBOORI; DATTA, 2019) the ensemble learning (MENDES-MOREIRA et al., 2012) is employed to enhance the prediction performance of the surrogate model. Bagging and boosting are two popular ensemble generation methods, and are also used in these works to outperforms a single surrogate model. The use of cross-validation (BROWNE, 2000) had brought benefits and security to the model selection of the papers in (GARCÍA-ORTEGA; GARCÍA-SÁNCHEZ; MERELO-GUERVÓS, 2020; FARFÁN; CEA, 2021; VOLZ, 2020; TIAN et al., 2019; GHORBEL et al., 2019; TENNE, 2012a). It applies a re-sampling procedure used to evaluate ML models on the data sample, encouraging the selection of a determined model that



Table 8 – Number of papers based on the classification of Machine Learning.

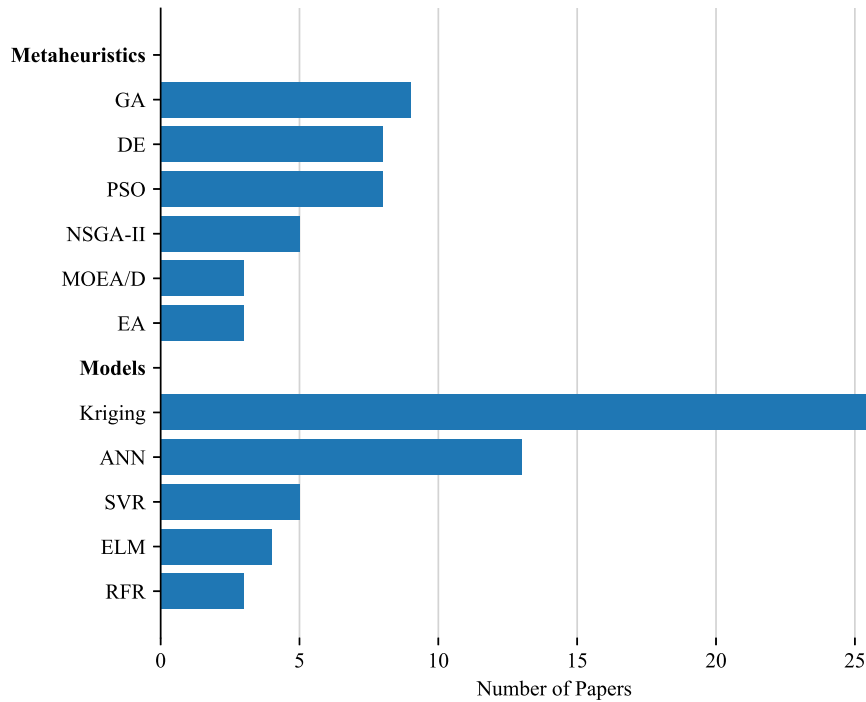
| Type                     | Papers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Number of Papers |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Supervised Learning      | (VIJAYACHANDRAN; DAVIDSON; WAAS, 2020; WENG et al., 2020; GARCÍA-ORTEGA; GARCÍA-SÁNCHEZ; MERELO-GUERVÓS, 2020; YIN; TSAI, 2020; PROBST et al., 2019; LI et al., 2019b; ZANGARI et al., 2017a; WANG; TANG, 2017; GUO; CHENG; WANG, 2017; LI et al., 2019a; LI et al., 2020; SIMONS; SMITH; WHITE, 2014; FARFÁN; CEA, 2021; LI et al., 2019; TANG et al., 2020; VOLZ, 2020; SONG et al., 2019; BLIEK; VERWER; WEERDT, 2020; STOEAN, 2020; HARADA; KAIDAN; THAWONMAS, 2020; PARK, 2019; MICHALAK, 2019; MICHALAK, 2018; DAI; WANG; YANG, 2020; PENNISI; RUSSO; PAPPALARDO, 2018; AL-DUOLI et al., 2018; PENG et al., 2019; PYLE; CHIMEH; RICHMOND, 2019; WU; WANG; HONG, 2020; GAO et al., 2020; HOSSEINZADEH et al., 2020; LI; XIANG; TAN, 2019; CAO et al., 2019; STANDER; WOOLWAY; ZYL, 2020; GATOPOULOS et al., 2020; TOSCANO; DEB, 2016; DU; RADA, 2012; KACEM et al., 2020; DEY et al., 2019; ABDESSALEM et al., 2018; LU; LI; JIN, 2013; JENNINGS et al., 2019; ZANGARI et al., 2017b; WANG et al., 2018; HILDEBRANDT; BRANKE, 2015; ZHOU; JIN; DING, 2020; NGUYEN; NGUYEN, 2021; JIANG et al., 2017; TIAN et al., 2019; LINDMARK; SERVIN, 2018; BACIGALUPO et al., 2021; PARQUE; MIYASHITA, 2017; SAIKIA; RAKSHIT et al., 2020; LIU; KOZIEL; ZHANG, 2016; XIE et al., 2021; MLAKAR et al., 2015; TIAN et al., 2020b; AL-JUBOORI; DATTA, 2019; GONZALEZ; JALALI; NIEUWENHUYSE, 2020; TIAN et al., 2020a; TENNE, 2012b; LIU et al., 2011; LIU et al., 2012; KUREK; BECKER; LUK, 2013; AZZOUZ; BECHIKH; SAID, 2014; RANA et al., 2017; OHTSUKA et al., 2018; MAHESWARANATHAN et al., 2019; MARTINEZ-CANTIN, 2018; GARRISON; PETTY, 2019; SHE et al., 2019; SANTOS; OLIVEIRA; BRITTO, 2019; GHORBEL et al., 2019; SHAHANE et al., 2020; SONG et al., 2018; WANG et al., 2012; TENNE, 2012a; AKINSOLU et al., 2019; LIU et al., 2014; ARIAS-MONTAÑO; COELLO; MEZURA-MONTES, 2012; CHENG et al., 2015; SHIM et al., 2013; ZULUAGA; KRAUSE; PÜSCHEL, 2016; VIRGOLIN et al., 2017; THIERENS; BOSMAN, 2011; HO et al., 2015; TINÓS, 2020) | 87               |
| Unsupervised Learning    | (MA et al., 2014; WANG; TANG, 2017; JØRGENSEN; GROVES; HAMMER, 2017; LI et al., 2019; SANTANA et al., 2012; VIRGOLIN et al., 2018; PANTULA; MITRA, 2019; AL-DUOLI et al., 2018; LE et al., 2020; ZHANG; SHAFI; ABBASS, 2016; GARCÍA et al., 2019; PLASTINO et al., 2014; WANG; YIN, 2018; SUN et al., 2018; PARQUE; MIYASHITA, 2017; LIU; KOZIEL; ZHANG, 2016; KOTINIS, 2014; NOURI; DRISS; GHÉDIRA, 2018; QI et al., 2012; LE et al., 2021; ZHANG; SHAFI; ABBASS, 2014)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 21               |
| Semi-supervised Learning | (SUN et al., 2013)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1                |
| Reinforcement Learning   | (CUI; TURAN; SAYER, 2012)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1                |

Source – Produced by authors.

yields a trustworthy calibration from the sample.

The regression task can also be employed in algorithms such as EDAs and CMAEs. The continuous values predicted by the model are used to support the population reproduction and

Figure 11 – Main models and metaheuristics used in at least three papers.



Source: Produced by authors.

variation on these algorithms, which allow an understanding of the problem and propose valuable areas of the search space. A different way of using Supervisioned Learning to extract knowledge from the problem are the ones proposed by (CHENG et al., 2015; SANTOS; OLIVEIRA; BRITTO, 2019). In the first paper, an EDA algorithm called IM-MOEA is proposed, where it uses the concept of inverse models, that map all found nondominated solutions from the objective space to the decision space. These models receive as input the values of the objective functions and learn their influence on the variables (or solutions). Like other EDAs, this model is used to predict offspring samples of the decision space. The second paper uses the same concept of modeling inversely. But, different from the EDAs, it uses the model to direct predict solution on the space that approximates to optimum values. Achieving in the evolution process both divergence and convergence.

Another well-known Supervisioned Learning task is the classification. It can be used in the SAEAs to model the constraints, as proposed in (GAO et al., 2020; KUREK; BECKER; LUK, 2013; TENNE, 2012a, 2012a). The classification model is trained in the same manner as the regression, and it is used to predict which solutions satisfy the constraints of the optimization problem. The classification model in (AL-DUOLI et al., 2018) extracts knowledge from the problem, making it possible to predict good solutions that can optimize the algorithm. In (PYLE; CHIMEH; RICHMOND, 2019) the surrogate is created in two different ways, by regression,

and by classification. An ANN is used for both models, and to accomplish the surrogate as a classification task, it uses discrete values of the objective function as the labels in the training step. In their experiments, the classifier achieves better results. A classifier model can also be used to add knowledge to the mutation operator as in (MICHALAK, 2018). The training of the model occurs offline, and during the search, it uses the informed mutation operator. In the proposed operator the classifier is used for determining which bits in a solution vector to consider for mutation.

From the Unsupervised Learning was found a total of 21 papers. Clustering is the most used task from this classification. The papers in (MA et al., 2014; WANG; TANG, 2017; VIRGOLIN et al., 2018; QI et al., 2012) are EDAs or variants that use probabilistic models to guide the solutions in the search space. The clustering is used in these papers to support the algorithm learning method. In (MA et al., 2014) for example, the clustering method is used to decompose the MOP into some scalar optimization subproblems. The learning strategy used, called Baldwin learning, is applied in each cluster to find the direction of the cluster in the search space. The algorithm proposed in (QI et al., 2012) uses a similar learning strategy. The clustering method is used to divide the population into K disjoint clusters. And a model is built for each cluster to sampling new solutions at each generation.

Some clustering strategies act directly on the objective modeling by speeding up the search time or reducing the number of evaluations. The cluster method proposed by (LE et al., 2020; LE et al., 2021; LI et al., 2019) evaluates only the solutions closest to the cluster center, the objective function value of the remaining solutions is calculated by an approximation to their distance to the evaluated solutions. These algorithms differ in the calculation strategy used to infer the objective function values of new solutions. Clustering can also be used to perform different kinds of searches. In (GARCÍA et al., 2019) a Db-Scan cluster operator is used to create different groups of the population. In each group is executed a different approach of search, based on the knowledge extract from the group.

Association rule learning is another kind of unsupervised learning task. We have found 4 papers that use association rules to enhance metaheuristics. The approaches found in (AL-DUOLI et al., 2018; WANG; YIN, 2018) uses association rules to enhance the effectiveness of the algorithm. Data Mining techniques are used to discover patterns associated with good solutions. This knowledge is used during the search to improve the convergence to better solutions. In (ZHANG; SHAFI; ABBASS, 2014), instead of search rules on the solutions, the association rules are discovered from optimal areas of the design space. The knowledge extracted improves the convergence of the algorithm by leading it to search for optimal areas. A new design of local search is proposed by (PLASTINO et al., 2014), where after a predefined number of iterations the local search is conducted by association rules extracted from elite solutions.

We have found only one paper that contributes by using a Semi-supervised Learning approach. In (SUN et al., 2013) the Cotraining Algorithm for Semi-supervised Learning (CSSL)

is used to handle the inherent uncertainties in fitness assignment by human users, where, interval-based fitness values are assigned to individuals. Also, only one paper found that uses Reinforcement Learning. Already noticed in previous sections, the algorithm proposed by (CUI; TURAN; SAYER, 2012) uses Q-learning (WATKINS; DAYAN, 1992), a model-free reinforcement learning technique. That uses the predicted Q-value calculation to replace the expensive objective function evaluation and also to improve the operation of the selection of offspring solutions, leading to a correction on the search direction and improving the quality of the solutions.

### 2.2.2.3 RQ3. Which strategies were used to validate the algorithm?

This question is to determine the validating techniques used to investigate the performance of metaheuristics algorithms enhanced by ML. The analyses performed on the optimization algorithms are mostly quantitative via validation metrics like mean error, standard deviation, and co-relations. Researchers have most commonly used benchmark test functions to validate the quality of optimization procedures. Details about these functions are available in (MOLGA; SMUTNICKI, 2005). Researchers also use state-of-art metaheuristics from literature for comparison with the new algorithm.

In most papers, the researchers have commonly used well-known benchmark test functions. Table 9 shows which kind of benchmark was used in each paper found. Some of them found in the papers are ZDT (ZITZLER; DEB; THIELE, 2000), DTLZ (DEB et al., 2002b), CF (ZHANG et al., 2008), F functions from the CEC 2005 (SUGANTHAN et al., 2005), the test functions in (JAMIL; YANG, 2013), and WFG (HUBAND et al., 2005). Beyond the test functions, some researchers used problem-specific benchmark problems, such as the benchmark data for FJSSP in (CHAMBERS; BARNES, 1996; DAUZÈRE-PÉRÈS; PAULLI, 1997), benchmark instances for PFSP (MINELLA; RUIZ; CIAVOTTA, 2008), CMT benchmark problems for the Capacitated Vehicle Routing Problem (CVRP), and the Mazda benchmark problem of Multiple Car Structures (KOHIRA et al., 2017).

While comparing the performance with other metaheuristics, the PSO and NSGA-II are the most used with 10 appearances each, as shown in Figure 12. The GA, MOEA/D and DE, appears right after with 8, 7 and 6 references each. If we use the nature of the metaheuristic for comparison, the algorithms based on GA takes the lead, since the NSGA-II is based on GA for problems with multiple objectives and the combination of these two algorithms appears in 15 papers.

Diverse validation criteria were used for comparison. These include mean, median, best, worst, and standard deviation of the objective function values or other metrics, such as the Hypervolume and Normalized Hypervolume, GD and IGD, c-metric, and e-indicator. Other than comparing with these methods, some statistical analysis techniques have also been utilized, such as the Wilcoxon SignedRank test, Friedman's test, t-test (ZIMMERMAN, 1997),

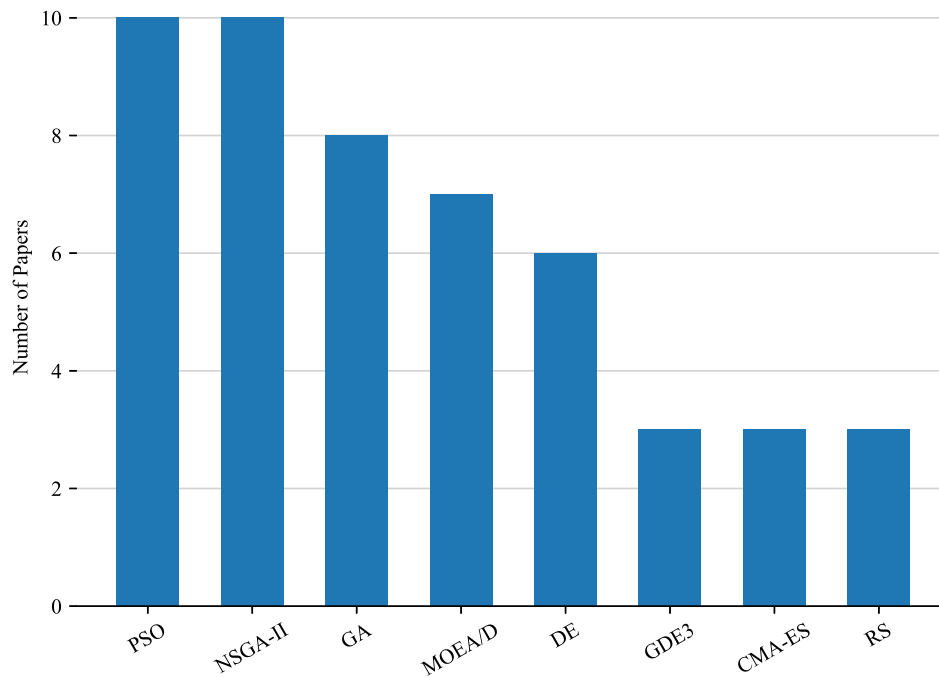
Table 9 – Papers classified by the use of benchmark functions.

| Kind of the Function | Papers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Number of Papers |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Test functions       | (MA et al., 2014; ELAZIZ; OLIVA; XIONG, 2017; LI et al., 2019a; LI et al., 2019; HARADA; KAIDAN; THAWONMAS, 2020; PENG et al., 2019; LI; XIANG; TAN, 2019; TOSCANO; DEB, 2016; ZHANG; SHAFI; ABBASS, 2016; ZANGARI et al., 2017b; WANG et al., 2018; WANG; YIN, 2018; SUN et al., 2018; JIANG et al., 2017; REMLI et al., 2017; LIU; KOZIEL; ZHANG, 2016; MLAKAR et al., 2015; TIAN et al., 2020b; GONZALEZ; JALALI; NIEUWENHUYSE, 2020; TIAN et al., 2020a; KOTINIS, 2014; AZZOUZ; BECHIKH; SAID, 2014; RANA et al., 2017; MARTINEZ-CANTIN, 2018; GARRISON; PETTY, 2019; SANTOS; OLIVEIRA; BRITTO, 2019; QI et al., 2012; WANG; WU; RAHNAMAYAN, 2011; CHIA et al., 2012; WANG; TANG, 2016; AKINSOLU et al., 2019; LIU et al., 2014; ZHANG; SHAFI; ABBASS, 2014; CHENG et al., 2015; ALI; SIARRY; PANT, 2012; THIERENS; BOSMAN, 2011; NGUYEN; NGUYEN, 2021; SUN et al., 2013)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 37               |
| Problem-specific     | (LI et al., 2019b; ZANGARI et al., 2017a; WANG; TANG, 2017; VOLZ, 2020; SONG et al., 2019; AL-DUOLI et al., 2018; TENNE, 2012b; NOURI; DRISS; GHÉDIRA, 2018; OHTSUKA et al., 2018; WANG et al., 2012; ARIAS-MONTAÑO; COELLO; MEZURA-MONTES, 2012; ZULUAGA; KRAUSE; PÜSCHEL, 2016; VIRGOLIN et al., 2017; TINÓS, 2020)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 14               |
| None                 | (VIJAYACHANDRAN; DAVIDSON; WAAS, 2020; WENG et al., 2020; GARCÍA-ORTEGA; GARCÍA-SÁNCHEZ; MERELO-GUERVÓS, 2020; YIN; TSAI, 2020; PROBST et al., 2019; CUI; TURAN; SAYER, 2012; JØRGENSEN; GROVES; HAMMER, 2017; GUO; CHENG; WANG, 2017; LI et al., 2020; SIMONS; SMITH; WHITE, 2014; FARFÁN; CEA, 2021; TANG et al., 2020; BLIEK; VERWER; WEERDT, 2020; STOEAN, 2020; PARK, 2019; SANTANA et al., 2012; MICHALAK, 2019; VIRGOLIN et al., 2018; MICHALAK, 2018; HO et al., 2015; PANTULA; MITRA, 2019; DAI; WANG; YANG, 2020; PENNISI; RUSSO; PAPPALARDO, 2018; LE et al., 2020; PYLE; CHIMEH; RICHMOND, 2019; WU; WANG; HONG, 2020; GAO et al., 2020; HOSSEINZADEH et al., 2020; CAO et al., 2019; STANDER; WOOLWAY; ZYL, 2020; GATOPOULOS et al., 2020; DU; RADA, 2012; KACEM et al., 2020; DEY et al., 2019; ABDESSALEM et al., 2018; LU; LI; JIN, 2013; JENNINGS et al., 2019; GARCÍA et al., 2019; HILDEBRANDT; BRANKE, 2015; PLASTINO et al., 2014; ZHOU; JIN; DING, 2020; TIAN et al., 2019; LINDMARK; SERVIN, 2018; BACIGALUPO et al., 2021; PARQUE; MIYASHITA, 2017; SAIKIA; RAKSHIT et al., 2020; XIE et al., 2021; AL-JUBOORI; DATTA, 2019; LIU et al., 2011; LIU et al., 2012; KUREK; BECKER; LUK, 2013; MAHESWARANATHAN et al., 2019; SHE et al., 2019; GHORBEL et al., 2019; SHAHANE et al., 2020; SONG et al., 2018; TENNE, 2012a; LE et al., 2021; SHIM et al., 2013) | 60               |

Source – Produced by authors.

Kruskal–Wallis test (DERRAC et al., 2011). The most used metric for comparison is the value of the objective function, as shown in Figure 13, mainly due to the problems from these papers being mono-objective. For multi-objectives, the Hypervolume and IGD metrics appear in a greater quantity of papers. Concerning the statistical analysis technique, the Wilcoxon SignedRank test is the most used.

Figure 12 – Metaheuristics used for comparison with at least three papers.



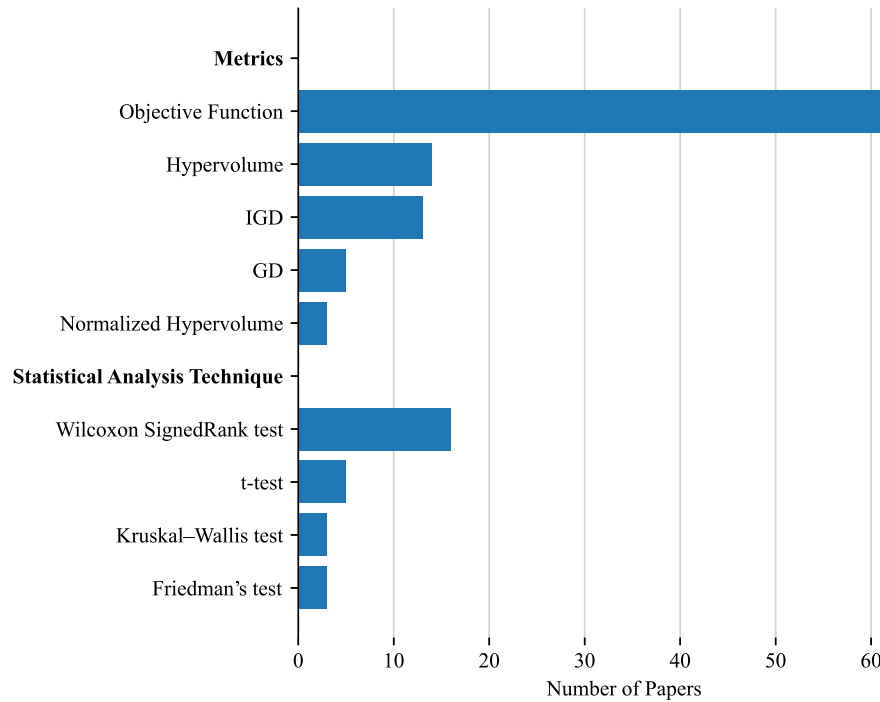
Source: Produced by authors.

#### 2.2.2.4 RQ4. In which real-world optimization problems the algorithms were applied?

Some aspects of real-world optimization problems make them very challenging. Some examples are the presence of an enormous number of local optima within the search space, the high number of the decision variable, and an expensive objective function. In this subsection we describe some aspects of these problems.

Most of the real-world optimization problems found are related to the set of problems that optimizes some aspect of the Machine Learning algorithm. The approaches found in the studies were used to optimizes the ML data preprocessing step, the ANN architecture, and the hyperparameters of the algorithm. Data preprocessing is a key step in a data mining application or ML process, many problems of this step, such as feature selection or instance selection can be modeled as an optimization problem (LE et al., 2020). In (LE et al., 2020; LE et al., 2021) they proposed optimization algorithms for the problem of undersampling in imbalanced classification problems. The use of advanced techniques and algorithms to automatically design the architecture of ANNs is a problem that has drawn attention in the literature, and some studies are achieving good results. The papers in (GATOPOULOS et al., 2020; ZHOU; JIN; DING, 2020; XIE et al., 2021; MAHESWARANATHAN et al., 2019; STOEAN, 2020) are studies found in our review that specifically deal with tuning the architecture of ANNs. Another way of increasing the performance of an ML model involves the tuning of hyperparameters. These hyperparameters

Figure 13 – Metrics used for validation with at least three papers.



Source: Produced by authors.

depend on the learning algorithm, which causes the characteristics of the search problem to depend on the learning technique. In (PENG et al., 2019; MARTINEZ-CANTIN, 2018) are proposed algorithms that cover the hyperparameters improvement in general.

A set of problems found requires the use of expensive simulators. Computational Fluid Dynamics (CFD) gives simulations to evaluate the performance of a given structure. These simulations can take over ten-hour to perform one calculation (JIN, 2005). The problems in (WANG et al., 2018; TENNE, 2012b; AZZOUZ; BECHIKH; SAID, 2014; TENNE, 2012a; ARIAS-MONTAÑO; COELLO; MEZURA-MONTES, 2012; GHORBEL et al., 2019; MARTINEZ-CANTIN, 2018) aim to resolve aerodynamic design problems, principally related to the shape and design of airfoils and wings. In some cases as in (TANG et al., 2020), the aerodynamic of the entire underwater vehicle's shape is an expensive problem to be optimized, which also makes use of the CFD simulation. In some problems as in (PARK, 2019), the simulation is used for computing the gas cyclone shape to evaluate the separation performance, separation efficiency, and pressure drop. The last problem found that uses CFD is the one proposed by (PROBST et al., 2019). The need for engine simulation makes use of CFD for modeling the combustion in the system. The Variable-fidelity Electromagnetic Model (EM) is another kind of computationally expensive simulation that is widely used in antenna array design optimization (KOZIEL; LEIFSSON; YANG, 2014). The papers in (WU; WANG; HONG, 2020; CAO et al., 2019; LIU; KOZIEL;



ZHANG, 2016; LIU et al., 2014) resolve this design problem with different surrogates strategies and achieve good results in their experiments.

Different areas have been studying and developing methods to resolve the scheduling problems (TAILLARD, 1993). Which basically is the allocation of a job to a machine in a determined time. This type of problem can be subdivided into different problems. We have encountered in (WANG et al., 2012; LI et al., 2019b; NOURI; DRISS; GHÉDIRA, 2018) approaches to resolve the Flexible Job-shop Scheduling Problem (FJSSP). Different from the classical job shop scheduling problem (JSSP), FJSSP is more general because the sequence of operations is allowed to be assigned to any machine of a given set instead of a prescribed one (LI et al., 2019b). In (WANG; TANG, 2017), the problem to resolve is a flowshop scheduling problem. Which is a strict of the JSSP where there is a strict order to perform the operations.

### 2.2.3 Research agenda

This section describes some of the related but potential areas of research that may build future literature. But before discussing the trends and research opportunities, we give a summary of the results found in this systematic review as well as the gaps identified.

#### 2.2.3.1 Summary

As mentioned in the previous section, a total of 111 works were found. None of them presents a novel kind of enhancement that was not in the taxonomies explained in our related works. The majority of these papers are based on SAEAs and EDAs, two different kinds of metaheuristic algorithms enhanced by ML. Based on the classification of when the knowledge is extracted, from the 111 papers, 96 papers use online strategies. Based on the impact of the enhancement, 70 papers use the Efficiency strategy. By the Involved Component, 72 papers enhance the Fitness Evaluation and Selection step. Supervised learning was the most found kind of machine learning problem, a total of 87 papers, most of them modeled as regression. Over half of the papers, precisely 60 studies, do not use benchmark instances to validate the problem, and only 14 papers use real benchmark instances. We found several types of real-world applications optimized by these papers, some of them are: enhance machine learning problems, aerodynamic design problems, and scheduling problems.

#### 2.2.3.2 Gaps

During our research, we could identify some gaps in the primary studies. To prove the validity of a newly designed metaheuristic algorithm, the researchers employ some performance validation criteria. These criteria are used to compare to well-known algorithms to prove the effectiveness of an algorithm. The gaps found in these studies are concerning to the validation criteria used in their studies:



1. The comparison with state-of-the-art optimization algorithms is a common practice in the literature to validate a new algorithm implementation. Despite that, we identify in the resulting paper set that in the 111 studies, 32 don't compare with state-of-the-art algorithms, only validate their new algorithm with different settings in a specific optimization problem;
2. In general, theoretically evaluating the performance of an optimization algorithm can be a hard task. That is why artificially designed benchmark functions have been widely used. Besides facilitating the experimentation process and prove its performance in different situations, these functions allow the easy reproduction of the experiment by other researchers. In our resulting papers, 60 don't apply the algorithm in a benchmark test function, making it difficult to simulate and compare the presented results;
3. Statistical tests have become a widespread technique in computational intelligence, and several different test kind has proven to be robust to be used in metaheuristic ([DERRAC et al., 2011](#)). But only 26 papers make use of statistical tests to improve the evaluation process of the new method.
4. The last gap found is relative to the machine learning validation. Only a few papers compare the performance of different models or techniques. Most papers apply well-known methods that have a good performance in a certain kind of algorithm, such as Kriging and ANNs for metamodeling.

These validations gaps make us identify the need to improve the experimentation phase with robust validation criteria in the papers found in our review.

### 2.2.3.3 Trends and research opportunities

Based on the results found in this review, we observed some research gaps and limitations. In the next subsections, we discuss the research opportunities found, and present some trends in the investigation of the field.

#### 2.2.3.3.1 SAEAs

Despite several advantages of using MOEAs on expensive objective functions, there are some important limitations while dealing with it, such as the relatively high number of objective function evaluations. By this, there is a motivation in the research field to find different approaches to deal with the problem. In ([COELLO et al., 2020](#)) surrogates have already been pointed as a trend in this area. In our systematic review, we identified several different approaches of SAEAs that obtained a good performance in this kind of problem, but this research field stills a challenge for the area. There is a lack of experimentation and comparison of the performance of different metamodels in diverse optimization applications. Also, some new strategies proposed need to be strongly validated in benchmark instances and in real-world applications. Most SAEAs normally

can handle expensive optimization problems with a low number of variables efficiently, but when the number of the dimensions increases many challenges appears to the algorithm. The use of machine learning methods such as principal component analysis and feature selection can benefit the surrogate performance in such problems (GARCÍA et al., 2019). Only a few papers, proposed in (LIU et al., 2012; DAI; WANG; YANG, 2020; TIAN et al., 2020b), address this problem and obtains the benefits in applying ML methods in problems with a high number of dimensions. Due to the large number of methods that use or extend SAEAs, we identify that a systematic review or systematic mapping that analyzes this area can be of great importance to the optimization field.

#### 2.2.3.3.2 EDAs

Algorithms based on probabilistic modeling have been successfully applied for solving different optimization problems. We have found in the resulting papers diverse variants of EDAs with different design strategies to successfully obtain knowledge from the problem to enhance the convergence and divergence of the Pareto front. We understand that the trend in this area relies on continuing to explore and experiment with different models and novel designs to improve the knowledge during the search. Also, validate the performance of the proposed EDA in different real-world optimization problems to understand if the strategy is generally sufficient to be used in any kind of problem. As in the same manner with the SAEAs algorithms, a systematic review and/or systematic mapping for EDAs could also lead to future work. The researchers can identify unique knowledge from these papers that could not be identified in our research due to the scope covered in this review.

#### 2.2.3.3.3 Explore Machine learning techniques

As also stated by (WANG et al., 2018), the use of different and more advanced machine learning techniques could be employed to improve the performance of the algorithm. The first example is ensemble learning, which is a method that explores the combination of several models. This combination can enhance the performance of the algorithm as shown in (YIN; TSAI, 2020; LU; LI; JIN, 2013; WANG et al., 2018; AL-JUBOORI; DATTA, 2019). We identify that more exploration of this technique to improve the results could be a trend to research. Other methods such as data preprocessing, deep learning, transfer learning, and other advanced machine learning techniques could be also deep studied to enhancing metaheuristics algorithms.

By the ML classification, some areas that obtain good results could be the target for further research. Semi-supervised Learning, for example, that was found in only one paper in our research, is able to enhance the performance of the surrogate-assisted algorithm when there is uncertainty in the fitness evaluations or when there is data paucity in the optimization problem (SUN et al., 2013). Also, Reinforcement Learning was present in only one paper, and as we detailed, obtained better performance in terms of both speed and final results. More studies in

these areas could provide more knowledge to the research field as well as identifying new design alternatives in these classifications.

We have identified in this systematic review several novel methods, based on different ML techniques, that have been successfully applied in different components of the metaheuristic. But this also raises a question about the impact that this method has on different components of the algorithm. Experiments with that purpose were not identified in the studies. The understanding of the impact of the method on different components could lead to the validation of robust methods and also the discovery of new approaches and interest areas. An attractive research line is to investigate the combination of more than one ML method in the same metaheuristic. The approaches in (LI et al., 2019; AL-DUOLI et al., 2018; GAO et al., 2020; PARQUE; MIYASHITA, 2017; LIU; KOZIEL; ZHANG, 2016; TENNE, 2012a), uses more than one method in different areas or in the same component but as support for the other technique. Increasing the performance of the algorithm in comparison if only one method was adopted.

#### 2.2.3.3.4 Knowledge Discovery

During the studies of the resulting papers, we have identified a large set of works that increase the efficiency by using surrogates and a smaller portion that enhances the effectiveness with EDAs and variants. Within this group, principally the one who enhances the effectiveness, some of these papers call our attention to the results found and the opening of new research possibilities. Their studies provide different approaches for discovering forms of knowledge during the search process. An example is the Baldwin learning strategy proposed by (QI et al., 2012) and the deterministic Markov decision process, Q-learning proposed by (CUI; TURAN; SAYER, 2012).

Another interesting learning approach found is the inverse modeling. First proposed by (GIAGKIOZIS; FLEMING, 2012) as an approach to increase the number of non-dominant solutions, that maps the values of the objective functions to decision variables, in an inverse manner as the surrogates models. In this systematic review, we have encountered two approaches that use the inverse model that differs from the one proposed by (CHENG et al., 2015; SANTOS; OLIVEIRA; BRITTO, 2019). The knowledge extracted from the model in these works allowed a better final quality of the solutions, improving the performance of the metaheuristic. This knowledge extracted from an inverse model can be further well explored in further research. And also, the works found opened new paths for research in the area. First, the method proposed by (CHENG et al., 2015) can be improved with different approaches in finding the better samples for training the model, choose the better combination of models, and improving the generation of the new solutions. From the method in (SANTOS; OLIVEIRA; BRITTO, 2019), it can be further improved as a method for providing solutions next to the Pareto front in combination with an MOEA, taking advantage of the strengths of each algorithm. In these two methods, experimentation of different ML models and techniques could find the best approach to enhance

the performance of the algorithm. We believe that the enhancement of the effectiveness of metaheuristics with these open research areas brings a new trend to the studied field.

## 2.2.4 Threats to validity

In this section, we describe the possible threats to the validity of our systematic review. According to (WOHLIN et al., 2012), four types of threats to the validity of experimental research must be analyzed. We conduct the remaining of this section motivated by this classification.

### 2.2.4.1 Construct validity

This validity is concerned with the relation between the theory and the observation (WOHLIN et al., 2012). One possible threat to the construct validity is the exclusion of relevant studies. That is why, it is related to the search string, research question, and databases used. Focus on these steps is important to not miss important studies in the review. Because of this, in this systematic review, we aim to construct the search string as general as possible, even if the size of the search result increased. But we confirm that the search string will coverage the studied field. We also make use of a control group of relevant papers about machine learning enhancing metaheuristics algorithms to validate the search string, described in the subsection 2.1.3

The research questions may not address all aspects of the field. To reduce this threat, we formulated research questions that cover the main elements of the field found in the related works. To mitigate the threats related to the choice of databases, we used multiple databases to avoid eventual limitations implied by a particular database. Databases are selected by considering their relevance to the computer science area. Only one database searched did not return papers published in important venues of the field.

### 2.2.4.2 Internal validity

This validity refers to a causal relationship between treatment and outcome (WOHLIN et al., 2012). When the researcher is investigating if one factor affects other, there is a risk that the investigated factor is also affected by a third factor. If the researcher is not aware of the third factor it could create a threat to the internal validity. To reduce this threat, we were guided by the inclusion and exclusion criteria, applying them when analyzing the papers. Beyond that, we performed a snowballing reading in the selected studies and included relevant studies. Meetings were made between the researchers to discuss doubts in data extraction and re-readings were performed to clear existing doubts. Also, we did not include grey literature in our research, due to the possibility to arise a threat to internal validity from unpublished research that had undesired outcomes or proprietary literature that is not made available.

### 2.2.4.3 External validity

This validity is concerned with the generalization of results outside the scope of the study (WOHLIN et al., 2012). We can relate it to what it is possible to extend to generalize the findings, and to what extent the findings are of interest to other people outside the scope of the review. To reduce this threat the adoption of a review protocol helped to achieve a representative set of studies that have a greater extent. Also, the number of researchers found, from different countries and a wide list of application areas make us believe that the finds in our review can be generalized to many different areas.

### 2.2.4.4 Conclusion validity

This validity refers to problems that affect the ability to draw the correct conclusions from the paper (WOHLIN et al., 2012). One possible threat to this validity is the failure in applying a good quality assessment and data extraction. To minimize this threat, we explicitly define inclusion and exclusion criteria that are detailed enough to provide an assessment of how we reached the final set of papers for analysis. To assess the consistency of data extraction, the spreadsheet generated from the extraction was reviewed and analyzed several times to maintain its validity. Another threat is related to the ability to replicate this review and obtain the same results. We believe that using the protocol and the details presented in our conduction satisfies this threat, making it possible to be replicated following the steps provided and using the search string.

# 3

## Validating DVL Performance

As reviewed in chapter 2, the use of machine learning methods to enhance metaheuristics has called the attention of the research literature. The main reason is the ability to handle hard optimization problems such as MOPs and MaOPs. SAEAs have been used to metamodeling expensive objective functions and are capable of achieving good results in improving the performance of optimization algorithms. Motivated by the use of surrogates, some researchers propose inverse modeling, which maps the objective function to the decision variables. The application of this method demonstrated that inferences of Pareto's optimum front can be achieved by the use of machine learning models. Recently, the DVL algorithm has been proposed, where the inverse model is used to directly estimate the Pareto front, differently from the previous studies. To advance in the improvement of the algorithm, it is necessary to validate the following hypothesis: inverse models can have good performance in estimating the Pareto optimal set. We believe that the research done in (SANTOS; OLIVEIRA; BRITTO, 2019) needs to be experimentally improved to confirm this hypothesis. Also, we need to validate the DVL as an algorithm for Many-objective Optimization in comparison with a state-of-the-art MOEA, such as the NSGA-III. For this reason, in this chapter, we propose to test the inverse model and the DVL algorithm in a larger set of experiments in MOP and MaOPs, using a variety set of benchmark functions and machine learning models for validation.

In the following sections, we describe the DVL algorithm, present the benchmark test problems used in the experiments as well as the learning models used to learn the decision variables. Two different test configurations are described and analyzed. The first one is called the regression experiment, and the second is the optimization experiment. The parameter settings and the performance indicators used in each test are described at the beginning of each test subsection.

### 3.1 Decision Variable Learning algorithm (DVL)

The Decision Variable Learning algorithm ([SANTOS; OLIVEIRA; BRITTO, 2019](#)) aims to use machine learning models to generate an approximated Pareto optimal set using information from samples of the objective space and a set of reference points. Differently from the Pareto Estimation (PE), the DVL is not a method used after the MOEA, instead, it is an algorithm that uses the inverse model approach into the algorithm to solve multi-objective problems. Although in the elaboration of the DVL the authors were not aware of the studies relative to the inverse model, the DVL has the theoretical basis of the PE method. But, instead of using the inverse model to increase the number of non-dominated solutions, the DVL directly infers an approximation of the Pareto optimal set.

DVL is based on three main steps, the first one is the creation of an initial sample dataset, that contains solutions with their respective objective functions. This initial sample generation is provided by the LHS method, which generates well-distributed samples over the search space. The second step is the training of the inverse method, a machine learning model is used for this purpose, which is fed with the dataset generate previously. The idea that supports this modeling is that, for continuous MOPs, it can be deduced from the Karush-Kuhn-Tucker optimality conditions that the Pareto optimal set is a piecewise continuous in the decision variables ([GIAGKIOZIS; FLEMING, 2014](#)). Depending on the machine learning model,  $M_n$  models are created, and each one estimates only one variable, so each solution generated in this process will be composed of variables estimated by different models. Another strategy is to use only one model  $M$  that can be used to estimate the entire decision variable set. In the original implementation ([SANTOS; OLIVEIRA; BRITTO, 2019](#)),  $M_n$  linear Ridge models are used, however, in our experiments we use a variety of models with different strategies to validate the impacts that different models have on learning decision variables. The last step is the prediction of new solutions next to the Pareto optimal set. The objectives set used as input for this step is the same as the PE method, which is based on the method in ([DAS; DENNIS, 1996](#)), that places points on a normalized hyper-plane equally inclined to all objective axes and has an intercept of one on each axis. The algorithm was projected to work iteratively, where, at each iteration, the training of the machine learning model and prediction of a new population is executed. These steps are performed using a set of solutions that is incremented at each iteration.

The algorithm was projected to work iteratively, where, at each iteration, the training of the machine learning model and prediction of a new population is executed. These steps are performed using a set of solutions that is incremented at each iteration. The pseudocode of the DVL is described in Algorithm 1.

As shown in the algorithm, the process is started from the creation of an initial dataset  $P$  with several solutions from the LHS. With the dataset created, machine learning models  $M_n$  are trained and new solutions  $P_{rp}$  are estimated from the reference points  $RP$ , it was defined that each model will estimate only one variable, so each solution generated in this process will



## Algorithm 1 – The DVL algorithm pseudocode

**Data:** the maximal number of iterations  $i$ , the  $\epsilon$  value for the difference between the hypervolume, the number of closest solutions used for training  $c$  and the  $k$  value for the size of the initial dataset;

**Result:** the population with the best hypervolume  $P_{best}$ ;

```

1 begin
2    $P \leftarrow LHS(k)$ ;
3    $Evaluate(P)$ ;
4    $RP \leftarrow GenerateReferencePoints()$ ;
5    $P_{best} \leftarrow \emptyset$ ;
6    $HV_{best} \leftarrow 0.0$ ;
7   repeat
8      $P_{rp} \leftarrow \emptyset$ ;
9     for  $r \in RP$  do
10       $P_{near} \leftarrow FindClosestSolutions(r, P, c)$ ;
11       $M_n \leftarrow TrainModels(P_{near})$ ;
12       $s \leftarrow M_n.Predict(r)$ ;
13       $P_{rp} \leftarrow P_{rp} \cup \{s\}$ ;
14    end
15     $HV \leftarrow NormalizedHypervolume(P_{rp})$ ;
16    if  $HV > HV_{best}$  then
17       $HV_{best} \leftarrow HV$ ;
18       $P_{best} \leftarrow P_{rp}$ ;
19    end
20     $P \leftarrow P \cup P_{rp}$ ;
21  until reach the maximum number of iterations  $i$  or the difference between the
    hypervolumes is less than  $\epsilon$ ;
22 end

```

be composed of variables estimated by different models. This training and estimation phase is repeated until reaches a stop condition. In the original implementation (SANTOS; OLIVEIRA; BRITTO, 2019), the linear Ridge model is used, however, in our experiments we use a variety of models to validate the impacts that different models have on learning decision variables.

The purpose of this iterative approach is to improve the quality of the dataset and thereby improve future estimations since the dataset will have solutions closer to the reference points. The details of each step, defined in Algorithm 1, will be further discussed in the next subsections.

### 3.1.1 Dataset and environment variables preparation

Initially, on line 2, the solution set  $P$  is created. This initial set can be interpreted as a dataset with  $m + n$  columns and  $k$  lines, where  $m$  is the number of objectives of the problem,  $n$  is the number of variables, and  $k$  is the size of the initial solutions set. Therefore, each row of this dataset corresponds to a pair  $(X, Y)$ , where  $X = [x_1, x_2, \dots, x_m]$  is the vector of objectives,  $Y = [y_1, y_2, \dots, y_n]$  is the vector of decision variables and  $x_i = f_i(Y)$ , where  $f_i$  is the objective



function  $i$ . Each model will use the subset of data that will contain the pairs  $(X, y_i)$ . In the original paper (SANTOS; OLIVEIRA; BRITTO, 2019), this initial dataset can be obtained in two ways: from sampling techniques or using individuals generated during the execution of an MOEA. However, it is demonstrated that the use of the LHS (IMAN; DAVENPORT; ZEIGLER, 1980) achieved better results on the DVL algorithm. The purpose of this methodology is to create a set of solutions that explore the entire space of variables and thereby explore all objective space. After generating the initial dataset, on line 3, each solution is evaluated using the real objective function and added to the  $P$  set.

After the dataset creation, some environment variables need to be initialized to enter the iterative process. First, the reference points set  $RP$  that is used to estimate the new solutions are created in the function *GenerateReferencePoints()* based on the method in (DAS; DENNIS, 1996). The reference points created by this method are placed on a normalized hyperplane which is also inclined to each objective and has an intersection on each axis. In a problem with 3 objectives, for example, the reference points are created on a triangle with the apex at  $(1, 0, 0)$ ,  $(0, 1, 0)$ , and  $(0, 0, 1)$ , with several points evenly divided within its area. Before the iterative process begins it is necessary to create two variables to maintain the best set of solutions and the hypervolume of this solution set, the  $P_{best}$  and  $HV_{best}$  respectively.

### 3.1.2 Iterative process

After the initial generation of the data and preparation of the environment variables, the iterative process is started. This process aims to train models of machine learning and from these models, to estimate solutions. The idea behind this proposal is that, for each decision variable of the problem, models will be used to learn the relationship between objective functions and the decision variable and from the information learned by the models estimate solutions that generate points in the space of objectives close to reference points that are considered optimal.

A set  $P_{rp}$  is initialized, on line 8, to maintain the new solutions generated at each iteration. Line 9 begins the process to estimate the points closer to the reference points, the following approach was defined: for each reference point,  $n$  models will be used, these  $n$  models will be trained with a dataset that contains the solutions closest to the reference point  $r$ . Lines 10 and 11 correspond to two main methods of the algorithm described in the next subsections:

#### 3.1.2.1 FindClosestSolutions

To train the models it was defined in (SANTOS; OLIVEIRA; BRITTO, 2019) that different datasets would be used. These datasets are subsets of the entire solutions set generated, where the solutions of each subset correspond to the solutions closest to the reference point that the model should estimate. The function receives as parameters the reference point  $r$ , the entire solutions set  $P$  and the size  $c$  of the closest solutions subset. The result of this function is the subset  $P_{near}$  with the solutions closest to the reference point  $r$ . The distance measure for this

calculus is the Euclidean norm between  $r$  and the points in  $P$ . The motivation behind choosing this approach is that in many problems certain regions of the objective space are difficult to achieve. So training models with the whole set of solutions located in these regions is not very effective because the models will be biased to estimate solutions in more common regions.

### 3.1.2.2 TrainModels

In this step, for a problem with  $n$  variables,  $n$  supervised learning models will be used, where each model  $M_i$ , for  $0 < i \leq n$ , will learn the function  $g_i(Y) = x_i$ , where  $Y$  is a vector of objectives and  $x_i$  is the  $i$ -th decision variable of solution  $X$ , such that  $Y$  is the objective vector of solution  $X$ . The models will use the set of closest points  $P_{near}$  for training. In this work, several models will be further compared in the experiments section to analyze the impacts of different models in learning decision variables. On line 12 a solution  $s$  is predicted from the reference point  $r$  defined as  $s = [M_1(r), M_2(r), \dots, M_n(r)]$ . On line 13 the new solution  $s$  is added to the  $P_{rp}$  set. After the new solutions are generated based on the reference points through the for loop, the solutions from  $P_{rp}$  are evaluated and the normalized Hypervolume  $HV$  of these solutions is calculated. If this hypervolume value is better than the best hypervolume encountered  $HV_{best}$  then the best solution set is saved and maintained on the  $P_{best}$  variable.

In the end, if a stop condition is not reached, the population is added to the set of solutions  $P$ , and the process is repeated, otherwise, the best population  $P_{best}$  generated during the process is returned as a result. The proposal of adding the individuals on the population  $P$  was used considering that the new individuals generated can add information that did not exist in the original dataset since the trend is that the individuals generated will be closer to the reference points. The stop condition of the DVL is reached when a maximum number of iterations is exceeded or when the difference between the hypervolume of the population generated in the current iteration and the hypervolume of the population generated in the previous iteration is less than a value  $\epsilon$ .

In the DVL experiments, the algorithm is tested in comparison with NSGA-III in problems with 3 and 10 objectives. In some of these problems, the DVL performs better than the NSGA-III, mainly in situations where the number of evaluations was small. This makes us believe that further exploration of the algorithm can lead to better results and open research areas for investigation. We found that some limitations of the algorithm can be better explored. Firstly, it is well-known that different ML models could lead to better results in different function approximations, it is needed to explore the performance of different models and machine learning techniques in estimate the Pareto optimal set in a wide variety of MOPs and MaOPs. Secondly, the strategy used for choosing the decision variables for training each model may vary according to the problem in question. In (CHENG et al., 2015) for example, it uses random grouping to increase the probability of allocating correlated decision variables into different sub-problems.

Thirdly, we also note that the iterations of the algorithm do not always lead to improvement

of the Pareto set, there is a lack of an exploratory process inside the algorithm to search for neighboring regions. Without this process the algorithm tends to become stagnant after a certain number of iterations, resulting in approximate hypervolume values. Attached to that we verify that the use of the reference point as input for the inverse model achieves good results in the initial iterations but it tends to lose strength because it does not represent the shape of the real PF. Finally, the calculation of the hypervolume of a front inside the iteration process makes the algorithm have an expensive computational complexity, exponential in the number of objectives in the worst case, as demonstrated in (WHILE et al., 2006).

## 3.2 Experiments and Results

In this section, we evaluate the DVL algorithm through empirical experiments that have two main motivations: first, it is necessary to validate the possibility to learn decision variables from objective functions, for this, we propose the Regression experiment. Secondly, in the Optimization experiment, we validate the impact that different learning models have on the algorithm. Also, we validate the DVL as an algorithm for Many-objective Optimization in comparison with a state-of-the-art MOEA, the NSGA-III.

In our experiments, we used the first seven test problems from the DTLZ benchmark test suite (DEB et al., 2005): DTLZ1 to DTLZ7, which are multiobjective problems for optimization that have different configurations and shapes of the Pareto-optimal front. The problems are parameterized from a variable  $k$ , where  $k = n - m + 1$ , being  $n$  the number of variables and  $m$  the number of objectives of the problem. Two different instances were defined for each problem, one with 3 objectives and 12 variables and the other with 10 objectives and 12 variables. By the DTLZ definition, the complexity of 10 objectives is lower than the complexity of 3 objectives. But this variation of the objective number allows the identification of the influence of the objectives in the learning model. In initial experiments, the variation in decision variables showed no impact on algorithm performance, so the number of variables is set to 12 in both cases. All the algorithms in the experiments were subjected to the same conditions, and our experiments aim to make a comparative analysis between them.

Two different test configurations are described and analyzed. The first one is called the regression experiment, and the second is the optimization experiment. The parameter settings and the performance indicators used in each test are described at the beginning of each test subsection. In all experiments, the algorithms are executed independently 20 times for each configuration. Different visual comparisons are used to better demonstrate the results of each experiment. In both experiments, the Friedman's test followed by the Wilcoxon rank-sum test is adopted to compare the results between each configuration of the experiments with a value of 0.05 of the significance level. The best results in the following tables will be highlighted in bold, if more than one value is highlighted in bold it means that there is no statistically significant

difference between the result and the best value. In the default implementation of the DVL algorithm,  $n$  models are created to estimate each decision variable. Different from the strategy used in (GIAGKIOZIS; FLEMING, 2014) where a single model is used to estimate the entire decision variable set. Because of the dependence between the input and the outputs that can occur in optimization problems, the two strategies are applied in our experiments.

### 3.2.1 Learning Models

A set of different regression machine learning models are used in our experiments to evaluate the impacts of these models in learning the decision variables. The Linear model, an artificial neural network, more precisely a multilayer perceptron (MLP) (GARDNER; DORLING, 1998), a random forest regression (RFR) (SVETNIK et al., 2003), and a support vector regression (SVR) (SMOLA; SCHÖLKOPF, 2004). Each one has unique characteristics that make them perform well in certain types of problems. The implementation and the parameters of each model were defined by Scikit-learn (PEDREGOSA et al., 2011), a python module that integrates a wide range of state-of-the-art machine learning algorithms. Different from the default implementation of each model in the Scikit-learn, the parameters used are described in Table 10. The values of the parameters described were the best found by experimentation in the diverse problems of the benchmark test suit.

Table 10 – Machine learning models used in the experiment and each parameter configuration.

| Model     | Description                                                                               | Parameter                                                                                                          | Value        |
|-----------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------|
| Linear    | A ordinary least squares linear model for each decision variable.                         | None different                                                                                                     |              |
| MO-Linear | A unique ordinary least squares linear model that has support for multioutput regression. | None different                                                                                                     |              |
| MLP       | A multilayer perceptron model for each decision variable.                                 | <b>hidden_layer_sizes</b> : Number of neurons in the $i$ th hidden layer, not counting the input and output layers | (11, 11, 11) |
| MO-MLP    | A unique multilayer perceptron model that has support for multioutput regression.         | <b>hidden_layer_sizes</b> : Number of neurons in the $i$ th hidden layer, not counting the input and output layers | (11, 11, 11) |
| RFR       | A random forest regression model for each decision variable.                              | <b>n_estimators</b> : Number of trees in the forest                                                                | 100          |
| MO-RFR    | A unique random forest regression model that has support for multioutput regression.      | <b>n_estimators</b> : Number of trees in the forest                                                                | 100          |
| SVR       | A support vector regression model for each decision variable.                             | <b>C</b> : Regularization parameter<br><b>gamma</b> : Kernel coefficient                                           | 0.1<br>auto  |

Source – Produced by authors.

With few exceptions, Machine Learning algorithms don't perform well when the input has numerical attributes with very different scales (GÉRON, 2019). It is the case of the objective values of the DTLZ benchmark test suit. Therefore, for each model was used the standardization

transformation which is an algorithm of feature scaling (KIM et al., 2018). The standardization subtracts the mean value (so standardized values always have a zero mean), and then it divides by the variance so that the resulting distribution has unit variance. This transformation causes a better performance in our experiments.

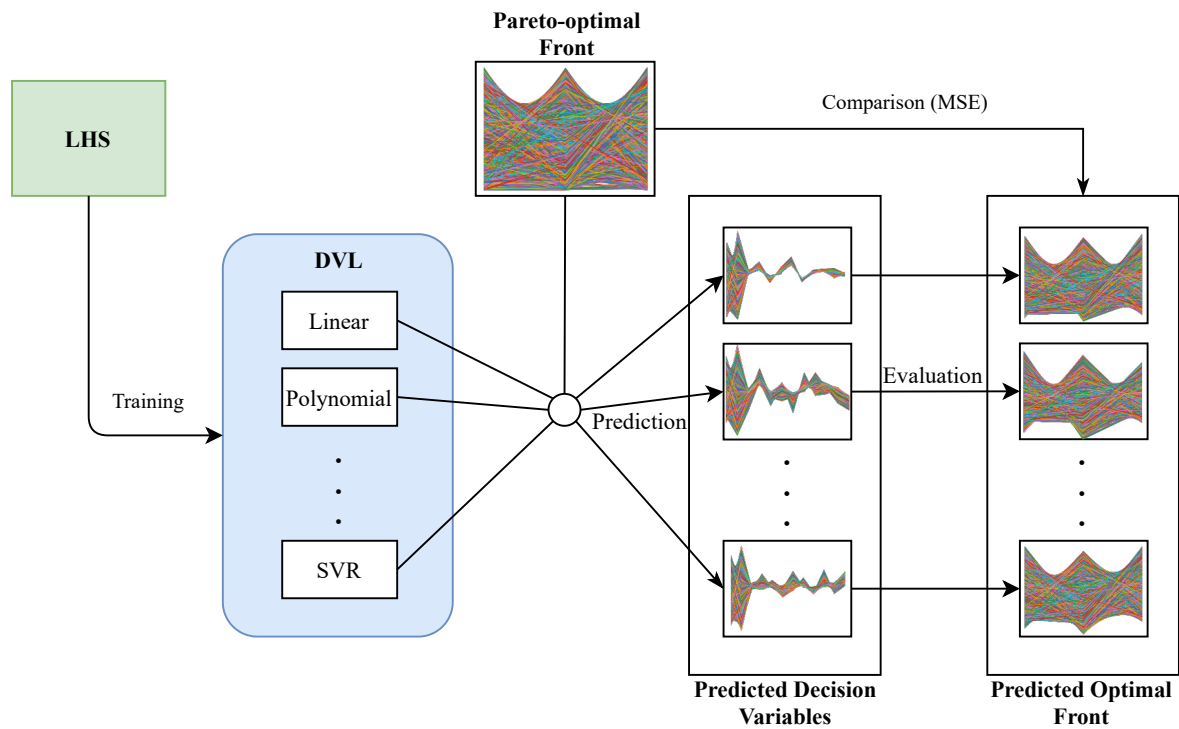
To validate and analyze the strategy of using one model to predict the entire decision variable set or using one model to each decision variable, in our experiments, we applied the multioutput regression in the models to obtain knowledge from the dependence between inputs and outputs. The exception is the SVR model, which implementation in Scikit-learn does not have support to multioutput regression. The models with this strategy are denoted with an "MO" before the model name in our results.

### 3.2.2 Regression Experiment

The first experiment is motivated to answer if it is possible to learn decision variables from objective functions. The task of understanding and predicting the variables of the DTLZ benchmark test problems can be classified as supervised learning since we have the input and output for the learning algorithm and can be classified as a regression task since we want to predict a numerical value. This experiment is based on two phases: training and testing. First, the model is trained by the execution of the DVL algorithm. The generated samples and modeling strategy follow the default flow of the algorithm. Secondly, in the testing step, a subset of the Pareto-optimal front of each DTLZ problem serves as input to the model for prediction. The DTLZ benchmark test suit allows the generation of the Pareto-optimal front of each problem analytically. The key idea of this experiment is to analyze if the inverse modeling of the DVL algorithm can correctly predict the optimal set of a problem. In a real optimization problem, the optimal front is unknown, but the purpose of this experiment is to validate the inverse modeling strategy trained with solutions far from the optimal front.

The overall workflow of the regression experiment is demonstrated in Figure 14. Each model in this experiment is trained with an initial number of solutions generated through the LHS algorithm. The size of the initial sample solutions is defined as 10000. For each DTLZ problem, approximately 5000 solutions of the Pareto-optimal front were generated for the experiment. The values of these Pareto-optimal front were obtained through the execution of diverse MOEAs without restriction and a post-processing routine implemented to eliminate duplicates and choose a diversified set of solutions. The objective function values of these generated solutions are computed and passed as input for each model to predict the decision variables. The predicted values of each model are evaluated to the objective space. The mean-squared error (MSE) (WANG; BOVIK, 2009) metric is used to make empirical comparisons between the results of each model. The MSE is the average squared difference between the predicted optimal front and the Pareto-optimal front values. The model with the lower MSE value was more successful in predicting decision variables.

Figure 14 – Regression experiment workflow.



Source: Produced by authors.

The results of the regression experiment for 3 and 10 objectives are shown in Table 11. The model with the lower MSE value was more successful in predicting decision variables. The result obtained in this experiment show that different models have better performance on different types of problems. Also, the number of objectives has interference in the results of MSE on each model for each problem. Although the MSE obtained has relatively low values, it is necessary to evaluate the predicted decision variables and the objectives produced by these variables in terms of divergence and convergence produced. For that, we made a visual comparison with the results obtained in the experiment and the Pareto-optimal front values of each problem.

Figures 15 to 21 show the visual comparison of the regression experiment. The generated approximated Pareto Front will be analyzed using parallel coordinates charts (Li; Zhen; Yao, 2017). Parallel coordinates display multidimensional data in a two-dimensional graph. The  $x$ -axis represents the number of objectives and the  $y$ -axis the value of each objective function. Thus, each obtained solution is represented as a polyline with vertices on the axes. With this chart, it can be observed some characteristics like convergence, coverage, and uniformity of each approximated Pareto front. For each problem, the model that achieves the best value of MSE is used to compare with the Pareto-optimal front. Different models were used for comparison which indicates that the use of a specific model is linked to the problem that will be solved. Each pair of charts represent the comparison for 3 and 10 objectives respectively. For example, Figure 16a



Table 11 – Statistical results (mean and standard deviation) of the MSE values for 3 and 10 objectives.

| Objectives | Problems | MO-Linear             | SVR                   | MO-MLP                | MO-RFR         | Linear                | MLP            | RFR                   |
|------------|----------|-----------------------|-----------------------|-----------------------|----------------|-----------------------|----------------|-----------------------|
| 3          | DTLZ1    | (1.42±0.01)e-2        | <b>(1.41±0.00)e-2</b> | (1.84±0.26)e-2        | (3.60±0.87)e-2 | (1.42±0.01)e-2        | (2.44±0.63)e-2 | (3.62±0.21)e-2        |
| 3          | DTLZ2    | (4.49±0.02)e-3        | <b>(2.20±0.14)e-3</b> | (3.97±0.99)e-3        | (7.91±1.96)e-3 | (4.50±0.02)e-3        | (4.37±1.35)e-3 | (3.69±0.37)e-3        |
| 3          | DTLZ3    | (1.43±0.00)e-2        | <b>(1.41±0.00)e-2</b> | (1.98±0.39)e-2        | (4.59±0.95)e-2 | (1.43±0.01)e-2        | (2.75±0.66)e-2 | (3.80±0.30)e-2        |
| 3          | DTLZ4    | (1.20±0.04)e-2        | (1.31±0.16)e-2        | (3.67±1.27)e-2        | (9.17±1.01)e-3 | (1.19±0.05)e-2        | (5.89±2.32)e-2 | <b>(9.39±1.26)e-3</b> |
| 3          | DTLZ5    | (8.58±0.02)e-3        | <b>(8.21±0.11)e-3</b> | (9.98±1.67)e-3        | (1.42±0.14)e-2 | (8.59±0.02)e-3        | (1.14±0.11)e-2 | (1.03±0.03)e-2        |
| 3          | DTLZ6    | (1.11±0.03)e-1        | (2.68±0.03)e-1        | <b>(9.41±3.83)e-2</b> | (1.10±0.07)e-1 | (1.11±0.04)e-1        | (1.17±0.67)e-1 | (1.20±0.08)e-1        |
| 3          | DTLZ7    | <b>(2.91±0.75)e-4</b> | (2.83±0.08)e-1        | (5.08±1.97)e-2        | (7.65±0.26)e-2 | (3.00±0.75)e-4        | (6.17±1.89)e-2 | (6.77±0.72)e-2        |
| 10         | DTLZ1    | (6.39±0.00)e-2        | (6.54±0.13)e-2        | (6.45±0.07)e-2        | (5.95±0.10)e-2 | (6.39±0.00)e-2        | (5.80±0.14)e-2 | <b>(1.76±0.33)e-2</b> |
| 10         | DTLZ2    | (3.53±0.01)e-2        | (3.47±0.01)e-2        | (3.07±0.21)e-2        | (1.80±0.03)e-2 | (3.52±0.01)e-2        | (7.07±0.39)e-3 | <b>(1.32±0.04)e-3</b> |
| 10         | DTLZ3    | (6.39±0.00)e-2        | (6.41±0.04)e-2        | (6.44±0.04)e-2        | (6.47±0.15)e-2 | (6.39±0.00)e-2        | (5.87±0.13)e-2 | <b>(1.90±0.19)e-2</b> |
| 10         | DTLZ4    | (5.41±0.01)e-2        | (5.37±0.03)e-2        | (6.87±0.16)e-2        | (4.33±0.06)e-2 | (5.41±0.02)e-2        | (5.93±0.12)e-2 | <b>(3.91±0.04)e-2</b> |
| 10         | DTLZ5    | <b>(5.69±0.00)e-2</b> | <b>(5.69±0.00)e-2</b> | (5.75±0.03)e-2        | (6.14±0.05)e-2 | <b>(5.69±0.00)e-2</b> | (5.82±0.05)e-2 | (6.17±0.04)e-2        |
| 10         | DTLZ6    | <b>(6.81±0.07)e-5</b> | (3.01±0.08)e-1        | (1.45±0.29)e-1        | (9.41±0.48)e-2 | (6.83±0.08)e-2        | (1.01±0.22)e-1 | (8.25±0.18)e-2        |
| 10         | DTLZ7    | <b>(4.99±0.59)e-5</b> | (9.73±0.80)e-2        | (3.10±1.35)e-2        | (4.30±0.05)e-2 | <b>(4.74±0.64)e-5</b> | (4.97±1.54)e-3 | (1.98±0.30)e-3        |

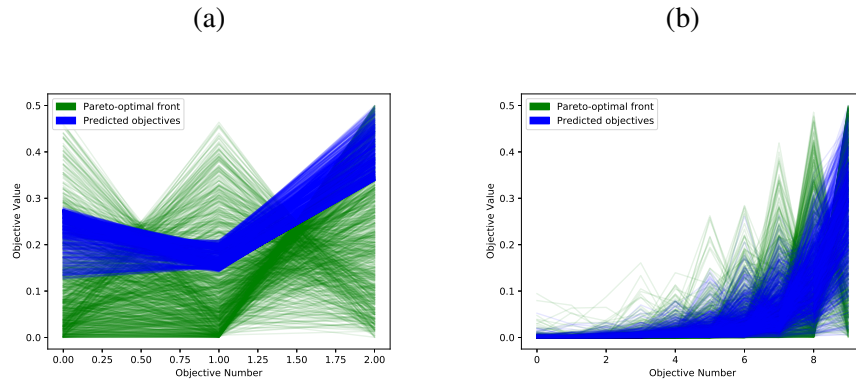
Source – Produced by authors.

shows the comparison between the predicted approximated Pareto Front and the Pareto-optimal front of the DTLZ2 for 3 objectives. The  $x$ -axis shows the objectives, 0 is the first one, and 2 is the last objective. The value of each objective is on the  $y$ -axis. Each line represents an objective vector, the green area represents the Pareto-optimal front expected to be found by the algorithm, and the blue area is the approximated Pareto Front. It can be observed in the DTLZ2 problem with 3 objectives that the two areas approach both in terms of convergence and divergence.

In all charts, the models used achieved a high convergence, i.e. the values found are very close to the ideal. In the majority of the problems the predicted values obtained have performed well in both divergence and convergence values. In these problems, the shape of the region formed by the predicted values is quite similar to the shape of the Pareto-optimal front, for example, DTLZ2 and DTLZ7 problems, Figures 16 and 21, respectively. In Figures 15a, and 17a, respectively, DTLZ1, and DTLZ3 problems, the models could not find all parts of the Pareto-optimal for 3 objectives. It occurred mainly due to the fact of the high search space presented in these problems. Also, they have many local Pareto-optimal fronts, which makes it difficult to obtain a good divergence on the predicted values. It is important to note that the models had a greater divergence of values in the problems with 10 objectives, demonstrating a shape closer to the ideal. This is mainly due to the increased number of features used for training which makes the models perform better than with 3 objectives. The DTLZ6 problem, Figure 20, is the case that the predicted objectives are a little far away from the ideal. DTLZ6 has a degenerate Pareto-optimal front like DTLZ5, besides that, the DTLZ6 has many local Pareto fronts which makes it difficult to converge to the Pareto-optimal front. Concerning the strategy of using 1 or  $n$  models to estimate each decision variable, the results show that both strategies can be used. In different optimization problems, each one gets the best results.

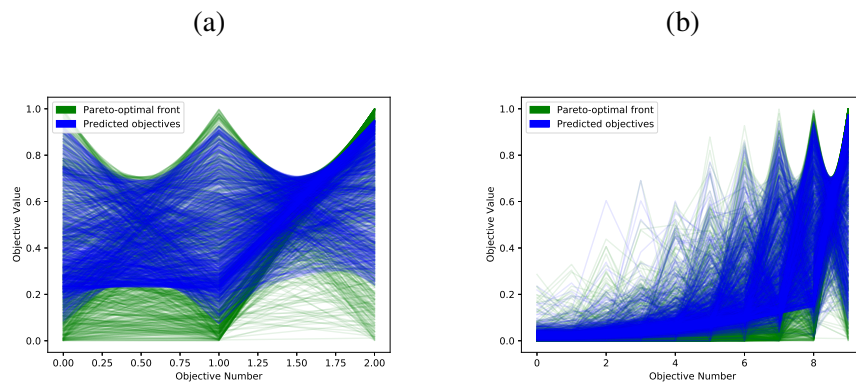
The results found by the regression experiment are quite promising. Different models could learn the decision variables by a set of objective function values. Even in more restricted scenarios, the models achieved good results. For the scope of this work, the models and techniques used were enough to validate the learning of decision variables.

Figure 15 – Visual comparisons for the DTLZ1 benchmark. (a) and (b) are the objectives from the predicted decision variables by the SVR model for 3 objectives and the RFR model for 10 objectives respectively.



Source: Produced by authors.

Figure 16 – Visual comparisons for the DTLZ2 benchmark. (a) and (b) are the objectives from the predicted decision variables by the SVR model for 3 objectives and the RFR model for 10 objectives respectively.



Source: Produced by authors.

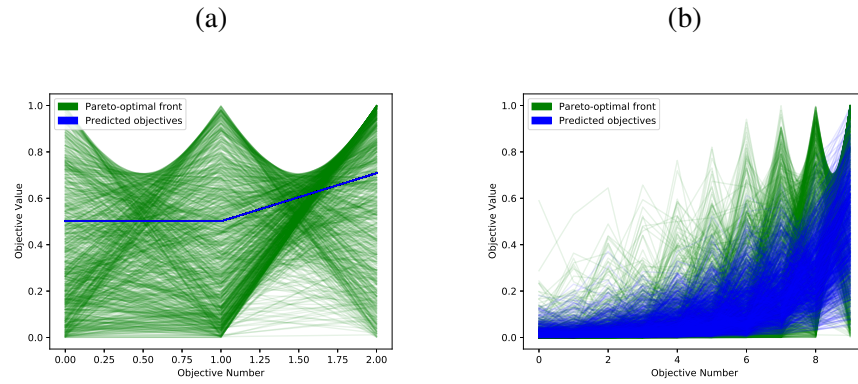
### 3.2.3 Optimization Experiment

In this experiment, we validate the DVL as an algorithm for Many-objective optimization. First, the same models are used to test the impact of different regression algorithms to optimize the DTLZ benchmark test suit. Then, the models that achieve goods results will be compared to the NSGA-III, a robust and well-known algorithm considered a state-of-the-art of MOEAs.

Each algorithm runs until it reaches a stop condition, in the experiments we use the number of evaluations of the objective function as a stop condition. A set of a maximum number of evaluations to our experiment is defined as (250, 500, 1000, 1500, and 10000), all algorithms run with the same number of calls to the objective function. For each maximum number of evaluations, the algorithms are executed 20 times independently to later perform the statistical comparison. This validation set is used to simulate different scenarios of restrictions in the use of the objective function. In this set, we classify as a higher restriction level a number of

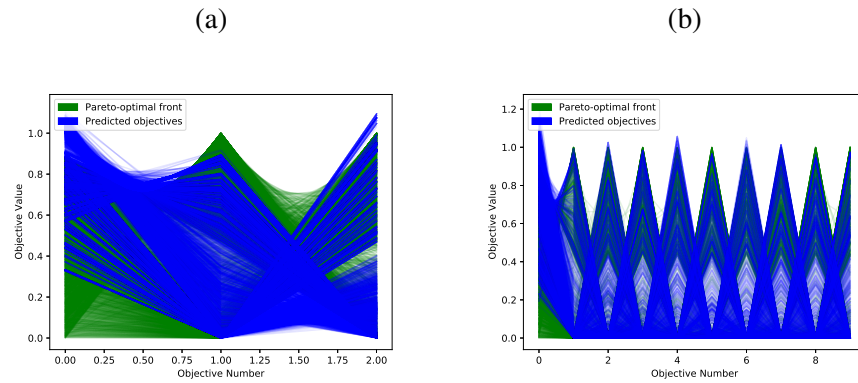


Figure 17 – Visual comparisons for the DTLZ3 benchmark. (a) and (b) are the objectives from the predicted decision variables by the SVR model for 3 objectives and the RFR model for 10 objectives respectively.



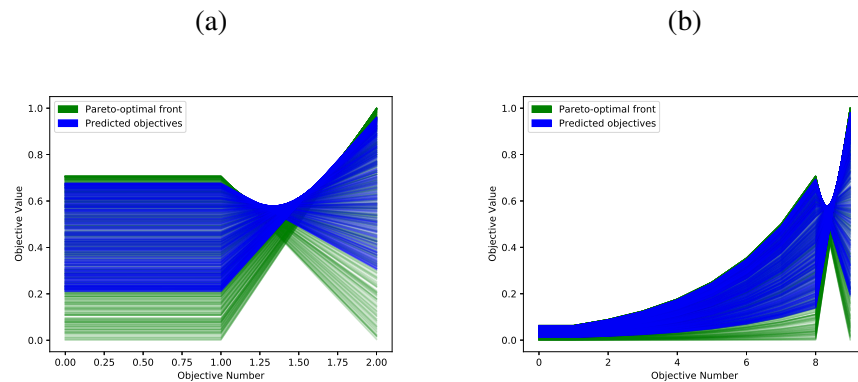
Source: Produced by authors.

Figure 18 – Visual comparisons for the DTLZ4 benchmark. (a) and (b) are the objectives from the predicted decision variables by the RFR model for 3 and 10 objectives respectively.



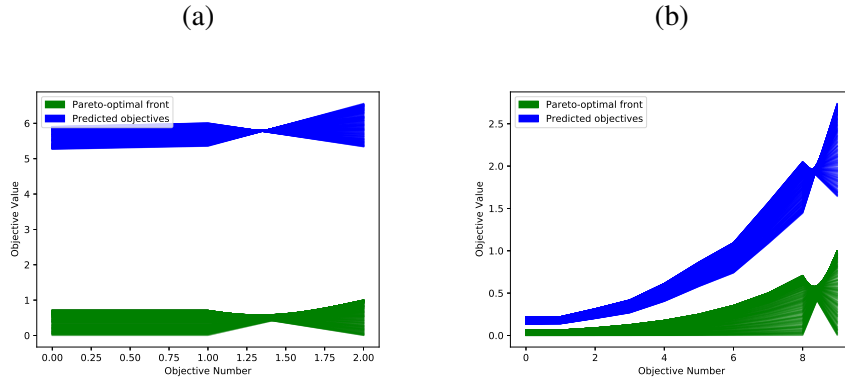
Source: Produced by authors.

Figure 19 – Visual comparisons for the DTLZ5 benchmark. (a) and (b) are the objectives from the predicted decision variables by the SVR model for 3 objectives and the RFR model for 10 objectives respectively.



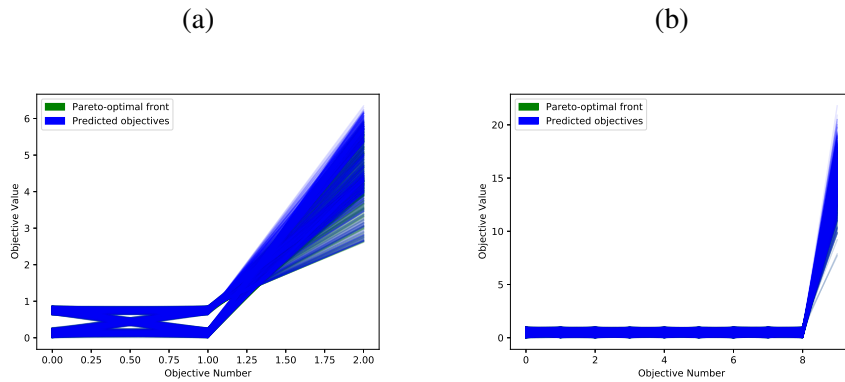
Source: Produced by authors.

Figure 20 – Visual comparisons for the DTLZ6 benchmark. (a) and (b) are the objectives from the predicted decision variables by the MO-MLP model for 3 objectives and the MO-Linear model for 10 objectives respectively.



Source: Produced by authors.

Figure 21 – Visual comparisons for the DTLZ7 benchmark. (a) and (b) are the objectives from the predicted decision variables by the MO-Linear model for 3 and 10 objectives respectively.



Source: Produced by authors.

validations lower or equal to 1000. The empirical comparisons between the results are made by the use of Hypervolume (HV) (WHILE et al., 2006), a performance indicator that assesses both convergence and divergence of the solutions. Let  $y^*$  be a reference point dominated by all the Pareto front solutions, and  $P$  the population resulting of a MOEA. The HV of  $P$  is the calculus of the volume of the region that dominates  $y^*$  and is dominated by  $P$ . The HV values presented in this work are all normalized to  $[0, 1]$  by dividing  $\prod_{i=1}^m y_i^*$ . Table 12 shows the reference points set that are used for each DTLZ problem. Generally, the reference points have values close to zero, well below the values used in this experiment. But, due to the restriction that we imposed on the maximum number of evaluations, the reference point must have higher values. In the literature, for example, the optimization algorithms execute around 100000 to 1 million objective function evaluations. While in our experiment, the maximum number of evaluations is 10000.

The DVL algorithm need four parameters to be predefined for execution. The value  $c$

Table 12 – Reference points used for each DTLZ problem.

| Database | Reference points set            |
|----------|---------------------------------|
| DTLZ1    | (300.0, 300.0, ..., 300.0)      |
| DTLZ2    | (2.0, 2.0, ..., 2.0)            |
| DTLZ3    | (700.0, 700.0, ..., 700.0)      |
| DTLZ4    | (2.0, 2.0, ..., 2.0)            |
| DTLZ5    | (4.0, 6.0, ..., $2(m + 1)$ )    |
| DTLZ6    | (10.0, 18.0, ..., $2(4m + 1)$ ) |
| DTLZ7    | (10.0, 18.0, ..., $2(4m + 1)$ ) |

for  $m$  number of objectives.

Source – Produced by authors.

of the number of closest solutions of the reference points used for training. A  $k$  value for the initial number of population used by the LHS. A value  $i$  of the maximum number of iterations and a value  $\epsilon$  of the difference between the HV of the population generated. In this experiment the  $c$  value is fixed in 200 and the  $\epsilon$  is set to 0.001, the  $i$  and the  $k$  has different values for each configuration of number objectives vs. number of evaluations. These different configurations for the optimization experiment are shown in Table 13.

Table 13 – DVL parameter configuration for the optimization experiment.

| Number of Objectives | Number of Evaluations | $i$ | $k$  |
|----------------------|-----------------------|-----|------|
| 3                    | 250                   | 1   | 159  |
|                      | 500                   | 3   | 227  |
|                      | 1000                  | 7   | 363  |
|                      | 1500                  | 9   | 681  |
|                      | 10000                 | 14  | 8726 |
| 10                   | 250                   | 1   | 50   |
|                      | 500                   | 2   | 112  |
|                      | 1000                  | 4   | 132  |
|                      | 1500                  | 5   | 464  |
|                      | 10000                 | 12  | 7388 |

Source – Produced by authors.

The parameters used in NSGA-III were defined by the pymoo (Blank; Deb, 2020), which is a python framework for optimization. For the simulated binary crossover (DEB; AGRAWAL et al., 1995), the distribution index is set to  $\eta c = 30$ , and the crossover probability  $pc = 1.0$ . For the polynomial mutation (DEB; GOYAL, 1996), the distribution index and the mutation probability are set to  $\eta m = 20$  and  $pm = 1/n$ , respectively, as recommended in (DEB, 2001). The population size used was 91 and 220 for the problem with 3 and 10 objectives, respectively, and the individual has a length of 12 for both objective sizes.

The results of the optimization experiment for 3 and 10 objectives are shown in Tables 14 and 15, respectively. Each line represents the values of the mean and standard deviation of the

20 executions of each algorithm per scenario of restriction. The best results are highlighted in bold if more than one value is highlighted, then it means that there is no statistically significant difference between these results. In comparison to the regression experiment, is possible to note that the models that have the best MSE not always achieved the best results of hypervolume. For 3 objectives, only the SVR and the MO-Linear models achieved good results of hypervolume in DTLZ1, DTLZ3, and DTLZ7, just as they obtained in the regression experiment. For 10 objectives, no model that got the best result in the regression experiment achieved the best HV mean. What draws the most attention is the performance of the SVR and the two MLP versions, MLP and MO-MLP, who achieved good results in 3 and 10 objectives. Mainly because although the MLP does not obtain the lowest MSE value, the generated solutions have a good diversity which implies a better HV. Besides that, the architecture of the DVL algorithm allows new training at each iteration with objectives closer to the ideal. As the model is reused during iterations, this task can be seen as transfer learning and many studies have demonstrated the superior performance of neural networks compared to statistical methods in transfer learning (LU et al., 2015).

Table 14 – Statistical results (mean and standard deviation) of the Hypervolume values for 3 objectives.

| Probl. | Eval. | MO- Linear            | SVR                   | MO-MLP                | MO-RFR         | NSGA-III              | Linear                | MLP                   | RFR                   |
|--------|-------|-----------------------|-----------------------|-----------------------|----------------|-----------------------|-----------------------|-----------------------|-----------------------|
| DTLZ1  | 250   | (7.38±7.22)e-2        | <b>(9.98±0.00)e-1</b> | (5.81±0.46)e-1        | (3.36±0.96)e-1 | (9.80±0.00)e-1        | (1.06±0.94)e-1        | (8.83±0.43)e-1        | (3.53±0.87)e-1        |
|        | 500   | (2.08±1.46)e-1        | (9.98±0.00)e-1        | (5.86±0.52)e-1        | (3.54±1.03)e-1 | <b>(1.36±0.00)e+0</b> | (1.58±0.87)e-1        | (9.05±0.21)e-1        | (4.68±0.96)e-1        |
|        | 1000  | (2.22±1.81)e-1        | (9.98±0.00)e-1        | (5.90±0.43)e-1        | (3.53±1.22)e-1 | <b>(2.16±0.00)e+0</b> | (2.36±1.66)e-1        | (9.14±0.15)e-1        | (5.09±1.04)e-1        |
|        | 1500  | (2.08±1.79)e-1        | (9.98±0.00)e-1        | (5.88±0.52)e-1        | (3.42±1.37)e-1 | <b>(2.85±0.00)e+0</b> | (1.65±1.22)e-1        | (9.26±0.11)e-1        | (5.15±0.60)e-1        |
|        | 10000 | (5.29±1.11)e-1        | (9.98±0.00)e-1        | (6.61±1.18)e-1        | (3.20±1.17)e-1 | <b>(4.17±0.00)e+0</b> | (1.56±1.11)e-1        | (9.33±0.18)e-1        | (5.61±0.87)e-1        |
| DTLZ2  | 250   | (5.63±0.20)e-1        | (5.54±0.07)e-1        | (7.40±0.25)e-1        | (5.95±0.32)e-1 | (7.12±0.00)e-1        | (5.66±0.19)e-1        | (1.38±0.58)e-1        | <b>(7.80±0.07)e-1</b> |
|        | 500   | (5.78±0.05)e-1        | (5.65±0.07)e-1        | (7.34±0.37)e-1        | (6.60±0.49)e-1 | (8.18±0.00)e-1        | (5.87±0.19)e-1        | (2.12±0.50)e-1        | <b>(8.26±0.14)e-1</b> |
|        | 1000  | (6.00±0.09)e-1        | (5.75±0.06)e-1        | (7.43±0.36)e-1        | (7.87±0.22)e-1 | <b>(8.74±0.00)e-1</b> | (8.28±0.27)e-1        | (2.60±0.43)e-1        | (8.49±0.22)e-1        |
|        | 1500  | (6.96±0.06)e-1        | (6.12±0.08)e-1        | (8.05±0.18)e-1        | (8.21±0.11)e-1 | <b>(8.98±0.00)e-1</b> | (8.41±0.18)e-1        | (2.40±0.34)e-1        | (8.71±0.22)e-1        |
|        | 10000 | (8.30±0.05)e-1        | (7.15±0.05)e-1        | (8.60±0.09)e-1        | (8.79±0.05)e-1 | <b>(9.26±0.00)e-1</b> | (8.74±0.13)e-1        | (2.42±0.45)e-1        | (9.14±0.02)e-1        |
| DTLZ3  | 250   | (1.19±2.54)e-2        | <b>(9.97±0.00)e-1</b> | (4.19±1.03)e-1        | (2.16±0.84)e-1 | (6.26±0.00)e-1        | (3.70±5.93)e-2        | (9.25±0.17)e-1        | (2.82±0.78)e-1        |
|        | 500   | (2.07±2.99)e-1        | <b>(9.97±0.00)e-1</b> | (3.65±1.30)e-1        | (2.62±1.00)e-1 | (9.21±0.00)e-1        | (3.31±3.75)e-2        | (9.31±0.16)e-1        | (3.79±0.93)e-1        |
|        | 1000  | (1.74±2.47)e-1        | <b>(9.98±0.00)e-1</b> | (3.54±1.01)e-1        | (2.32±0.63)e-1 | (9.97±0.00)e-1        | (1.04±1.65)e-1        | (9.39±0.15)e-1        | (3.95±0.65)e-1        |
|        | 1500  | (6.66±9.11)e-2        | (9.98±0.00)e-1        | (3.75±0.60)e-1        | (2.39±0.68)e-1 | <b>(1.26±0.00)e+0</b> | (4.53±7.41)e-2        | (9.44±0.10)e-1        | (4.54±0.66)e-1        |
|        | 10000 | (4.48±1.27)e-1        | (9.98±0.00)e-1        | (4.67±0.99)e-1        | (3.60±1.12)e-1 | <b>(1.95±0.00)e+0</b> | (2.64±3.78)e-2        | (9.50±0.07)e-1        | (4.49±0.77)e-1        |
| DTLZ4  | 250   | (5.45±1.06)e-1        | (4.89±0.06)e-1        | <b>(6.99±0.51)e-1</b> | (4.89±0.17)e-1 | (2.64±0.00)e-1        | (6.03±1.13)e-1        | (1.66±3.33)e-2        | (5.47±0.35)e-1        |
|        | 500   | (6.84±1.21)e-1        | (5.16±0.80)e-1        | (6.65±0.67)e-1        | (5.45±0.37)e-1 | (3.98±0.00)e-1        | <b>(7.30±0.90)e-1</b> | (7.75±4.80)e-2        | (6.58±0.44)e-1        |
|        | 1000  | <b>(7.73±0.65)e-1</b> | (6.36±1.50)e-1        | (7.08±0.44)e-1        | (6.15±0.32)e-1 | <b>(7.83±0.00)e-1</b> | (7.42±0.73)e-1        | (7.79±8.97)e-2        | (7.07±0.57)e-1        |
|        | 1500  | (7.79±0.90)e-1        | (7.63±1.46)e-1        | (7.38±0.43)e-1        | (6.95±0.35)e-1 | <b>(8.64±0.00)e-1</b> | (7.31±1.15)e-1        | (5.71±5.25)e-2        | (7.56±0.33)e-1        |
|        | 10000 | (6.66±1.70)e-1        | (8.99±0.02)e-1        | (7.95±0.48)e-1        | (8.09±0.14)e-1 | <b>(9.26±0.00)e-1</b> | (7.86±0.75)e-1        | (9.41±6.58)e-2        | (8.58±0.16)e-1        |
| DTLZ5  | 250   | (8.22±0.06)e-1        | (8.26±0.04)e-1        | <b>(9.51±0.19)e-1</b> | (8.52±0.25)e-1 | <b>(9.43±0.00)e-1</b> | (8.22±0.06)e-1        | (9.06±0.07)e-1        | (7.90±0.26)e-1        |
|        | 500   | (8.30±0.04)e-1        | (8.30±0.04)e-1        | <b>(9.44±0.20)e-1</b> | (8.98±0.18)e-1 | <b>(9.52±0.00)e-1</b> | (8.43±0.10)e-1        | (9.16±0.05)e-1        | (8.86±0.62)e-1        |
|        | 1000  | (8.45±0.02)e-1        | (8.35±0.05)e-1        | (9.59±0.09)e-1        | (9.21±0.14)e-1 | (9.60±0.00)e-1        | (9.18±0.05)e-1        | (9.18±0.05)e-1        | <b>(9.60±0.38)e-1</b> |
|        | 1500  | (8.93±0.03)e-1        | (8.59±0.04)e-1        | (9.65±0.02)e-1        | (9.35±0.11)e-1 | (9.66±0.00)e-1        | (9.25±0.06)e-1        | (9.21±0.04)e-1        | <b>(9.70±0.01)e-1</b> |
|        | 10000 | (9.42±0.02)e-1        | (8.98±0.02)e-1        | (9.65±0.02)e-1        | (9.57±0.05)e-1 | <b>(9.71±0.00)e-1</b> | (9.44±0.03)e-1        | (9.16±0.07)e-1        | (9.71±0.01)e-1        |
| DTLZ6  | 250   | (2.81±0.43)e-1        | (2.10±0.02)e-1        | (8.79±0.18)e-1        | (3.08±0.45)e-1 | (8.41±0.00)e-1        | (2.86±0.44)e-1        | <b>(9.83±0.04)e-1</b> | (4.38±0.28)e-1        |
|        | 500   | (3.18±0.63)e-1        | (2.07±0.02)e-1        | (8.71±0.25)e-1        | (3.07±0.50)e-1 | (8.56±0.00)e-1        | (4.70±1.46)e-1        | <b>(9.84±0.04)e-1</b> | (4.54±0.36)e-1        |
|        | 1000  | (4.20±1.50)e-1        | (2.34±0.04)e-1        | (8.77±0.18)e-1        | (4.62±0.42)e-1 | (8.78±0.00)e-1        | (7.53±1.17)e-1        | <b>(9.87±0.03)e-1</b> | (4.88±0.28)e-1        |
|        | 1500  | (5.86±1.35)e-1        | (2.93±0.11)e-1        | (8.96±0.23)e-1        | (5.75±0.36)e-1 | (8.99±0.00)e-1        | (8.51±0.63)e-1        | <b>(9.88±0.02)e-1</b> | (4.95±0.28)e-1        |
|        | 10000 | (6.69±0.59)e-1        | (3.82±0.16)e-1        | (9.24±0.16)e-1        | (7.04±0.13)e-1 | (9.81±0.00)e-1        | (8.91±1.00)e-1        | <b>(9.89±0.02)e-1</b> | (5.39±0.13)e-1        |
| DTLZ7  | 250   | <b>(8.34±0.12)e-1</b> | (2.31±0.11)e-1        | (6.28±0.58)e-1        | (4.66±0.21)e-1 | (5.87±0.00)e-1        | <b>(8.35±0.06)e-1</b> | (7.48±0.33)e-1        | (5.22±0.23)e-1        |
|        | 500   | <b>(8.46±0.00)e-1</b> | (2.27±0.13)e-1        | (6.35±0.39)e-1        | (5.07±0.23)e-1 | (6.76±0.00)e-1        | <b>(8.46±0.00)e-1</b> | (7.82±0.24)e-1        | (5.38±0.18)e-1        |
|        | 1000  | <b>(8.46±0.00)e-1</b> | (3.02±0.08)e-1        | (6.73±0.50)e-1        | (5.25±0.20)e-1 | (7.68±0.00)e-1        | <b>(8.44±0.05)e-1</b> | (8.09±0.16)e-1        | (5.58±0.19)e-1        |
|        | 1500  | <b>(8.46±0.00)e-1</b> | (3.70±0.08)e-1        | (7.24±0.50)e-1        | (5.44±0.28)e-1 | (8.17±0.00)e-1        | <b>(8.46±0.00)e-1</b> | (7.95±0.21)e-1        | (5.76±0.24)e-1        |
|        | 10000 | (8.46±0.00)e-1        | (4.88±0.08)e-1        | (8.31±0.11)e-1        | (6.01±0.12)e-1 | <b>(8.92±0.00)e-1</b> | (8.39±0.20)e-1        | (8.08±0.22)e-1        | (6.17±0.16)e-1        |

Source – Produced by authors.

In comparison with the NSGA-III algorithm, for 3 objectives, in the problems DTLZ1-4 that have simpler Pareto-optimal fronts, all of them triangular with DTLZ1 being plan and DTLZ2, DTLZ3 and DTLZ4 a quarter of a sphere, the DVL algorithm was superior only when there is a higher restriction in the number of evaluations. In a less restrictive scenario, such as

Table 15 – Statistical results (mean and standard deviation) of the Hypervolume values for 10 objectives.

| Probl. | Eval. | MO-Linear             | SVR            | MO-MLP                | MO-RFR         | NSGA-III              | Linear                | MLP                   | RFR            |
|--------|-------|-----------------------|----------------|-----------------------|----------------|-----------------------|-----------------------|-----------------------|----------------|
| DTLZ1  | 250   | (8.80±0.81)e-1        | (9.94±0.05)e-1 | (9.87±0.09)e-1        | (9.67±0.21)e-1 | (9.44±0.00)e-1        | (9.08±0.72)e-1        | <b>(9.99±0.01)e-1</b> | (9.68±0.16)e-1 |
|        | 500   | (9.00±0.76)e-1        | (9.91±0.21)e-1 | (9.86±0.08)e-1        | (9.73±0.17)e-1 | (9.49±0.00)e-1        | (9.05±0.71)e-1        | <b>(9.99±0.01)e-1</b> | (9.79±0.10)e-1 |
|        | 1000  | (9.10±0.74)e-1        | (9.98±0.01)e-1 | (9.86±0.07)e-1        | (9.76±0.12)e-1 | (9.80±0.00)e-1        | (9.71±0.23)e-1        | <b>(1.00±0.00)e-1</b> | (9.80±0.09)e-1 |
|        | 1500  | (9.01±0.65)e-1        | (9.95±0.05)e-1 | (9.80±0.12)e-1        | (9.84±0.10)e-1 | (9.84±0.00)e-1        | (9.08±1.43)e-1        | <b>(1.00±0.00)e-1</b> | (9.82±0.06)e-1 |
|        | 10000 | (9.62±0.31)e-1        | (9.40±0.90)e-1 | (9.97±0.01)e-1        | (9.89±0.06)e-1 | <b>(9.99±0.00)e-1</b> | (9.25±0.92)e-1        | <b>(1.00±0.00)e-1</b> | (9.91±0.05)e-1 |
| DTLZ2  | 250   | (7.73±0.46)e-1        | (3.91±0.06)e-1 | (9.36±0.10)e-1        | (5.24±0.36)e-1 | (9.15±0.00)e-1        | (8.11±0.52)e-1        | <b>(9.90±0.02)e-1</b> | (7.74±0.36)e-1 |
|        | 500   | (7.93±0.41)e-1        | (4.62±0.07)e-1 | (9.32±0.11)e-1        | (6.00±0.26)e-1 | (9.23±0.00)e-1        | (8.13±0.60)e-1        | <b>(9.93±0.01)e-1</b> | (8.49±0.23)e-1 |
|        | 1000  | (7.99±0.41)e-1        | (4.79±0.05)e-1 | (9.27±0.08)e-1        | (6.47±0.24)e-1 | (9.40±0.00)e-1        | (9.42±0.26)e-1        | <b>(9.94±0.01)e-1</b> | (9.28±0.15)e-1 |
|        | 1500  | (8.21±0.18)e-1        | (5.53±0.03)e-1 | (9.29±0.09)e-1        | (7.10±0.18)e-1 | (9.57±0.00)e-1        | (9.88±0.11)e-1        | <b>(9.94±0.01)e-1</b> | (9.75±0.04)e-1 |
|        | 10000 | (9.74±0.03)e-1        | (7.23±0.02)e-1 | (9.59±0.07)e-1        | (8.30±0.06)e-1 | <b>(9.99±0.00)e-1</b> | <b>(9.98±0.01)e-1</b> | (9.95±0.01)e-1        | (9.96±0.00)e-1 |
| DTLZ3  | 250   | (4.92±1.86)e-1        | (9.91±0.04)e-1 | (9.75±0.14)e-1        | (9.48±0.33)e-1 | (5.35±0.00)e-1        | (4.88±1.99)e-1        | <b>(1.00±0.00)e-1</b> | (9.45±0.24)e-1 |
|        | 500   | (5.05±1.76)e-1        | (9.96±0.00)e-1 | (9.76±0.13)e-1        | (9.56±0.30)e-1 | (5.78±0.00)e-1        | (5.33±1.57)e-1        | <b>(1.00±0.00)e-1</b> | (9.54±0.29)e-1 |
|        | 1000  | (6.38±1.89)e-1        | (9.96±0.00)e-1 | (9.70±0.13)e-1        | (9.68±0.19)e-1 | (6.56±0.00)e-1        | (6.56±1.98)e-1        | <b>(1.00±0.00)e-1</b> | (9.63±0.14)e-1 |
|        | 1500  | (6.33±2.01)e-1        | (9.97±0.00)e-1 | (9.42±0.28)e-1        | (9.81±0.10)e-1 | (7.44±0.00)e-1        | (5.68±2.26)e-1        | <b>(1.00±0.00)e-1</b> | (9.70±0.13)e-1 |
|        | 10000 | (8.73±0.81)e-1        | (9.97±0.00)e-1 | (9.76±0.17)e-1        | (9.92±0.03)e-1 | (9.85±0.00)e-1        | (4.83±1.92)e-1        | <b>(1.00±0.00)e-1</b> | (9.82±0.11)e-1 |
| DTLZ4  | 250   | (9.75±0.16)e-1        | (5.00±0.00)e-1 | <b>(9.97±0.01)e-1</b> | (5.00±0.00)e-1 | (8.81±0.00)e-1        | (9.78±0.11)e-1        | (9.89±0.02)e-1        | (5.57±0.40)e-1 |
|        | 500   | (9.89±0.05)e-1        | (5.00±0.00)e-1 | <b>(9.97±0.01)e-1</b> | (5.00±0.01)e-1 | (8.88±0.00)e-1        | (9.92±0.06)e-1        | (9.93±0.01)e-1        | (7.37±0.24)e-1 |
|        | 1000  | (9.91±0.05)e-1        | (5.00±0.00)e-1 | <b>(9.97±0.01)e-1</b> | (5.14±0.35)e-1 | (9.66±0.00)e-1        | (9.92±0.06)e-1        | (9.94±0.01)e-1        | (7.91±0.42)e-1 |
|        | 1500  | <b>(9.96±0.02)e-1</b> | (5.00±0.00)e-1 | (9.93±0.05)e-1        | (7.64±0.50)e-1 | (9.89±0.00)e-1        | (9.89±0.09)e-1        | (9.94±0.01)e-1        | (8.70±0.19)e-1 |
|        | 10000 | (9.99±0.00)e-1        | (9.99±0.00)e-1 | (9.63±0.26)e-1        | (9.40±0.17)e-1 | <b>(1.00±0.00)e-1</b> | (5.23±0.69)e-1        | (9.95±0.01)e-1        | (9.70±0.02)e-1 |
| DTLZ5  | 250   | (9.71±0.17)e-1        | (9.06±0.02)e-1 | <b>(9.96±0.00)e-1</b> | (9.52±0.06)e-1 | (9.52±0.00)e-1        | (9.72±0.16)e-1        | (9.96±0.00)e-1        | (9.80±0.06)e-1 |
|        | 500   | (9.62±0.16)e-1        | (9.23±0.01)e-1 | <b>(9.96±0.00)e-1</b> | (9.60±0.04)e-1 | (9.53±0.00)e-1        | (9.64±0.17)e-1        | (9.96±0.00)e-1        | (9.90±0.02)e-1 |
|        | 1000  | (9.55±0.10)e-1        | (9.25±0.01)e-1 | <b>(9.96±0.00)e-1</b> | (9.62±0.04)e-1 | (9.55±0.00)e-1        | (9.67±0.11)e-1        | (9.96±0.00)e-1        | (9.92±0.01)e-1 |
|        | 1500  | (9.68±0.03)e-1        | (9.41±0.01)e-1 | <b>(9.97±0.00)e-1</b> | (9.68±0.02)e-1 | (9.55±0.00)e-1        | (9.79±0.04)e-1        | (9.96±0.00)e-1        | (9.93±0.00)e-1 |
|        | 10000 | (9.93±0.01)e-1        | (9.72±0.01)e-1 | <b>(9.97±0.00)e-1</b> | (9.87±0.01)e-1 | (9.95±0.00)e-1        | (9.95±0.01)e-1        | (9.96±0.00)e-1        | (9.94±0.00)e-1 |
| DTLZ6  | 250   | (9.54±0.12)e-1        | (8.52±0.05)e-1 | (9.99±0.00)e-1        | (9.26±0.07)e-1 | (9.97±0.00)e-1        | (9.54±0.13)e-1        | <b>(1.00±0.00)e-1</b> | (9.34±0.07)e-1 |
|        | 500   | (9.56±0.10)e-1        | (8.51±0.06)e-1 | (9.99±0.00)e-1        | (9.21±0.12)e-1 | (9.97±0.00)e-1        | (9.60±0.07)e-1        | <b>(1.00±0.00)e-1</b> | (9.44±0.07)e-1 |
|        | 1000  | (9.61±0.12)e-1        | (8.53±0.05)e-1 | (9.99±0.00)e-1        | (9.24±0.12)e-1 | (9.98±0.00)e-1        | (9.73±0.19)e-1        | <b>(1.00±0.00)e-1</b> | (9.49±0.03)e-1 |
|        | 1500  | (9.70±0.01)e-1        | (8.80±0.06)e-1 | (9.99±0.00)e-1        | (9.66±0.06)e-1 | (9.98±0.00)e-1        | (9.74±0.04)e-1        | <b>(1.00±0.00)e-1</b> | (9.52±0.04)e-1 |
|        | 10000 | (9.83±0.03)e-1        | (9.62±0.00)e-1 | (9.99±0.00)e-1        | (9.84±0.02)e-1 | (9.99±0.00)e-1        | (9.84±0.02)e-1        | <b>(1.00±0.00)e-1</b> | (9.70±0.14)e-1 |
| DTLZ7  | 250   | (7.61±0.16)e-1        | (1.79±0.19)e-1 | (7.49±0.37)e-1        | (3.21±0.32)e-1 | (5.40±0.00)e-1        | (7.68±0.01)e-1        | <b>(8.29±0.08)e-1</b> | (5.31±0.33)e-1 |
|        | 500   | (7.68±0.00)e-1        | (1.76±0.11)e-1 | (7.46±0.48)e-1        | (3.65±0.62)e-1 | (5.54±0.00)e-1        | (7.68±0.00)e-1        | <b>(8.33±0.08)e-1</b> | (5.60±0.23)e-1 |
|        | 1000  | (7.68±0.00)e-1        | (1.80±0.09)e-1 | (7.47±0.42)e-1        | (3.67±0.32)e-1 | (5.79±0.00)e-1        | (7.68±0.00)e-1        | <b>(8.32±0.08)e-1</b> | (6.03±0.33)e-1 |
|        | 1500  | (7.68±0.00)e-1        | (3.41±0.07)e-1 | (7.78±0.26)e-1        | (4.44±0.58)e-1 | (6.60±0.00)e-1        | (7.68±0.00)e-1        | <b>(8.41±0.07)e-1</b> | (6.70±0.30)e-1 |
|        | 10000 | (7.68±0.00)e-1        | (5.10±0.03)e-1 | (8.37±0.12)e-1        | (6.18±0.16)e-1 | (7.81±0.00)e-1        | (7.68±0.00)e-1        | <b>(8.48±0.07)e-1</b> | (7.32±0.15)e-1 |

Source – Produced by authors.

1500 and 10000 number of evaluations, the NSGA-III got better results in these problems. For the problems DTLZ5-7 the scenario is quite different, with the DVL being superior in most cases. These problems are characterized by having Pareto-optimal fronts with different shapes from the previous ones. The case of DTLZ5 and DTLZ6 both have degenerated Pareto-optimal fronts, and DTLZ7 has a disconnected front. These kinds of problems make it difficult for MOEAs to find the ideal frontier.

The results of the experiment with 10 objectives show that the DVL algorithm was superior in the vast majority of cases, except for the case of the DTLZ4 problem with 10000 evaluations, in which the NSGA-III is superior to the DVL with any model. Table 16 summarizes the results of the statistical comparison in terms of wins, draws, and losses. A win means that the algorithm has the best-isolated result, no other is statistically comparable to it. A draw means that the algorithm has the best result and it is statistically comparable with others. And a loss means that the algorithm has not achieved the best value and it is not statistically comparable with the best result. The heat maps of the statistical comparison of this experiment can be seen in Appendix-Section A. It is possible to note the DVL superiority in problems with 10 objectives by winning 33 of 35 scenarios. With the MLP model having 23 wins in comparison to the others. In the experimentation with 3 objectives, the NSGA-III wins a total of 13 in 35, and the combination of all DVL different models wins 15. This shows that the DVL can also be used for optimization in problems with a lower number of objectives. We can deduce from the experiments that the

DVL has a good performance in different kinds of problems with different Pareto-optimal shapes, search space sizes, with different local and global optimal, and a different number of decision variables. According to the experiments, this last characteristic, the number of decision variables, is a key factor for the greater performance of the DVL in comparison to the NSGA-III. The results also show that the strategy used for learning as well as the model is problem-dependent. In different scenarios, different strategies got better results, with the MO-MLP, and MLP having the best overall between these strategies.

Table 16 – Wins/Ties/Loses of the Statistical Comparison.

|           | 3 objectives | 10 objectives |
|-----------|--------------|---------------|
| MO-Linear | 0 / 5 / 30   | 1 / 0 / 34    |
| SVR       | 4 / 0 / 31   | 0 / 0 / 35    |
| MO-MLP    | 1 / 2 / 32   | 8 / 0 / 27    |
| MO-RFR    | 0 / 0 / 35   | 0 / 0 / 35    |
| NSGA-III  | 13 / 3 / 19  | 1 / 2 / 32    |
| Linear    | 1 / 4 / 30   | 0 / 1 / 34    |
| MLP       | 5 / 0 / 30   | 23 / 1 / 11   |
| RFR       | 4 / 0 / 31   | 0 / 0 / 35    |

Source – Produced by authors.

The overall result found is quite promising, the DVL produced a valuable performance against the NSGA-III in many scenarios, even though the DVL algorithm is recent and also it needs greater maturity for optimization tasks. It is observed that different from the NSGA-III, the DVL does not behave like a search algorithm for optimization. The hypervolume of the solutions generated by the DVL is not always been improved by the increase of iterations. This means that there is a lack of local search to improve the solutions found near the Pareto-optimal front. Another point is that the reference points used by the DVL may not be close to the Pareto-optimal front which can lead to unwanted values of decision variables. Research on the algorithm architecture to improve the DVL can lead to even better results.

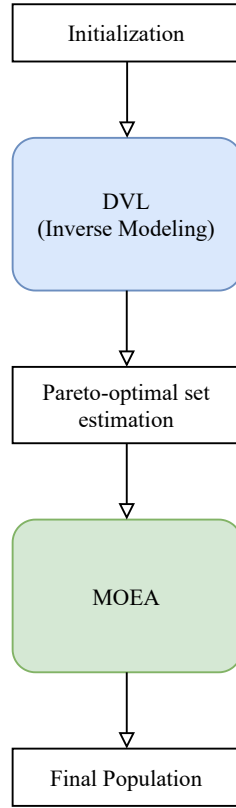
# 4

## DVL Framework

In the previous chapter, it was demonstrated that the inverse model is experimentally capable of estimating the Pareto front in diverse optimization problems. So it is possible to proceed to the next step, in this chapter, we propose a DVL variation that can be used in integration with MOEAs, and design a framework that abstracts this integration. We believe that this approach can lead to the improvement of both algorithms, producing better results in solving optimization problems. The DVL algorithm can benefit from the MOEA due to its global search strategies and adaptation to the Pareto front of the problem. The MOEA can benefit from the DVL due to its high capacity for a global understanding of the problem, achieving a high convergence to the PF with a low number of evaluations, as we showed in Chapter 3. We also compare the proposed DVL Framework integrated with some well-known MOEAs and apply it to solve several multi-objective optimization problems, from benchmark to real optimization problems. The goal is to validate the proposed framework and evaluate its ability to improve the performance of well-known algorithms in the literature.

To improve the algorithm limitations and leverage its strengths, we propose a DVL Framework that uses a DVL variation that can be executed in cooperation with an MOEA. The main reason to create a framework instead of an algorithm is due to the capacity to abstract the implementation, integrate with other algorithms, and provide a generic functionality, in this case, providing an initial set of solutions to any MOEA of the literature. Figure 22 presents the overall algorithm framework. The DVL Framework is based on two main steps, the first one is the use of DVL inverse modeling. The key idea is to use the good performance in restrictive scenarios of the DVL algorithm to estimate an initial set of solutions to the MOEA. The second step is the use of the initial set, which is an approximation of the Pareto-optimal provided by the DVL, to the MOEA to continue the optimization process until it reaches a pre-determined time or number of evaluations. In the end, a final population set, which represents the predicted set of the Pareto-optimal front, is outputted by the framework.

Figure 22 – DVL Framework.



Source: Produced by authors.

## 4.1 Algorithms

We separate the implementation details of the DVL Framework into two parts to facilitate the analysis of the algorithms. The first one has a higher abstraction level and provides the overall pseudocode of the framework. The second one references the new implementation of the DVL inverse modeling, which has been modified to work in the DVL Framework. The pseudocode of the framework is described in Algorithm 2.

Algorithm 2 – The DVL Framework pseudocode.

**Data:** the  $k$  value for the size of the dataset used for training, the  $e$  value of the maximum number of evaluations and the  $m$  MOEA algorithm used to continue the optimization process.

**Result:** the population  $P_f$  generated by the framework;

```

1 begin
2    $P_{est} \leftarrow ExecuteDVL(k);$ 
3    $d \leftarrow e - (k + Size(P_{est}));$ 
4    $P_f \leftarrow ExecuteMOEA(P_{est}, m, d);$ 
5 end
  
```



The DVL Framework receives as parameters the  $k$  value that represents the size of the dataset used by the DVL inverse model to generate the population set by the LHS and train the models. The  $e$  value is a variable responsible to indicate when the optimization process ends containing the maximum number of evaluations. And the MOEA algorithm  $m$  used inside the framework to continue the optimization process, this MOEA can be any optimization algorithm and have any implementation as long as it accepts the given parameters and returns the final population at the end of the optimization process. The design of the framework facilitates the use and replication of the framework with different algorithms and in different environments. Benefiting from the strengths of different MOEAs have in solving optimization problems.

On line 2, the framework executes the DVL inverse modeling algorithm passing as parameters the  $k$  size of the dataset used by the DVL for training the model. The specification of this DVL inverse model algorithm is better described below. The DVL inverse modeling estimates the  $P_{est}$  population as close as possible to the Pareto-optimal front. After this step, on line 3, a variable  $d$  is declared that represents the total amount of the number of objective function evaluations left to complete the framework. The value  $d$  is calculated by the difference between the total  $e$  evaluations and the number of evaluations that the algorithm already performed. On line 4, the framework executes the MOEA, passing as a parameter the population  $P_{est}$  estimated by the DVL, as an initial population for the MOEA, the  $d$  maximum number of evaluations that the MOEA has to perform, and the  $m$  MOEA that will continue the optimization process. The final set  $P_f$ , which is the best population found by the MOEA based on the initial set  $P_{est}$ , is used as the output of the framework.

The *ExecuteDVL* function demonstrated in Algorithm 2 executes a different DVL algorithm version from the one proposed in Santos, Oliveira & Britto (2019). We preserve the strengths points of the DVL algorithm, more precisely, the inverse modeling used to estimate the Pareto-optimal. Also, we understand that the iteration process could be removed and changed by the search process of the MOEA executed in the framework, taking advantage of the strategy to perform and adapt to global and local searches, improving the estimation of the final Pareto set. Doing such alteration, the necessity of using the hypervolume inside the algorithm is eliminated, being exchanged by the selection operator of the MOEA. Algorithm 3 describes the pseudocode of the new DVL variation.

As shown in the algorithm, the number of input data passed to the algorithm decreased to only one, the  $k$  value of the size of the population generated by the LHS algorithm used for training the inverse model. The algorithm starts, on line 2, creating the solution set  $P$ . This set contains  $m + n$  columns and  $k$  lines, where  $m$  is the number of objectives of the problem,  $n$  is the number of variables, and  $k$  is the size of the initial solutions set. The LHS algorithm is used to generate  $k$  solutions, resulting in  $n$  columns of the set, and on line 3, the evaluation of these solutions generates the  $m$  columns, representing the vector of objectives. The purpose of this methodology is to create a set of solutions that explore the entire space of variables and thereby

## Algorithm 3 – The DVL inverse modeling pseudocode.

**Data:** the  $k$  value for the size of the dataset used for training;  
**Result:** the population  $P_{est}$  generated by the model estimation;

```

1 begin
2    $P \leftarrow LHS(k)$ ;
3    $Evaluate(P)$ ;
4    $RP \leftarrow GenerateReferencePoints()$ ;
5    $P_{est} \leftarrow \emptyset$ ;
6    $M_i \leftarrow TrainModels(P)$ ;
7   for  $r \in RP$  do
8      $P_{est} \leftarrow P_{est} \cup M_i.Predict(r)$ ;
9   end
10   $Evaluate(P_{est})$ ;
11 end

```

explore all objective space.

On line 4, the reference points set  $RP$  that is used to estimate the new solutions are created in the function *GenerateReferencePoints()* based on the method in (DAS; DENNIS, 1996). The reference points are placed on a normalized hyperplane which is also inclined to each objective and has an intersection on each axis. The population set  $P_{est}$  is initialized on line 5. After the initial generation of the data and preparation of the environment variables, the estimation process is started on line 6. This process aims to train models of machine learning and from these models, to estimate solutions. Two different strategies can be used in this step, for each decision variable of the problem, models will be trained to learn the relationship between objective functions and the decision variable or only one model can be trained with the entire dataset  $P$ . In the first strategy, for a problem with  $n$  variables,  $n$  supervised learning models will be used, where each model  $M_i$ , for  $0 < i \leq n$ , will learn the function  $g_i(Y) = x_i$ , where  $Y$  is a vector of objectives and  $x_i$  is the  $i$ -th decision variable of solution  $X$ , such that  $Y$  is the objective vector of solution  $X$ . For the second strategy, only one model  $M_1$  is generated to learn the function  $g(Y) = X$ .

From the information learned by the model, it can estimate solutions that generate points in the space of objectives close to reference points that are considered optimal. This estimation is made on lines 7 to 9, where for each reference point  $r$  from the  $RP$  set, a new solution is generated and added to the  $P_{est}$  set. We maintain the use of the reference point as input for prediction because despite not representing precisely the optimal front, it is a good starting set, especially when used at the beginning of the optimization process as proposed by the framework. Finally, on line 10, the estimated Pareto-optimal set  $P_{est}$  is evaluated and used as the output of the DVL algorithm.

## 4.2 Experiments and Results

In this section, we evaluate the new strategy of using the DVL as a framework for inverse modeling in combination with an MOEA. The motivation is to validate the DVL algorithm as a strategy for producing initial solutions as close as possible to the Pareto-optimal front. Due to its capacity of global understanding of the problem with a low number of evaluations. The performance of the DVL Framework is compared to the default implementation of the DVL algorithm, and also, with state-of-the-art MOEAs used in the literature. We use different MOEA approaches to deal with optimization problems, such as the NSGA-III as a candidate for Pareto-based approach, the MOEA/D as a decomposition-based algorithm, and the RVEA as a reference vector-guided algorithm. Also, these MOEAs are used inside the DVL Framework in our experiments. The DVL algorithm and the DVL Framework are tested in the Traffic Lights Signaling optimization problem described in Section 4.2.1.1. Different from the previous experiments, this real optimization problem has an expensive objective function and constraints which provides a new scenario of study the DVL inverse modeling.

### 4.2.1 Optimization Problems

Two different kinds of optimization problems are used in the experiments of the DVL Framework. The first set of problems is the first seven test problems from the DTLZ benchmark test: DTLZ1 to DTLZ7, already described in our previous experiments in Section 3.2. We also maintain the configuration of the DTLZ instances for this experiment, one with 3 objectives and 12 variables and the other with 10 objectives and 12 variables. This kind of optimization problem gives us support in the initial analysis of the DVL Framework.

The second type of optimization problem used in the DVL Framework experiment is the problem of traffic light signaling modeled as an MaOP, a real-world expensive many-objective optimization problem. This problem is characterized by having many objectives, a costly objective function, constraints, and an unknown Pareto-optimal set. This problem is well described below to improve the understanding for later use in our experiments. The use of an experiment of this kind is important to analyze the performance of the DVL Framework and also the DVL in a scenario never tested before.

#### 4.2.1.1 Traffic Lights Signaling: A Many-Objective Optimization Problem

The Intelligent Transportation Systems (ITS) aim to improve transportation through the use of technologies. One of its areas of improvement is traffic management, where the efficiency can be improved by using data from the roads and the traffic to optimize the efficiency, safety, and quality of the traffic flow, for example, by reducing traffic congestion and travel time. Traffic light signaling optimization is one way of improving traffic efficiency. It relies on optimizing the traffic-light cycle, which is the sequence of states that a traffic light has (SÁNCHEZ-

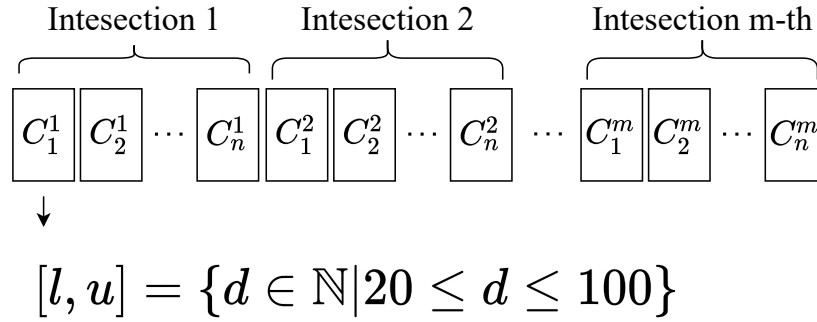
MEDINA; GALÁN-MORENO; RUBIO-ROYO, 2009). To obtain the data of the traffic flow a traffic simulator can be used. A well-known traffic simulator is SUMO (Simulation of Urban Mobility) (KRAJZEWICZ; BONERT; WAGNER, 2006) which provides all the data necessary to simulate a traffic flow, manage the traffic-light cycle, and obtain several traffic quality measures such as travel and stopped time, the number of vehicles that arrive at their destination, the overall average speed, and traffic flow, carbon dioxide emission, among others.

In traffic lights signaling the synchronization is achieved when two or more traffic lights are running in a manner that allows a vehicle to pass the traffic lights synchronized without stops. Traffic light signal optimization in general is a very complex problem, even for a single intersection can be no obvious optimal solution. The synchronization will not always ensure that vehicles will pass through open traffic lights. In this problem, it is possible to use several traffic quality measures as objective functions. But, the majority of the related work only selects a small set of objective functions (often two). This limitation on the number of objective functions lies in the incapacity of traditional MOEA to deal with a high number of objective functions. In this work, we modeled this problem as a MaOP with 6 objective functions to approximate the modeling of real situations through the use of simulators, since it is important to consider different quality measures when defining traffic-light cycles.

In the representation of the problem of traffic light synchronization modeled as a MaOP, it is necessary to give a combination of traffic lights to build a computational representation of the duration of each phase for each traffic light. The duration may assume a finite interval, usually with few options, e.g, in our study, it assumes a range from four to one hundred and twenty. The decision variables are the time of each phase of each traffic light, and the objective functions are the traffic-related quality measures calculated by the simulator. The computational representation of the duration time of the traffic light phases can assume a mathematical representation using a solution vector,  $\mathbf{x}$ , of integer variables, such that:  $\mathbf{x} = (C_1^1, C_2^1, \dots, C_n^1, C_1^2, \dots, C_n^2, C_1^m, \dots, C_n^m)$ , where  $C_j^k$  represents the duration,  $d$ , of the  $j^{th}$  phase (optimizable) of the  $k^{th}$  intersection and  $20 \leq d \leq 100, d \in \mathbb{N}$ . From each solution  $\mathbf{x}$ , the simulator calculates the values of the  $m$ -objective functions to form the objective vector  $f(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))$  which will be later evaluated by the search algorithm.

Thus, it is possible to treat the traffic light synchronization problem as an optimization problem, representing a computationally possible combination of the phase times of the traffic lights through a vector. Figure 23 shows all elements of the vector. These elements must be within the range contained in the lower ( $l$ ) and upper ( $u$ ) limits of the “green phases” duration. For each intersection (ranging from 1 to  $m$ ), the size of the vector to be optimized will be formed by the phases (ranging from 1 to  $n$ ) where, in at least one direction, one has a green state, a “green phase”. This implies that the phase in which the directions are formed only by red states, or only by yellow and red states, will have its time set to 4 seconds and will not be used in the vector.

Figure 23 – Vector of the phases of all intersections.



Source: (MATOS; CARVALHO, 2016)

#### 4.2.2 Algorithms, Parameters and quality metrics

In this experiment, the same procedures of the optimization experiment in Section 3.2.3 are followed. Each algorithm runs until reaches a maximum number of evaluations, defined as (250, 500, 1000, 1500, 10000, 100000). Different from the previous experiment, the value of 100000 maximum number of evaluations was added to validate the DVL and the DVL Framework in scenarios where the MOEAs have good results. The empirical comparisons between the results are made by the use of Hypervolume (HV) (WHILE et al., 2006), a performance indicator that assesses both convergence and divergence of the solutions. Different from the previous experiment, a new reference points set is used with values closer to the Optimal-front. Table 17 shows the reference points set used in this experiment for each DTLZ problem. The motivation to use this new set is to achieve more precision and detail in the comparison of the results.

Table 17 – Reference points used for each DTLZ problem.

| Database | Reference points set for 3 objectives | Reference points set for 10 objectives |
|----------|---------------------------------------|----------------------------------------|
| DTLZ1    | (1.0, 1.0, 1.0)                       | (5.0, 5.0, ..., 5.0)                   |
| DTLZ2    | (2.0, 2.0, 2.0)                       | (1.5, 1.5, ..., 1.5)                   |
| DTLZ3    | (10.0, 10.0, 10.0)                    | (50.0, 50.0, ..., 50.0)                |
| DTLZ4    | (2.0, 2.0, 2.0)                       | (1.5, 1.5, ..., 1.5)                   |
| DTLZ5    | (3.0, 5.0, 7.0)                       | (3.0, 5.0, ..., 21.0)                  |
| DTLZ6    | (6.0, 10.0, 14.0)                     | (6.0, 10.0, ..., 42.0)                 |
| DTLZ7    | (10.0, 18.0, 26.0)                    | (10.0, 18.0, ..., 82.0)                |

Source – Produced by authors.

The DVL Framework needs three parameters to be predefined for execution. A  $k$  value for the initial number of the population used by the LHS. A value  $e$  of the maximum number of evaluations. And the MOEA algorithm  $m$  to execute after the DVL inverse model. The  $k$  and the  $e$  parameters have different values for each configuration of DTLZ problem vs. number of objectives vs. number of evaluations. The  $m$  parameter value corresponds to the three MOEAs that will be

tested in the experiment: NSGA-III, MOEA/D, and RVEA. These different configurations of  $k$  and  $e$  for the optimization experiment are shown in Table 18, the  $k$  values were found empirically during our experiments.

Table 18 – DVL parameter configuration for the optimization experiment.

| Problem | Number of Evaluations = $e$ | $k$ for 3 Objectives | $k$ for 10 Objectives |
|---------|-----------------------------|----------------------|-----------------------|
| DTLZ1   | 250                         | 159                  | 50                    |
|         | 500                         | 227                  | 112                   |
|         | 1000                        | 250                  | 200                   |
|         | 1500                        | 300                  | 200                   |
|         | 10000                       | 300                  | 300                   |
| DTLZ2   | 250                         | 159                  | 50                    |
|         | 500                         | 227                  | 280                   |
|         | 1000                        | 600                  | 300                   |
|         | 1500                        | 600                  | 300                   |
|         | 10000                       | 600                  | 300                   |
| DTLZ3   | 250                         | 159                  | 50                    |
|         | 500                         | 227                  | 112                   |
|         | 1000                        | 250                  | 200                   |
|         | 1500                        | 300                  | 200                   |
|         | 10000                       | 300                  | 300                   |
| DTLZ4   | 250                         | 159                  | 50                    |
|         | 500                         | 410                  | 280                   |
|         | 1000                        | 410                  | 300                   |
|         | 1500                        | 410                  | 300                   |
|         | 10000                       | 500                  | 300                   |
| DTLZ5   | 250                         | 159                  | 50                    |
|         | 500                         | 227                  | 280                   |
|         | 1000                        | 300                  | 300                   |
|         | 1500                        | 300                  | 300                   |
|         | 10000                       | 300                  | 600                   |
| DTLZ6   | 250                         | 159                  | 50                    |
|         | 500                         | 227                  | 100                   |
|         | 1000                        | 300                  | 100                   |
|         | 1500                        | 600                  | 100                   |
|         | 10000                       | 600                  | 100                   |
| DTLZ7   | 250                         | 159                  | 50                    |
|         | 500                         | 400                  | 100                   |
|         | 1000                        | 600                  | 200                   |
|         | 1500                        | 600                  | 300                   |
|         | 10000                       | 600                  | 600                   |

Source – Produced by authors.

The parameters used and the implementations of the NSGA-III and MOEA/D algorithms were defined by the pymoo (Blank; Deb, 2020). For the NSGA-III, using simulated binary crossover (DEB; AGRAWAL et al., 1995), the distribution index is set to  $\eta c = 30$ , and the crossover probability  $pc = 1.0$ . For the polynomial mutation (DEB; GOYAL, 1996), the distribution index and the mutation probability are set to  $\eta m = 20$  and  $pm = 1/n$ , respectively, as recommended in (DEB, 2001). For the MOEA/D algorithm, the number of the weight vectors in the neighborhood  $n = 15$  and the mating probability of two solutions in the neighborhood is set

to  $T = 0.7$ . For the RVEA algorithm, was used the python library pyRVEA (SAINI, 2019), which is a python implementation of the original version of the RVEA based in Cheng et al. (2016). The parameters used were, for the simulated binary crossover, the same as the NSGA-III algorithm, the distribution index is set to  $\eta_c = 30$ , and the crossover probability  $p_c = 1.0$ . And also the same as the NSGA-III for the polynomial mutation, the distribution index, and the mutation probability are set to  $\eta_m = 20$  and  $p_m = 1/n$ , respectively. For all the algorithms the population size used was 91 and 220 for the problem with 3 and 10 objectives, respectively, and the individual has a length of 12 for both objective sizes.

### 4.2.3 Results and discussion

In this subsection, we analyze the results of the DVL Framework experiment on the DTLZ benchmark test suit. All the algorithms were executed 20 times independently for each restriction configuration in the number of evaluations. One of the parameters of the DVL Framework is the MOEA  $m$  used in the optimization process after the DVL inverse modeling, in the experiments we use the same MOEAs running isolated and with the DVL Framework: NSGA-III, MOEA/D, and RVEA. With this, in addition to analyzing the performance of the framework competing with these algorithms, it is also possible to analyze the impact that the framework has on each one. In the results, the nomenclature DVL+NSGA-III represents the framework running with the NSGA-III as a parameter, while the nomenclature NSGA-III means the algorithm running by the default implementation.

The result of the DVL Framework experiment is separated into two parts. First, we analyze only the impact of the framework over the three MOEAs: NSGA-III, MOEA/D, and RVEA. Figures 24 and 25 show the percentage of the difference in the mean of hypervolume between the result of the MOEA integrated into the DVL Framework and the result of the MOEA in standalone. The blue bar for example, in each plot, demonstrates the percentage in hypervolume mean of the DVL Framework using the NSGA-III decreased by the value of the hypervolume mean of the NSGA-III. A positive value means that the DVL Framework improves the MOEA and a negative value means that the MOEA has a better result. The x-axis represents the maximum number of evaluations, and the y-axis is the percentage of improvement in the hypervolume mean over 20 runs of the algorithm.

Analyzing the impact of the DVL Framework on the algorithms is possible to note that the framework improves substantially the NSGA-III algorithm in the DTLZ1 and DTLZ3 for both 3 and 10 objectives. These problems are characterized by having a high search space. And also, a substantial improvement in the DTLZ6 with 3 objectives, a degenerated problem. For the other degenerated problem, DTLZ5, the framework had a low impact. In all the other cases of the NSGA-III, the framework provided a good improvement, which comes close to zero improvement in the scenario of 100000 function evaluations. In some cases of this scenario, the algorithm standalone wins the framework, but the values are very close to each other. The results show that



the DVL Framework with the NSGA-III can be used instead of the NSGA-III standalone with a great improvement in performance. The MOEA/D is improved substantially in the DTLZ1 and DTLZ3 for 3 and 10 objectives. The DTLZ4 with 10 objectives is an interesting example that the MOEA/D had more improvements than the other algorithms. But in comparison to the others, in the rest of the problems, the MOEA/D has the lower values in impact. The MOEA/D has some examples where the DVL Framework has a loss in hypervolume, mainly the case of the DTLZ3 with 10 objectives, and some cases of the DTLZ2 and DTLZ5 with 3 objectives, where is better to use the MOEA/D in standalone. But, in the scenarios where the DVL Framework loses, the values are very close to the MOEA, showing the good performance of the DVL Framework over the MOEA/D.

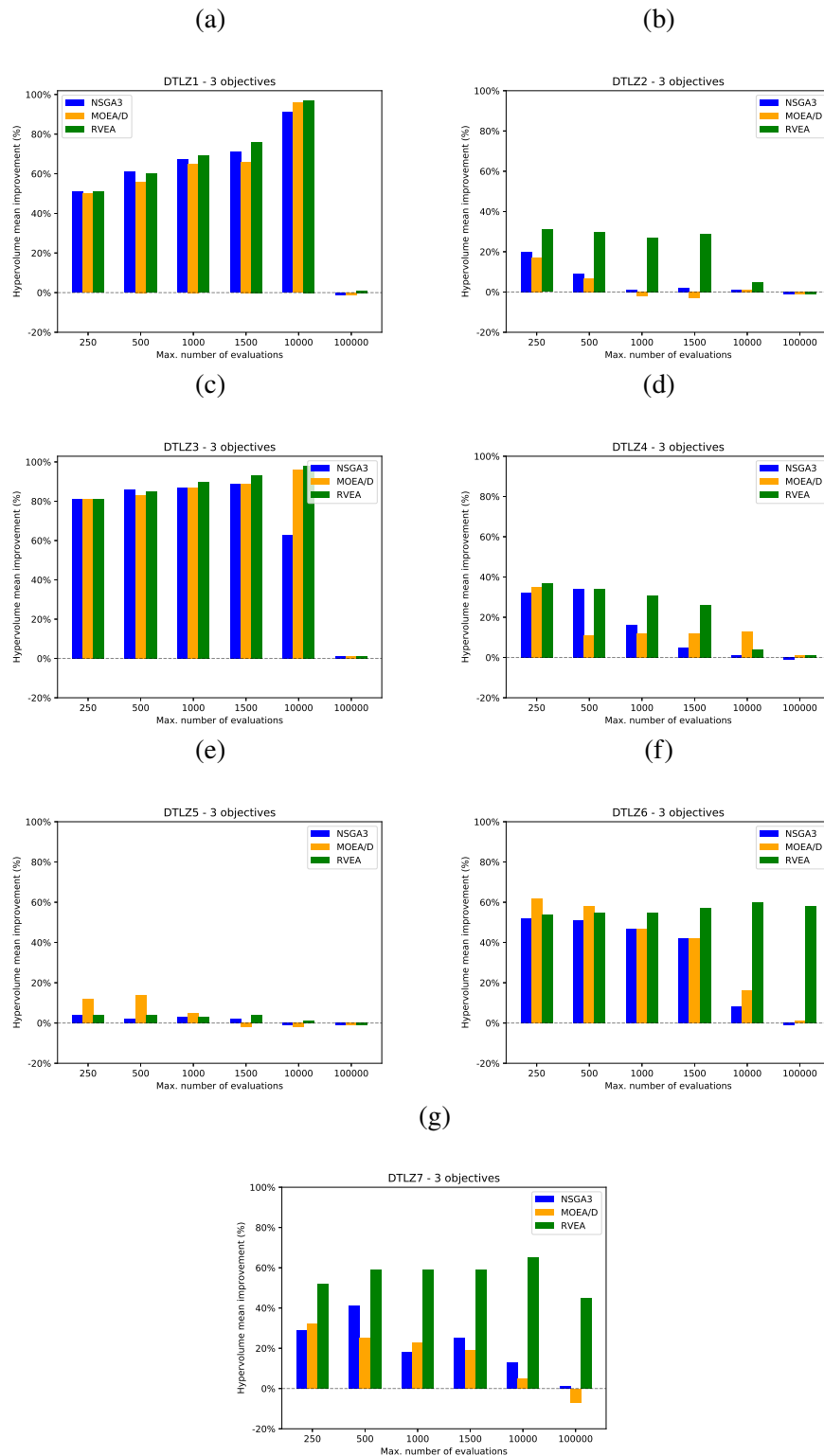
From the algorithms tested, the RVEA is the one that most benefits from the framework. In scenarios where the other algorithms have lower improvements, like DTLZ7 with 3 and 10 objectives, the framework had a great impact on the RVEA. The RVEA is the unique algorithm that obtains gains in the scenario of 100000 function evaluations, in the DTLZ6 and DTLZ7 with 3 objectives, and DTLZ2, DTLZ6, and DTLZ7 with 10 objectives. These are examples of problems with different shapes and search spaces. For the others cases of the RVEA, the framework provided a good improvement. Our hypothesis of the high impact of the DVL Framework in the RVEA is mainly due to the approximation of the estimated solution set provided by the DVL inverse model and the reference vectors used by the RVEA. The method that generates the reference points for the prediction of new solutions in the DVL inverse model is similar to the method that produces the reference vectors of the RVEA. With the initial solution set already closed to the reference vectors, the RVEA can better improve the search in terms of divergence and convergence. Another algorithm that uses the same method that generates reference points is the NSGA-III, but in this case, the algorithm uses the reference points only to ensure diversity in the obtained solutions. Because of this, the impact of the framework over the NSGA-III may have been smaller than on RVEA.

The results show that the DVL Framework has a positive impact on the MOEAs, in the vast majority of cases, the DVL Framework outperformed the MOEAs especially in the initial evaluations, approaching zero in the difference of hypervolume when it reaches 100000 objective function evaluations. The few exceptions that an MOEA surpassed the DVL Framework are the ones with the MOEA/D, these results are in some cases of the DTLZ2 and DTLZ5 with 3 objectives and DTLZ3 with 10 objectives. This leads to greater security in the use of the framework, regardless of the scenario, the impact on the gain in hypervolume is greater than the loss. Making the framework a good choice to use instead of the MOEA standalone.

The second part of the analysis consists of the statistical comparison for 3 and 10 objectives shown in Tables 19 and 20, respectively. The heat maps of the statistical comparison of this experiment can be seen in Appendix-Section B. This experiment leads to an understanding in which situations the DVL, the DVL Framework, or the MOEA can be used. Analyzing the



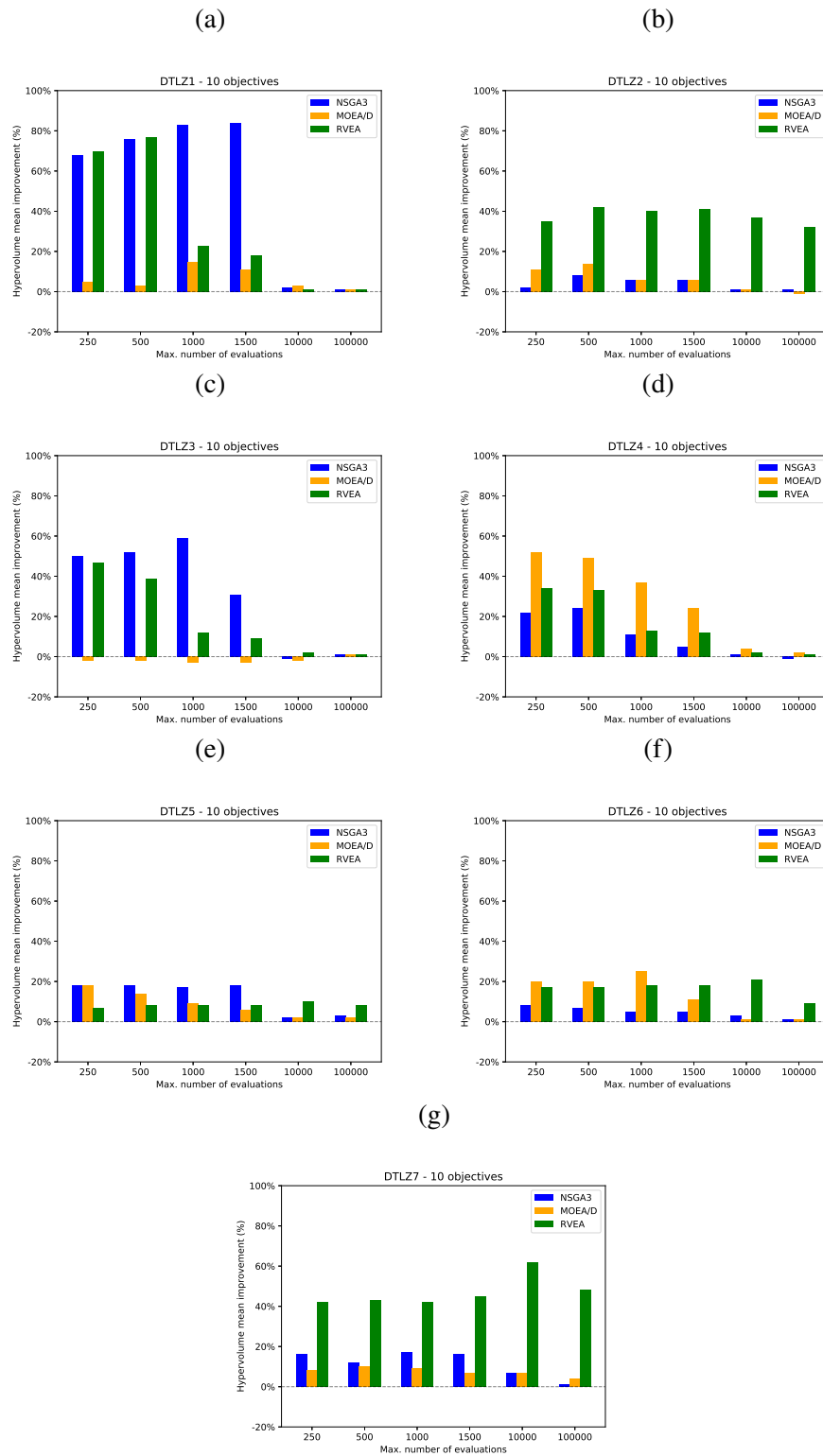
Figure 24 – Impact of the DVL Framework over the MOEAs for 3 objectives.



Source: Produced by authors.

comparison between all algorithms in each problem for each number of objectives it is possible to note that the DVL Framework is a good choice of algorithm, but the approach used in the MOEA impacts the final result. Different combinations of the DVL Framework obtain the best

Figure 25 – Impact of the DVL Framework over the MOEAs for 10 objectives.



result in different problems. For example, the DVL Framework with RVEA got the best results for the DTLZ1 problem with 3 and 10 objectives. The DVL Framework with NSGA-III has high hypervolume values in DTLZ2, DTLZ4, DTLZ5, and DTLZ6 with 3 and 10 objectives, and

also in DTLZ7 with 10 objectives. For 3 objectives the DVL algorithm outstands in the DTLZ7 problem, performing better than the others in this kind of problem. In DTLZ2, DTLZ5, and DTLZ6 it is comparable to the DVL Framework with NSGA-III in most cases where there is a higher restriction in the number of validations of the objective function. For 10 objectives the DVL algorithm has good results in the DTLZ5 and DTLZ6 problems, being comparable to the DVL Framework with NSGA-III. The DVL Framework with MOEA/D obtains the best results in the DTLZ4 and DTLZ6 for 10 objectives. Some differences in the DVL algorithm can be seen in this experiment in comparison with the previous, mainly due to the reference point set used. In this experiment, the reference set is more close to the ideal, leading to greater distances in the values of the results found. There are few cases where an MOEA wins over the DVL or the DVL Framework. For 3 objectives these cases are in the restriction scenario of 100000 evaluations, and for 10 objectives are mainly in the DTLZ3 problem where the MOEA/D wins in different scenarios. This shows that the MOEA/D has a high performance in this kind of problem, converging fast to a value next to the reference point adopted.

Table 19 – Statistical results (mean and standard deviation) of the Hypervolume values for 3 objectives.

| Problems | Eval.  | DVL                   | NSGA-III              | DVL+NSGA-III          | MOEA/D                | DVL+MOEA/D            | RVEA                  | DVL+RVEA              |
|----------|--------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| DTLZ1    | 250    | <b>(5.16±0.29)e-1</b> | (0.00±0.00)e+0        | <b>(5.08±0.21)e-1</b> | (0.00±0.00)e+0        | (4.91±0.29)e-1        | (0.00±0.00)e+0        | (5.02±0.26)e-1        |
|          | 500    | (5.48±0.11)e-1        | (0.00±0.00)e+0        | <b>(6.07±0.05)e-1</b> | (0.00±0.00)e+0        | (5.52±0.01)e-1        | (0.00±0.00)e+0        | (6.00±0.29)e-1        |
|          | 1000   | (5.59±0.08)e-1        | (0.00±0.00)e+0        | (6.63±0.27)e-1        | (0.00±0.00)e+0        | (6.44±0.02)e-1        | (0.00±0.00)e+0        | <b>(6.90±0.31)e-1</b> |
|          | 1500   | (5.56±0.07)e-1        | (0.00±0.00)e+0        | (7.02±0.20)e-1        | (0.00±0.00)e+0        | (6.50±0.01)e-1        | (0.00±0.00)e+0        | <b>(7.54±0.47)e-1</b> |
|          | 10000  | (5.56±0.20)e-1        | (0.00±0.00)e+0        | (9.05±0.32)e-1        | (0.00±0.00)e+0        | (9.58±0.00)e-1        | (0.00±0.00)e+0        | <b>(9.63±0.13)e-1</b> |
| DTLZ2    | 100000 | (5.58±0.24)e-1        | (9.65±0.00)e-1        | (9.61±0.05)e-1        | (9.70±0.00)e-1        | (9.70±0.00)e-1        | (9.73±0.00)e-1        | <b>(9.74±0.00)e-1</b> |
|          | 250    | <b>(8.39±0.16)e-1</b> | (6.40±0.00)e-1        | <b>(8.34±0.23)e-1</b> | (6.69±0.00)e-1        | <b>(8.37±0.22)e-1</b> | (5.33±0.61)e-1        | <b>(8.38±0.25)e-1</b> |
|          | 500    | <b>(8.78±0.12)e-1</b> | (7.73±0.00)e-1        | (8.62±0.12)e-1        | (7.74±0.00)e-1        | (8.36±0.02)e-1        | (5.40±0.52)e-1        | (8.39±0.21)e-1        |
|          | 1000   | <b>(9.11±0.04)e-1</b> | (8.63±0.00)e-1        | (8.73±0.19)e-1        | (8.57±0.00)e-1        | (8.44±0.08)e-1        | (5.85±0.46)e-1        | (8.53±0.17)e-1        |
|          | 1500   | <b>(9.13±0.04)e-1</b> | (8.94±0.00)e-1        | (9.08±0.11)e-1        | (8.98±0.00)e-1        | (8.75±0.00)e-1        | (5.76±0.75)e-1        | (8.60±0.16)e-1        |
| DTLZ3    | 10000  | (9.20±0.03)e-1        | (9.26±0.00)e-1        | <b>(9.26±0.00)e-1</b> | (9.25±0.00)e-1        | (9.25±0.00)e-1        | (8.42±0.19)e-1        | (8.83±0.12)e-1        |
|          | 100000 | (9.20±0.02)e-1        | (9.27±0.00)e-1        | <b>(9.27±0.00)e-1</b> | (9.27±0.00)e-1        | (9.27±0.00)e-1        | (9.02±0.01)e-1        | (9.02±0.01)e-1        |
|          | 250    | <b>(8.15±0.09)e-1</b> | (0.00±0.00)e+0        | (8.05±0.10)e-1        | (0.00±0.00)e+0        | (8.10±0.10)e-1        | (0.00±0.00)e+0        | (8.04±0.15)e-1        |
|          | 500    | (8.28±0.04)e-1        | (0.00±0.00)e+0        | <b>(8.52±0.01)e-1</b> | (0.00±0.00)e+0        | (8.30±0.00)e-1        | (0.00±0.00)e+0        | (8.45±0.10)e-1        |
|          | 1000   | (8.34±0.02)e-1        | (0.00±0.00)e+0        | (8.63±0.09)e-1        | (0.00±0.00)e+0        | (8.63±0.01)e-1        | (0.00±0.00)e+0        | <b>(8.93±0.14)e-1</b> |
| DTLZ4    | 1500   | (8.34±0.02)e-1        | (0.00±0.00)e+0        | (8.87±0.10)e-1        | (0.00±0.00)e+0        | (8.89±0.01)e-1        | (0.00±0.00)e+0        | <b>(9.20±0.17)e-1</b> |
|          | 10000  | (8.34±0.02)e-1        | (3.67±0.00)e-1        | <b>(9.89±0.06)e-1</b> | (3.95±0.00)e-2        | <b>(9.91±0.00)e-1</b> | (1.23±4.19)e-2        | (9.85±0.21)e-1        |
|          | 100000 | (8.35±0.03)e-1        | (9.99±0.00)e-1        | <b>(9.99±0.00)e-1</b> | (9.97±0.00)e-1        | (9.98±0.00)e-1        | (9.98±0.00)e-1        | (9.98±0.00)e-1        |
|          | 250    | <b>(7.82±0.45)e-1</b> | (4.77±0.00)e-1        | <b>(7.89±0.31)e-1</b> | (4.56±0.00)e-1        | <b>(8.00±0.35)e-1</b> | (4.27±0.64)e-1        | <b>(7.94±0.35)e-1</b> |
|          | 500    | <b>(8.68±0.10)e-1</b> | (5.24±0.00)e-1        | (8.62±0.08)e-1        | (7.57±0.00)e-1        | <b>(8.62±0.08)e-1</b> | (5.27±1.07)e-1        | (8.58±0.14)e-1        |
| DTLZ5    | 1000   | (8.81±0.07)e-1        | (7.35±0.00)e-1        | (8.87±0.13)e-1        | (7.92±0.00)e-1        | <b>(9.10±0.00)e-1</b> | (5.63±0.77)e-1        | (8.63±0.10)e-1        |
|          | 1500   | (8.89±0.05)e-1        | (8.69±0.00)e-1        | <b>(9.11±0.03)e-1</b> | (7.97±0.00)e-1        | (9.10±0.01)e-1        | (6.05±0.99)e-1        | (8.60±0.10)e-1        |
|          | 10000  | (8.98±0.05)e-1        | (9.26±0.00)e-1        | <b>(9.26±0.00)e-1</b> | (8.01±0.00)e-1        | (9.26±0.00)e-1        | (8.55±0.28)e-1        | (8.93±0.10)e-1        |
|          | 100000 | (9.07±0.04)e-1        | <b>(9.27±0.00)e-1</b> | (9.27±0.00)e-1        | (9.27±0.00)e-1        | (9.27±0.00)e-1        | (8.76±0.02)e-1        | (8.82±0.05)e-1        |
|          | 250    | <b>(9.43±0.15)e-1</b> | (9.09±0.00)e-1        | <b>(9.45±0.12)e-1</b> | (8.27±0.00)e-1        | <b>(9.43±0.15)e-1</b> | (9.04±0.09)e-1        | <b>(9.42±0.23)e-1</b> |
| DTLZ6    | 500    | <b>(9.53±0.03)e-1</b> | (9.32±0.00)e-1        | <b>(9.49±0.14)e-1</b> | (8.19±0.00)e-1        | <b>(9.54±0.03)e-1</b> | (9.08±0.10)e-1        | (9.40±0.21)e-1        |
|          | 1000   | <b>(9.55±0.02)e-1</b> | (9.20±0.00)e-1        | <b>(9.50±0.15)e-1</b> | (9.06±0.00)e-1        | <b>(9.52±0.13)e-1</b> | (9.11±0.10)e-1        | (9.40±0.17)e-1        |
|          | 1500   | <b>(9.57±0.01)e-1</b> | (9.44±0.00)e-1        | <b>(9.55±0.07)e-1</b> | <b>(9.57±0.00)e-1</b> | (9.43±0.03)e-1        | (9.16±0.09)e-1        | (9.48±0.07)e-1        |
|          | 10000  | (9.57±0.01)e-1        | <b>(9.59±0.00)e-1</b> | <b>(9.59±0.00)e-1</b> | (9.58±0.00)e-1        | (9.48±0.01)e-1        | (9.51±0.03)e-1        | (9.58±0.03)e-1        |
|          | 100000 | (9.58±0.00)e-1        | (9.59±0.00)e-1        | (9.59±0.00)e-1        | (9.59±0.00)e-1        | (9.59±0.00)e-1        | <b>(9.60±0.00)e-1</b> | (9.60±0.00)e-1        |
| DTLZ7    | 250    | <b>(9.57±0.17)e-1</b> | (4.39±0.00)e-1        | <b>(9.55±0.19)e-1</b> | (3.47±0.00)e-1        | <b>(9.62±0.18)e-1</b> | (4.11±0.22)e-1        | <b>(9.48±0.28)e-1</b> |
|          | 500    | (9.76±0.10)e-1        | (4.63±0.00)e-1        | (9.67±0.08)e-1        | (4.04±0.00)e-1        | <b>(9.81±0.07)e-1</b> | (4.28±0.21)e-1        | (9.75±0.11)e-1        |
|          | 1000   | <b>(9.82±0.07)e-1</b> | (5.12±0.00)e-1        | <b>(9.81±0.05)e-1</b> | (5.02±0.00)e-1        | (9.70±0.02)e-1        | (4.33±0.23)e-1        | <b>(9.80±0.09)e-1</b> |
|          | 1500   | (9.35±0.33)e-1        | (5.68±0.00)e-1        | <b>(9.87±0.04)e-1</b> | (5.69±0.00)e-1        | (9.80±0.02)e-1        | (4.18±0.17)e-1        | (9.81±0.10)e-1        |
|          | 10000  | (9.30±0.26)e-1        | (9.17±0.00)e-1        | <b>(9.89±0.00)e-1</b> | (8.33±0.00)e-1        | (9.89±0.02)e-1        | (3.92±0.11)e-1        | (9.82±0.13)e-1        |
| DTLZ7    | 100000 | (8.54±0.33)e-1        | (9.89±0.00)e-1        | (9.89±0.00)e-1        | (9.89±0.00)e-1        | <b>(9.89±0.00)e-1</b> | (4.10±0.01)e-1        | (9.86±0.05)e-1        |
|          | 250    | (5.31±0.52)e-1        | (2.42±0.00)e-1        | (5.26±0.52)e-1        | (2.36±0.00)e-1        | <b>(5.55±0.57)e-1</b> | (5.74±5.65)e-2        | <b>(5.69±0.50)e-1</b> |
|          | 500    | <b>(7.03±0.15)e-1</b> | (2.26±0.00)e-1        | (6.28±0.17)e-1        | (3.93±0.00)e-1        | (6.36±0.15)e-1        | (4.39±4.80)e-2        | (6.32±0.35)e-1        |
|          | 1000   | <b>(7.47±0.20)e-1</b> | (4.97±0.00)e-1        | (6.70±0.58)e-1        | (5.04±0.00)e-1        | (7.24±0.05)e-1        | (5.76±7.31)e-2        | (6.47±0.36)e-1        |
|          | 1500   | <b>(7.45±0.23)e-1</b> | (4.44±0.00)e-1        | (6.89±0.35)e-1        | (5.53±0.00)e-1        | <b>(7.40±0.06)e-1</b> | (7.22±5.72)e-2        | (6.62±0.45)e-1        |
| DTLZ7    | 10000  | <b>(7.65±0.12)e-1</b> | (6.33±0.00)e-1        | <b>(7.62±0.16)e-1</b> | (7.15±0.00)e-1        | (7.60±0.04)e-1        | (0.00±0.00)e+0        | (6.46±0.31)e-1        |
|          | 100000 | <b>(8.69±0.16)e-1</b> | (8.39±0.00)e-1        | (8.48±0.17)e-1        | <b>(8.62±0.00)e-1</b> | (8.02±0.00)e-1        | (3.70±0.00)e-1        | (8.13±0.15)e-1        |

Source – Produced by authors.

The overall result of this experiment shows that the DVL Framework can be used instead of the MOEA to improve the results of the algorithm principally in scenarios that have restrictions

Table 20 – Statistical results (mean and standard deviation) of the Hypervolume values for 10 objectives.

| Problems | Eval.  | DVL                   | NSGA-III              | DVL+NSGA-III          | MOEA/D                | DVL+MOEA/D            | RVEA            | DVL+RVEA              |
|----------|--------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------|-----------------------|
| DTLZ1    | 250    | <b>(7.25±1.66)e-1</b> | (0.00±0.00)e+0        | <b>(6.71±2.01)e-1</b> | (6.66±0.00)e-1        | <b>(7.07±1.92)e-1</b> | (2.16±4.29)e-2  | <b>(7.12±1.90)e-1</b> |
|          | 500    | (7.36±2.65)e-1        | (0.00±0.00)e+0        | (7.57±0.29)e-1        | (7.29±0.00)e-1        | (7.50±0.01)e-1        | (8.02±12.60)e-2 | <b>(8.47±0.75)e-1</b> |
|          | 1000   | (8.46±0.66)e-1        | (6.13±0.00)e-3        | (8.29±2.02)e-1        | (7.87±0.00)e-1        | <b>(9.30±0.00)e-1</b> | (6.83±1.62)e-1  | (9.12±0.27)e-1        |
|          | 1500   | (8.23±1.82)e-1        | (5.03±0.00)e-2        | (8.84±0.36)e-1        | (8.17±0.00)e-1        | (9.22±0.05)e-1        | (7.89±1.19)e-1  | <b>(9.60±0.20)e-1</b> |
|          | 10000  | (8.21±0.72)e-1        | (9.24±0.00)e-1        | (9.43±0.26)e-1        | (9.56±0.00)e-1        | (9.79±0.02)e-1        | (9.99±0.00)e-1  | <b>(1.00±0.00)e+0</b> |
| DTLZ2    | 100000 | (8.39±0.96)e-1        | (1.00±0.00)e+0        | (1.00±0.00)e+0        | (1.00±0.00)e+0        | (1.00±0.00)e+0        | (1.00±0.00)e+0  | <b>(1.00±0.00)e+0</b> |
|          | 250    | <b>(8.20±0.25)e-1</b> | (8.04±0.00)e-1        | <b>(8.18±0.28)e-1</b> | (7.24±0.00)e-1        | <b>(8.29±0.31)e-1</b> | (4.83±0.37)e-1  | <b>(8.26±0.23)e-1</b> |
|          | 500    | <b>(8.85±0.19)e-1</b> | (8.23±0.00)e-1        | <b>(8.96±0.17)e-1</b> | (7.64±0.00)e-1        | <b>(8.96±0.17)e-1</b> | (4.74±0.36)e-1  | <b>(8.93±0.19)e-1</b> |
|          | 1000   | (8.93±0.17)e-1        | (8.65±0.00)e-1        | <b>(9.19±0.13)e-1</b> | (8.47±0.00)e-1        | (9.02±0.08)e-1        | (5.03±0.41)e-1  | (8.98±0.14)e-1        |
|          | 1500   | (9.10±0.13)e-1        | (8.93±0.00)e-1        | <b>(9.52±0.10)e-1</b> | (8.82±0.00)e-1        | (9.37±0.05)e-1        | (5.00±0.45)e-1  | (9.00±0.10)e-1        |
| DTLZ3    | 10000  | (8.64±0.16)e-1        | (9.94±0.00)e-1        | <b>(9.97±0.00)e-1</b> | (9.84±0.00)e-1        | (9.93±0.00)e-1        | (5.38±0.59)e-1  | (8.99±0.20)e-1        |
|          | 100000 | (8.68±0.21)e-1        | (9.98±0.00)e-1        | <b>(9.99±0.00)e-1</b> | <b>(9.99±0.00)e-1</b> | (9.99±0.00)e-1        | (5.57±0.40)e-1  | (8.75±0.16)e-1        |
|          | 250    | (8.53±0.62)e-1        | (3.61±0.00)e-1        | (8.56±0.74)e-1        | <b>(9.02±0.00)e-1</b> | <b>(8.86±0.48)e-1</b> | (4.06±1.78)e-1  | (8.68±0.55)e-1        |
|          | 500    | (9.46±0.08)e-1        | (4.30±0.00)e-1        | (9.50±0.02)e-1        | <b>(9.60±0.00)e-1</b> | (9.40±0.00)e-1        | (5.60±1.51)e-1  | (9.47±0.04)e-1        |
|          | 1000   | (9.50±0.03)e-1        | (3.81±0.00)e-1        | (9.64±0.01)e-1        | <b>(9.90±0.00)e-1</b> | (9.65±0.00)e-1        | (8.55±0.76)e-1  | (9.70±0.06)e-1        |
| DTLZ4    | 1500   | (9.52±0.01)e-1        | (6.59±0.00)e-1        | (9.67±0.02)e-1        | <b>(9.96±0.00)e-1</b> | (9.70±0.01)e-1        | (8.98±0.34)e-1  | (9.83±0.06)e-1        |
|          | 10000  | (9.52±0.03)e-1        | (9.98±0.00)e-1        | (9.93±0.04)e-1        | <b>(9.99±0.00)e-1</b> | (9.89±0.01)e-1        | (9.87±0.10)e-1  | <b>(9.98±0.04)e-1</b> |
|          | 100000 | (9.53±0.01)e-1        | <b>(1.00±0.00)e+0</b> | (1.00±0.00)e+0        | (0.00±0.00)e+0        | (1.00±0.00)e+0        | (9.90±0.04)e-1  | (9.95±0.05)e-1        |
|          | 250    | <b>(9.46±0.16)e-1</b> | (7.32±0.00)e-1        | <b>(9.51±0.13)e-1</b> | (4.31±0.00)e-1        | <b>(9.49±0.15)e-1</b> | (6.12±1.02)e-1  | <b>(9.50±0.10)e-1</b> |
|          | 500    | (9.71±0.07)e-1        | (7.41±0.00)e-1        | <b>(9.77±0.05)e-1</b> | (4.88±0.00)e-1        | <b>(9.77±0.04)e-1</b> | (6.53±1.29)e-1  | <b>(9.75±0.05)e-1</b> |
| DTLZ5    | 1000   | (9.82±0.05)e-1        | (8.83±0.00)e-1        | <b>(9.87±0.02)e-1</b> | (6.16±0.00)e-1        | (9.85±0.01)e-1        | (8.51±0.72)e-1  | (9.76±0.05)e-1        |
|          | 1500   | (9.86±0.01)e-1        | (9.46±0.00)e-1        | <b>(9.93±0.02)e-1</b> | (7.39±0.00)e-1        | (9.76±0.03)e-1        | (8.58±1.32)e-1  | (9.78±0.05)e-1        |
|          | 10000  | (9.90±0.01)e-1        | (9.98±0.00)e-1        | (9.98±0.00)e-1        | (9.62±0.00)e-1        | <b>(9.98±0.01)e-1</b> | (9.74±0.23)e-1  | (9.91±0.04)e-1        |
|          | 100000 | (9.89±0.01)e-1        | (9.99±0.00)e-1        | (9.99±0.00)e-1        | (9.88±0.00)e-1        | <b>(9.99±0.00)e-1</b> | (9.77±0.05)e-1  | (9.85±0.01)e-1        |
|          | 250    | <b>(9.20±0.03)e-1</b> | (7.39±0.00)e-1        | <b>(9.19±0.03)e-1</b> | (7.43±0.00)e-1        | (9.18±0.03)e-1        | (8.56±0.15)e-1  | <b>(9.19±0.02)e-1</b> |
| DTLZ6    | 500    | <b>(9.22±0.02)e-1</b> | (7.43±0.00)e-1        | <b>(9.23±0.01)e-1</b> | (7.85±0.00)e-1        | <b>(9.22±0.02)e-1</b> | (8.45±0.27)e-1  | <b>(9.23±0.01)e-1</b> |
|          | 1000   | (9.21±0.02)e-1        | (7.54±0.00)e-1        | (9.22±0.02)e-1        | (8.34±0.00)e-1        | <b>(9.22±0.01)e-1</b> | (8.47±0.21)e-1  | (9.22±0.02)e-1        |
|          | 1500   | <b>(9.24±0.01)e-1</b> | (7.50±0.00)e-1        | (9.22±0.02)e-1        | (8.73±0.00)e-1        | <b>(9.24±0.00)e-1</b> | (8.45±0.25)e-1  | (9.23±0.02)e-1        |
|          | 10000  | <b>(9.28±0.01)e-1</b> | (9.05±0.00)e-1        | (9.21±0.02)e-1        | (9.02±0.00)e-1        | (9.22±0.00)e-1        | (8.27±0.33)e-1  | (9.21±0.02)e-1        |
|          | 100000 | <b>(9.28±0.00)e-1</b> | (8.97±0.00)e-1        | (9.22±0.02)e-1        | (9.05±0.00)e-1        | (9.20±0.01)e-1        | (8.48±0.28)e-1  | (9.20±0.03)e-1        |
| DTLZ7    | 250    | <b>(9.85±0.06)e-1</b> | (9.11±0.00)e-1        | <b>(9.85±0.04)e-1</b> | (7.89±0.00)e-1        | (9.83±0.04)e-1        | (8.15±0.20)e-1  | (9.81±0.06)e-1        |
|          | 500    | (9.76±0.08)e-1        | (9.22±0.00)e-1        | <b>(9.85±0.03)e-1</b> | (7.91±0.00)e-1        | <b>(9.85±0.01)e-1</b> | (8.07±0.32)e-1  | (9.75±0.08)e-1        |
|          | 1000   | (9.72±0.10)e-1        | (9.30±0.00)e-1        | (9.78±0.07)e-1        | (7.43±0.00)e-1        | <b>(9.85±0.00)e-1</b> | (8.05±0.37)e-1  | (9.80±0.07)e-1        |
|          | 1500   | (9.63±0.12)e-1        | (9.32±0.00)e-1        | (9.76±0.07)e-1        | (8.84±0.00)e-1        | <b>(9.85±0.00)e-1</b> | (7.99±0.21)e-1  | (9.79±0.08)e-1        |
|          | 10000  | (9.51±0.16)e-1        | (9.58±0.00)e-1        | <b>(9.80±0.06)e-1</b> | (9.77±0.00)e-1        | <b>(9.78±0.01)e-1</b> | (7.69±0.50)e-1  | <b>(9.78±0.06)e-1</b> |
| DTLZ7    | 100000 | (9.46±0.14)e-1        | (9.74±0.00)e-1        | (9.82±0.02)e-1        | (9.78±0.00)e-1        | <b>(9.84±0.00)e-1</b> | (8.04±0.36)e-1  | (8.88±0.20)e-1        |
|          | 250    | <b>(6.04±0.64)e-1</b> | (4.42±0.00)e-1        | <b>(5.93±0.73)e-1</b> | (5.34±0.00)e-1        | <b>(6.04±0.62)e-1</b> | (1.67±1.11)e-1  | <b>(5.81±0.80)e-1</b> |
|          | 500    | <b>(6.01±0.44)e-1</b> | (4.91±0.00)e-1        | <b>(6.01±0.06)e-1</b> | (5.04±0.00)e-1        | (5.98±0.07)e-1        | (1.64±1.23)e-1  | <b>(5.89±0.57)e-1</b> |
|          | 1000   | <b>(6.29±0.29)e-1</b> | (4.44±0.00)e-1        | (6.11±0.24)e-1        | (5.21±0.00)e-1        | (6.09±0.02)e-1        | (1.92±1.27)e-1  | (6.10±0.25)e-1        |
|          | 1500   | <b>(6.24±0.38)e-1</b> | (4.50±0.00)e-1        | (6.05±0.21)e-1        | (5.37±0.00)e-1        | (6.06±0.02)e-1        | (1.61±0.79)e-1  | (6.04±0.30)e-1        |
| DTLZ7    | 10000  | (6.06±0.16)e-1        | (5.53±0.00)e-1        | <b>(6.16±0.25)e-1</b> | (5.35±0.00)e-1        | (6.03±0.03)e-1        | (0.00±0.00)e+0  | <b>(6.15±0.24)e-1</b> |
|          | 100000 | (7.96±0.16)e-1        | (8.16±0.00)e-1        | <b>(8.17±0.01)e-1</b> | (7.56±0.00)e-1        | (7.90±0.05)e-1        | (3.20±0.08)e-1  | (7.97±0.12)e-1        |

Source – Produced by authors.

in the number of validations and also when the number of objective functions increases. The results also show a good performance in a restriction of 100000 objective function validations. The choice of what MOEA uses in the DVL Framework is dependent on the problem. In our results, different combinations outperformed in different kinds of problems. The DVL algorithm can also be an alternative in some scenarios, for example, it outstanding the other algorithms in problems that have degenerated Pareto-optimal fronts such as the DTLZ5 with 3 and 10 objectives and the DTLZ6 with 3 objectives. And also, in disconnected Pareto-optimal front such as the DTLZ7 with 3 and 10 objectives.

#### 4.2.4 Traffic Lights Signaling Experiment

The preview experiments demonstrate that both the DVL and the DVL Framework have good performance in a wide set of benchmark problems. But, during our analysis of these experiments, some research questions arose:

1. What is the cost of training a machine learning model during a real expensive optimization problem?

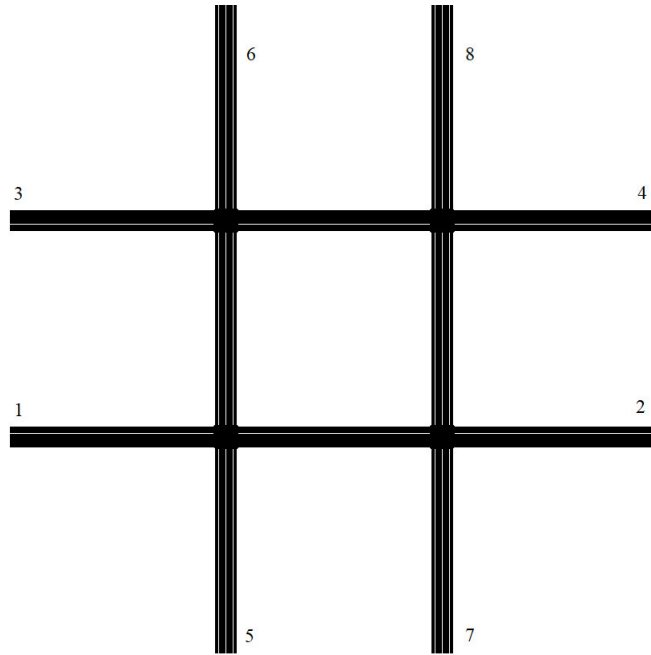
## 2. How the algorithm will perform in face of an unknown Pareto-optimal front?

The use of the DTLZ benchmark test suite helps in the agility to produce experiments due to the easy usability of the suite. But these objective functions run in an insignificant processing time in comparison to the total time of running the algorithm which can cause differences in the results even when we use the maximum number of validations to compare the algorithms. In this section, we propose to compare the DVL algorithm, the DVL Framework, and an MOEA in a real expensive many-objective optimization problem, the Traffic Lights Signaling. The stop condition used in this experiment will be the total processing time of the algorithms. It will allow us to identify if there is a cost in training the models during the optimization, and also validate the DVL inverse model in a different scenario, where the Pareto-optimal front is unknown.

In the optimization problem of traffic light synchronization, the decision variables are the time of each phase of each traffic light and the objective functions are the traffic-related quality measures calculated by the simulator. The scenario used in this experiment is represented by Figure 26, based on the experiments proposed by Matos & Carvalho (2016). Each intersection is configured with a two-phase traffic light. The phases schema and number of roads are the same at all intersections. Each intersection is crossed with 2 two-way roads, where each road has 6 lanes, i.e. 3 lanes for each way (direction). The solution is represented by an eight-position integer vector ( $n = 8$ ), since each of the four intersections has two “green” phases, totalizing 2 decision variables for each intersection. Each value represents the duration of the phase assuming a finite interval in the range from twenty to one hundred and twenty. For the simulation, 21000 vehicles were used. The vehicles are inserted into the simulation through the injectors, located at the extremities, numbered by 1 to 8 in Figure 26. The distribution of the vehicle flows in the injectors was defined at random before initiating any simulation and thus maintained in all experiments. If the distribution between the injectors were not maintained for the different experiments, the results would be different in the end. The total time to be simulated, for each execution, was 1 hour, but in the SUMO simulator, it is possible to speed up this time to approximately 10 seconds of clock time or 0.23 of processing time. This time represents one execution of the objective function evaluation. At the end the simulator outputs the six objective functions:

- *Depart delay*: The time the vehicle had to wait before it could start his journey;
- *Trip duration*: The time the vehicle needed to accomplish the route;
- *Wait steps*: The number of steps in which the vehicle speed was below 0.1m/s;
- *Time loss*: The time lost due to driving below the ideal speed (60km/h);
- *CO abs*: complete amount of CO (in milligram) emitted by the vehicle during the trip;
- *CO<sub>2</sub> abs*: The complete amount of CO<sub>2</sub> emitted by the vehicle during the trip;

Figure 26 – Traffic light synchronization scenario.



Source: Produced by authors.

In this experiment three algorithms will be tested: the NSGA-II algorithm (DEB et al., 2002a), which has good results in this kind of problem in comparison to the NSGA-III algorithm (MATOS; CARVALHO, 2016), the DVL algorithm, and the DVL Framework with the NSGA-II algorithm. The algorithms are executed independently 20 times for each configuration and the Friedman's test followed by the Wilcoxon rank-sum test is adopted to compare the results between each configuration of the experiments with a value of 0.05 of the significance level. The best results in the following table will be highlighted in bold, if more than one value is highlighted in bold it means that there is no statistically significant difference between the result and the best value. Each algorithm runs until it reaches a stop condition, different from the previous experiments, in this experiment we use the processing time as a stop condition. This time represents the sum of the system and user CPU time of the current process. A set of a maximum number of processing time is defined as (70, 220, and 1100) in fractional seconds. This validation set is used to simulate different scenarios of restrictions in the use of the processing time. The empirical comparisons between the results are made by the use of Hypervolume, which is calculated by considering a nadir point as the reference point. In this experiment, the nadir was obtained considering all values obtained in all executions. The overall execution of this experiment took a total of approximately 44 days to complete.

The DVL algorithm and the DVL Framework were modified to attend to this experiment. A parameter  $s$  were added that represents a stop condition, the maximum process time that the algorithm can run, this parameter has different values for each configuration of the experiment.

For the other parameters, the DVL algorithm is configured as the  $c$  value is fixed in 200, the  $\epsilon$  is set to 0.001, and the  $k$  value is set to 250. For the DVL Framework the value of the parameter  $k$  is 250, and the MOEA  $m$  used is the NSGA-II. The implementation and the parameters used in the NSGA-II were defined by the pymoo (Blank; Deb, 2020). For the simulated binary crossover, the distribution index is set to  $\eta c = 15$ , and the crossover probability  $pc = 0.9$ . For the polynomial mutation, the distribution index and the mutation probability are set to  $\eta m = 20$  and  $pm = 1/n$ , respectively.

In Table 21 is demonstrated the difference in number of evaluations of each algorithm per configuration of the experiment. It can be seen that the DVL algorithm and the DVL Framework have a lower number of evaluations in comparison to the NSGA-II. In more restrictive scenarios this difference is smaller but increases in less restrictive scenarios, with the increase of the processing time. This difference in the number of calls of the function evaluation is mainly due to the time spent the DVL has to train the models and predicting new solutions. Table 22 presents the results of the Traffic Lights Signaling Experiment by showing the mean and the standard deviation of the hypervolume obtained during the 20 runs. The result of this experiment demonstrates a similarity with the previous experiments, where the DVL algorithm shows good performance in more restrictive scenarios while the DVL Framework has a better overall performance, even with a smaller number of objective function calls.

Also, the results found indicates a greater safety and reliability of the results obtained in the previous experiments that used the DTLZ benchmark test suit. In the restriction scenario of 70 processing time, the DVL algorithm and the DVL Framework have the better result and both are statistically comparable. For 220 and 1100, the DVL Framework has the best results. It is possible to note the efficiency of the DVL inverse model in finding good solutions as close as possible to the optimal values even in a restricted scenario. The difference between the hypervolume mean in the 70 process time and 1100 is not so far, but the DVL Framework has good results by using the efficiency of the MOEA in finding good local optima improving the overall hypervolume. The statistical comparison that results in this experiment can be seen in Figure 27.

Table 21 – Mean number of evaluations of each algorithm of the Traffic Lights Signaling Experiment.

| Process Time | DVL   | DVL+NSGA-II | NSGA-II |
|--------------|-------|-------------|---------|
| 70           | 296   | 296         | 302     |
| 220          | 898   | 972         | 1009    |
| 1100         | 4.157 | 4.924       | 5045    |

Source – Produced by authors.

Table 22 – Statistical results (mean and standard deviation) of the Hypervolume values of the Traffic Lights Signaling Experiment.

| Process Time | Mean Clock Time | DVL             | DVL+NSGA-II     | NSGA-II  |
|--------------|-----------------|-----------------|-----------------|----------|
| 70           | 3117            | <b>7.03E+38</b> | <b>7.02E+38</b> | 5.47E+38 |
|              |                 | <b>8.54E+36</b> | <b>8.26E+36</b> | 2.37E+18 |
| 220          | 9734            | 7.04E+38        | <b>7.10E+38</b> | 6.27E+38 |
|              |                 | 6.08E+36        | <b>5.58E+36</b> | 1.51E+23 |
| 1100         | 51455           | 7.04E+38        | <b>7.08E+38</b> | 6.64E+38 |
|              |                 | 6.26E+36        | <b>5.07E+36</b> | 8.94E+36 |

Source – Produced by authors.

Figure 27 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration.



Source: Produced by authors.



# 5

## Conclusion

Machine Learning enhancing metaheuristic algorithms is an open topic of the literature that can produce knowledge and open research topic in the major areas of Machine Learning and Optimization. In this work, we explore this area in a systematic review that presents relevant results to guide future research in the field. We show that the number of publications has increased over the last ten years, the majority in journals, which shows the maturity of the research area. Most approaches are focused on decreasing the number of objective function validations during the search. And the most used type of algorithms is the SAEAs and EDAs.

Within this area, our topic of research was specifically the inverse models, an approach that uses an inverse functional mapping from the Pareto Front to the Pareto Set. The DVL algorithm was our main topic of work, it is an approach of inverse modeling designed to generate solutions as close as possible to the Pareto-optimal front. Due to the study of the DVL being recent and the necessity of proving the algorithm, we conduct a set of experiments, based on previously established hypotheses, to validate the algorithm and find its strengths and weaknesses. So in this work, we showed experimentally that the DVL algorithm and its strategy of inverse modeling can compete with well-known MOEAs of the literature, such as the NSGA-III in the optimization of benchmark problems. Experimental results demonstrate that the inverse model can estimate the Pareto-optimal front of the DTLZ benchmark test suit problems, and also, the DVL algorithm can outperform the NSGA-III algorithm in restrictive scenarios, i.e. with restrictions on the number of evaluations, and problems with a higher number of objectives.

Due to the good performance of the DVL algorithm in restrictive scenarios, we propose a DVL Framework that uses a DVL variation that can be executed with an MOEA. The key idea of the framework is to use the DVL to estimate an initial solution set as close as possible to the Pareto-optimal set. After this step, the MOEA continues the search to find better local optima. With the implementation of the DVL Framework in hand, we evaluate it in the optimization experimentation. Where different combinations of the framework are compared to the DVL

algorithm and different MOEAs of the literature. The results show that the DVL Framework can improve the MOEAs in a wide range of scenarios, according to our experiments, up to a limit of 100000 objective function evaluations. The choice of which MOEA uses in the DVL Framework is dependent on the problem. Also, our results shows that the DVL algorithm can be used in some scenarios of higher restriction in the number of validations and in problems that have degenerated or disconnected Pareto-optimal fronts.

Finally, we evaluate the two algorithms in a real-world optimization expensive problem. A problem of this kind can show different results from the previous experiments mainly due to its constraints in the problem, the expensive time to evaluate the objective function, and the unknown Pareto-optimal front which can have a shape different from the previous problems. The experiment is conducted on the optimization problem of traffic light synchronization. The algorithms are compared with the NSGA-II algorithm, which has good results in this problem with the same scenario used. The results show that, even if the DVL inverse model spent time in training and predicting solutions, this time lost is minimal compared to the hypervolume gain with the solutions set proposed by the algorithm. In a more restrictive scenario, with a processing time of 70, the DVL algorithm and the DVL Framework have the best result, and in the other scenarios, the DVL Framework outperformed the other algorithms.

The results found in this work demonstrate a promising area of research in inverse modeling. The DVL algorithm and the DVL Framework show good results in diverse scenarios of different problems. For future works, we suggest the improvement of the weak points of the DVL algorithm, such as the deterioration of the search performance, other forms of improving the iterative process of the DVL algorithm can be analyzed to improve the local optima. Also, the investigation of better shapes of the reference points or using a dynamic reference point set. Different problems have different formats of the Pareto-optimal front, the shape of the reference point can greatly influence the result obtained from the DVL. For the DVL Framework, we suggest a wide comparison with other MOEAs, and a full investigation of the restriction point that the inverse model has no more improvements over the MOEA.

We also believe that exploring the DVL inverse modeling in a different manner could lead to future works. The DVL Framework proposed in this work uses the inverse model in the Population Initialization component, as detailed in Chapter 2. But the DVL inverse model could also be used for Population Reproduction and Variation, where the inverse model could extract knowledge to improve the generation of new solutions. And in the Local Search, which could explore the inverse modeling to improve local populations. Future works can also explore, in a deeper way, models with more robust techniques of machine learning such as ensemble learning, deep learning, and others.

# Bibliography

ABDESSALEM, R. B. et al. Testing vision-based control systems using learnable evolutionary algorithms. In: IEEE. *2018 IEEE/ACM 40th International Conference on Software Engineering (ICSE)*. [S.l.], 2018. p. 1016–1026. Citado 6 vezes nas páginas 42, 43, 46, 47, 48, and 52.

AKINSOLU, M. O. et al. A parallel surrogate model assisted evolutionary algorithm for electromagnetic design optimization. *IEEE Transactions on Emerging Topics in Computational Intelligence*, v. 3, n. 2, p. 93–105, 2019. Citado 6 vezes nas páginas 42, 43, 46, 47, 48, and 52.

AL-DUOLI, F. et al. Hybridizing meta-raps with machine learning algorithms. In: IEEE. *2018 IEEE Technology and Engineering Management Conference (TEMSCON)*. [S.l.], 2018. p. 1–6. Citado 9 vezes nas páginas 42, 43, 46, 47, 48, 49, 50, 52, and 58.

AL-JUBOORI, M.; DATTA, B. Optimum design of hydraulic water retaining structures incorporating uncertainty in estimating heterogeneous hydraulic conductivity utilizing stochastic ensemble surrogate models within a multi-objective multi-realisation optimisation model. *Journal of Computational Design and Engineering*, Oxford University Press, v. 6, n. 3, p. 296–315, 2019. Citado 7 vezes nas páginas 42, 43, 46, 47, 48, 52, and 57.

ALI, M.; SIARRY, P.; PANT, M. An efficient differential evolution based algorithm for solving multi-objective optimization problems. *European Journal of Operational Research*, v. 217, n. 2, p. 404–416, 2012. ISSN 0377-2217. Disponível em: <<https://www.sciencedirect.com/science/article/pii/S0377221711008538>>. Citado 6 vezes nas páginas 42, 43, 44, 45, 46, and 52.

ANDERSON, J. R. *Machine learning: An artificial intelligence approach*. [S.l.]: Morgan Kaufmann, 1990. v. 3. Citado na página 24.

ARIAS-MONTAÑO, A.; COELLO, C. A. C.; MEZURA-MONTES, E. Multi-objective airfoil shape optimization using a multiple-surrogate approach. In: *2012 IEEE Congress on Evolutionary Computation*. [S.l.: s.n.], 2012. p. 1–8. Citado 7 vezes nas páginas 42, 43, 46, 47, 48, 52, and 54.

AZZOUZ, N.; BECHIKH, S.; SAID, L. B. Steady state ibea assisted by mlp neural networks for expensive multi-objective optimization problems. In: *Proceedings of the 2014 Annual Conference on Genetic and Evolutionary Computation*. [S.l.: s.n.], 2014. p. 581–588. Citado 7 vezes nas páginas 42, 43, 45, 46, 48, 52, and 54.

BACIGALUPO, A. et al. Computational design of innovative mechanical metafilters via adaptive surrogate-based optimization. *Computer Methods in Applied Mechanics and Engineering*, Elsevier, v. 375, p. 113623, 2021. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

BALUJA, S.; CARUANA, R. Removing the genetics from the standard genetic algorithm. In: *Machine Learning Proceedings 1995*. [S.l.]: Elsevier, 1995. p. 38–46. Citado na página 42.

BENGIO, Y.; LODI, A.; PROUVOST, A. Machine learning for combinatorial optimization: a methodological tour d’horizon. *European Journal of Operational Research*, Elsevier, 2020. Citado na página 27.

Blank, J.; Deb, K. pymoo: Multi-objective optimization in python. *IEEE Access*, v. 8, p. 89497–89509, 2020. Citado 3 vezes nas páginas 74, 85, and 94.

BLIEK, L.; VERWER, S.; WEERDT, M. de. Black-box combinatorial optimization using models with integer-valued minima. *Annals of Mathematics and Artificial Intelligence*, Springer, p. 1–15, 2020. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

BOUSSAÏD, I.; LEPAGNOT, J.; SIARRY, P. A survey on optimization metaheuristics. *Information sciences*, Elsevier, v. 237, p. 82–117, 2013. Citado 2 vezes nas páginas 23 and 24.

BROWNE, M. W. Cross-validation methods. *Journal of mathematical psychology*, Elsevier, v. 44, n. 1, p. 108–132, 2000. Citado na página 47.

CALVET, L. et al. Learnheuristics: hybridizing metaheuristics with machine learning for optimization with dynamic inputs. *Open Mathematics*, De Gruyter Open, v. 15, n. 1, p. 261–280, 2017. Citado 2 vezes nas páginas 18 and 26.

CAO, Y. et al. Design of broadband reflectarray antenna using machine-learning-assisted optimization method. In: IEEE. *2019 International Symposium on Antennas and Propagation (ISAP)*. [S.l.], 2019. p. 1–3. Citado 7 vezes nas páginas 42, 43, 46, 48, 52, 54, and 55.

CHAMBERS, J. B.; BARNES, J. W. Flexible job shop scheduling by tabu search. *The University of Texas, Austin, TX, Technical Report Series ORP96-09, Graduate Program in Operations Research and Industrial Engineering*, 1996. Citado na página 51.

CHENG, R. et al. A multiobjective evolutionary algorithm using gaussian process-based inverse modeling. *IEEE Transactions on Evolutionary Computation*, v. 19, n. 6, p. 838–856, 2015. Citado 12 vezes nas páginas 19, 31, 41, 42, 43, 45, 46, 48, 49, 52, 58, and 65.

CHENG, R. et al. A reference vector guided evolutionary algorithm for many-objective optimization. *IEEE Transactions on Evolutionary Computation*, IEEE, v. 20, n. 5, p. 773–791, 2016. Citado na página 86.

CHIA, J. Y. et al. A data mining approach to evolutionary optimisation of noisy multi-objective problems. *International Journal of Systems Science*, Taylor & Francis, v. 43, n. 7, p. 1217–1247, 2012. Citado 4 vezes nas páginas 42, 43, 46, and 52.

COELLO, C. A. C. et al. Evolutionary multiobjective optimization: open research areas and some challenges lying ahead. *Complex & Intelligent Systems*, Springer, v. 6, n. 2, p. 221–236, 2020. Citado na página 56.

COELLO, C. A. C. et al. *Evolutionary algorithms for solving multi-objective problems*. [S.l.]: Springer, 2007. v. 5. Citado 3 vezes nas páginas 17, 23, and 24.

CUI, H.; TURAN, O.; SAYER, P. Learning-based ship design optimization approach. *Computer-Aided Design*, Elsevier, v. 44, n. 3, p. 186–195, 2012. Citado 8 vezes nas páginas 42, 43, 44, 46, 48, 51, 52, and 58.

DAI, Z.; WANG, L.; YANG, S. Data mining-based model simplification and optimization of an electrical power generation system. *IEEE Transactions on Transportation Electrification*, IEEE, v. 6, n. 4, p. 1665–1678, 2020. Citado 6 vezes nas páginas 42, 43, 46, 48, 52, and 57.

DAS, I.; DENNIS, J. *Normal-Boundary Intersection: An Alternate Method for Generating Pareto Optimal Points in Multicriteria Optimization Problems*. [S.l.], 1996. Citado 3 vezes nas páginas 62, 64, and 81.

DAUZÈRE-PÉRÈS, S.; PAULLI, J. An integrated approach for modeling and solving the general multiprocessor job-shop scheduling problem using tabu search. *Annals of Operations Research*, Springer, v. 70, p. 281–306, 1997. Citado na página 51.

DEB, K. *Multi-objective optimization using evolutionary algorithms*. [S.l.]: John Wiley & Sons, 2001. v. 16. Citado 2 vezes nas páginas 74 and 85.

DEB, K.; AGRAWAL, R. B. et al. Simulated binary crossover for continuous search space. *Complex systems*, Citeseer, v. 9, n. 2, p. 115–148, 1995. Citado 2 vezes nas páginas 74 and 85.

DEB, K.; GOYAL, M. A combined genetic adaptive search (geneas) for engineering design. *Computer Science and informatics*, COMPUTER SOCIETY OF INDIA, v. 26, p. 30–45, 1996. Citado 2 vezes nas páginas 74 and 85.

DEB, K. et al. A taxonomy for metamodeling frameworks for evolutionary multiobjective optimization. *IEEE Transactions on Evolutionary Computation*, IEEE, v. 23, n. 1, p. 104–116, 2019. Citado na página 30.

DEB, K.; JAIN, H. An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part i: Solving problems with box constraints. *IEEE Transactions on Evolutionary Computation*, IEEE, v. 18, n. 4, p. 577–601, 2014. Citado na página 19.

DEB, K. et al. A fast and elitist multiobjective genetic algorithm: Nsga-ii. *IEEE transactions on evolutionary computation*, IEEE, v. 6, n. 2, p. 182–197, 2002. Citado na página 93.

DEB, K. et al. Scalable multi-objective optimization test problems. In: IEEE. *Proceedings of the 2002 Congress on Evolutionary Computation. CEC'02 (Cat. No. 02TH8600)*. [S.l.], 2002. v. 1, p. 825–830. Citado na página 51.

DEB, K. et al. Scalable test problems for evolutionary multiobjective optimization. In: *Evolutionary multiobjective optimization*. [S.l.]: Springer, 2005. p. 105–145. Citado na página 66.

DERRAC, J. et al. A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms. *Swarm and Evolutionary Computation*, Elsevier, v. 1, n. 1, p. 3–18, 2011. Citado 2 vezes nas páginas 52 and 56.

DEY, A. Machine learning algorithms: a review. *International Journal of Computer Science and Information Technologies*, v. 7, n. 3, p. 1174–1179, 2016. Citado 2 vezes nas páginas 24 and 25.

DEY, A. et al. Population based study on arsenic induced blood samples employing hybrid metaheuristic optimization based ml approach. In: IEEE. *2019 IEEE Region 10 Symposium (TENSYP)*. [S.l.], 2019. p. 599–604. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

DU, J.; RADA, R. Knowledge-guided genetic algorithm for financial forecasting. In: IEEE. *2012 IEEE Conference on Computational Intelligence for Financial Engineering & Economics (CIFer)*. [S.l.], 2012. p. 1–8. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

EL-BELTAGY, M.; KEANE, A. A comparison of various optimization algorithms on a multilevel problem. *Engineering Applications of Artificial Intelligence*, v. 12, n. 5, p. 639–654, 1999. ISSN 0952-1976. Disponível em: <<https://www.sciencedirect.com/science/article/pii/S0952197699000330>>. Citado na página 44.

ELAZIZ, M. A.; OLIVA, D.; XIONG, S. An improved opposition-based sine cosine algorithm for global optimization. *Expert Systems with Applications*, Elsevier, v. 90, p. 484–500, 2017. Citado 6 vezes nas páginas 42, 43, 44, 45, 46, and 52.

FAN, Z. et al. Multi-objective evolutionary algorithms embedded with machine learning—a survey. In: IEEE. *2016 IEEE Congress on Evolutionary Computation (CEC)*. [S.l.], 2016. p. 1262–1266. Citado 3 vezes nas páginas 18, 23, and 27.

FARFÁN, J. F.; CEA, L. Coupling artificial neural networks with the artificial bee colony algorithm for global calibration of hydrological models. *Neural Computing and Applications*, Springer, p. 1–16, 2021. Citado 6 vezes nas páginas 42, 43, 46, 47, 48, and 52.

FLEMING, P. J.; PURSHOUSE, R. C.; LYGOE, R. J. Many-objective optimization: An engineering design perspective. In: SPRINGER. *International conference on evolutionary multi-criterion optimization*. [S.l.], 2005. p. 14–32. Citado na página 24.

GAO, Y. et al. Filter design and optimization of electromechanical actuation systems using search and surrogate algorithms for more-electric aircraft applications. *IEEE Transactions on Transportation Electrification*, IEEE, v. 6, n. 4, p. 1434–1447, 2020. Citado 7 vezes nas páginas 42, 43, 46, 48, 49, 52, and 58.

GARCÍA, J. et al. A db-scan binarization algorithm applied to matrix covering problems. *Computational intelligence and neuroscience*, Hindawi, v. 2019, 2019. Citado 8 vezes nas páginas 42, 43, 46, 47, 48, 50, 52, and 57.

GARCÍA-ORTEGA, R. H.; GARCÍA-SÁNCHEZ, P.; MERELO-GUERVÓS, J. J. Startroper, a film trope rating optimizer using machine learning and evolutionary algorithms. *Expert Systems*, Wiley Online Library, v. 37, n. 6, p. e12525, 2020. Citado 7 vezes nas páginas 40, 42, 43, 46, 47, 48, and 52.

GARDNER, M. W.; DORLING, S. Artificial neural networks (the multilayer perceptron)—a review of applications in the atmospheric sciences. *Atmospheric environment*, Elsevier, v. 32, n. 14-15, p. 2627–2636, 1998. Citado 2 vezes nas páginas 40 and 67.

GARRISON, W. T.; PETTY, M. D. A comparison of simulation optimization algorithm performance. In: IEEE. *2019 SoutheastCon*. [S.l.], 2019. p. 1–7. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

GATOPOULOS, I. et al. Evolutionary algorithm with non-parametric surrogate model for tensor program optimization. In: IEEE. *2020 IEEE Congress on Evolutionary Computation (CEC)*. [S.l.], 2020. p. 1–8. Citado 6 vezes nas páginas 42, 43, 46, 48, 52, and 53.

GEE, S. B.; TAN, K. C.; ALIPPI, C. Solving multiobjective optimization problems in unknown dynamic environments: An inverse modeling approach. *IEEE transactions on cybernetics*, IEEE, v. 47, n. 12, p. 4223–4234, 2016. Citado na página 31.

GÉRON, A. *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow: Concepts, tools, and techniques to build intelligent systems*. [S.l.]: O'Reilly Media, 2019. Citado na página 67.



- GHOLAMNEZHAD, P.; BROUMANDNIA, A.; SEYDI, V. An inverse model-based multiobjective estimation of distribution algorithm using random-forest variable importance methods. *Computational Intelligence*, Wiley Online Library, 2020. Citado na página 31.
- GHORBEL, H. et al. Smart adaptive run parameterization (sarp): enhancement of user manual selection of running parameters in fluid dynamic simulations using bio-inspired and machine-learning techniques. *Soft Computing*, Springer, v. 23, n. 22, p. 12031–12047, 2019. Citado 8 vezes nas páginas 40, 42, 43, 46, 47, 48, 52, and 54.
- GIAGKIOZIS, I.; FLEMING, P. J. Increasing the density of available pareto optimal solutions. *Automatic Control and Systems Engineering*, University of Sheffield, 2012. Citado 3 vezes nas páginas 18, 29, and 58.
- GIAGKIOZIS, I.; FLEMING, P. J. Pareto front estimation for decision making. *Evolutionary computation*, MIT Press, v. 22, n. 4, p. 651–678, 2014. Citado 5 vezes nas páginas 18, 19, 29, 62, and 67.
- GONZALEZ, S. R.; JALALI, H.; NIEUWENHUYSE, I. V. A multiobjective stochastic simulation optimization algorithm. *European Journal of Operational Research*, Elsevier, v. 284, n. 1, p. 212–226, 2020. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- GUO, P.; CHENG, W.; WANG, Y. Hybrid evolutionary algorithm with extreme machine learning fitness function evaluation for two-stage capacitated facility location problems. *Expert Systems with Applications*, Elsevier, v. 71, p. 57–68, 2017. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- HANSEN, N.; OSTERMEIER, A. Completely derandomized self-adaptation in evolution strategies. *Evolutionary computation*, MIT Press, v. 9, n. 2, p. 159–195, 2001. Citado na página 44.
- HARADA, T.; KAIDAN, M.; THAWONMAS, R. Comparison of synchronous and asynchronous parallelization of extreme surrogate-assisted multi-objective evolutionary algorithm. *Natural Computing*, Springer, p. 1–31, 2020. Citado 6 vezes nas páginas 42, 43, 46, 47, 48, and 52.
- HILDEBRANDT, T.; BRANKE, J. On using surrogates with genetic programming. *Evolutionary computation*, MIT Press, v. 23, n. 3, p. 343–367, 2015. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- HO, S. L. et al. A real coded population-based incremental learning for inverse problems in continuous space. *IEEE Transactions on Magnetics*, IEEE, v. 51, n. 3, p. 1–4, 2015. Citado 7 vezes nas páginas 41, 42, 43, 45, 46, 48, and 52.
- HOSSEINZADEH, M. et al. A hybrid service selection and composition model for cloud-edge computing in the internet of things. *IEEE Access*, IEEE, v. 8, p. 85939–85949, 2020. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- HUBAND, S. et al. A scalable multi-objective test problem toolkit. In: SPRINGER. *International Conference on Evolutionary Multi-Criterion Optimization*. [S.l.], 2005. p. 280–295. Citado na página 51.
- IMAN, R. L.; DAVENPORT, J. M.; ZEIGLER, D. K. *Latin hypercube sampling (program user's guide)*. [S.l.]: Department of Energy, Sandia Laboratories, 1980. Citado 2 vezes nas páginas 45 and 64.

- ISHIBUCHI, H. et al. Modified distance calculation in generational distance and inverted generational distance. In: SPRINGER. *International conference on evolutionary multi-criterion optimization*. [S.l.], 2015. p. 110–125. Citado na página 45.
- JAMIL, M.; YANG, X.-S. A literature survey of benchmark functions for global optimisation problems. *International Journal of Mathematical Modelling and Numerical Optimisation*, Inderscience Publishers Ltd, v. 4, n. 2, p. 150–194, 2013. Citado na página 51.
- JENNINGS, P. C. et al. Genetic algorithms for computational materials discovery accelerated by machine learning. *npj Computational Materials*, Nature Publishing Group, v. 5, n. 1, p. 1–6, 2019. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- JIANG, Q. et al. Multi-objective differential evolution with dynamic covariance matrix learning for multi-objective optimization problems with variable linkages. *Knowledge-Based Systems*, Elsevier, v. 121, p. 111–128, 2017. Citado 7 vezes nas páginas 42, 43, 44, 45, 46, 48, and 52.
- JIN, Y. A comprehensive survey of fitness approximation in evolutionary computation. *Soft computing*, Springer, v. 9, n. 1, p. 3–12, 2005. Citado na página 54.
- JIN, Y. Surrogate-assisted evolutionary computation: Recent advances and future challenges. *Swarm and Evolutionary Computation*, Elsevier, v. 1, n. 2, p. 61–70, 2011. Citado 3 vezes nas páginas 18, 40, and 47.
- JØRGENSEN, M. S.; GROVES, M. N.; HAMMER, B. Combining evolutionary algorithms with clustering toward rational global structure optimization at the atomic scale. *Journal of chemical theory and computation*, ACS Publications, v. 13, n. 3, p. 1486–1493, 2017. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- JOURDAN, L.; DHAENENS, C.; TALBI, E.-G. Using datamining techniques to help metaheuristics: A short survey. In: SPRINGER. *International Workshop on Hybrid Metaheuristics*. [S.l.], 2006. p. 57–69. Citado na página 26.
- KACEM, A. B. et al. A hybrid algorithm to size the hospital resources in the case of a massive influx of victims. *International Journal of Electrical & Computer Engineering (2088-8708)*, v. 10, n. 1, 2020. Citado 7 vezes nas páginas 40, 42, 43, 45, 46, 48, and 52.
- KIM, T. et al. A multimodal deep learning method for android malware detection using various features. *IEEE Transactions on Information Forensics and Security*, IEEE, v. 14, n. 3, p. 773–788, 2018. Citado na página 68.
- KITCHENHAM, B. Procedures for performing systematic reviews. *Keele, UK, Keele University*, v. 33, n. 2004, p. 1–26, 2004. Citado na página 32.
- KITCHENHAM, B. et al. Systematic literature reviews in software engineering—a systematic literature review. *Information and software technology*, Elsevier, v. 51, n. 1, p. 7–15, 2009. Citado na página 32.
- KOHIRA, T. et al. Proposal of simultaneous design optimization benchmark problem of multiple car structures using response surface method. *Transaction of the Japanese Society for Evolutionary Computation*, v. 8, n. 1, p. 11–21, 2017. Citado na página 51.



KOTINIS, M. Improving a multi-objective differential evolution optimizer using fuzzy adaptation and

-medoids clustering. *Soft computing*, Springer, v. 18, n. 4, p. 757–771, 2014. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

KOZIEL, S.; LEIFSSON, L.; YANG, X.-S. *Solving computationally expensive engineering problems: methods and applications*. [S.l.]: Springer, 2014. v. 97. Citado na página 54.

KRAJZEWICZ, D.; BONERT, M.; WAGNER, P. The open source traffic simulation package sumo. *RoboCup 2006*, 2006. Citado na página 83.

KUDIKALA, R.; GIAGKIOZIS, I.; FLEMING, P. Estimation of pareto optimal solutions for multi-objective multi-modal problems. In: *19th International Conference on Soft Computing, MENDEL*. [S.l.: s.n.], 2013. Citado 2 vezes nas páginas 19 and 31.

KUDIKALA, R.; GIAGKIOZIS, I.; FLEMING, P. Increasing the density of multi-objective multi-modal solutions using clustering and pareto estimation techniques. In: *The 2013 World Congress in Computer Science Computer Engineering and Applied Computing*. [S.l.: s.n.], 2013. Citado 2 vezes nas páginas 19 and 31.

KUREK, M.; BECKER, T.; LUK, W. Parametric optimization of reconfigurable designs using machine learning. In: SPRINGER. *International Symposium on Applied Reconfigurable Computing*. [S.l.], 2013. p. 134–145. Citado 6 vezes nas páginas 42, 43, 46, 48, 49, and 52.

LARRAÑAGA, P.; LOZANO, J. A. *Estimation of distribution algorithms: A new tool for evolutionary computation*. [S.l.]: Springer Science & Business Media, 2001. v. 2. Citado 2 vezes nas páginas 18 and 41.

LE, H. L. et al. A hybrid surrogate model for evolutionary undersampling in imbalanced classification. In: IEEE. *2020 IEEE Congress on Evolutionary Computation (CEC)*. [S.l.], 2020. p. 1–8. Citado 7 vezes nas páginas 42, 43, 46, 48, 50, 52, and 53.

LE, H. L. et al. Eusc: A clustering-based surrogate model to accelerate evolutionary undersampling in imbalanced classification. *Applied Soft Computing*, Elsevier, v. 101, p. 107033, 2021. ISSN 1568-4946. Disponível em: <<https://www.sciencedirect.com/science/article/pii/S1568494620309728>>. Citado 7 vezes nas páginas 42, 43, 46, 48, 50, 52, and 53.

LI, B. et al. Many-objective evolutionary algorithms: A survey. *ACM Computing Surveys (CSUR)*, Acm New York, NY, USA, v. 48, n. 1, p. 1–35, 2015. Citado na página 17.

LI, K.; XIANG, Z.; TAN, K. C. Which surrogate works for empirical performance modelling? a case study with differential evolution. In: IEEE. *2019 IEEE Congress on Evolutionary Computation (CEC)*. [S.l.], 2019. p. 1988–1995. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

LI, L. et al. Boost particle swarm optimization with fitness estimation. *Natural Computing*, Springer, v. 18, n. 2, p. 229–247, 2019. Citado 6 vezes nas páginas 43, 46, 48, 50, 52, and 58.

Li, M.; Zhen, L.; Yao, X. How to read many-objective solution sets in parallel coordinates [educational forum]. *IEEE Computational Intelligence Magazine*, v. 12, n. 4, p. 88–100, 2017. Citado na página 69.

- LI, X. et al. A multi-constraint failure-pursuing sampling method for reliability-based design optimization using adaptive kriging. *Engineering with Computers*, Springer, p. 1–14, 2020. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- LI, Y. et al. A sequential kriging method assisted by trust region strategy for proxy cache size optimization of the streaming media video data due to fragment popularity distribution. *Multimedia Tools and Applications*, Springer, v. 78, n. 20, p. 28737–28756, 2019. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- LI, Z. et al. An elitist nondominated sorting hybrid algorithm for multi-objective flexible job-shop scheduling problem with sequence-dependent setups. *Knowledge-Based Systems*, Elsevier, v. 173, p. 83–112, 2019. Citado 8 vezes nas páginas 41, 42, 43, 45, 46, 48, 52, and 55.
- LINDMARK, D. M.; SERVIN, M. Computational exploration of robotic rock loading. *Robotics and Autonomous Systems*, Elsevier, v. 106, p. 117–129, 2018. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- LIU, B. et al. An efficient method for antenna design optimization based on evolutionary computation and machine learning techniques. *IEEE Transactions on Antennas and Propagation*, v. 62, n. 1, p. 7–18, 2014. Citado 8 vezes nas páginas 42, 43, 45, 46, 48, 52, 54, and 55.
- LIU, B. et al. An efficient high-frequency linear rf amplifier synthesis method based on evolutionary computation and machine learning techniques. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, IEEE, v. 31, n. 7, p. 981–993, 2012. Citado 6 vezes nas páginas 42, 43, 46, 48, 52, and 57.
- LIU, B.; KOZIEL, S.; ZHANG, Q. A multi-fidelity surrogate-model-assisted evolutionary algorithm for computationally expensive optimization problems. *Journal of computational science*, Elsevier, v. 12, p. 28–37, 2016. Citado 10 vezes nas páginas 42, 43, 44, 45, 46, 48, 52, 54, 55, and 58.
- LIU, B. et al. Synthesis of integrated passive components for high-frequency rf ics based on evolutionary computation and machine learning techniques. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, IEEE, v. 30, n. 10, p. 1458–1468, 2011. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- LU, J. et al. Transfer learning using computational intelligence: A survey. *Knowledge-Based Systems*, Elsevier, v. 80, p. 14–23, 2015. Citado na página 75.
- LU, J.; LI, B.; JIN, Y. An evolution strategy assisted by an ensemble of local gaussian process models. In: *Proceedings of the 15th annual conference on Genetic and evolutionary computation*. [S.l.: s.n.], 2013. p. 447–454. Citado 7 vezes nas páginas 42, 43, 46, 47, 48, 52, and 57.
- LUKE, S. *Essentials of Metaheuristics*. second. [S.l.]: Lulu, 2013. Available for free at <http://cs.gmu.edu/~sean/book/metaheuristics/>. Citado 2 vezes nas páginas 17 and 24.
- MA, X. et al. Moea/d with baldwinian learning inspired by the regularity property of continuous multiobjective problem. *Neurocomputing*, Elsevier, v. 145, p. 336–352, 2014. Citado 8 vezes nas páginas 41, 42, 43, 45, 46, 48, 50, and 52.
- MAHESWARANATHAN, N. et al. Guided evolutionary strategies: Augmenting random search with surrogate gradients. In: PMLR. *International Conference on Machine Learning*. [S.l.], 2019. p. 4264–4273. Citado 6 vezes nas páginas 42, 43, 46, 48, 52, and 53.

MARTÍ, L. et al. On current model-building methods for multi-objective estimation of distribution algorithms: Shortcomings and directions for improvement. *Department of Informatics, Universidad Carlos III de Madrid, Madrid, Spain, Tech. Rep. GIAA2010E001*, 2010. Citado na página 17.

MARTINEZ-CANTIN, R. Funneled bayesian optimization for design, tuning and control of autonomous systems. *IEEE transactions on cybernetics*, IEEE, v. 49, n. 4, p. 1489–1500, 2018. Citado 6 vezes nas páginas 42, 43, 46, 48, 52, and 54.

MATOS, S. A.; CARVALHO, A. B. Traffic lights signaling optimization as a many-objective optimization problem. *XIII Encontro Nacional de Inteligência Artificial e Computacional*, p. 421–432, 2016. Citado 3 vezes nas páginas 84, 92, and 93.

MENDES-MOREIRA, J. et al. Ensemble approaches for regression: A survey. *Acm computing surveys (csur)*, ACM New York, NY, USA, v. 45, n. 1, p. 1–40, 2012. Citado na página 47.

MICHALAK, K. Informed mutation operator using machine learning for optimization in epidemics prevention. In: *Proceedings of the Genetic and Evolutionary Computation Conference*. [S.l.: s.n.], 2018. p. 1294–1301. Citado 6 vezes nas páginas 42, 43, 46, 48, 50, and 52.

MICHALAK, K. Surrogate-based optimization for reduction of contagion susceptibility in financial systems. In: *Proceedings of the Genetic and Evolutionary Computation Conference*. [S.l.: s.n.], 2019. p. 1266–1274. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

MINELLA, G.; RUIZ, R.; CIAVOTTA, M. A review and evaluation of multiobjective algorithms for the flowshop scheduling problem. *INFORMS Journal on Computing*, INFORMS, v. 20, n. 3, p. 451–471, 2008. Citado na página 51.

MLAKAR, M. et al. Gp-demo: differential evolution for multiobjective optimization based on gaussian process models. *European Journal of Operational Research*, Elsevier, v. 243, n. 2, p. 347–361, 2015. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

MOLGA, M.; SMUTNICKI, C. Test functions for optimization needs. *Test functions for optimization needs*, v. 101, p. 48, 2005. Citado na página 51.

NGUYEN, D. D.; NGUYEN, L. An adaptive control for surrogate assisted multi-objective evolutionary algorithms. In: *ICT Systems and Sustainability*. [S.l.]: Springer, 2021. p. 123–132. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

NOURI, H. E.; DRISS, O. B.; GHÉDIRA, K. Solving the flexible job shop problem by hybrid metaheuristics-based multiagent model. *Journal of Industrial Engineering International*, Springer, v. 14, n. 1, p. 1–14, 2018. Citado 7 vezes nas páginas 42, 43, 46, 47, 48, 52, and 55.

OHTSUKA, H. et al. Evolutionary algorithm using surrogate assisted model for simultaneous design optimization benchmark problem of multiple car structures. In: *Proceedings of the Genetic and Evolutionary Computation Conference Companion*. [S.l.: s.n.], 2018. p. 55–56. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

PANTULA, P. D.; MITRA, K. An evolutionary machine learning approach towards less conservative robust optimization. In: IEEE. *2019 IEEE Congress on Evolutionary Computation (CEC)*. [S.l.], 2019. p. 2990–2997. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

PARK, D. Research of cyclone optimization based on cfd, gmdh-type neural network and genetic algorithm. In: *Proceedings of the 2019 3rd International Conference on Virtual and Augmented Reality Simulations*. [S.l.: s.n.], 2019. p. 90–96. Citado 6 vezes nas páginas 42, 43, 46, 48, 52, and 54.

PARQUE, V.; MIYASHITA, T. A method to learn high-performing and novel product layouts and its application to vehicle design. *Neurocomputing*, Elsevier, v. 248, p. 41–56, 2017. Citado 7 vezes nas páginas 40, 42, 43, 46, 48, 52, and 58.

PEDREGOSA, F. et al. Scikit-learn: Machine learning in python. *the Journal of machine Learning research*, JMLR. org, v. 12, p. 2825–2830, 2011. Citado na página 67.

PELIKAN, M.; GOLDBERG, D. E.; LOBO, F. G. A survey of optimization by building and using probabilistic models. *Computational optimization and applications*, Springer, v. 21, n. 1, p. 5–20, 2002. Citado na página 27.

PENG, C. et al. A surrogate model assisted quantum-inspired evolutionary algorithm for hyperparameter optimization in machine learning. In: IEEE. *2019 IEEE Congress on Evolutionary Computation (CEC)*. [S.l.], 2019. p. 1060–1067. Citado 6 vezes nas páginas 42, 43, 46, 48, 52, and 54.

PENNISI, M.; RUSSO, G.; PAPPALARDO, F. Combining parallel genetic algorithms and machine learning to improve the research of optimal vaccination protocols. In: IEEE. *2018 26th Euromicro International Conference on Parallel, Distributed and Network-based Processing (PDP)*. [S.l.], 2018. p. 399–405. Citado 6 vezes nas páginas 42, 43, 46, 47, 48, and 52.

PLASTINO, A. et al. Adaptive and multi-mining versions of the dm-grasp hybrid metaheuristic. *Journal of Heuristics*, Springer, v. 20, n. 1, p. 39–74, 2014. Citado 7 vezes nas páginas 42, 43, 46, 47, 48, 50, and 52.

PROBST, D. M. et al. Evaluating optimization strategies for engine simulations using machine learning emulators. *Journal of Engineering for Gas Turbines and Power*, American Society of Mechanical Engineers Digital Collection, v. 141, n. 9, 2019. Citado 7 vezes nas páginas 40, 42, 43, 46, 48, 52, and 54.

PROVOST, F.; KOHAVI, R. Glossary of terms. *Journal of Machine Learning*, v. 30, n. 2-3, p. 271–274, 1998. Citado na página 24.

PYLE, J.; CHIMEH, M. K.; RICHMOND, P. Surrogate modelling for efficient discovery of emergent population dynamics. In: IEEE. *2019 International Conference on High Performance Computing & Simulation (HPCS)*. [S.l.], 2019. p. 99–106. Citado 7 vezes nas páginas 40, 42, 43, 46, 48, 49, and 52.

QI, Y. et al. Multi-objective immune algorithm with baldwinian learning. *Applied Soft Computing*, Elsevier, v. 12, n. 8, p. 2654–2674, 2012. Citado 9 vezes nas páginas 41, 42, 43, 45, 46, 48, 50, 52, and 58.

RANA, S. et al. High dimensional bayesian optimization with elastic gaussian process. In: PMLR. *International conference on machine learning*. [S.l.], 2017. p. 2883–2891. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

- REMLI, M. A. et al. An enhanced scatter search with combined opposition-based learning for parameter estimation in large-scale kinetic models of biochemical systems. *Engineering Applications of Artificial Intelligence*, Elsevier, v. 62, p. 164–180, 2017. Citado 4 vezes nas páginas 42, 43, 46, and 52.
- SAIKIA, P.; RAKSHIT, D. et al. Designing a clean and efficient air conditioner with ai intervention to optimize energy-exergy interplay. *Energy and AI*, Elsevier, v. 2, p. 100029, 2020. Citado 6 vezes nas páginas 40, 42, 43, 46, 48, and 52.
- SAINI, B. *The python version reference vector guided evolutionary algorithm*. [S.l.]: GitHub, 2019. <<https://github.com/industrial-optimization-group/pyRVEA>>. Citado na página 86.
- SÁNCHEZ-MEDINA, J. J.; GALÁN-MORENO, M. J.; RUBIO-ROYO, E. Traffic signal optimization in "la almozara" district in saragossa under congestion conditions, using genetic algorithms, traffic microsimulation, and cluster computing. *IEEE Transactions on Intelligent Transportation Systems*, IEEE, v. 11, n. 1, p. 132–141, 2009. Citado na página 83.
- SANTANA-QUINTERO, L. V.; MONTANO, A. A.; COELLO, C. A. C. A review of techniques for handling expensive functions in evolutionary multi-objective optimization. In: *Computational intelligence in expensive optimization problems*. [S.l.]: Springer, 2010. p. 29–59. Citado 2 vezes nas páginas 28 and 29.
- SANTANA, R. et al. Introducing the use of model-based evolutionary algorithms for eeg-based motor imagery classification. In: *Proceedings of the 14th annual conference on Genetic and evolutionary computation*. [S.l.: s.n.], 2012. p. 1159–1166. Citado 7 vezes nas páginas 42, 43, 44, 45, 46, 48, and 52.
- SANTOS, M.; OLIVEIRA, J. A. de; BRITTO, A. Decision variable learning. In: IEEE. *2019 8th Brazilian Conference on Intelligent Systems (BRACIS)*. [S.l.], 2019. p. 497–502. Citado 14 vezes nas páginas 19, 42, 43, 44, 46, 48, 49, 52, 58, 61, 62, 63, 64, and 80.
- SCHUTZE, O.; LARA, A.; COELLO, C. A. C. On the influence of the number of objectives on the hardness of a multiobjective optimization problem. *IEEE Transactions on Evolutionary Computation*, IEEE, v. 15, n. 4, p. 444–455, 2010. Citado 2 vezes nas páginas 17 and 24.
- SHAHANE, S. et al. Optimization of solidification in die casting using numerical simulations and machine learning. *Journal of Manufacturing Processes*, Elsevier, v. 51, p. 130–141, 2020. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- SHE, D. et al. Neuzz: Efficient fuzzing with neural program smoothing. In: IEEE. *2019 IEEE Symposium on Security and Privacy (SP)*. [S.l.], 2019. p. 803–817. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- SHI, L.; RASHEED, K. A survey of fitness approximation methods applied in evolutionary algorithms. In: *Computational intelligence in expensive optimization problems*. [S.l.]: Springer, 2010. p. 3–28. Citado 3 vezes nas páginas 17, 28, and 29.
- SHIM, V. A. et al. Enhancing the scalability of multi-objective optimization via restricted boltzmann machine-based estimation of distribution algorithm. *Information Sciences*, Elsevier, v. 248, p. 191–213, 2013. ISSN 0020-0255. Disponível em: <<https://www.sciencedirect.com/science/article/pii/S0020025513004726>>. Citado 7 vezes nas páginas 41, 42, 43, 45, 46, 48, and 52.



SIMONS, C. L.; SMITH, J.; WHITE, P. Interactive ant colony optimization (iaco) for early lifecycle software design. *Swarm Intelligence*, Springer, v. 8, n. 2, p. 139–157, 2014. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

SMOLA, A. J.; SCHÖLKOPF, B. A tutorial on support vector regression. *Statistics and computing*, Springer, v. 14, n. 3, p. 199–222, 2004. Citado na página 67.

SONG, J. et al. Adaptive surrogate model based multiobjective optimization for coastal aquifer management. *Journal of hydrology*, Elsevier, v. 561, p. 98–111, 2018. Citado 6 vezes nas páginas 42, 43, 45, 46, 48, and 52.

SONG, K. et al. Reliability-based design optimization using adaptive surrogate model and importance sampling-based modified sora method. *Engineering with Computers*, Springer, p. 1–20, 2019. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

STANDER, L.; WOOLWAY, M.; ZYL, T. van. Extended surrogate assisted continuous process optimisation. In: IEEE. *2020 7th International Conference on Soft Computing & Machine Intelligence (ISCMI)*. [S.l.], 2020. p. 275–279. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

STOEAN, R. Analysis on the potential of an ea-surrogate modelling tandem for deep learning parametrization: an example for cancer classification from medical images. *Neural Computing and Applications*, Springer, v. 32, n. 2, p. 313–322, 2020. Citado 6 vezes nas páginas 42, 43, 46, 48, 52, and 53.

SUGANTHAN, P. et al. Problem definitions and evaluation criteria for the cec 2005 special session on real-parameter optimization. *Natural Computing*, v. 341-357, 01 2005. Citado na página 51.

SUN, J. et al. Learning from a stream of nonstationary and dependent data in multiobjective evolutionary optimization. *IEEE Transactions on Evolutionary Computation*, IEEE, v. 23, n. 4, p. 541–555, 2018. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

SUN, X. et al. A new surrogate-assisted interactive genetic algorithm with weighted semisupervised learning. *IEEE Transactions on Cybernetics*, IEEE, v. 43, n. 2, p. 685–698, 2013. Citado 7 vezes nas páginas 42, 43, 46, 48, 50, 52, and 57.

SVETNIK, V. et al. Random forest: a classification and regression tool for compound classification and qsar modeling. *Journal of chemical information and computer sciences*, ACS Publications, v. 43, n. 6, p. 1947–1958, 2003. Citado na página 67.

TAILLARD, E. Benchmarks for basic scheduling problems. *European journal of operational research*, Elsevier, v. 64, n. 2, p. 278–285, 1993. Citado na página 55.

TALBI, E.-G. Combining metaheuristics with mathematical programming, constraint programming and machine learning. *Annals of Operations Research*, Springer, v. 240, n. 1, p. 171–215, 2016. Citado 3 vezes nas páginas 18, 26, and 27.

TANG, Q. et al. Optimal shape design of an autonomous underwater vehicle based on multi-objective particle swarm optimization. *Natural Computing*, Springer, v. 19, n. 4, p. 733–742, 2020. Citado 7 vezes nas páginas 40, 42, 43, 46, 48, 52, and 54.

- TENNE, Y. A computational intelligence algorithm for expensive engineering optimization problems. *Engineering Applications of Artificial Intelligence*, v. 25, n. 5, p. 1009–1021, 2012. ISSN 0952-1976. Disponível em: <<https://www.sciencedirect.com/science/article/pii/S0952197612000656>>. Citado 9 vezes nas páginas 42, 43, 46, 47, 48, 49, 52, 54, and 58.
- TENNE, Y. A computational intelligence algorithm for simulation-driven optimization problems. *Advances in Engineering Software*, Elsevier, v. 47, n. 1, p. 62–71, 2012. Citado 6 vezes nas páginas 42, 43, 46, 48, 52, and 54.
- THIERENS, D.; BOSMAN, P. A. Optimal mixing evolutionary algorithms. In: *Proceedings of the 13th Annual Conference on Genetic and Evolutionary Computation*. New York, NY, USA: Association for Computing Machinery, 2011. (GECCO '11), p. 617–624. ISBN 9781450305570. Disponível em: <<https://doi.org/10.1145/2001576.2001661>>. Citado 7 vezes nas páginas 41, 42, 43, 45, 46, 48, and 52.
- TIAN, J. et al. Granularity-based surrogate-assisted particle swarm optimization for high-dimensional expensive optimization. *Knowledge-Based Systems*, Elsevier, v. 187, p. 104815, 2020. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- TIAN, K. et al. Enhanced variable-fidelity surrogate-based optimization framework by gaussian process regression and fuzzy clustering. *Computer Methods in Applied Mechanics and Engineering*, Elsevier, v. 366, p. 113045, 2020. Citado 6 vezes nas páginas 42, 43, 46, 48, 52, and 57.
- TIAN, L. et al. Multi-objective surrogate modeling for real-time energy-efficient station grouping in iee 802.11 ah. *Pervasive and Mobile Computing*, Elsevier, v. 57, p. 33–48, 2019. Citado 6 vezes nas páginas 42, 43, 46, 47, 48, and 52.
- TINÓS, R. Artificial neural network based crossover for evolutionary algorithms. *Applied Soft Computing*, Elsevier, v. 95, p. 106512, 2020. Citado 6 vezes nas páginas 42, 43, 45, 46, 48, and 52.
- TIZHOOSH, H. R. Opposition-based learning: a new scheme for machine intelligence. In: IEEE. *International conference on computational intelligence for modelling, control and automation and international conference on intelligent agents, web technologies and internet commerce (CIMCA-IAWTIC'06)*. [S.l.], 2005. v. 1, p. 695–701. Citado na página 44.
- TOSCANO, G.; DEB, K. Study of the approximation of the fitness landscape and the ranking process of scalarizing functions for many-objective problems. In: IEEE. *2016 IEEE congress on evolutionary computation (CEC)*. [S.l.], 2016. p. 4358–4365. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- VIJAYACHANDRAN, A. A.; DAVIDSON, P.; WAAS, A. M. Optimal fiber paths for robotically manufactured composite structural panels. *International Journal of Non-Linear Mechanics*, Elsevier, v. 126, p. 103567, 2020. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.
- VIRGOLIN, M. et al. Symbolic regression and feature construction with gp-gomea applied to radiotherapy dose reconstruction of childhood cancer survivors. In: *Proceedings of the Genetic and Evolutionary Computation Conference*. [S.l.: s.n.], 2018. p. 1395–1402. Citado 9 vezes nas páginas 41, 42, 43, 44, 45, 46, 48, 50, and 52.

VIRGOLIN, M. et al. Scalable genetic programming by gene-pool optimal mixing and input-space entropy-based building-block learning. In: *Proceedings of the Genetic and Evolutionary Computation Conference*. New York, NY, USA: Association for Computing Machinery, 2017. (GECCO '17), p. 1041–1048. ISBN 9781450349208. Disponível em: <<https://doi.org/10.1145/3071178.3071287>>. Citado 8 vezes nas páginas 41, 42, 43, 44, 45, 46, 48, and 52.

VOLZ, V. Uncertainty handling in surrogate assisted optimisation of games. *KI-Künstliche Intelligenz*, Springer, v. 34, n. 1, p. 95–99, 2020. Citado 6 vezes nas páginas 42, 43, 46, 47, 48, and 52.

WANG, H. et al. Offline data-driven evolutionary optimization using selective surrogate ensembles. *IEEE Transactions on Evolutionary Computation*, IEEE, v. 23, n. 2, p. 203–216, 2018. Citado 9 vezes nas páginas 40, 42, 43, 46, 47, 48, 52, 54, and 57.

WANG, H.; WU, Z.; RAHNAMAYAN, S. Enhanced opposition-based differential evolution for solving high-dimensional continuous optimization problems. *Soft Computing*, Springer, v. 15, n. 11, p. 2127–2140, 2011. Citado 6 vezes nas páginas 42, 43, 44, 45, 46, and 52.

WANG, L. et al. A bi-population based estimation of distribution algorithm for the flexible job-shop scheduling problem. *Computers & Industrial Engineering*, Elsevier, v. 62, n. 4, p. 917–926, 2012. Citado 8 vezes nas páginas 41, 42, 43, 45, 46, 48, 52, and 55.

WANG, S.; YIN, Y. Polygene-based evolutionary algorithms with frequent pattern mining. *Frontiers of Computer Science*, Springer, v. 12, n. 5, p. 950–965, 2018. Citado 6 vezes nas páginas 42, 43, 46, 48, 50, and 52.

WANG, S.-C. Artificial neural network. In: *Interdisciplinary computing in java programming*. [S.l.]: Springer, 2003. p. 81–100. Citado na página 25.

WANG, X.; TANG, L. An adaptive multi-population differential evolution algorithm for continuous multi-objective optimization. *Information Sciences*, v. 348, p. 124–141, 2016. ISSN 0020-0255. Disponível em: <<https://www.sciencedirect.com/science/article/pii/S002002551600102X>>. Citado 5 vezes nas páginas 42, 43, 46, 47, and 52.

WANG, X.; TANG, L. A machine-learning based memetic algorithm for the multi-objective permutation flowshop scheduling problem. *Computers & Operations Research*, Elsevier, v. 79, p. 60–77, 2017. Citado 7 vezes nas páginas 42, 43, 46, 48, 50, 52, and 55.

WANG, Z.; BOVIK, A. C. Mean squared error: Love it or leave it? a new look at signal fidelity measures. *IEEE signal processing magazine*, IEEE, v. 26, n. 1, p. 98–117, 2009. Citado na página 68.

WATKINS, C. J.; DAYAN, P. Q-learning. *Machine learning*, Springer, v. 8, n. 3-4, p. 279–292, 1992. Citado 2 vezes nas páginas 25 and 51.

WENG, J. et al. A machine learning based approach for determining the stress-strain relation of grey cast iron from nanoindentation. *Mechanics of Materials*, Elsevier, v. 148, p. 103522, 2020. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

WHILE, L. et al. A faster algorithm for calculating hypervolume. *IEEE transactions on evolutionary computation*, IEEE, v. 10, n. 1, p. 29–38, 2006. Citado 4 vezes nas páginas 45, 66, 73, and 84.



WOHLIN, C. Guidelines for snowballing in systematic literature studies and a replication in software engineering. In: *Proceedings of the 18th international conference on evaluation and assessment in software engineering*. [S.l.: s.n.], 2014. p. 1–10. Citado na página 36.

WOHLIN, C. et al. *Experimentation in software engineering*. [S.l.]: Springer Science & Business Media, 2012. Citado 2 vezes nas páginas 59 and 60.

WU, Q.; WANG, H.; HONG, W. Multistage collaborative machine learning and its application to antenna modeling and optimization. *IEEE Transactions on Antennas and Propagation*, IEEE, v. 68, n. 5, p. 3397–3409, 2020. Citado 7 vezes nas páginas 42, 43, 46, 48, 52, 54, and 55.

XIE, W. et al. Surrogate network-based sparseness hyper-parameter optimization for deep expression recognition. *Pattern Recognition*, Elsevier, v. 111, p. 107701, 2021. Citado 6 vezes nas páginas 42, 43, 46, 48, 52, and 53.

YAN, Y.; GIAGKIOZIS, I.; FLEMING, P. J. Improved sampling of decision space for pareto estimation. In: *Proceedings of the 2015 Annual Conference on Genetic and Evolutionary Computation*. [S.l.: s.n.], 2015. p. 767–774. Citado 2 vezes nas páginas 19 and 31.

YIN, J.; TSAI, F. T.-C. Bayesian set pair analysis and machine learning based ensemble surrogates for optimal multi-aquifer system remediation design. *Journal of Hydrology*, Elsevier, v. 580, p. 124280, 2020. Citado 8 vezes nas páginas 40, 42, 43, 46, 47, 48, 52, and 57.

ZANGARI, M. et al. A decomposition-based binary aco algorithm for the multiobjective ubqp. *Neurocomputing*, Elsevier, v. 246, p. 58–68, 2017. Citado 7 vezes nas páginas 41, 42, 43, 45, 46, 48, and 52.

ZANGARI, M. et al. Not all pbils are the same: Unveiling the different learning mechanisms of pbil variants. *Applied Soft Computing*, Elsevier, v. 53, p. 88–96, 2017. Citado 7 vezes nas páginas 41, 42, 43, 45, 46, 48, and 52.

ZHANG, B.; SHAFI, K.; ABBASS, H. Online knowledge-based evolutionary multi-objective optimization. In: *2014 IEEE Congress on Evolutionary Computation (CEC)*. [S.l.: s.n.], 2014. p. 2222–2229. Citado 6 vezes nas páginas 42, 43, 46, 48, 50, and 52.

ZHANG, B.; SHAFI, K.; ABBASS, H. A. Hybrid knowledge-based evolutionary many-objective optimization. In: IEEE. *2016 IEEE Congress on Evolutionary Computation (CEC)*. [S.l.], 2016. p. 1007–1014. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

ZHANG, J. et al. Evolutionary computation meets machine learning: A survey. *IEEE Computational Intelligence Magazine*, IEEE, v. 6, n. 4, p. 68–75, 2011. Citado 2 vezes nas páginas 27 and 28.

ZHANG, Q. et al. Multiobjective optimization test instances for the cec 2009 special session and competition. Colchester, UK, 2008. Citado na página 51.

ZHANG, Z. et al. An enhanced multi-objective evolutionary optimization algorithm with inverse model. *Information Sciences*, Elsevier, v. 530, p. 128–147, 2020. Citado na página 31.

ZHOU, Y.; JIN, Y.; DING, J. Surrogate-assisted evolutionary search of spiking neural architectures in liquid state machines. *Neurocomputing*, Elsevier, v. 406, p. 12–23, 2020. Citado 6 vezes nas páginas 42, 43, 46, 48, 52, and 53.

ZIMMERMAN, D. W. Teacher's corner: A note on interpretation of the paired-samples t test. *Journal of Educational and Behavioral Statistics*, Sage Publications Sage CA: Los Angeles, CA, v. 22, n. 3, p. 349–360, 1997. Citado na página 51.

ZITZLER, E.; DEB, K.; THIELE, L. Comparison of multiobjective evolutionary algorithms: Empirical results. *Evolutionary computation*, MIT Press, v. 8, n. 2, p. 173–195, 2000. Citado na página 51.

ZULUAGA, M.; KRAUSE, A.; PÜSCHEL, M. e-pal: An active learning approach to the multi-objective optimization problem. *Journal of Machine Learning Research*, v. 17, n. 104, p. 1–32, 2016. Disponível em: <<http://jmlr.org/papers/v17/15-047.html>>. Citado 5 vezes nas páginas 42, 43, 46, 48, and 52.

# Appendix

## **APPENDIX A – Heatmap plots of the optimization experiment.**

Figure 28 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ1 with 3 objectives.

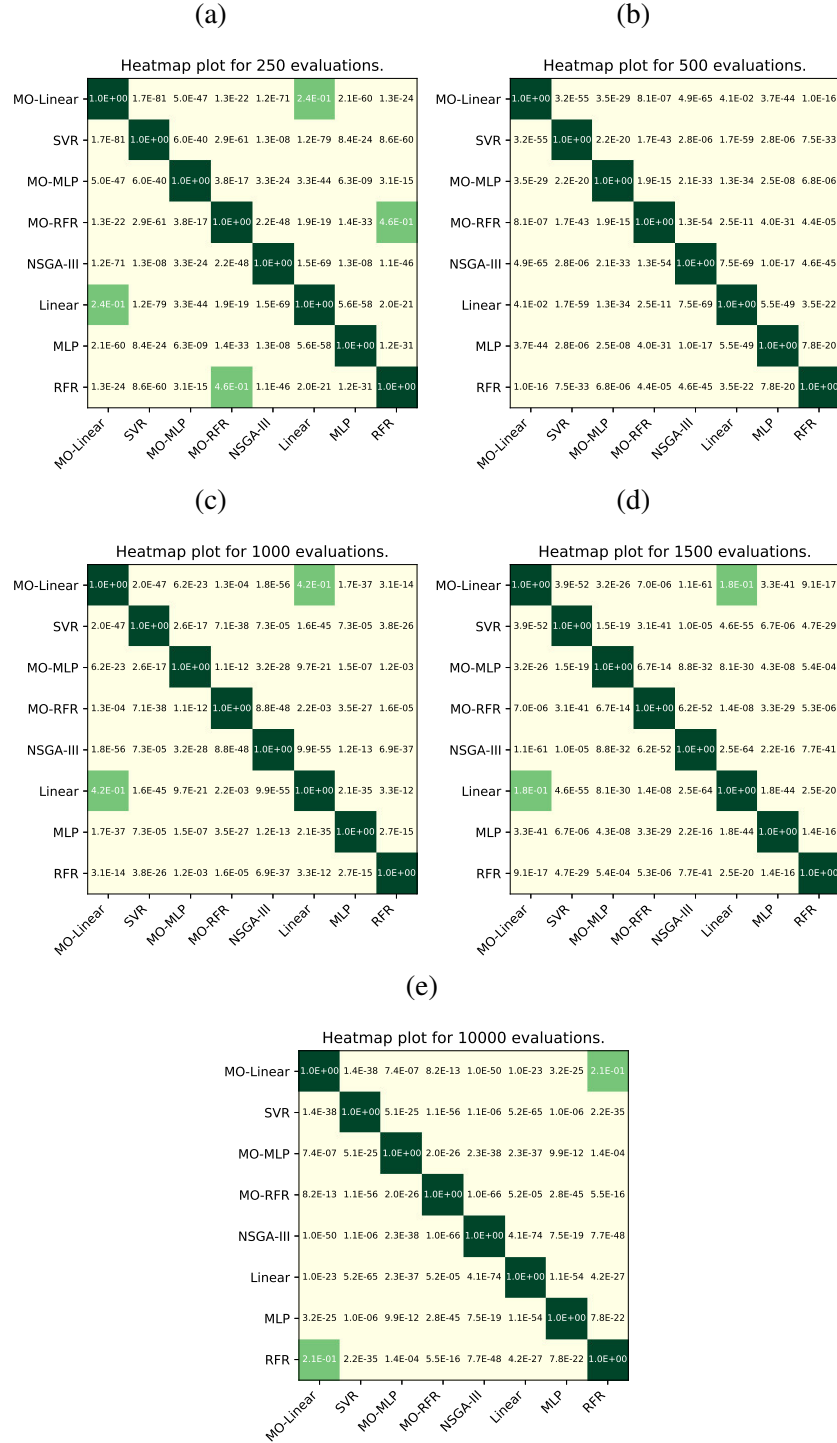
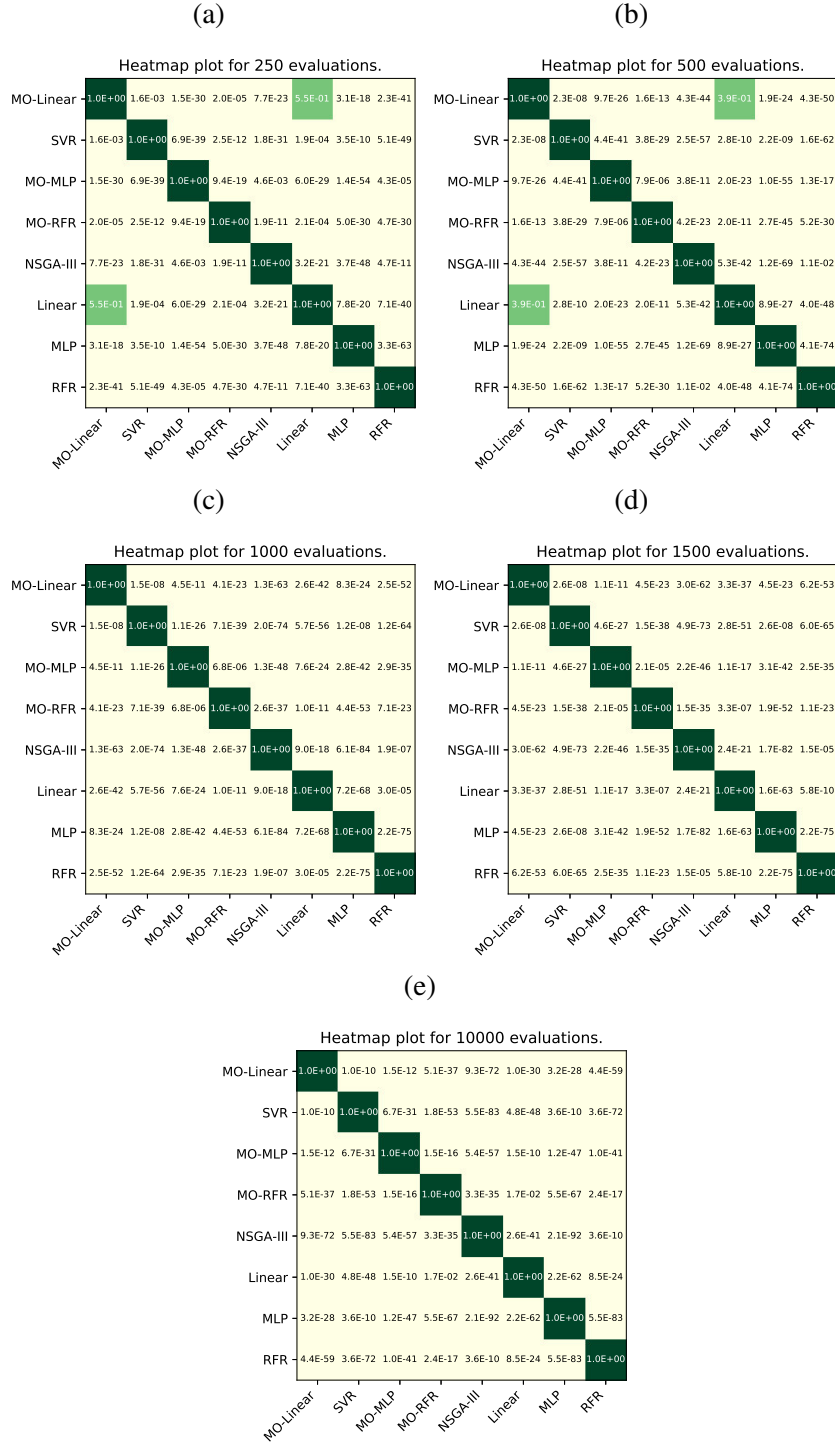
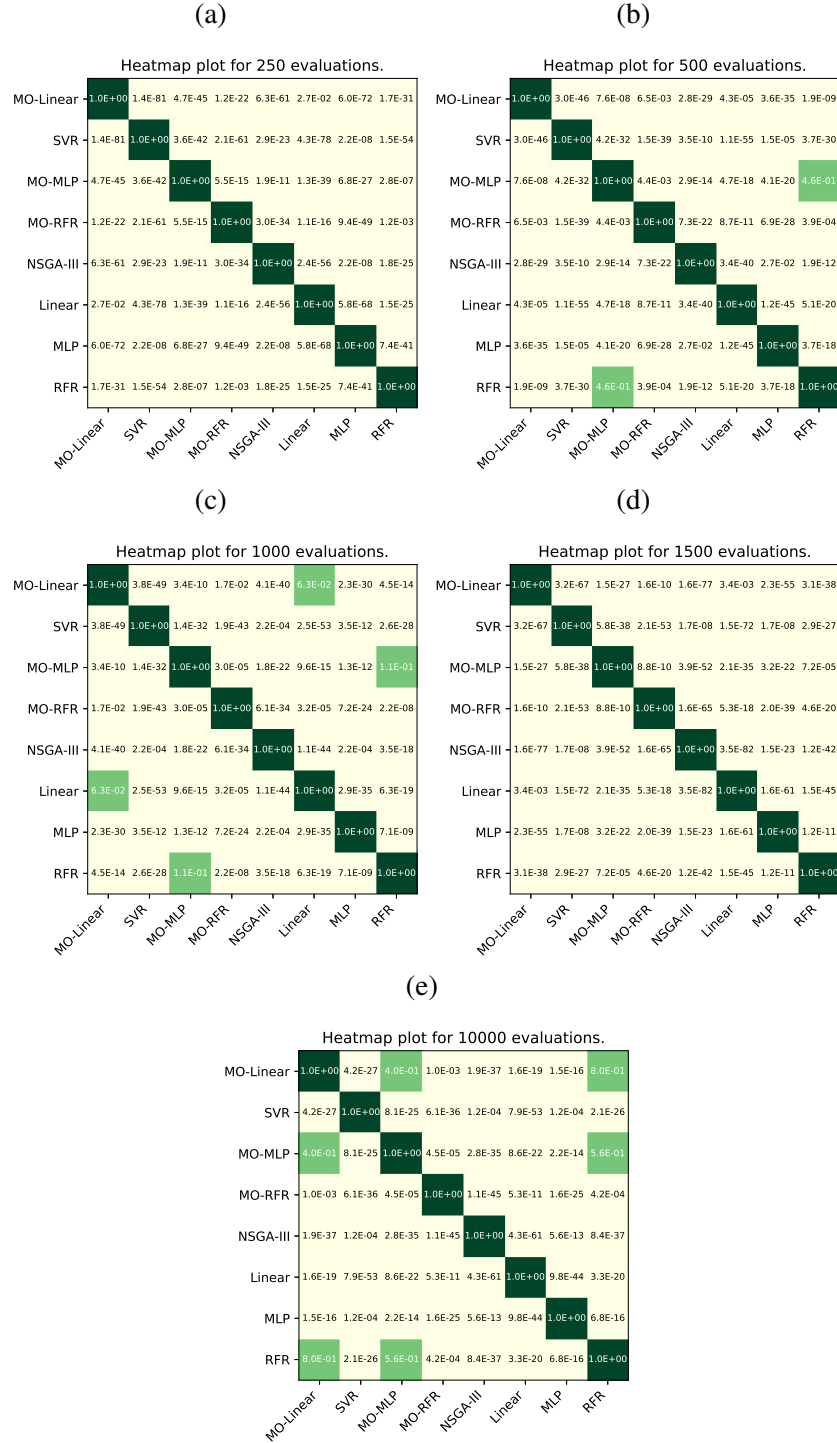


Figure 29 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ2 with 3 objectives.



Source: Produced by authors.

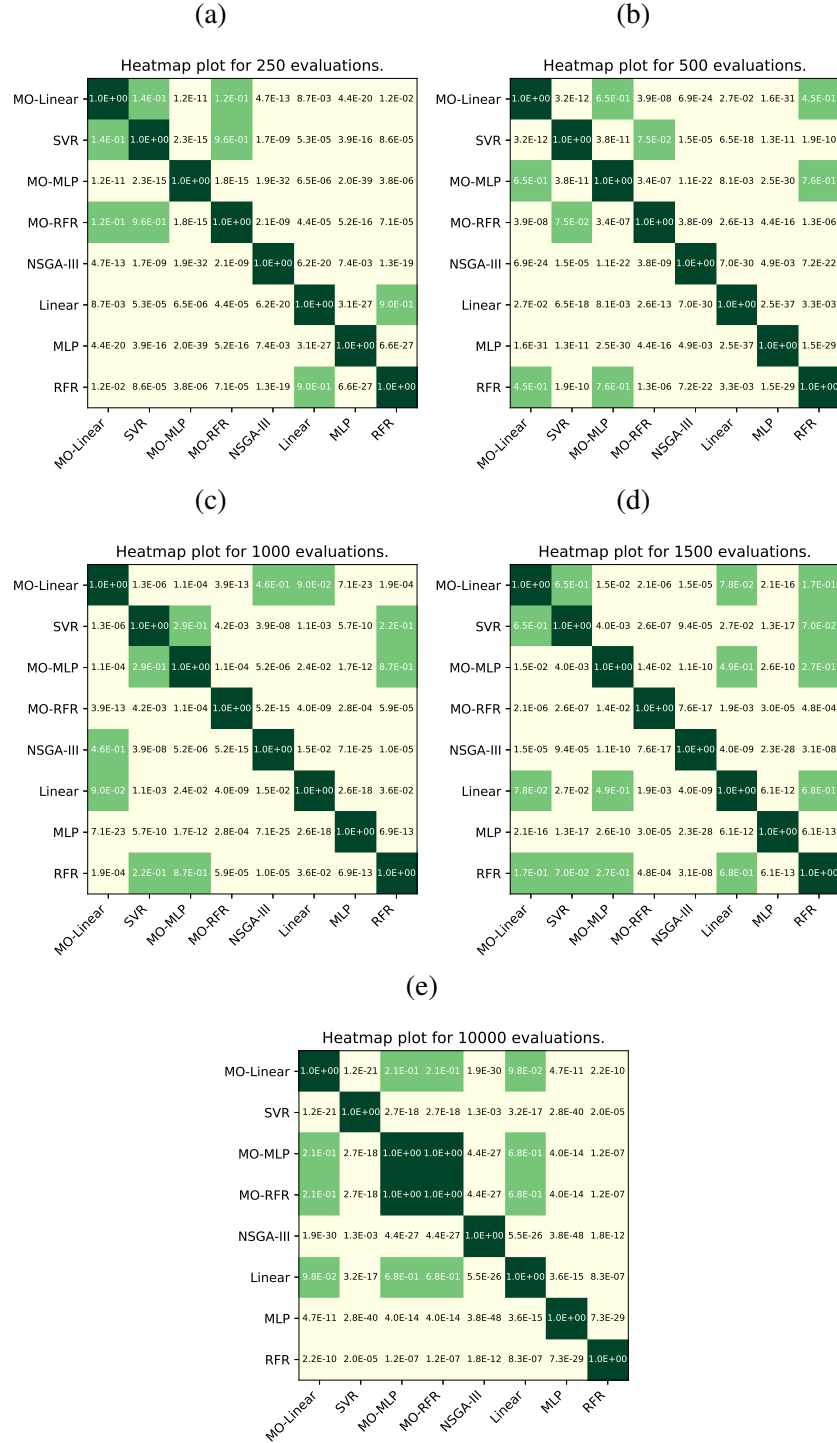
Figure 30 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ3 with 3 objectives.



Source: Produced by authors.

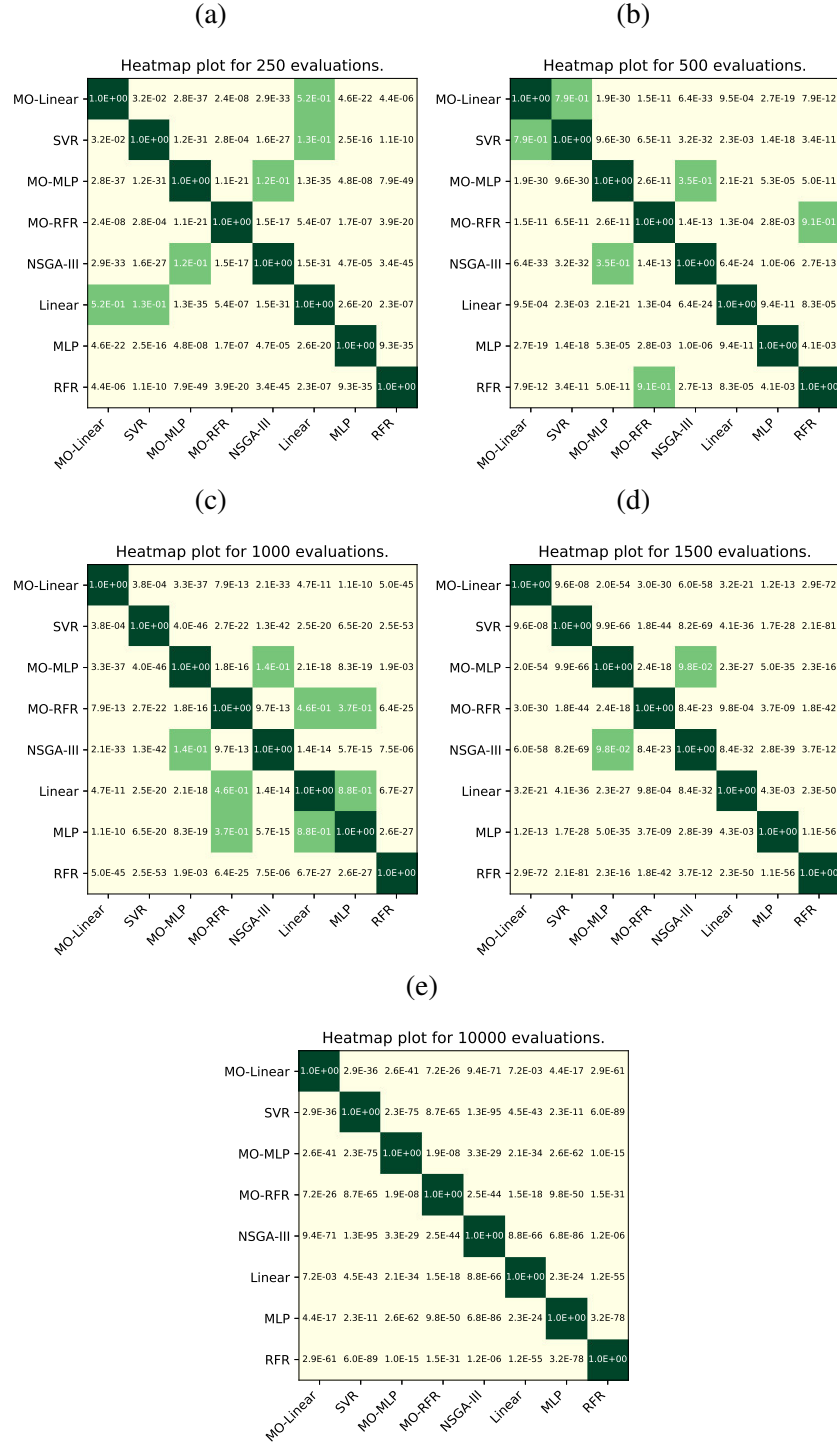


Figure 31 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ4 with 3 objectives.



Source: Produced by authors.

Figure 32 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ5 with 3 objectives.



Source: Produced by authors.

Figure 33 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ6 with 3 objectives.

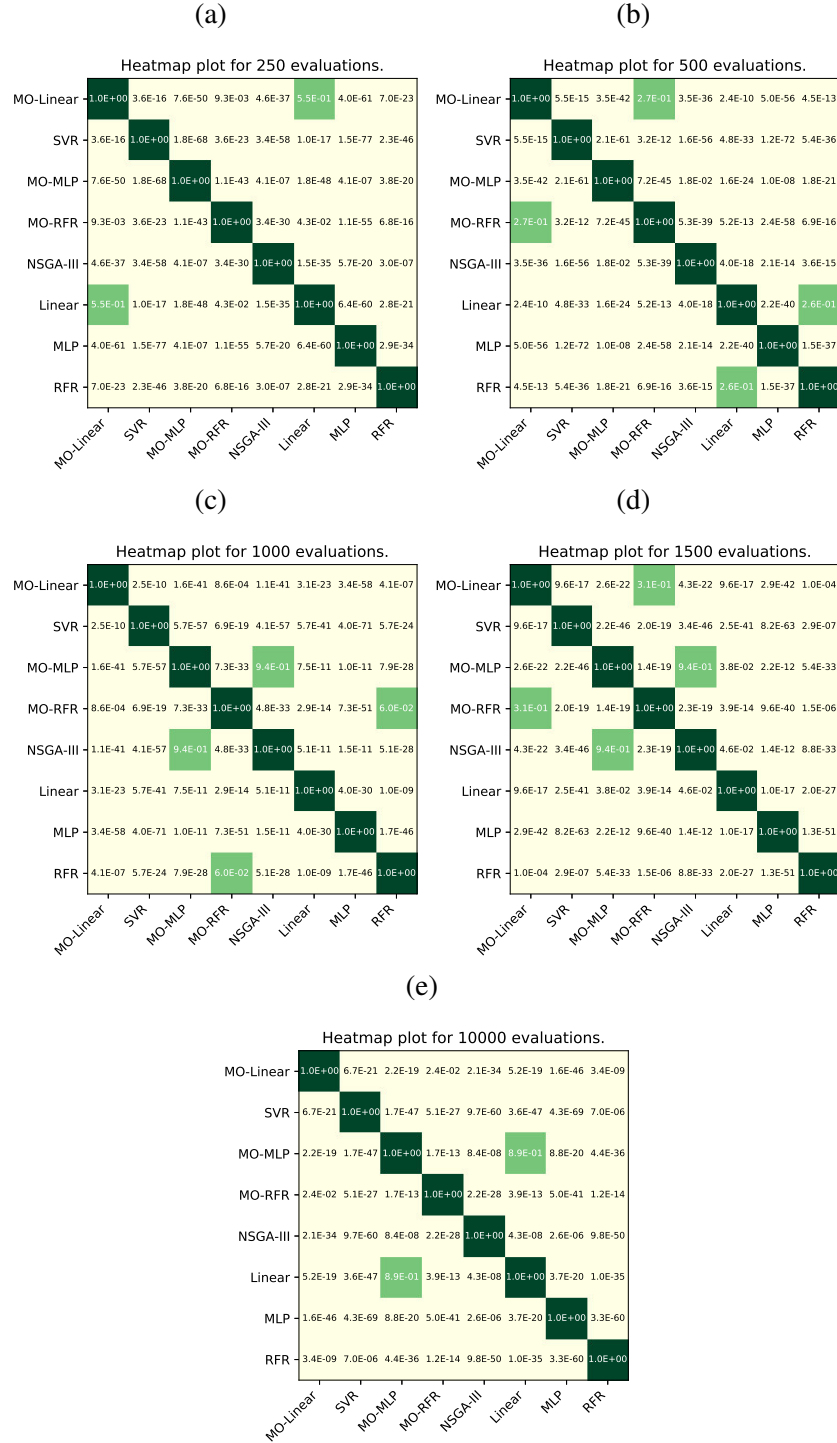
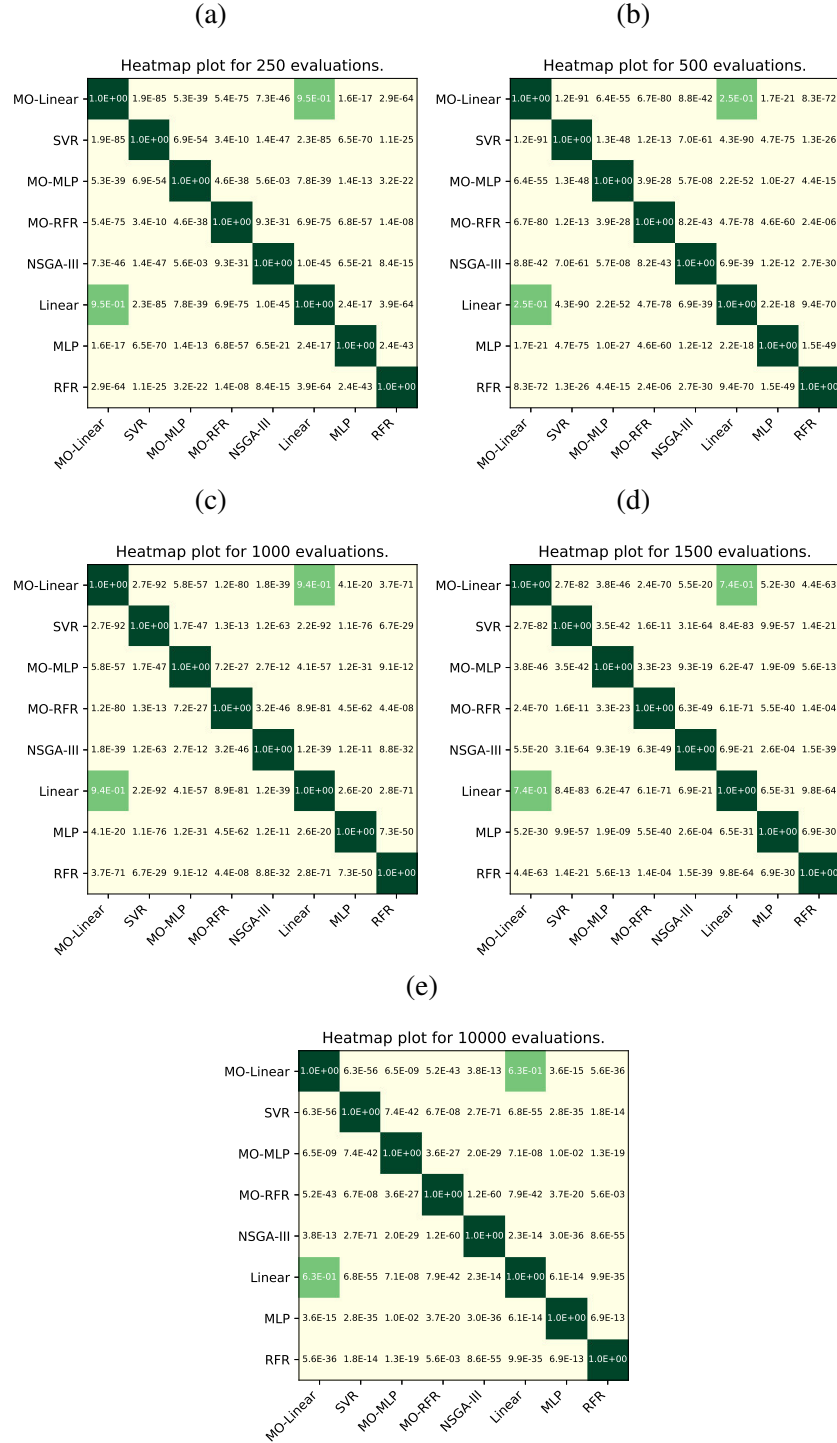
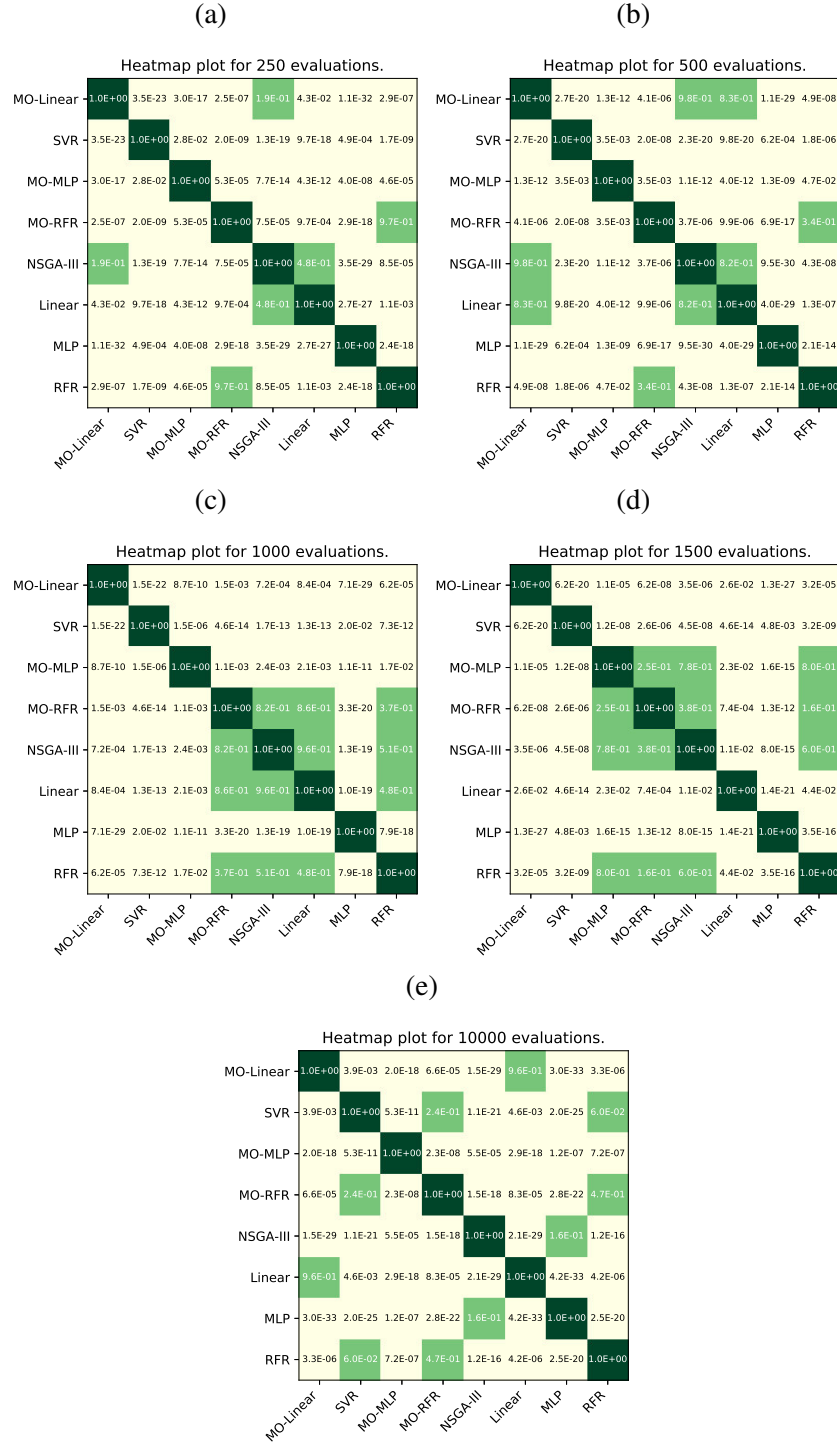


Figure 34 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ7 with 3 objectives.



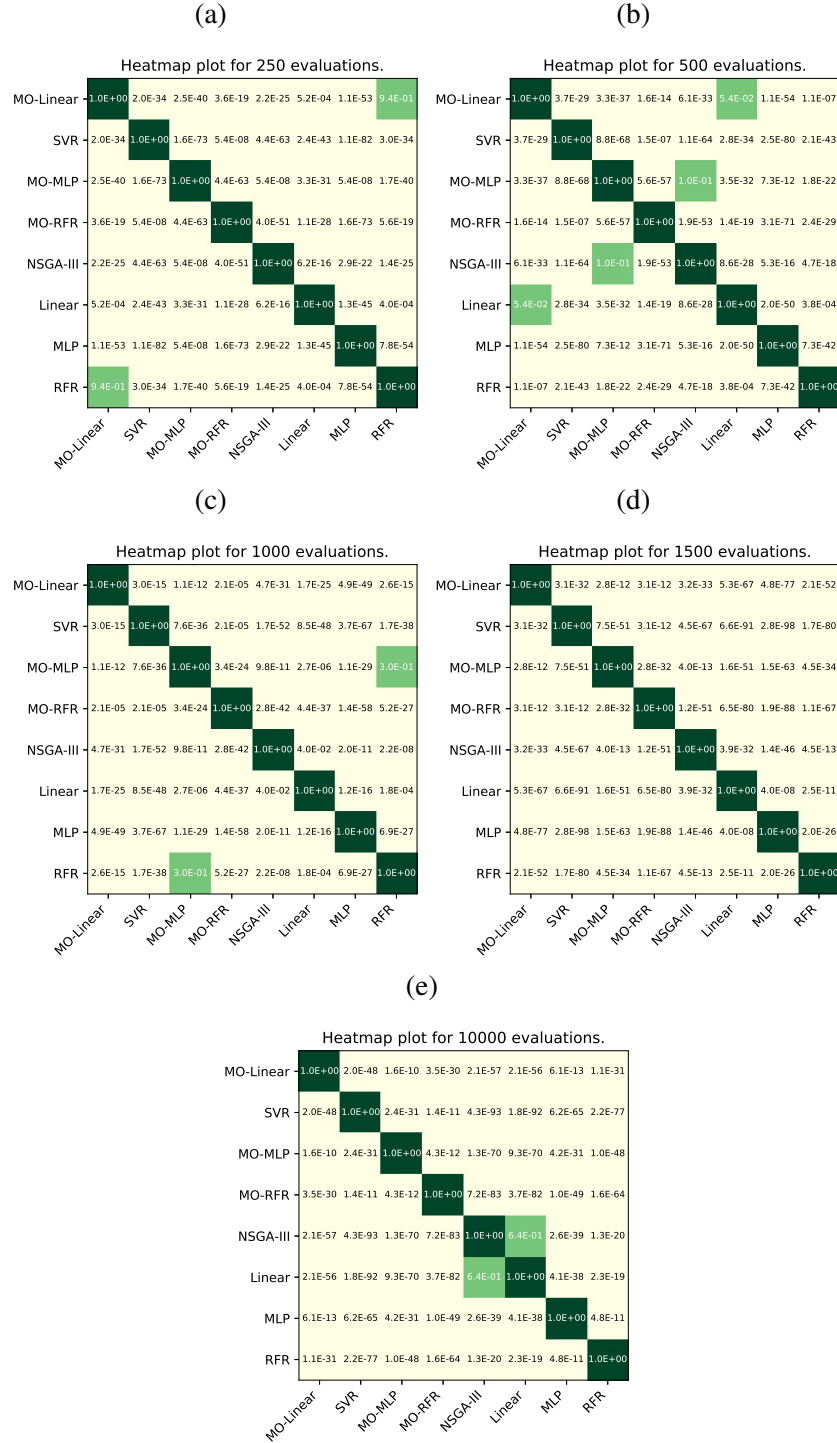
Source: Produced by authors.

Figure 35 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ1 with 10 objectives.



Source: Produced by authors.

Figure 36 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ2 with 10 objectives.



Source: Produced by authors.

Figure 37 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ3 with 10 objectives.

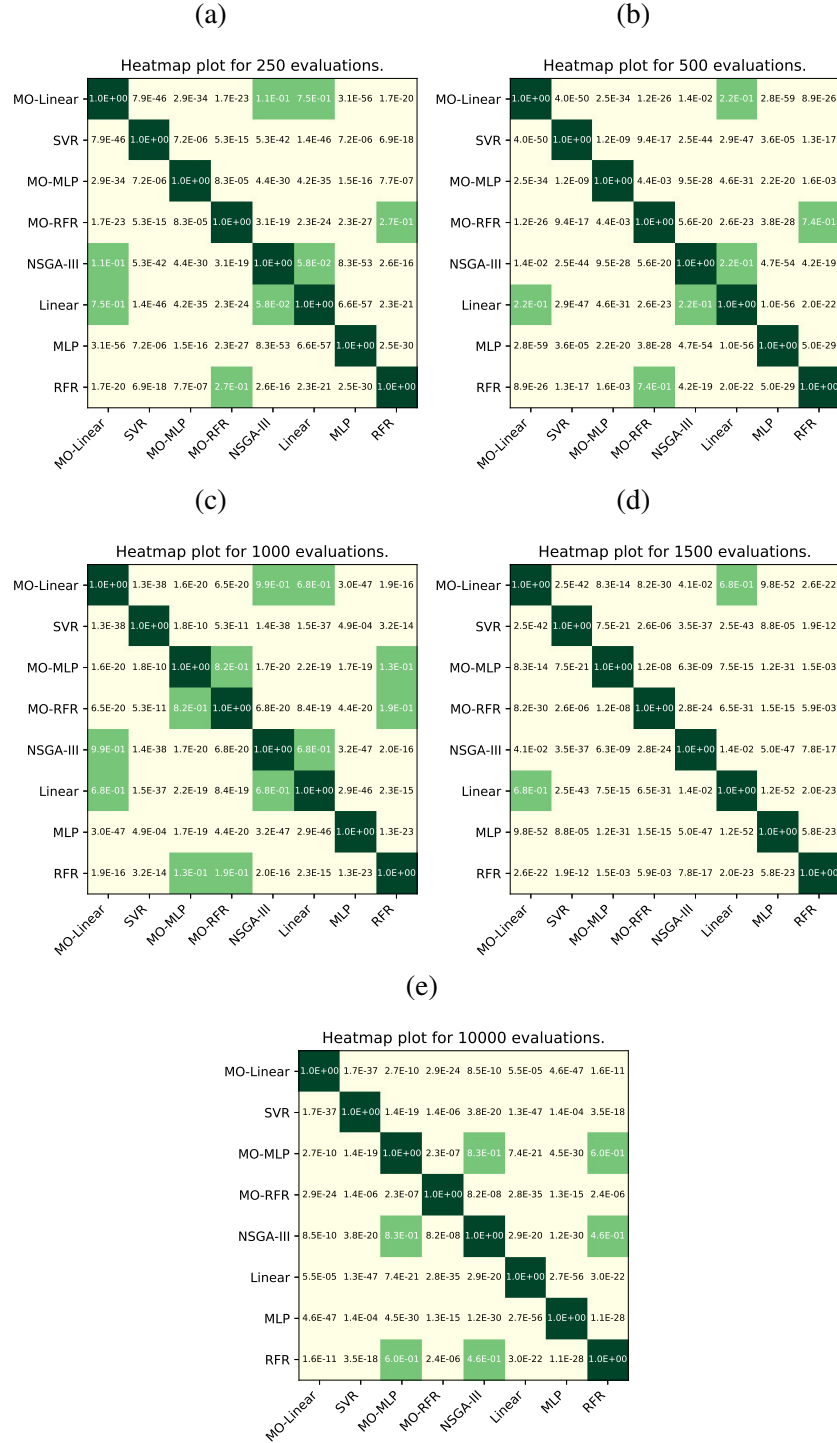


Figure 38 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ4 with 10 objectives.

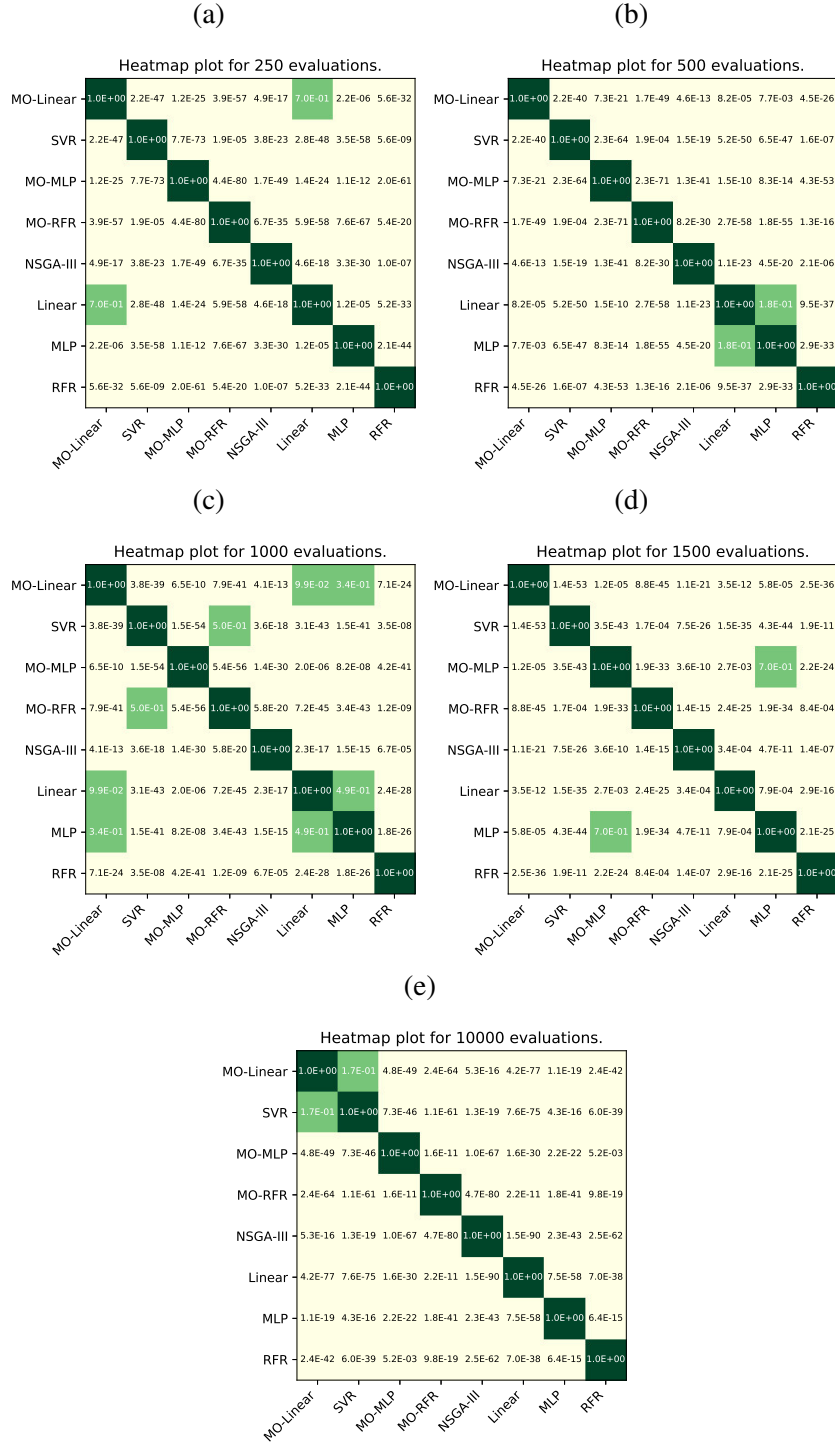
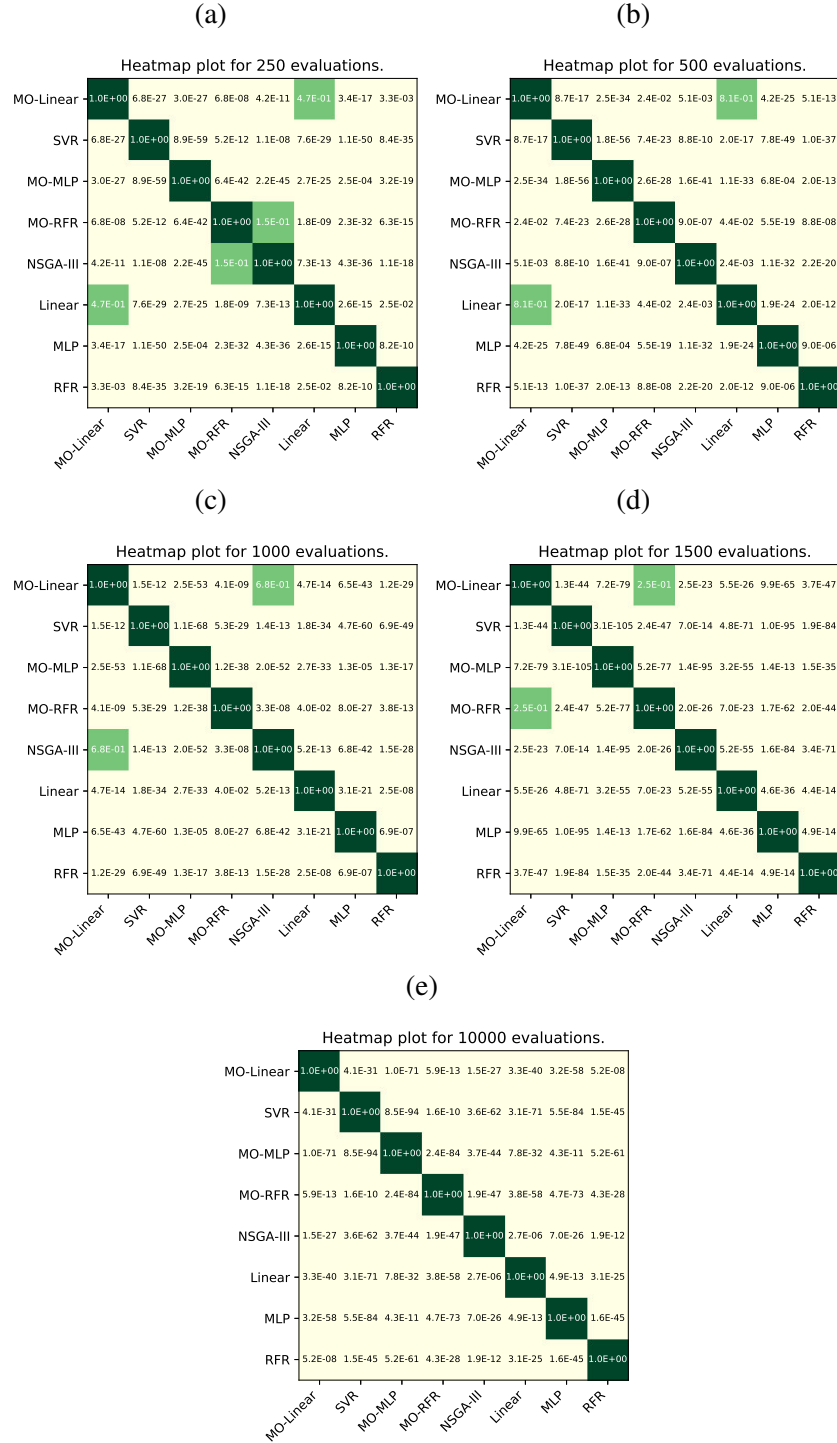


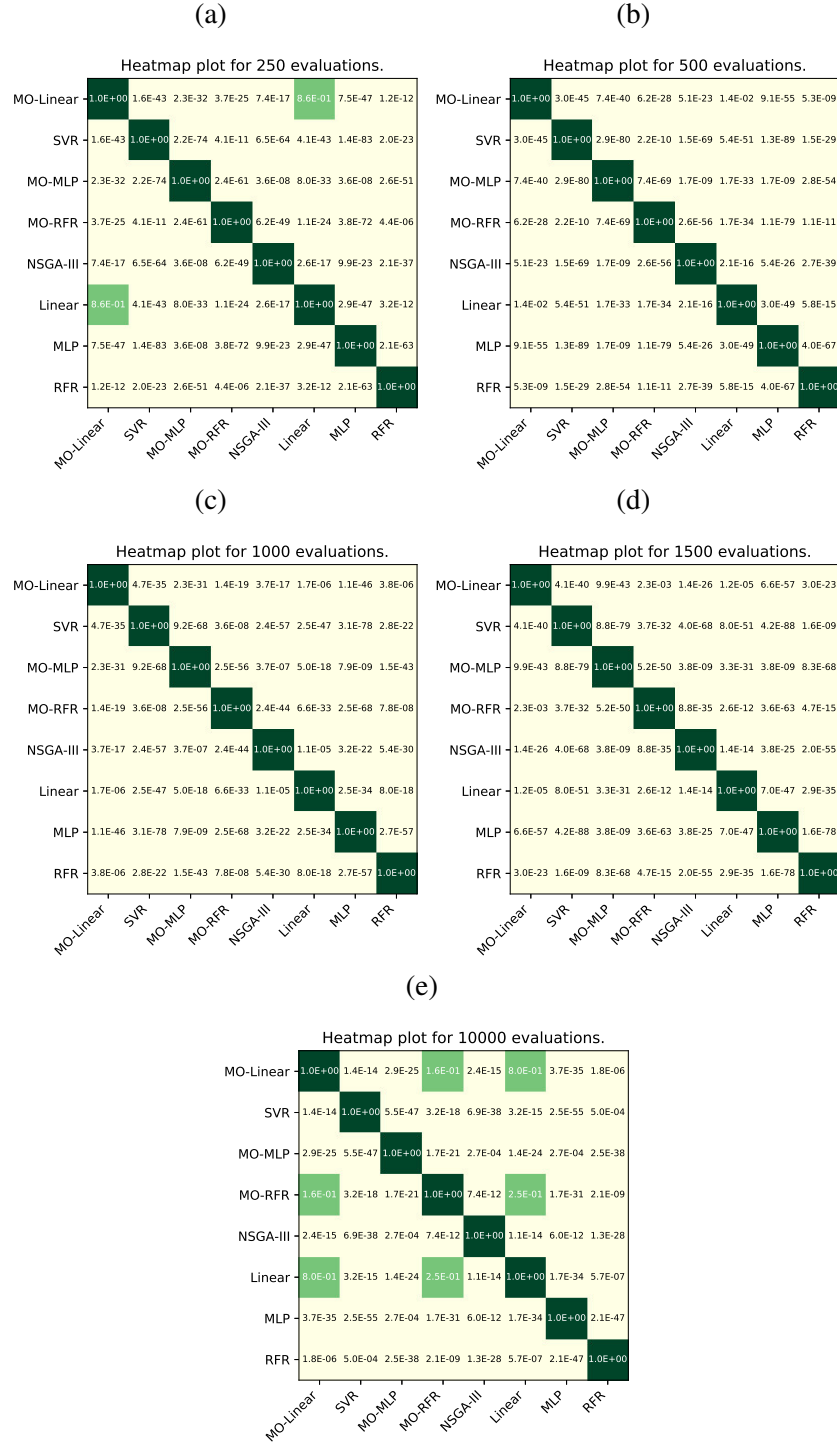


Figure 39 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ5 with 10 objectives.



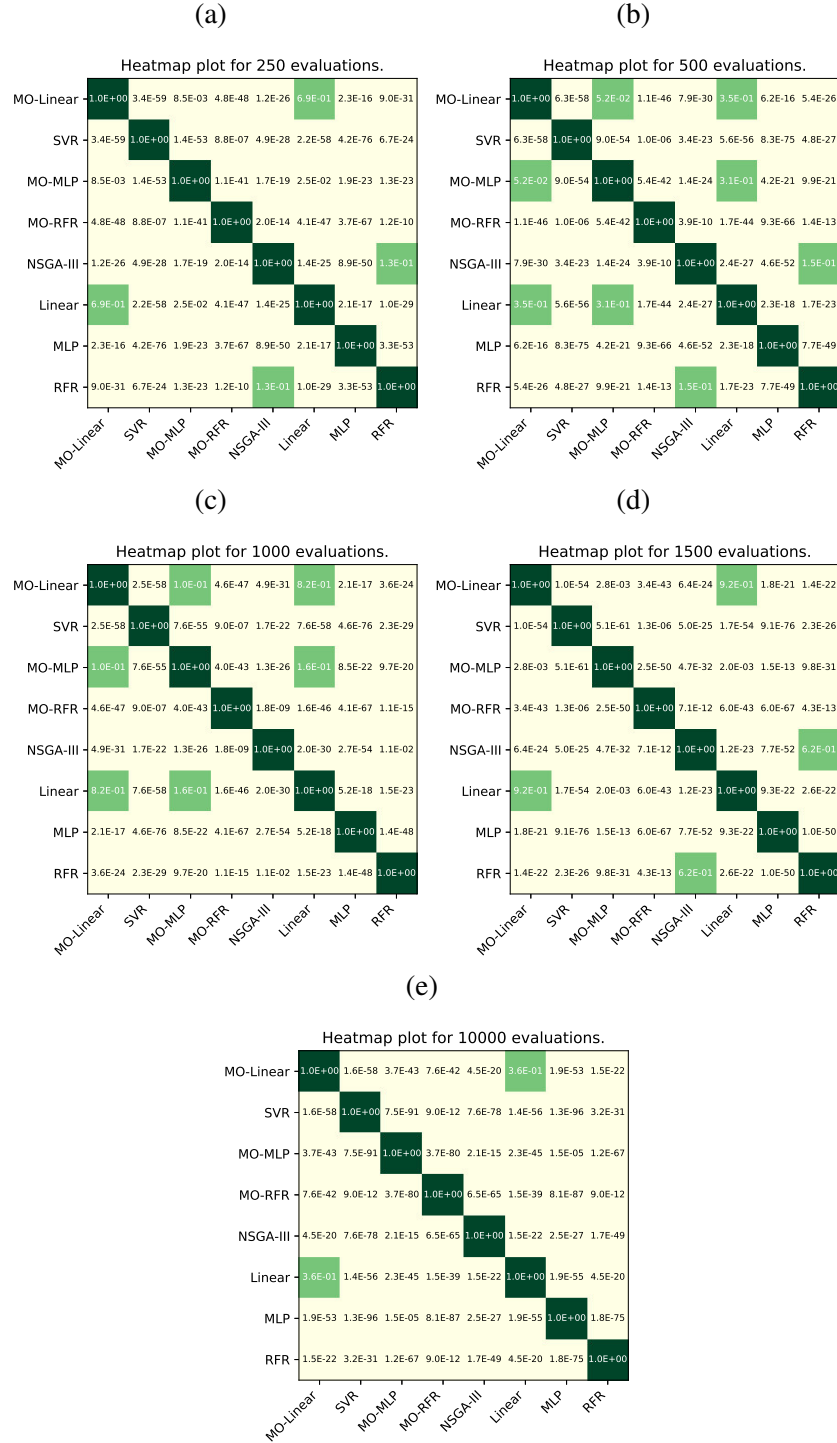
Source: Produced by authors.

Figure 40 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ6 with 10 objectives.



Source: Produced by authors.

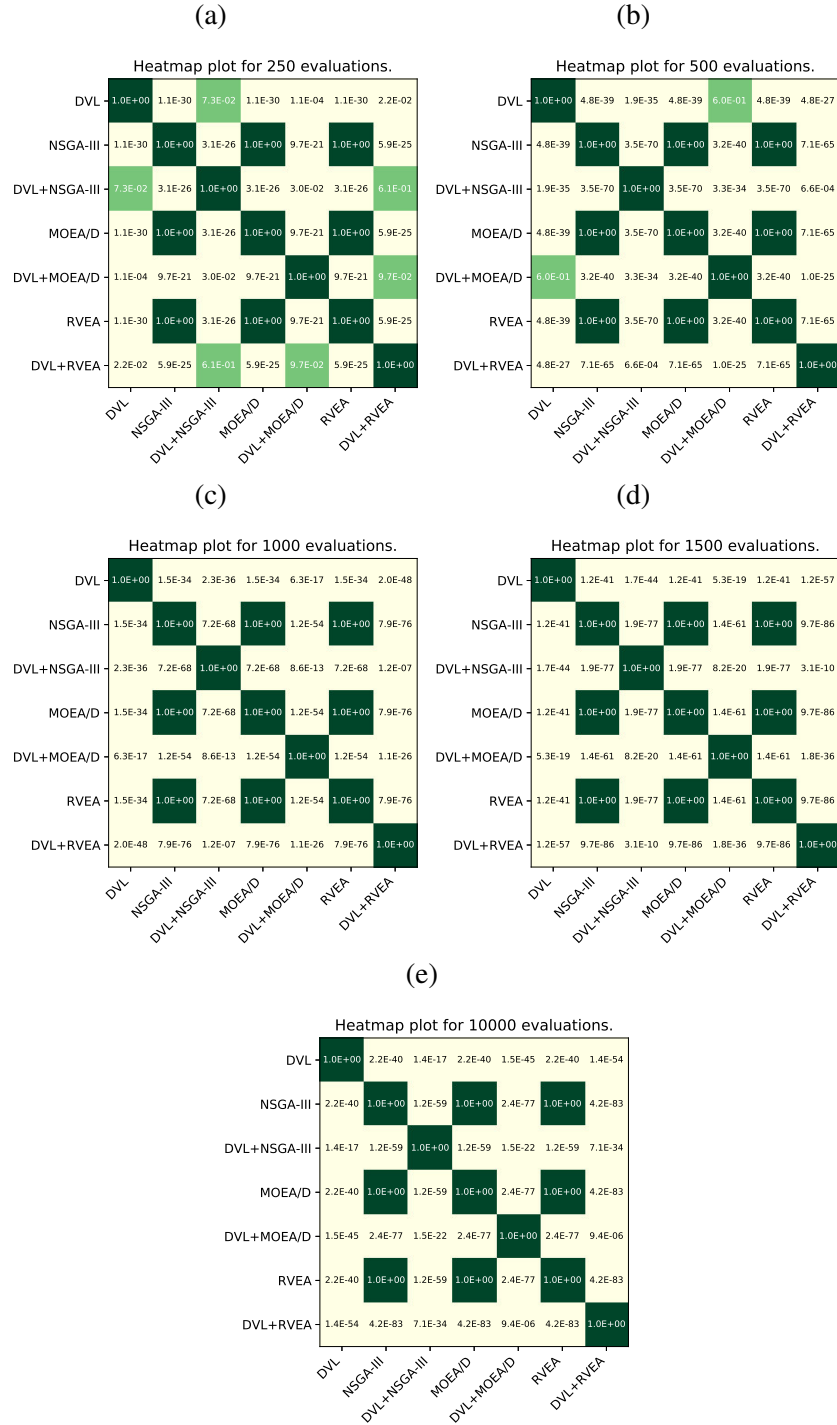
Figure 41 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ7 with 10 objectives.



Source: Produced by authors.

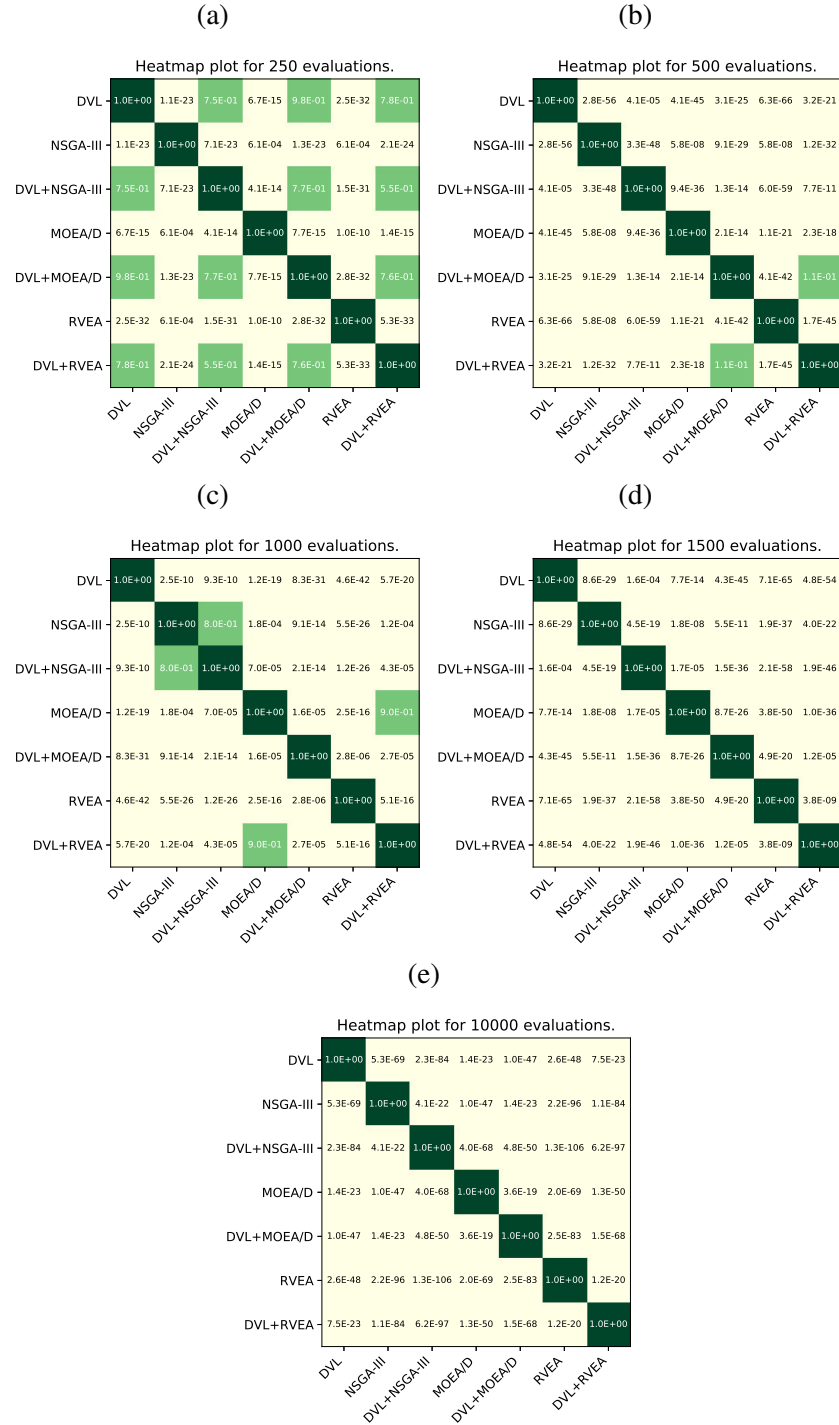
## **APPENDIX B – Heatmap plots of the DVL Framework experiment.**

Figure 42 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ1 with 3 objectives.



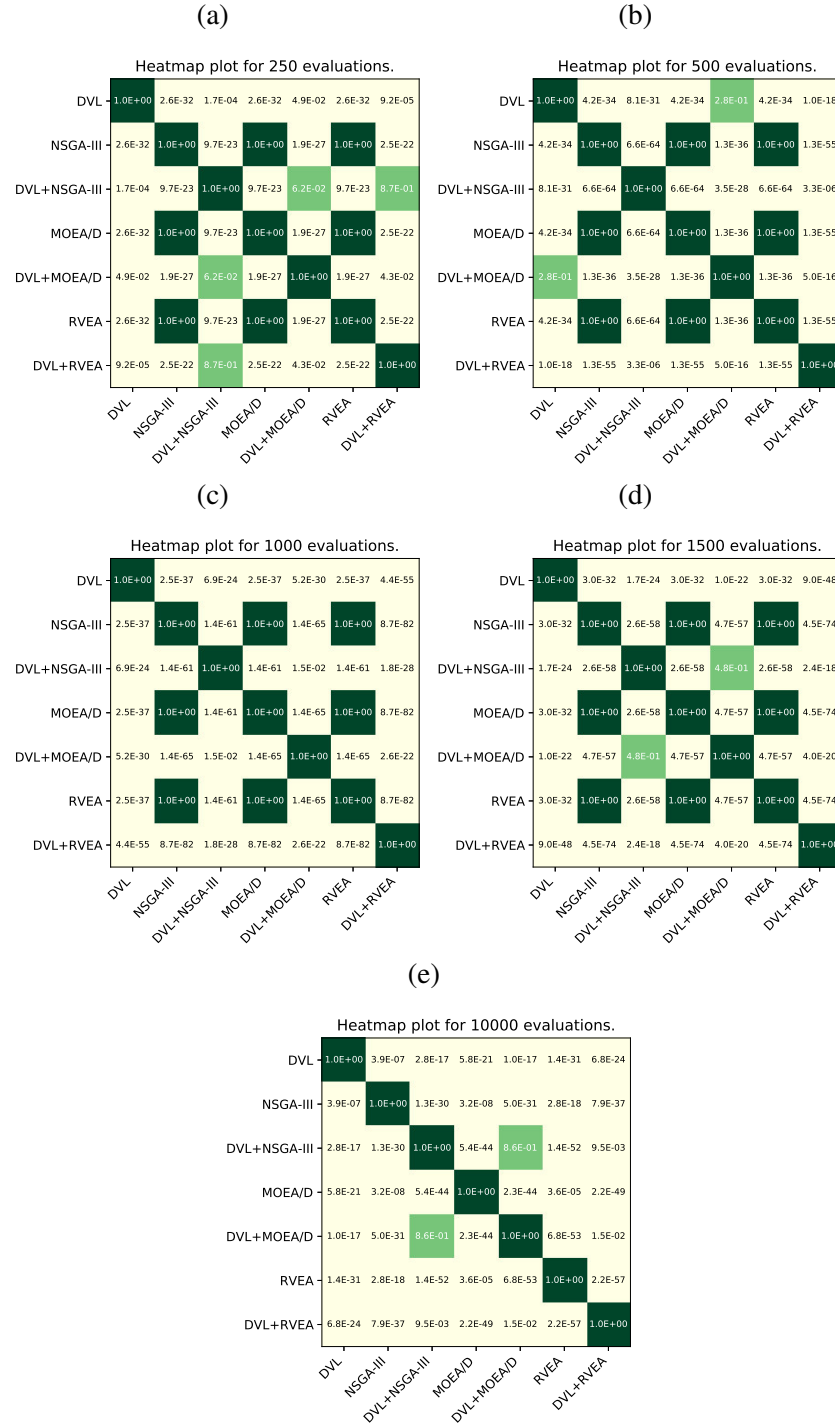
Source: Produced by authors.

Figure 43 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ2 with 3 objectives.



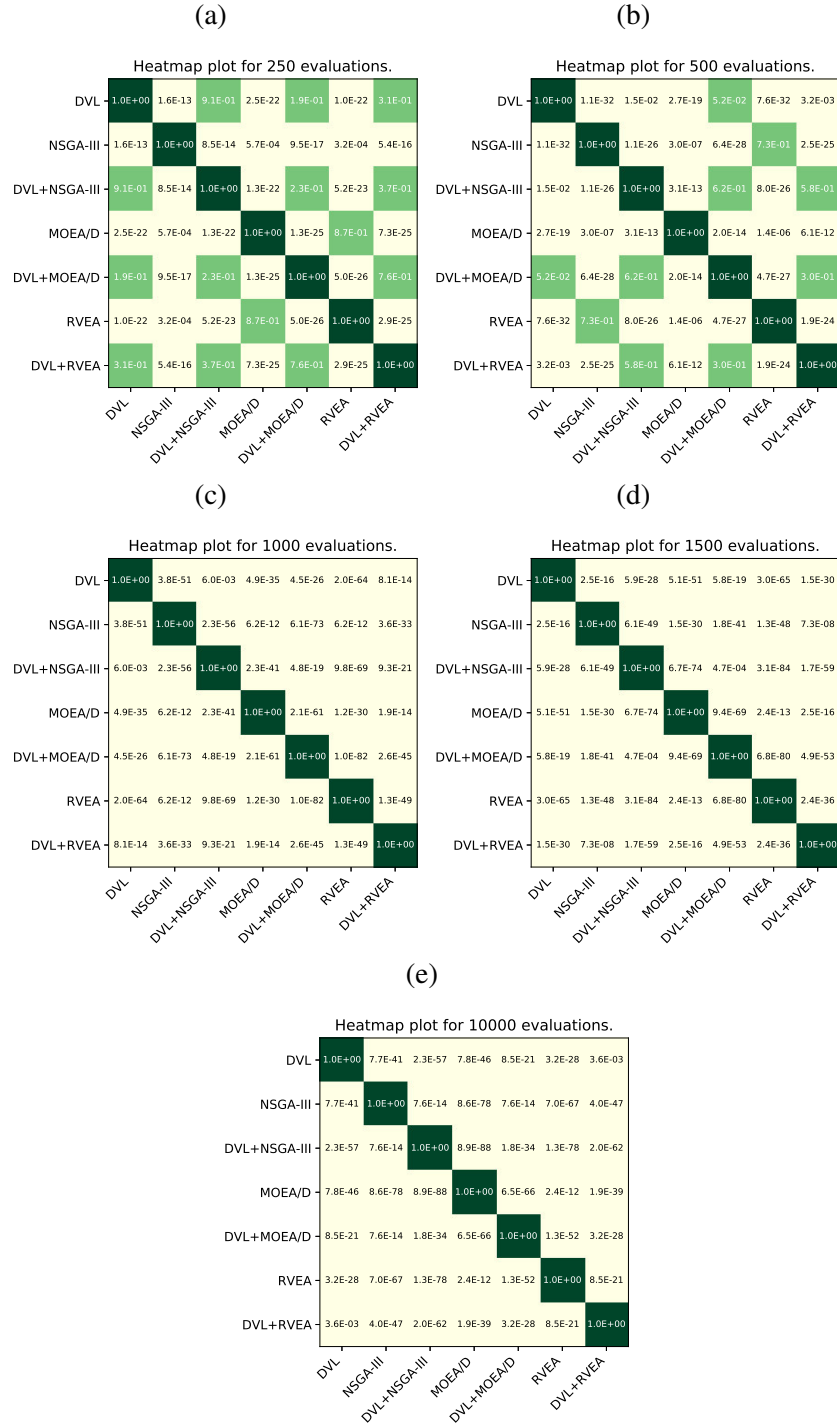
Source: Produced by authors.

Figure 44 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ3 with 3 objectives.



Source: Produced by authors.

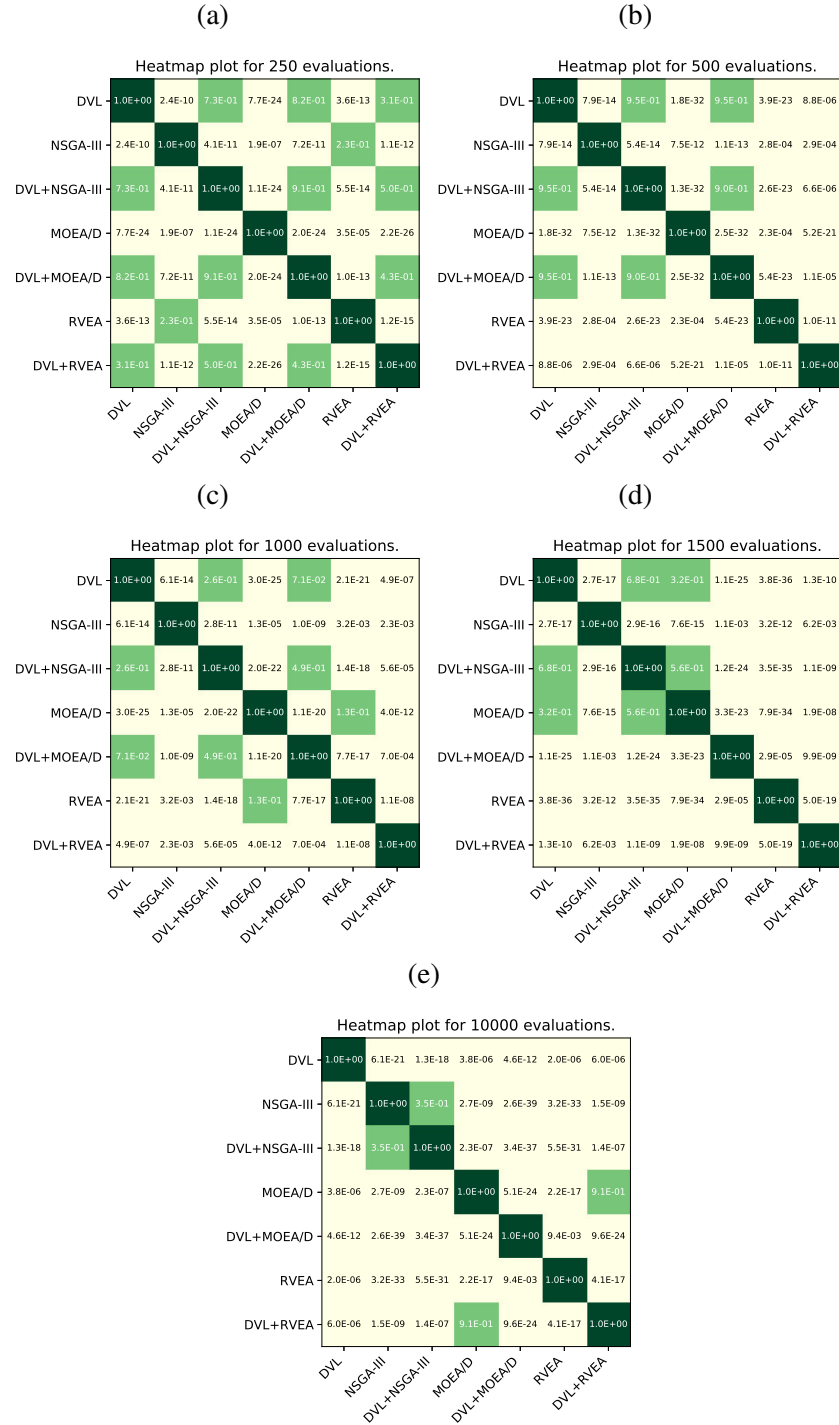
Figure 45 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ4 with 3 objectives.



Source: Produced by authors.

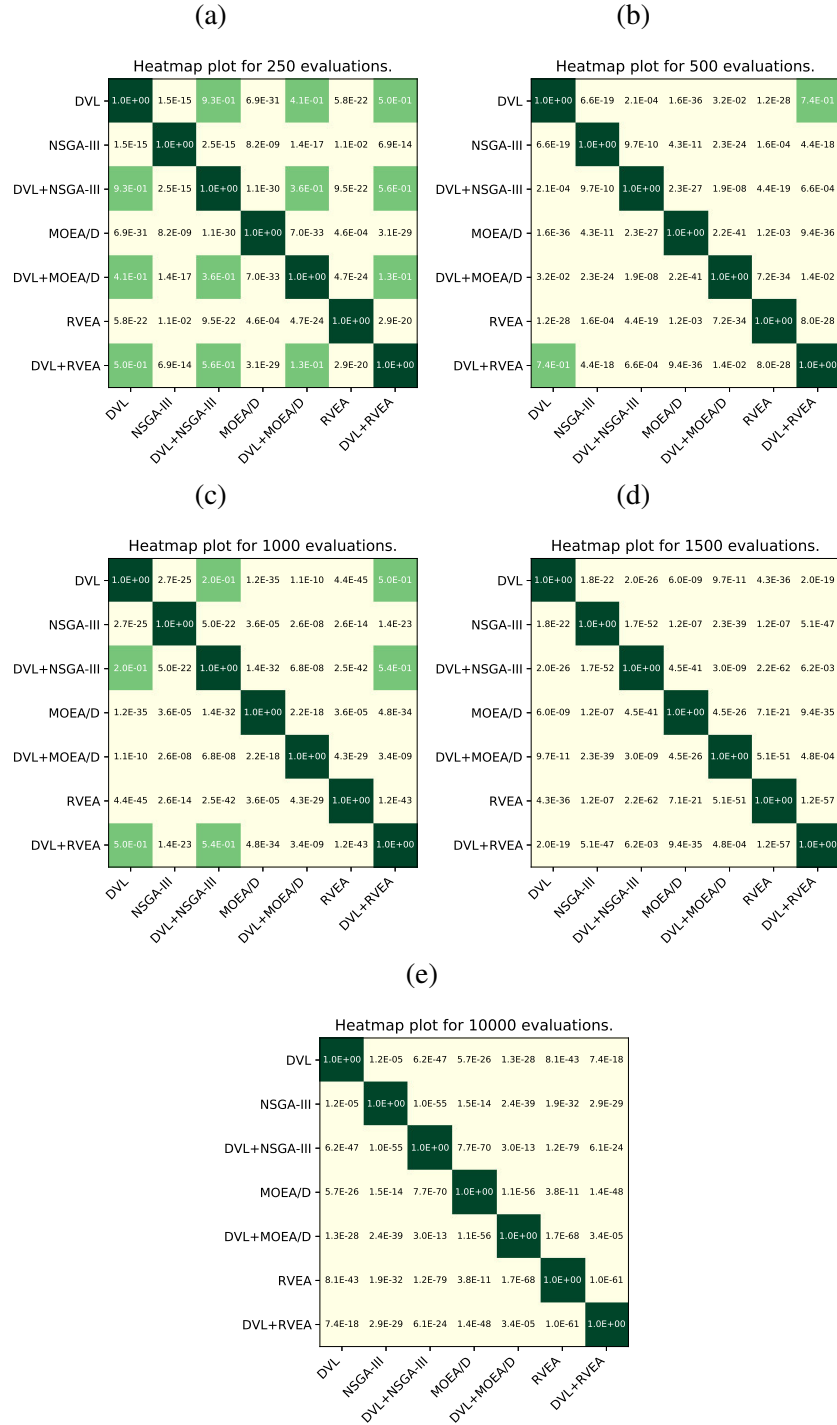


Figure 46 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ5 with 3 objectives.



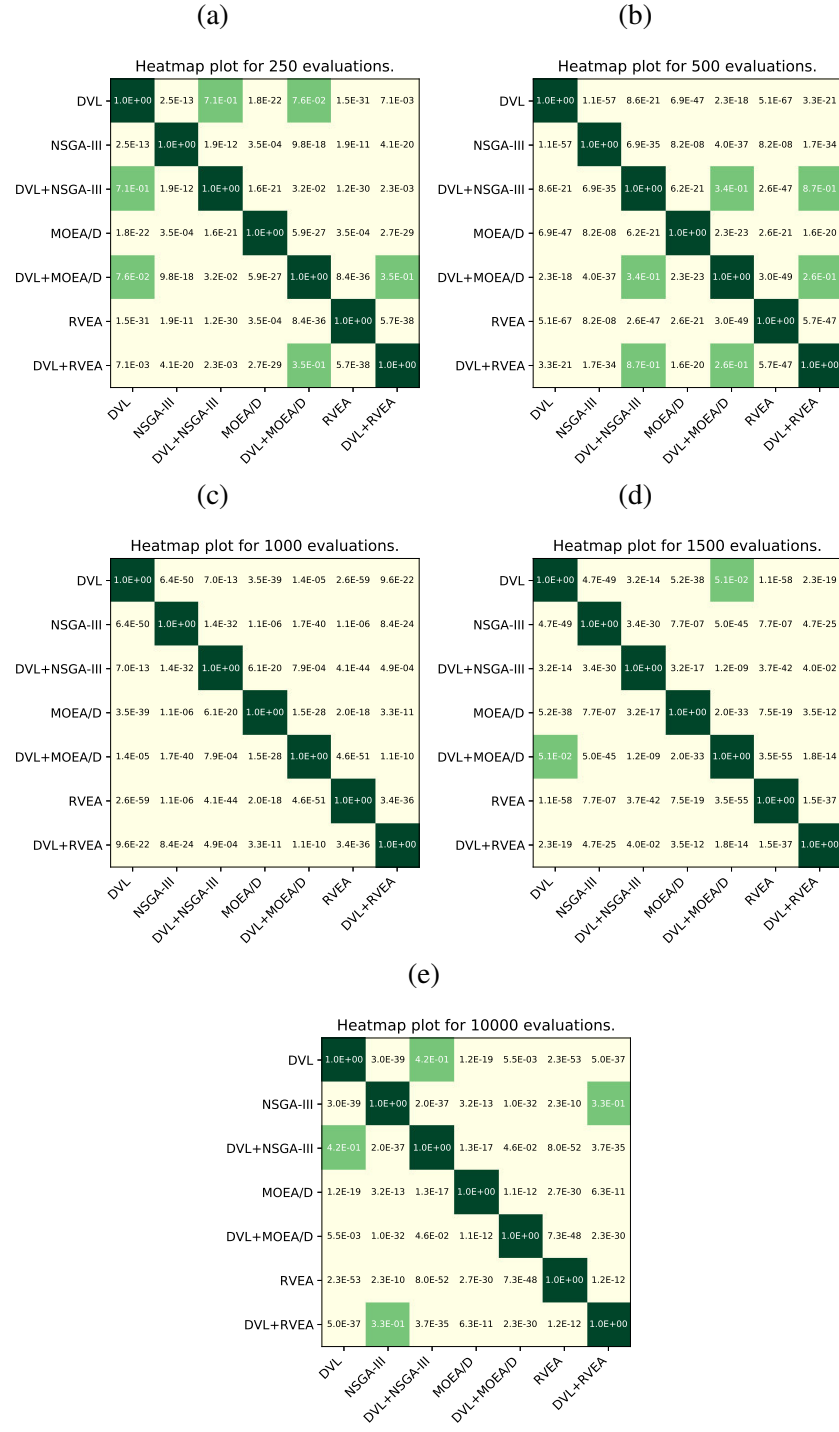
Source: Produced by authors.

Figure 47 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ6 with 3 objectives.



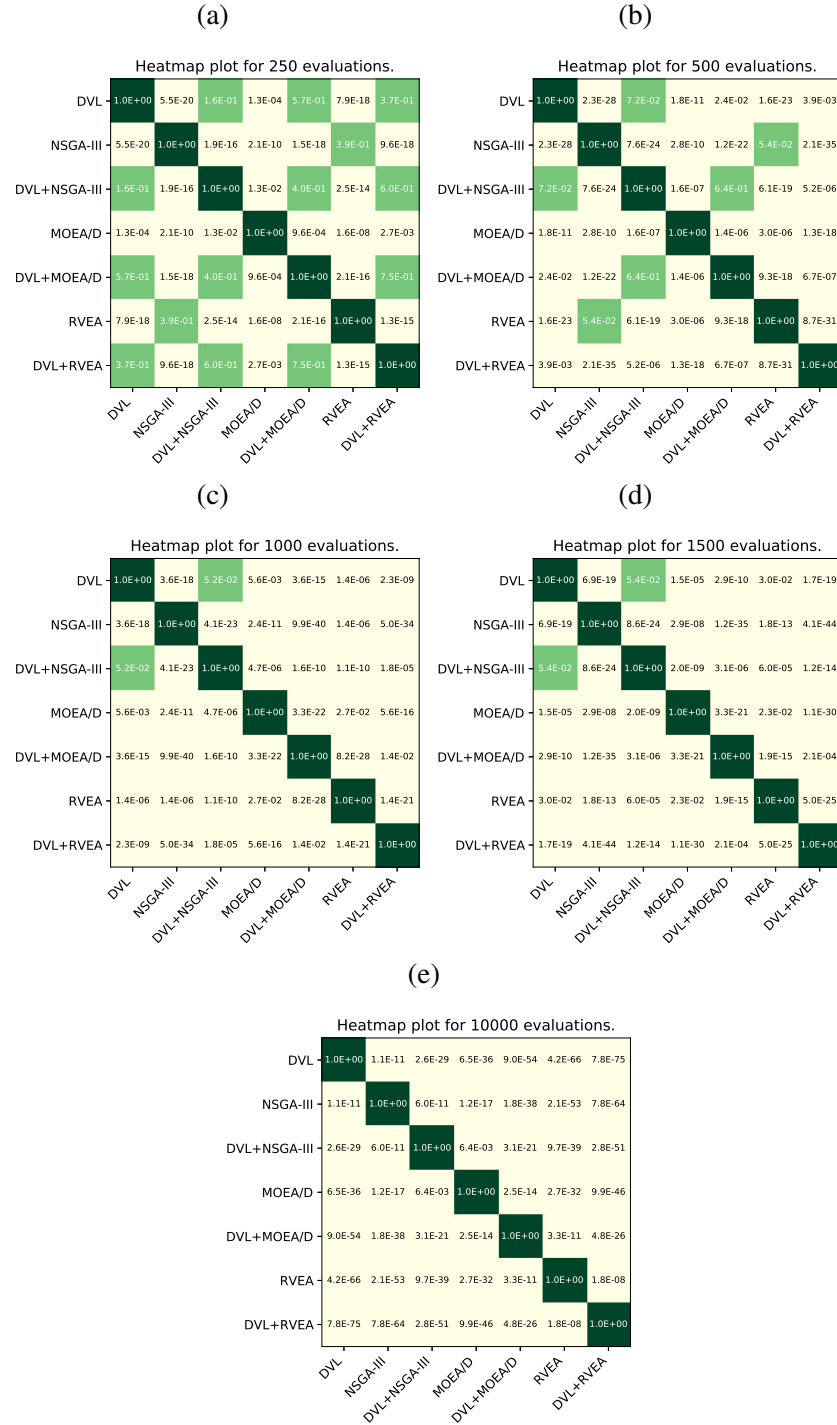
Source: Produced by authors.

Figure 48 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ7 with 3 objectives.



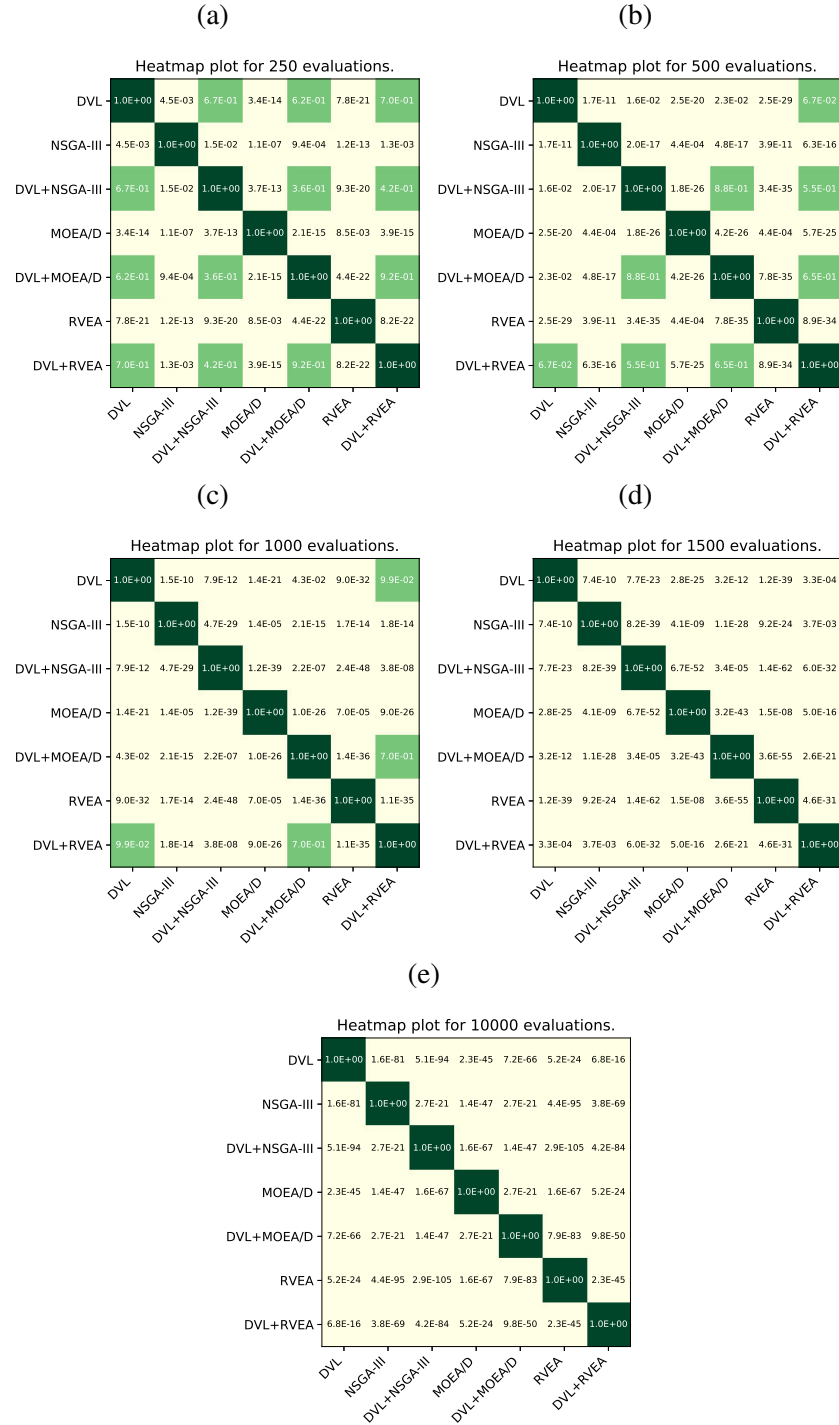
Source: Produced by authors.

Figure 49 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ1 with 10 objectives.



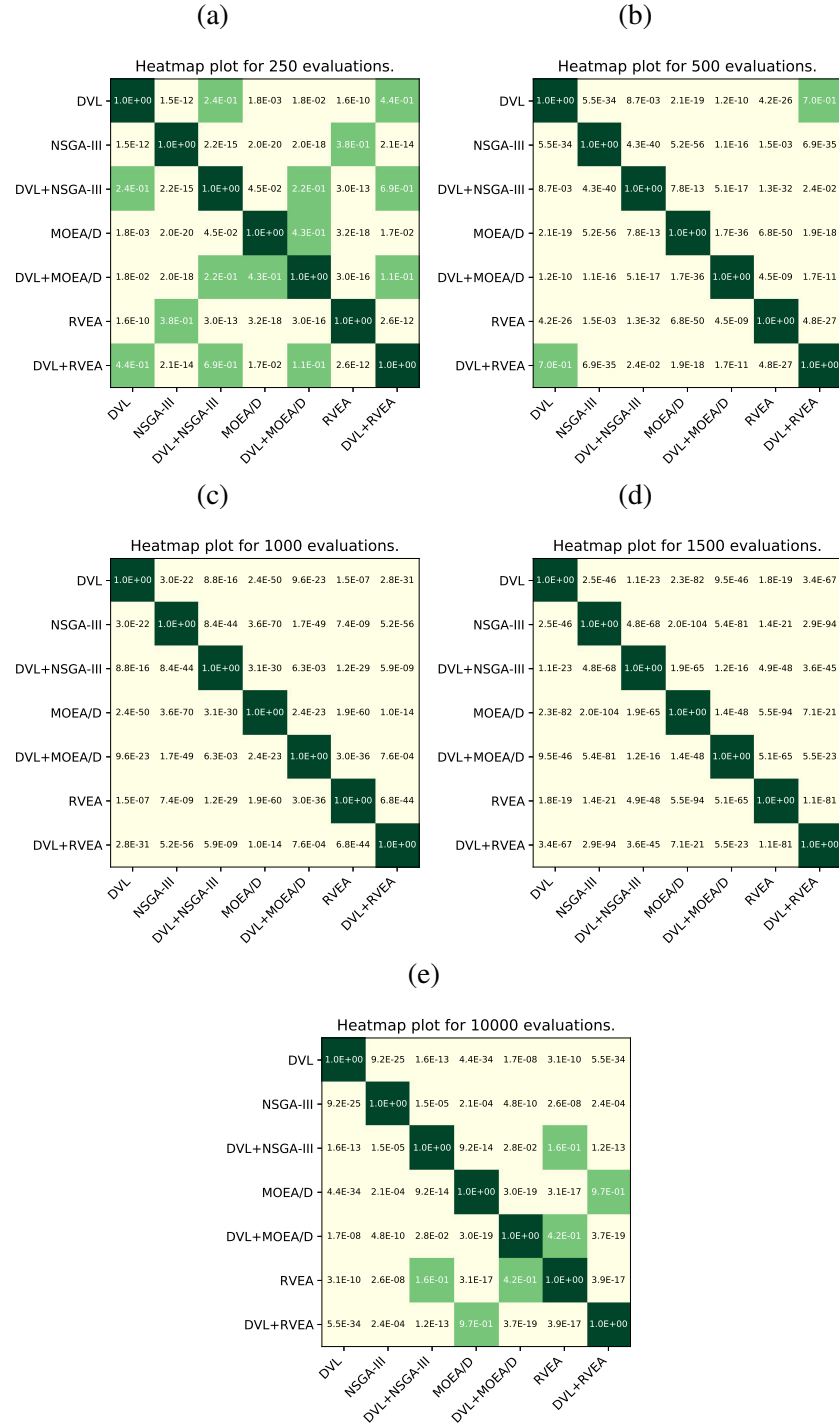
Source: Produced by authors.

Figure 50 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ2 with 10 objectives.



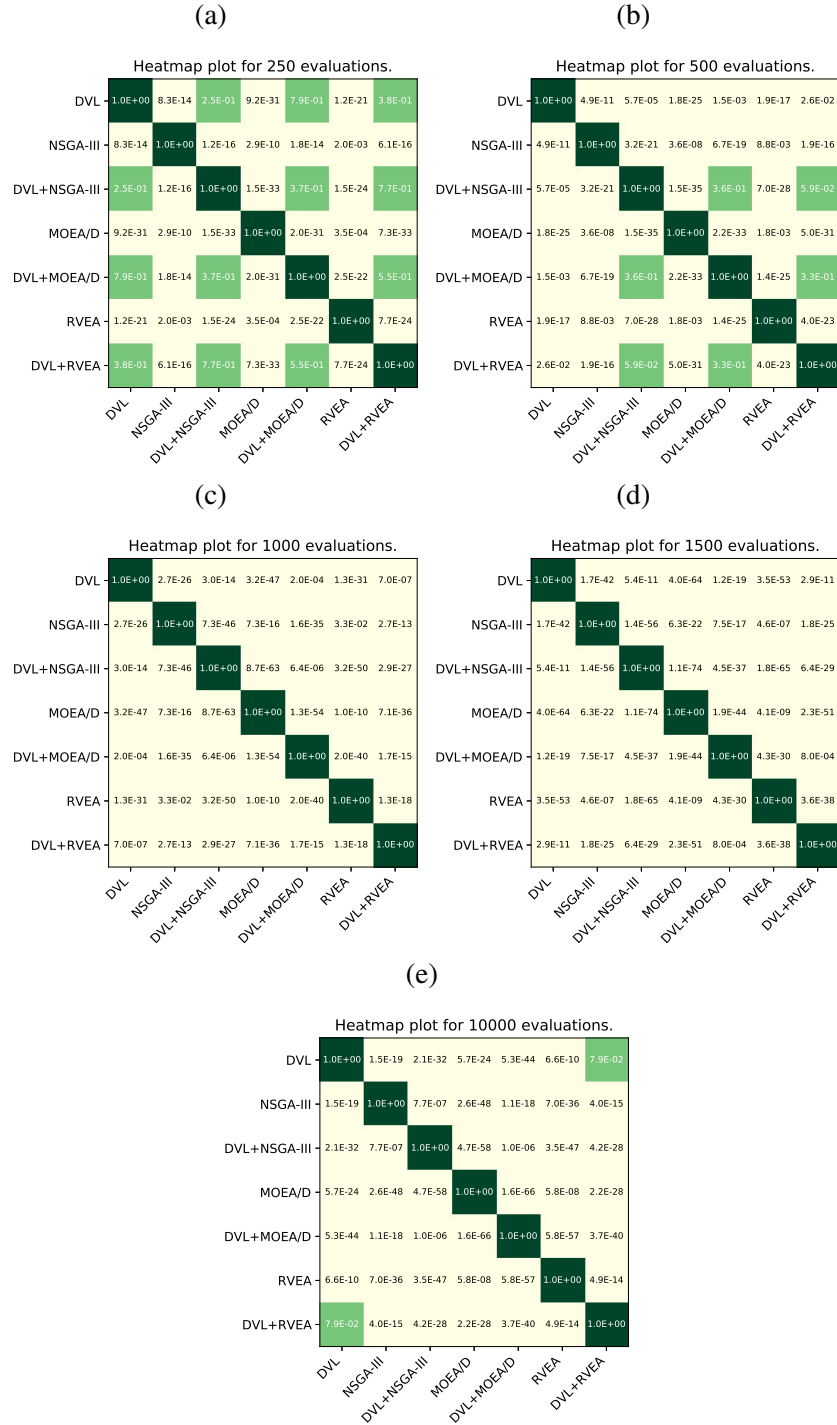
Source: Produced by authors.

Figure 51 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ3 with 10 objectives.



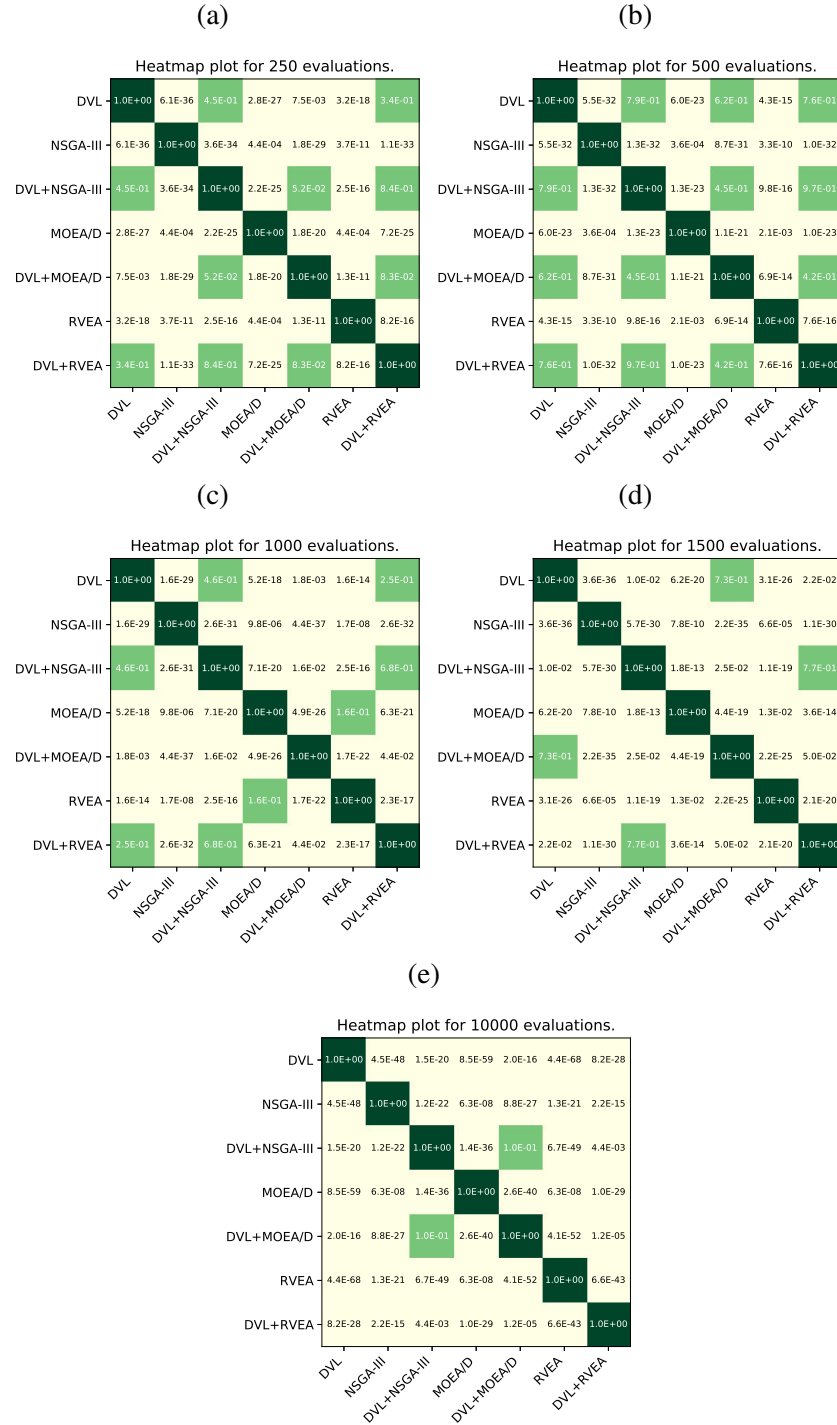
Source: Produced by authors.

Figure 52 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ4 with 10 objectives.



Source: Produced by authors.

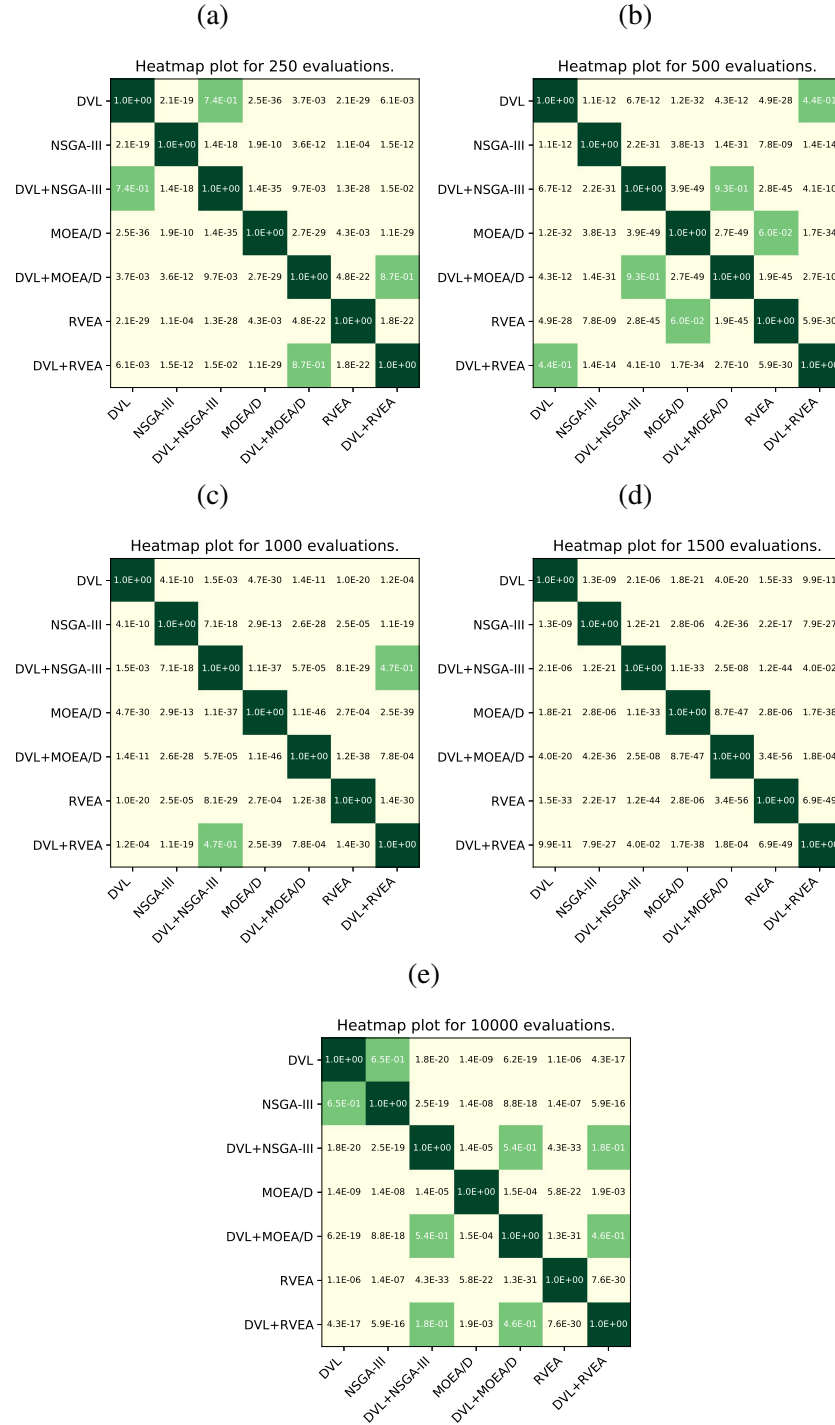
Figure 53 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ5 with 10 objectives.



Source: Produced by authors.

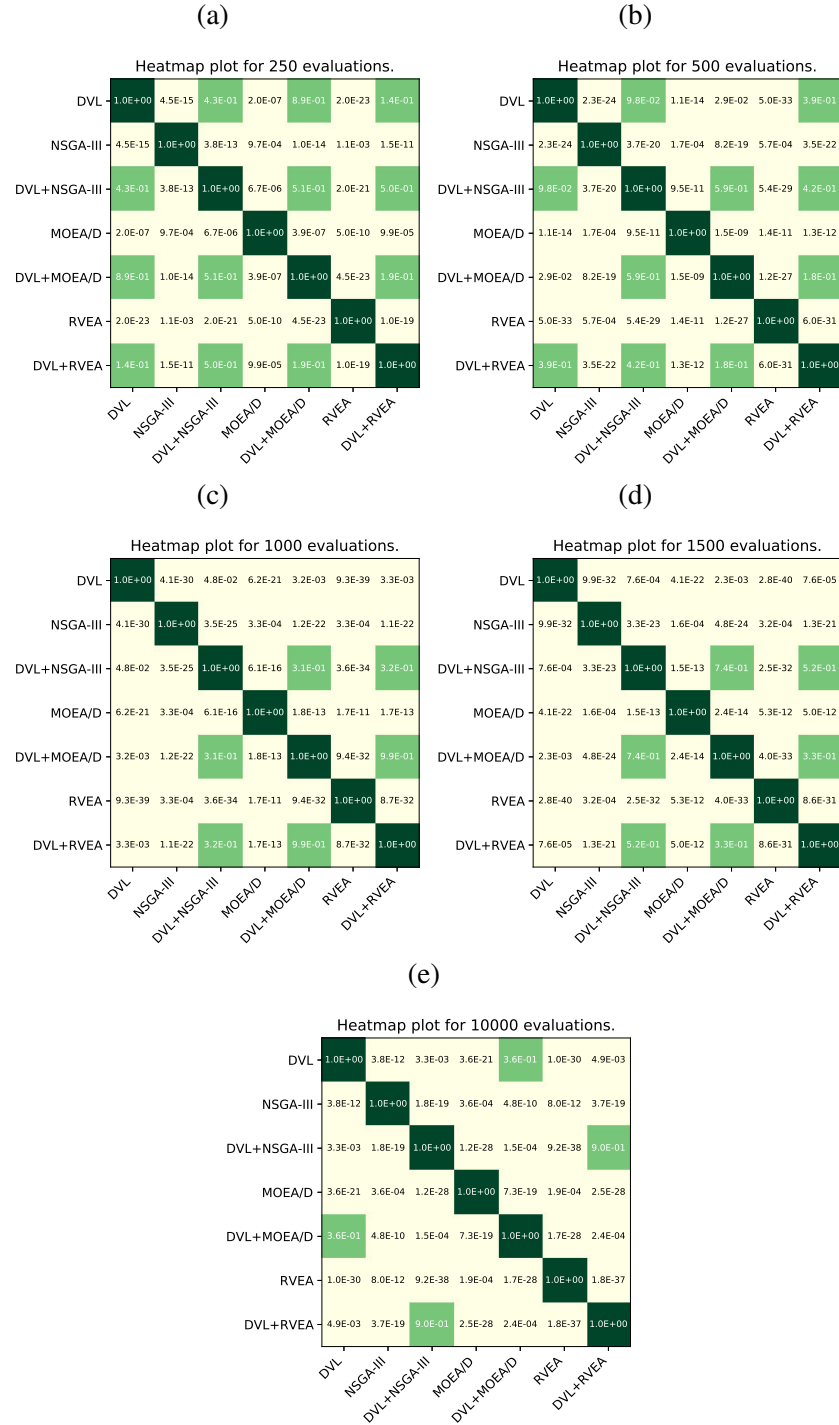


Figure 54 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ6 with 10 objectives.



Source: Produced by authors.

Figure 55 – Heatmap plots of the Wilcoxon rank-sum test with a value of 0.05 of the significance level for each configuration of the DTLZ7 with 10 objectives.



Source: Produced by authors.